

XANDER STEGEHUIS

SERVITIZATION IN  
INTER-ORGANIZATIONAL  
CONSTELLATIONS:  
A MULTI-PERSPECTIVE  
APPROACH

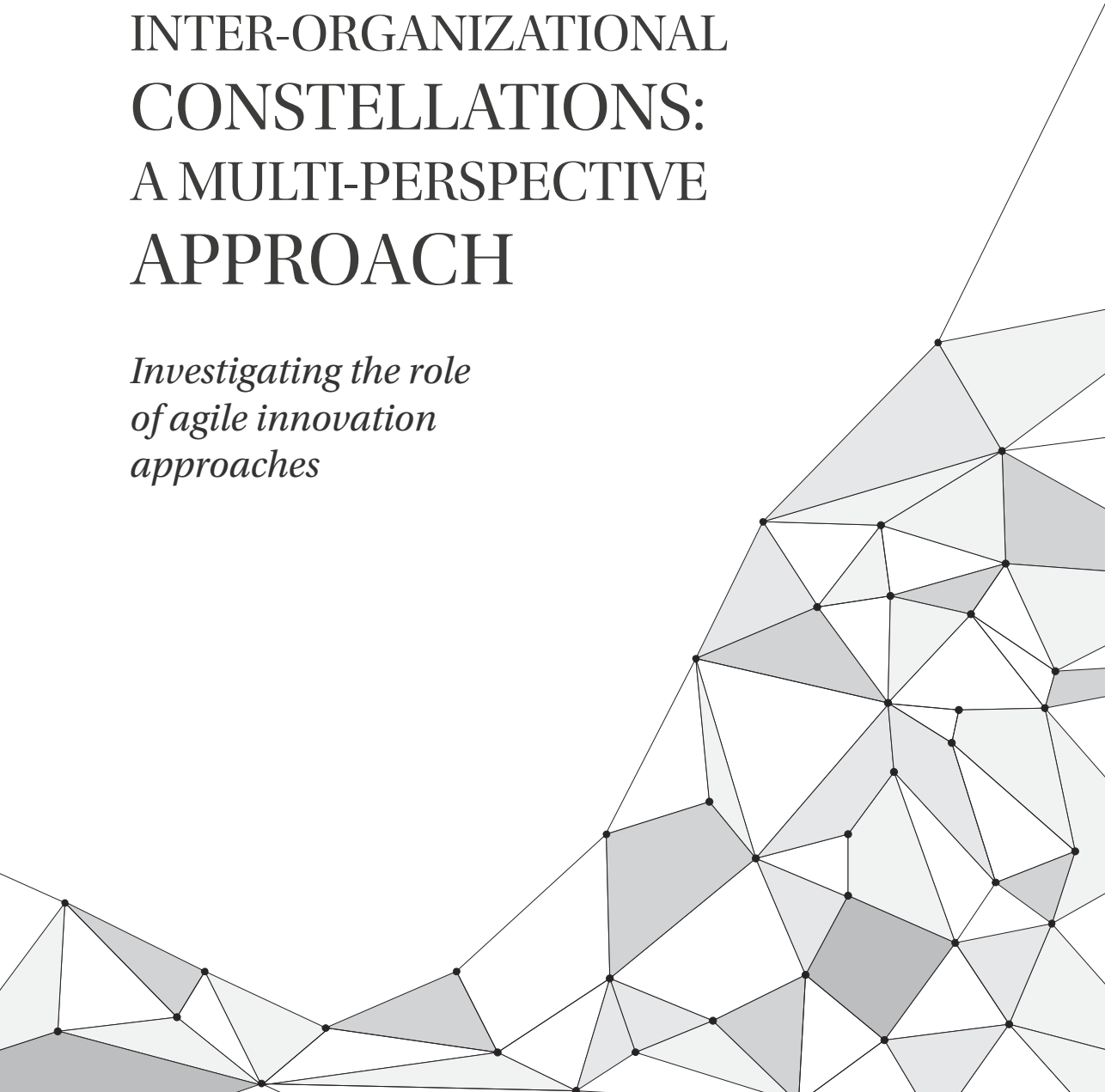
*Investigating the role  
of agile innovation  
approaches*



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**Investigating the role of agile innovation  
approaches**

**DISSERTATION**

to obtain  
the degree of doctor at the University of Twente,  
on the authority of the rector magnificus,  
prof. dr. ir. A. Veldkamp,  
on account of the decision of the Doctorate Board,  
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by

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*“Perfection is the enemy of progress.”*

Winston Churchill

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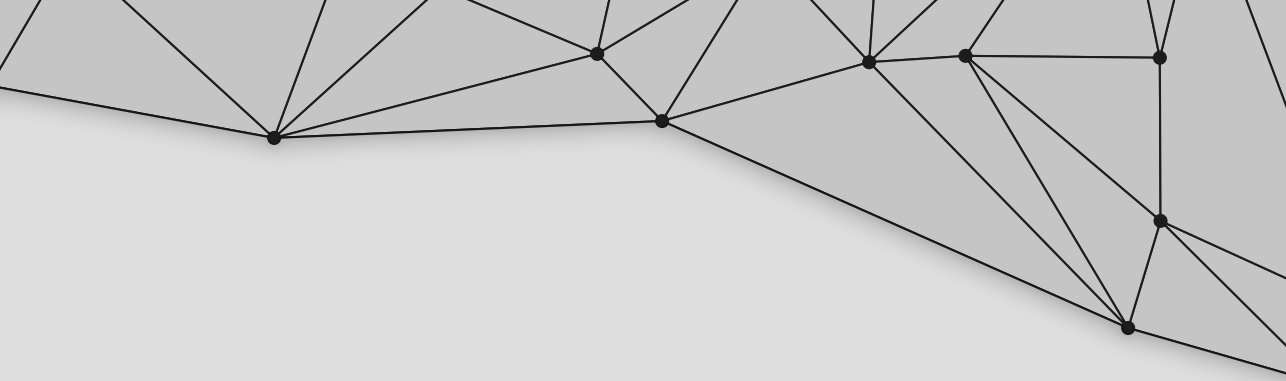
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Xander Stegehuis

Enschede, January 11th 2024



# 1

## INTRODUCTION

---

## 1.1 Setting the stage

This dissertation delves into the phenomenon of “servitization,” a transformative shift observed in manufacturing companies. Conceptualized as the transformation from traditional product offerings to hybrid product-service combinations, servitization represents a significant business model innovation (Forkmann et al., 2017; Raddats et al., 2019; Ulaga & Reinartz, 2011). At its core, servitization signifies the evolution of businesses from mere transactional product sales to fuller market packages or product-service bundles that cater to customers’ needs throughout the product’s lifecycle (Vandermerwe & Rada, 1988). An iconic example is Rolls Royce’s TotalCare® offering, which revolutionized the aerospace industry by selling “power-by-the-hour” rather than conventional jet engines (Ng et al., 2013). This example illustrates how servitization enables firms to redefine their role in the market and carve a competitive edge by extending their value proposition beyond product offerings.

Central to the servitization narrative is the strategic imperative for manufacturers to leverage their installed base and intimate product knowledge by broadening the horizon of their offerings through services (Oliva & Kallenberg, 2003). This move allows manufacturers to strategically ‘go downstream,’ assuming responsibilities and risks previously internalized by customers (Wise & Baumgartner, 1999). Manufacturers often have pre-existing product knowledge and expertise, so they are well-positioned to internalize these tasks. The momentum towards servitization is further fueled by challenges such as product commoditization, heightened competition from global players, and increasing demand from customers for services (Baines et al., 2009). Within this context, servitization emerges as a promising strategic response to the commonly observed commodity trap, which refers to reduced profit margins due to difficulties in differentiating offerings in the market. Notably, the margins associated with services are often considerably higher than those from new product sales (Kowalkowski, Gebauer, & Oliva, 2017), bolstering the financial case for this transformative approach.

In many Western economies, servitization has been prevalent in the last sixteen years (Neely, 2008) and has gained increasing attention from scholars, businesses, and policymakers. Between 2007 and 2016, the number of academic papers published per year about servitization has five folded, and this exponential trend has continued since then (Rabetino et al., 2018; Shen et al., 2023; J. Wang et al., 2023). Furthermore, prominent and influential firms such as McKinsey & Co, ABN AMRO, and Allen & Overy publish practitioner reports about servitization, suggesting that the concept is gaining widespread attention and interest among businesses (ABN AMRO, 2021; Coalition Circular Accounting, 2020; McKinsey & Company, 2023). Finally, governments are also keen to promote servitization, enabling firms to increase their employment and deal with competitive threats from foreign low-cost rivals (Crozet & Milet, 2017; Matthyssens & Vandenbempt, 2010).

The increasing importance of servitization is partially grounded in the strategic importance of digitalization for businesses and governments. For example, the Dutch government has been accelerating the servitization of manufacturing firms in

the last five years by setting up field labs and providing subsidies. These efforts are part of the policy on digitalizing the manufacturing sector in the Netherlands (*Smart Industry Roadmap*, 2018). While the role of digital technologies in servitization has been acknowledged from its inception (Vandermerwe & Rada, 1988), this link has become stronger since 2011, when governmental policies and corporate strategies put more emphasis on digital transformation under the banner of the “fourth industrial revolution” or “Industry 4.0”. Since then, manufacturers increasingly adopted sensors, big data analytics, and cloud computing technologies to integrate various objects and processes in the Industrial Internet of Things (Paschou et al., 2020; Shen et al., 2021). Because servitization and digitalization have become more intertwined, both trends have merged into the concept of digital servitization (Struyf et al., 2021; Vendrell-Herrero et al., 2017). Digital servitization is defined as “*the transformation in processes, capabilities, and offering within industrial firms and their associate ecosystem to progressively create, deliver, and capture increased service value arising from a broad range of enabling digital technologies*” (Sjödin et al., 2020, p. 478), and will be the main theoretical and empirical context of this dissertation.

Digital servitization underscores the reinforcing relationship between servitization and digitalization. Digitalization benefits from servitization because service offerings enable firms to recoup investments in digital technologies (Abou-foul et al., 2020; Gebauer et al., 2020; Visnjic et al., 2018). One way to benefit from digital transformation is by tapping into new resource pools through novel offerings and business models (Kiel et al., 2017), such as servitization. Conversely, digital technologies are fundamental to successful servitization. They streamline service delivery, reduce associated costs, and enhance the quality of service offerings (Coreynen et al., 2020; Vendrell-Herrero et al., 2017). For instance, predictive maintenance powered by IoT sensors and data analytics can preemptively detect equipment faults, reducing downtime and ensuring consistent service delivery (Porter & Heppelmann, 2014). Even though digital technologies are prevalent in almost all service offerings nowadays, their relative importance increases when they become more complex and outcome-focused (Marcon et al., 2022).

To distinguish between the complexity of different service offerings, they are often categorized as base, intermediate, and advanced (Baines & Lightfoot, 2013). These services can be differentiated in terms of their orientation and their revenue model. So, service offerings can be oriented toward the supplier’s good or customer’s process, and the revenue model can be input-based (e.g., hourly billing) or output-based (e.g., pay-per-use) (Ulaga & Reinartz, 2011). Base services are input-based and geared towards the supplier’s product. Base services include installation, spare parts provision, and technical support. Intermediate services are input-based and geared toward the supplier’s process. These services have a broader life cycle scope and include, for example, training services, maintenance contracts, and recycling services. Advanced services are outcome-based services geared towards the supplier product or customer process. These services are often based on a value-based or usage-based pricing strategy (Wirtz & Kowalkowski, 2023) and include maintenance contracts with an uptime guarantee and Product-as-a-Service offerings. When manufacturing firms offer intermediate or advanced services, they take on more activities, responsibilities, and

risks from the customer (Neely, 2008; Visnjic et al., 2018).

## 1.2 Implications of servitization

Servitization is associated with multiple benefits and challenges. A major advantage of servitization is increasing a firm's financial performance through revenue stability and higher profit margins (Baines et al., 2009). Notably, machinery and equipment manufacturers can benefit from more stable revenues by mitigating the cyclical nature of significant capital expenditures through long-term service contracts (Sawhney et al., 2004; Visnjic et al., 2019). Furthermore, servitization enables manufacturing firms to substantially increase their profit margins (Crozet & Milet, 2017). However, servitization is also associated with substantial financial risks, and many firms experience disappointing financial results when service investments do not provide positive financial returns (Benedettini et al., 2015; Gebauer et al., 2005; Korkeamäki et al., 2022). One reason for such disappointing results is the increasing complexity and costs that higher revenues cannot offset (Neely, 2008; Ostrom et al., 2015). Another reason is that servitization requires substantial investments that lead to an initial profitability dip, which is overcome when firms sufficiently scale their service business by increasing the total revenues from services (Baines et al., 2009; Fang et al., 2008; Oliva & Kallenberg, 2003; Visnjic Kastalli & Van Looy, 2013).

While the relationship between servitization and financial performance remains complex, servitization positively impacts non-financial performance (Wang et al., 2018). Non-financial performance refers to a firm's ability to satisfy customers and differentiate itself from competitors, that is, the strategic and marketing benefits of servitization (Baines et al., 2009). Integrated product-service offerings signal product quality, performance, and operational support with higher differentiation potential than a cost-leadership strategy (Ulaga & Eggert, 2006; Zimmer et al., 2020). As such, servitization enables firms to gain a competitive advantage. This advantage can be sustainable because service offerings are less visible and more socially complex than product offerings and thus challenging to imitate (Baines et al., 2009; Oliva & Kallenberg, 2003). Furthermore, servitization increases customer intimacy because labor-intensive services such as maintenance involve frequent customer visits (Visnjic Kastalli & Van Looy, 2013). Increased customer intimacy enables firms to understand complex and latent customer needs and tailor their product and service offerings to their needs (Martín-Peña et al., 2023; Visnjic et al., 2016; Y. Wang et al., 2022). These benefits enable firms to respond to significant trends such as product commoditization, digitalization, and sustainability (Matthyssens & Vandenbempt, 2008; Rabetino et al., 2018).

Another frequently discussed implication of servitization is reducing a manufacturer's ecological footprint, particularly for performance-based contracts that separate ownership and use of capital goods (Geissdoerfer et al., 2018; Tukker, 2015). In a product's design stage, servitization incentivizes manufacturers to design their products to last longer, be easily repairable, deployable, and have high recyclability value (Yang & Evans, 2019). In the use stage, servitization can reduce energy consumption and consumables by improving efficiency through optimization and

auditing services (Paschou et al., 2020; Ulaga & Reinartz, 2011). Furthermore, pay-per-use models can promote sustainable consumption patterns through pricing schemes, encouraging consumers to behave more sustainably (Bocken, Mugge, et al., 2018). In the end-of-life stage, servitization can facilitate closed resource loops through remanufacturing and reprocessing services (Benedettini, 2022). However, the ecological benefits of servitization are not undisputed. Separating ownership and use may result in reckless customer behavior, leading to more breakdowns, shorter physical resources' longevity, and a higher frequency of replacing resources (Tukker, 2004). Furthermore, while pay-per-use services such as car-sharing may give people access to previously unaffordable products, they may trigger the so-called "rebound effect" (Herring & Sorrel, 2009; Paschou et al., 2020). This effect refers to energy and resource savings from using the service, which are offset by the increased consumption of that service.

In summary, while servitization offers promising avenues for improved financial and non-financial performance and sustainability benefits, its multifaceted implications underscore the complexities firms navigate. Grasping these complexities requires a fundamental shift in how we perceive and understand value and value creation.

### **1.3 Servitization: moving toward a Service-Dominant Logic**

In addition to reshaping a firm's offerings, servitization underscores a more profound transition in how value and its emergence are conceived and interpreted. This shift aligns closely with the evolution in marketing literature, moving from a goods-dominant perspective to the more encompassing Service-Dominant Logic (SDL) posited by Vargo and Lusch (2004, 2008). Like servitization, SDL blurs the traditional boundaries between products and services. Instead, it emphasizes service as the primary unit of exchange, defining it as the application of specialized knowledge manifested through tangible products, intangible services, or a combination of both (Vargo & Lusch, 2008).

By focusing on service as the primary unit of exchange, SDL offers a conceptual lens for understanding value and the emergence of value. Regarding the nature of value, SDL negates the long-held belief that value is embedded in physical goods and can be exchanged (i.e., value-in-exchange). Instead, SDL conceptualizes value as materializing when an offering is used in the beneficiary context (i.e., value-in-use). Shifting from value-in-exchange to value-in-use implies moving away from an objective perspective on value reflected in market prices to a subjective and contextual view in which value is phenomenologically experienced by the beneficiary (Eggert et al., 2018; Vargo & Lusch, 2008). Regarding value emergence, SDL posits that value is co-created interactively between providers and beneficiaries. As such, SDL emphasizes joint resource integration between the service provider and the beneficiary, prioritizing intangible resources like knowledge and skills (i.e., operant resources) over static and tangible ones like physical goods (i.e., operand resources). This joint resource integration implies a processual and contextual nature of value co-creation (Chandler & Vargo, 2011).

Servitization reflects the service-dominant logic by emphasizing the shift from product

manufacturing to solution selling (Kowalkowski, Gebauer, Kamp, et al., 2017; Ulaga & Reinartz, 2011). For example, Kone, a Finnish elevator manufacturer, embraced servitization by moving from product manufacturing to providing ‘people flow’ solutions, integrating elevators, escalators, and doors (Salonen & Jaakkola, 2015). Solution selling implies that manufacturers go beyond selling customized products and service offerings in single transactions and provide support throughout the product life cycle through close ongoing customer interaction (Macdonald et al., 2016; Tuli et al., 2007). Thus, servitization goes beyond merely adding services to products; it represents a profound organizational transformation in how it perceives its role in the market and its approach to customers. The shift to SDL redirects the organizational focus from solely manufacturing products to value co-creation by applying specialized knowledge. Furthermore, the contextual nature of value co-creation also implies that firms should broaden their horizon beyond the customer-supplier dyad (Chandler & Vargo, 2011).

Shifting to a Service-Dominant Logic not only implies a different view on value co-creation within the firm and the provider-customer dyad, but its contextuality has implications in a broader context. By introducing a generic actor-to-actor view on value co-creation, Vargo and Lusch (2016) embed the value co-creation process in a service ecosystem context. Service ecosystems will be discussed in more detail later, but it refers to a broader network of relations in which value co-creation occurs. For example, value co-creation between service providers and customers is affected by other actors with whom customers co-create value. Similarly, other actors like suppliers often play an essential role in the value co-creation process, and their role must also be considered. In other words, adopting a multi-actor perspective and considering inter-organizational dynamics is crucial in understanding value co-creation and the complexities associated with the shift to service-oriented business models (Eggert et al., 2018; Vargo et al., 2017).

In essence, the shift towards a Service-Dominant Logic underscores the evolution towards a subjective and perceived nature of value, prioritizing specialized knowledge and skills over physical goods and advocating for a networked perspective on value-cocreation. In line with this theoretical understanding of value co-creation, the servitization literature increasingly goes beyond the customer-supplier dyad and focuses on inter-organizational constellations.

## **1.4 Servitization in inter-organizational constellations**

Scholars have increasingly acknowledged the importance of studying servitization through an inter-organizational lens (Raddats et al., 2019; Wang et al., 2023). Unlike pure product offerings, services require new resources and capabilities, such as IoT infrastructure, data analytics competencies, and a field service network (Paschou et al., 2020; Ulaga & Reinartz, 2011). Particularly, process-oriented advanced services require resources and capabilities distant from the manufacturer’s core competencies because of the increased scope and complexity (Eggert et al., 2014; Marcon et al., 2022). Some newly required resources and capabilities can be cost-efficiently developed in-house (Eggert et al., 2014). However, others cannot because they are too distant from

a manufacturer's core capabilities and are therefore sourced from external actors (Paiola et al., 2013; Salonen & Jaakkola, 2015). As such, servitization requires inter-organ collaboration and is influenced by factors beyond the organizational boundaries (Baines et al., 2020; Story et al., 2017; Struyf et al., 2021).

Involving external actors in servitization involves transforming existing business relationships and establishing new ones (Kolagar et al., 2022; Lütjen et al., 2019). For example, downstream actors such as agents and distributors must develop service delivery and development resources (Reim et al., 2019; Story et al., 2017). These actors must be able to identify local needs, have access well-trained technicians, and internalize competencies for data analysis (Reim et al., 2019; Ulaga & Reinartz, 2011). To align external actors with the envisioned servitization strategy, firms identify complementarities that can be leveraged through collaboration and create a win-win relationship (Parida & Jovanovic, 2021; Sjödin et al., 2022). Once potential complementarities are identified, firms must incentivize external actors to get involved by demonstrating the collective and individual benefits of the envisioned collaboration (Kamalaldin et al., 2021; Parida & Jovanovic, 2021). This requires a thorough understanding of the goals and needs of network counterparts and communicating a clear, compelling, and convincing vision that fits their goals and needs (Kolagar et al., 2021; Tronvoll et al., 2020). In other words, servitization is a network-embedded process that affects and is affected by existing network structures and dynamics (Sklyar et al., 2019).

Even though the servitization literature has increasingly adopted an inter-organizational perspective, it requires further investigation (Huikkola et al., 2020; J. Wang et al., 2023). Specifically, more research is needed into how digital servitization influences network relationships, processes of integrating external resources, and the challenges associated with network interactions (Favoretto et al., 2022; Raddats et al., 2019). To address this need, this dissertation positions itself within the multi-actor perspective of the servitization literature. The subsequent chapter delves deeper into this research gap and formulates the research questions guiding this dissertation.

## **1.5 Crafting research questions**

Servitization has long been perceived as a universal, linear transformation in which companies evolve their service portfolio from relatively simple product-oriented services to more sophisticated process-oriented ones, such as outcome-based contracts (Lütjen et al., 2017; Oliva & Kallenberg, 2003). This transformation, as traditionally understood, assumed a structured, predictable progression. However, the validity of such an approach has been contested, with the identification of diverse transformation trajectories like 'industrializers,' 'availability providers,' and 'performance providers,' suggesting a non-uniform servitization path (Kowalkowski et al., 2015). This perspective has been taken further by studies arguing for an idiosyncratic transformation model, wherein each company's servitization journey is unique and tailored to its circumstances (Kowalkowski et al., 2013; Martinez et al., 2017). This alternative perspective highlights the emergent, irregular, and unstructured nature of servitization, which requires firms to adapt to evolving opportunities and challenges as they present themselves (Martinez

et al., 2017). An integrated viewpoint recently offered by (Baines et al., 2020) reconciles these seemingly disparate perspectives. According to their model, firms sequentially progress through four macro-stages, albeit the servitization within these stages remains emergent and unstructured. This integrated view provides a nuanced understanding of the servitization process and implies that servitization is driven by two forces, namely, strategic intent and strategic emergence (Kohtamäki et al., 2021; Mintzberg & Waters, 1985).

Even though the understanding of the nature of the servitization journey has increased over the years, the service paradox — the struggle to reap benefits from servitization — persists as a salient and unresolved issue, with numerous firms failing to maximize the benefits of their servitization initiatives (Benedettini et al., 2015; Gebauer et al., 2005; Xing et al., 2023). This ongoing struggle warrants further research into the nature of the servitization process and how managers can navigate its challenges. The service paradox is amplified in the current landscape by the increasingly important role of digital technologies, which introduce additional layers of complexity and pace to the transformation from product-centric to product-service business models (Favoretto et al., 2022; Sjödin et al., 2020; Struyf et al., 2021). In this context, complexity refers to the multifaceted challenges in integrating new technologies, developing new competencies, and aligning these with strategic goals within and across the organization to offer comprehensive solutions beyond mere products (Tóth et al., 2022; J. Wang et al., 2023). Complexity is accompanied by an accelerating technological advancement and market evolution rate, demanding firms to adapt quickly, often in uncertain and competitive environments (Huikkola, Kohtamäki, Rabetino, et al., 2022; Tronvoll et al., 2020). This requires further research into the underlying mechanisms that drive the servitization process.

Particularly pressing is the need to investigate the emergent character of servitization concerning the need to involve external stakeholders in servitization. Despite the acknowledgment of the importance of inter-organizational dynamics in shaping the servitization process (Chen et al., 2022; Raddats et al., 2017) and acknowledging inter-organizational dynamics as a source of emergence (Baines et al., 2020), there is limited understanding of the underlying mechanisms that connect inter-organizational dynamics and emergence in servitization. Specifically, counterparts with potentially conflicting strategic goals and priorities might resist a firm's servitization initiative (Reim et al., 2019; Story et al., 2017). While the work of Galvani and Bocconcelli (2022) illustrates how inter-organizational tensions introduce uncertainty and complexity in servitization, it remains unclear how heterogeneity among network counterparts and the resulting inter-organizational dynamics shape emergence in the servitization process. A comprehensive examination of these tensions in relation to the servitization process could contribute significantly to understanding the emergent nature of servitization and provide guidance on managing such complexities, thus enhancing firms' ability to navigate the challenges of their servitization journeys.

The interplay between strategic emergence and intent presents another area requiring further scrutiny. The servitization literature has begun to unpack this interplay, albeit

primarily from a single firm perspective. For instance, Kohtamäki et al. (2021) shed light on the strategic use of intentional narratives by managers navigating the unpredictable service transformation. However, the dialogue remains limited regarding how this interplay between intent and emergence unfolds when servitizing firms consciously strategize in the broader context of industrial networks. Increasing our understanding of the interplay between strategic intent and emergence in inter-organizational interaction will offer valuable insights into how bottom-up transformations and top-down conditioning concurrently shape the servitization process (Möller et al., 2020). Furthermore, such exploration would increase understanding of the paradoxes inherent in business networks, such as the dichotomies of opportunities and limitations, influencing and being influenced, and trying to control counterparts while simultaneously giving counterparts space and freedom to pursue their own goals and priorities (Håkansson & Ford, 2002). These understandings could contribute significantly to our understanding of 'doing' strategy for companies undertaking the servitization journey.

Finally, further research is needed into effective management strategies to shape servitization in the face of complexity, uncertainty, and a turbulent environment. Even though many manufacturers adopt linear plan-based approaches stemming from traditional product development to structure the servitization process (Huikkola, Kohtamäki, & Ylimäki, 2022; Kowalkowski & Ulaga, 2017), these approaches lack the required flexibility to deal with the dynamic and uncertain environment in which servitization occurs (Bustinza et al., 2018; Struyf et al., 2021). To respond to this disparity between traditional management strategies and the nature of servitization, the literature has recently started to focus on agile co-creation as an alternative way to structure servitization (Sjödin et al., 2020). An agile co-creation approach is based on incremental service investments, sprint-based working, and learning by doing, and can deal with the non-linear and fast-paced character of servitization (Huikkola, Kohtamäki, Rabetino, et al., 2022; Sjödin et al., 2020). Despite the promising role of agile innovation approaches in servitization, the literature on this management approach is still nascent and requires further elaboration. More specifically, our understanding of the role of agile innovation approaches in servitization would benefit from more generalizable results about its effectiveness and a better understanding of the implications of agile innovation approaches in an inter-organizational context (Linde et al., 2023; Sjödin et al., 2020).

To sum up, this dissertation addresses three gaps in the servitization literature. First, there is limited understanding of the emergent character of the servitization process, particularly regarding the role of inter-organizational dynamics. Second, further research is needed into the interplay between strategic intent and emergence in inter-organizational interaction to understand better strategizing for servitization. Third, the servitization literature investigating the role of agile innovation approaches would benefit from more generalizable findings and a better understanding of the interplay between agile innovation approaches and inter-organizational dynamics. To address these gaps, this dissertation answers the following research question:

## **How does servitization unfold in inter-organizational constellations, and how do agile innovation approaches facilitate this transformation?**

To answer this question, this dissertation is structured around four underlying questions. These questions align with one of the four studies presented in this dissertation. Before delving into each sub-question and corresponding study, the subsequent section will explore the varied theoretical perspectives on inter-organizational collaborations. Grasping these perspectives is crucial. Inter-organizational constellations have been debated across various literature streams and traditions, offering a diversity of perspectives to analyze and comprehend the complex dynamics of servitization within inter-organizational settings.

### **1.5.1 Theoretical perspectives on inter-organizational constellations**

Studies focusing on inter-organizational collaboration for servitization mainly adopt a network or ecosystem perspective, making up 62% of all papers (J. Wang et al., 2023). While they overlap, the network and ecosystem perspectives also have distinct characteristics (Möller et al., 2020; Möller & Halinen, 2017). As these perspectives differ from each other, adopting a single-perspective approach risks a limited framing and sensemaking of the research phenomenon (Ojansivu et al., 2022). Therefore, this dissertation includes multiple perspectives to study servitization in inter-organizational constellations. Specifically, this dissertation includes three views on inter-organizational constellations: industrial networks, service ecosystems, and business ecosystems. These perspectives were included because they are widely used in the industrial marketing and strategy literature.

IMP's Industrial Network approach is a highly influential network perspective on business-to-business markets that helped shape the industrial marketing domain (Möller & Halinen, 2018). The industrial marketing domain is one of the major areas where the servitization narrative is situated (Rabetino et al., 2018), so including the industrial network perspective is essential. Furthermore, this dissertation includes and differentiates between service and business ecosystems because they have different theoretical origins and characteristics (Aarikka-Stenroos & Ritala, 2017). As such, both perspectives provide unique theoretical understandings of inter-organizational collaborations. Service ecosystems originate from the SDL and, as a significant theoretical perspective, are extensively adopted in the servitization literature (Sklyar et al., 2019). The business ecosystem perspective is dominant in the strategy literature and has been frequently adopted in servitization research (Jacobides et al., 2018; J. Wang et al., 2023). Hence, this perspective is also included in this dissertation. These perspectives shape our analytical approach to inter-organizational collaborations and provide a multi-faceted lens to study servitization in inter-organizational constellations.

For the scope of this dissertation, we have also excluded specific frequently used theoretical perspectives on inter-organizational collaborations. First, we excluded the strategic net perspective because it overlaps with the business ecosystem perspective

(Möller & Halinen, 2017). Even though strategic nets put forth a unique view on inter-organizational collaborations, their focus on strategically managed network organizations can be captured in the business ecosystem perspective (Adner, 2017; Möller et al., 2020). Second, innovation ecosystems are not separately included because they are considered specific business ecosystems geared towards innovation and, therefore, captured in the business ecosystem perspective (Adner & Kapoor, 2010). Third, platform ecosystems have been omitted in this dissertation because even though platforms and their associated dynamics are an essential topic in digital servitization (Chirumalla et al., 2023), their focus on technological infrastructures and multi-sided markets is beyond the scope of this dissertation. Fourth, entrepreneurship ecosystems and knowledge ecosystems have been omitted because these concepts consider policymakers and knowledge institutes such as universities as main actors (Isenberg, 2010; Järvi et al., 2018), which is not the scope of this dissertation.

Next, this dissertation will elaborate on the three inter-organizational perspectives and how they complement each other. The discussion concludes with a multi-perspective approach to studying digital servitization in inter-organizational constellations.

#### **1.5.1.1 Industrial network approach**

Industrial networks arise from the Industrial Marketing and Purchasing (IMP) tradition, which emerged in the 1960s and '70s to challenge the dominant atomistic and transactional perspective of markets (Håkansson & Gadde, 2018). Rather than viewing markets as a series of isolated transactions, IMP scholars observed that business activities typically unfold within enduring relationships in industrial networks (Håkansson & Snehota, 1989). Organizations interact due to interdependencies stemming from resource heterogeneity and the interconnected nature of activities (Håkansson et al., 2009). Over time, interaction forms observable structures of activities, resources, and actors (ARA) at the firm, dyadic relation, and network levels (Håkansson & Snehota, 1995). The packaging manufacturer Tetra Pak exemplifies this. They had to adapt their paper conversion equipment to fit the specific size and weight of paper rolls sourced from their suppliers and buy customized lifting forks to handle them (Prenekert et al., 2019). Such relationships offer access to resources, increase efficiency, and stabilize networks (Håkansson et al., 2009). However, at the same time, business relationships limit a firm's agency to act upon its goals and priorities (Håkansson & Ford, 2002).

The industrial network approach underscores that strategic initiatives are deeply embedded within an ever-evolving web of business relationships. From the IMP perspective, firms are not merely operating in isolation. Instead, they act within the boundaries of network structures, influenced heavily by the subjective perceptions of managers about their firm's position within the broader network (Håkansson & Ford, 2002; Henneberg et al., 2006). Initiatives like servitization arise from a firm's perceived network position, a construct shaped by its relationships and resources (Johanson & Mattsson, 1987). However, an important realization in this approach is that strategic repositioning within a network cannot be unilaterally decided. Repositioning efforts gain validity only when other network participants perceive and recognize it as such

(Håkansson & Snehota, 1989). For instance, while servitization mandates intermediaries to adapt and cultivate new capabilities (Reim et al., 2019), the servitizing firms cannot dictate these changes in isolation. Past commitments, such as investments in specific technologies or established routines, can cause partners to hesitate to embrace network repositioning initiatives (Baraldi et al., 2012; Butler & Purchase, 2021). So, servitization is not merely a single-firm-driven strategy that involves network repositioning (Huikkola et al., 2020; Wise & Baumgartner, 1999). Instead, it actively involves network counterparts and results from formulating, testing, and negotiating new perspectives on social, technical, and economic ties over time (Hoholm & Olsen, 2012).

With its foundation in networked and interdependent views of strategic actions, the industrial network approach fits well with the turn to a multi-actor perspective in the servitization literature (Story et al., 2017; Struyf et al., 2021). Beyond merely understanding intra-organization capability development and transformation, the IMP approach provides the tools to dissect the network-embedded nature of servitization: a dynamic interplay of multiple actors, each influencing and being influenced in return. IMP's ARA framework offers a way to dive deep into the embedded and interdependent nature of strategic actions like servitization (Håkansson et al., 2009). It helps to analyze how servitization initiatives evolve, considering that existing relationships, along with social and material factors, play a pivotal role in the direction of such trajectories (Vildåsen & Havenvid, 2018). With its intricate understanding of industrial networks, the IMP tradition offers a valuable lens for this dissertation. Embracing its insights enables us to delve deeper into the multi-actor dynamics of servitization, ensuring a comprehensive and nuanced exploration of this transformative strategic initiative.

### **1.5.1.2 Service ecosystems**

Like the industrial network approach, service ecosystems originate in the marketing literature but stem from the service-dominant logic (SDL). As discussed earlier, this meta-theory of the market opposes traditional economic thinking by stating that value-in-use is co-created through integrating skills and knowledge, i.e., operant resources (Vargo & Lusch, 2004, 2008). More recently, SDL shifted its attention to the networked character of value co-creation through service ecosystems (Vargo & Lusch, 2016). These ecosystems are “relatively self-contained, self-adjusting systems of resource-integrating actors linked by shared institutional arrangements and mutual value creation through service exchange” (Vargo & Lusch, 2016, p. 10-11). By recognizing the role of multiple resource-integrating actors, SDL acknowledges the complexity involved in the emergence of value-in-use. For example, optimizing the uptime of industrial equipment through predictive maintenance depends on numerous integrated resources. It consists of the equipment manufacturer's knowledge about the machine and its state, the customer's technical knowledge for operation, and third-party resources like sensors for data collection, internet connectivity, and cloud computing technology combined with data analytics software for data processing (Porter & Heppelmann, 2014). This example illustrates the complex and systemic nature of value co-creation, which is enabled and constrained by institutional arrangements (Edvardsson et al., 2014).

With its focus on institutional arrangements, service ecosystems emphasize the importance of social structures in innovation and strategic action. Institutional arrangements exist of legislative, normative, and socio-cognitive dimensions and coordinate resource integration among actors through the practices, symbols, and organizing principles they carry (Akaka et al., 2014; Koskela-Huotari & Vargo, 2016; Scott, 2014). These dimensions can be illustrated by building on the previous example of optimizing the uptime of industrial equipment through predictive maintenance. Our empirical experience is that manufacturers are often wary of sharing strategically sensitive data beyond firm boundaries. This reluctance is not merely a product of internal company policies; broader societal concerns about data privacy also influence it. European legislation, like the General Data Protection Regulation (GDPR), exemplifies how external institutional arrangements can further shape and constrain the sharing and utilization of data, thereby influencing resource integration in the context of IoT-driven services. Through its focus on the role of social structures in resource integration, service ecosystems provide a unique lens to study dynamic and multi-actor settings.

Despite some overlaps, SDL's service ecosystem perspective offers a distinct lens compared to IMP's industrial network approach. Both agree on the embedded, interactive nature of business and the limited control firms have. Yet, they diverge in their understanding of the interactive nature of business in two important ways. First, IMP primarily leans on the observable ARA elements to conceptualize structure, focusing on interactions between specific entities. In contrast, service ecosystems perceive structure through unobservable societal-level forces (i.e., institutional arrangements) that shape actor perceptions and behaviors. This distinction is essential as it directs our attention: while IMP zooms in on the immediate interactions between specific entities, service ecosystems give a bird 's-eye view, focusing on the broader societal context influencing multiple actors. However, it is essential to note that observable ARA elements can be considered manifestations of unobservable institutional arrangements. Second, in the IMP tradition, behavior stems from an actor's perceived network position, designated role, and resulting power dynamics (Håkansson & Ford, 2002).

In contrast, SDL's service ecosystem departs from fixed actor roles (Vargo & Lusch, 2016). Instead, through a generic actor-to-actor perspective, SDL posits that all actors and their behavior as resource integrators are shaped by their unique institutional contexts. Rather than a single network structure as in IMP, SDL emphasizes that actors' behaviors arise from converging the multiple, often diverse, institutional arrangements in which they are embedded (Sitaloppi et al., 2016; Vargo & Lusch, 2011). This difference is essential as service ecosystems provide a more abstract but also a broader and multifaceted perspective on explaining actor behavior and interaction.

In summary, service ecosystems offer a broad theoretical lens for understanding actor behavior and their interactions. However, it is essential to recognize that neither the IMP's industrial network approach nor the SDL's service ecosystem perspective holds inherent superiority over the other. Instead, they complement each other by directing our attention to different explanatory mechanisms for inter-organizational constellations. Service ecosystems take a broader view, while industrial networks

emphasize reciprocity, interdependence, and power dynamics, spotlighting specific interaction episodes and their resultant structures. They focus on overarching social structures, with legislative, normative, and socio-cognitive institutions as principal explanatory mechanisms. As such, each perspective and its distinctive emphasis offer valuable insights and contribute meaningfully to this dissertation by enhancing our understanding of servitization within inter-organizational constellations.

### 1.5.1.3 Business ecosystems

Business ecosystems have gained substantial traction in strategic management literature, emerging as a nuanced perspective that extends beyond conventional industry boundaries and the traditional buyer-seller competitive dynamics (Adner, 2017; Jacobides et al., 2018). The business ecosystem perspective unveils complex dynamics where firms engage simultaneously in collaborative and competitive interactions with ecosystem actors. They collaborate to enhance the ecosystem's competitive position relative to external ecosystems, while internally, actors try to increase their value capture within the ecosystem (Iansiti & Levien, 2004; Moore, 1993). Take, for example, the symbiotic relationship between Samsung and Google within the Android ecosystem. Both firms collaborate in the Android ecosystem to rival Apple's iOS ecosystem, yet they also compete for supremacy within the same ecosystem, aiming to maximize their value capture (Gawer & Cusumano, 2014). Thus, a central managerial challenge within business ecosystems lies in balancing the collective pursuit of a shared vision with a firm's intrinsic self-interest (Adner, 2006; Kolagar et al., 2022).

Like IMP's industrial network approach, but unlike SDL's service ecosystem perspective, business ecosystems adopt a structural approach to inter-organizational constellations. This approach is mainly reflected in the ecosystem-as-structure approach, which defines business ecosystems as *"the alignment structure of the multilateral set of partners that need to interact in order for a focal value proposition to materialize"* (Adner, 2017 p. 42). Structure is conceptualized as the activity links between actors who are uniquely positioned in the ecosystem (Adner, 2017). Even though business ecosystems consider resources as the basis for value-creation activities (Talmar et al., 2018), its view on structure is more restricted than IMP's approach. Industrial networks also explicitly consider resource interfaces and actor bonds as important forces affecting interaction (Baraldi et al., 2012; Håkansson et al., 2009). As such, IMP's industrial network approach is a more elaborate perspective for studying the role of structure.

Even though business and service ecosystems differ in their approach to structure, they both take value (co-)creation as the starting point. In the business ecosystem perspective, structure emerges around a value proposition to materialize (Adner, 2017), whereas service ecosystems focus on value creation through mutual exchange (Vargo & Lusch, 2016). In this sense, both approaches can be distinguished from the industrial network approach, emphasizing interdependence and previous investments as drivers of network development (Håkansson & Ford, 2002). Even though both business and service ecosystems highlight the importance of value creation, SDL's view on value-in-use and the phenomenologically experienced nature is a better fit for studying digital

servitization, especially because digital servitization is about shifting to solutions, which' value is subjectively judged by the customer (Macdonald et al., 2016; Tuli et al., 2007).

In contrast to IMP's industrial network approach and SDL's service ecosystem perspective, business ecosystems distinctly emphasize the role of the focal firm. This unique orientation is articulated through the concept of ecosystem strategy, defined as *"the way in which a focal firm approaches the alignment of partners and secures its role in a competitive ecosystem"* (Adner, 2017, p. 47). Noteworthy is the adaptability of the ecosystem strategy concept, which remains agnostic about the specific roles played by various actors within the ecosystem. For instance, an ecosystem strategy may embody a keystone actor's approach to solidify its central position by advocating a modular platform design (Gawer & Cusumano, 2014; Jacobides et al., 2018). Conversely, it could represent the tactics of a complementary actor aligning with the standards and interfaces orchestrated by the ecosystem's leader, thereby securing a specialized niche and fostering competition within it (Adner, 2017; Iansiti & Levien, 2004). Dynamic interplays in business ecosystems are fueled by interactions among various actors' ecosystem strategies through innovation and the evolution of business models (Adner & Kapoor, 2010; Massa et al., 2017).

Dynamic capabilities is an essential theory from the strategic management domain augmenting the discourse on business ecosystems (Foss et al., 2023). Dynamic capabilities refer to a firm's ability to adapt to changing environments by sensing threats and opportunities in the external environment, seizing these opportunities through timely decision-making, and reconfiguring internal and external structures and resources accordingly (Teece, 2007; Teece et al., 1997). For leading keystone actors, dynamic capabilities are instrumental in consolidating their central roles by facilitating a shared vision among ecosystem partners, securing the required investments from ecosystem partners, and ensuring ecosystem stability through joint problem-solving (Foss et al., 2023; Lütjen et al., 2019). Complementing follower firms, conversely, leverage dynamic capabilities to identify niche opportunities and align with counterpart strategies in which it is expected to play a part (Iansiti & Levien, 2004; Qiu et al., 2022). Thus, dynamic capabilities emerge as a pivotal asset for leaders and followers, enhancing their strategic positioning and adaptability within business ecosystems.

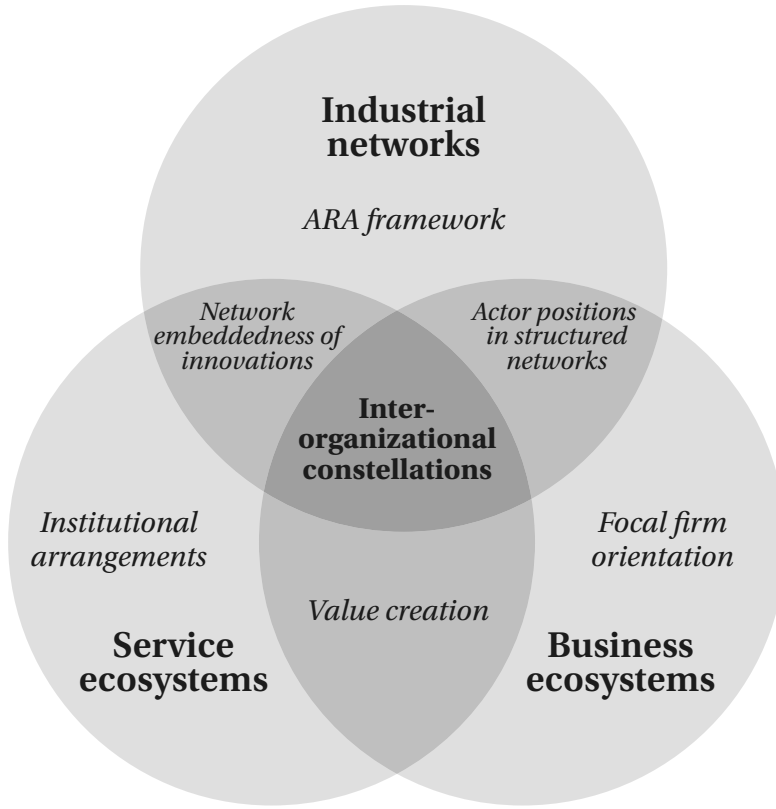
With its focal firm orientation, the business ecosystem perspective holds a distinctive advantage in analyzing digital servitization within inter-organizational constellations. It brings pragmatic clarity to the complexities of partner alignment and interdependence, providing an actionable perspective on managing in inter-organizational constellations (Kamalaldin et al., 2021). This perspective stands out by enriching IMP's industrial network approach, which, while focusing on actors' embedded behaviors, often falls short of providing articulate managerial recommendations (Möller, 2013; Waluszewski et al., 2019). Moreover, through its actionable approach to partner alignment, the business ecosystem perspective complements the SDL's service ecosystem perspective, which needs mid-range theories yielding normative insights (Ford, 2011; Vargo et al., 2023). The business ecosystem perspective emerges as a powerful theoretical lens, offering insightful strategies and tools for servitizing firms. It enables these firms to

engage and align external actors deliberately.

#### **1.5.1.4 A multi-perspective approach to studying inter-organizational constellations**

Based on the three perspectives on inter-organizational constellations and their distinct advantages, this dissertation brings forth a multi-perspective model for studying digital servitization in inter-organizational constellations (Figure 1.1). This model illustrates that the industrial network approach, service ecosystem perspective, and business ecosystem perspective have substantial overlap while simultaneously bringing a unique lens to the table. The center illustrates that all three perspectives focus on the same issue. Namely, firms do not operate in isolation but in a constellation of multiple economic and non-economic actors. Moreover, they also partly overlap in some aspects. Both industrial networks and service ecosystems emphasize the network-embedded nature of innovation and the limited agency of individual actors. Industrial networks overlap with business ecosystems in their assumption that inter-organizational constellations are structured and actors have specific positions in these constellations. Finally, service ecosystems and business ecosystems share their starting point, which is value creation and the materialization of value propositions.

Despite their conceptual overlap, all three perspectives provide a unique lens for studying digital servitization in inter-organizational constellations. IMP is powerful for studying the role of inter-organizational structures in digital servitization. Through its ARA framework, the industrial network approach provides an elaborate analytical lens for studying inter-organizational structure. Service ecosystems emphasize the role of social structures in value creation, which allows for incorporating forces and dynamics beyond a firm's direct interaction. For example, the behavior of actors in business relations may be influenced by industry "best-practices," which are forces on a higher abstraction level than inter-organizational interactions. This focus on institutional arrangements presents a suitable lens for studying the role of broader and more abstract social structures in inter-organizational collaboration in digital servitization. Business ecosystems bring a focal firm perspective and infuse prominent theories from strategic management, like dynamic capabilities in the discourse on inter-organizational constellations. This perspective helps provide actionable insights into how managers can approach actor alignment in inter-organizational constellations (Kamalaldin et al., 2021).



**Figure 1.1** Multi-perspective approach for studying servitization in inter-organizational constellations

### 1.5.2 Sub-questions and Research Framework

Based on the multi-perspective approach developed in the previous section, this section defines specific sub-questions through which this dissertation answers the main research question. The sub-questions represent different perspectives on inter-organizational constellations. Study one is built on the industrial network approach, study two on the service ecosystem perspective, and studies three and four on the business ecosystem perspective. This dissertation provides a complete but nuanced understanding of digital servitization in inter-organizational constellations by answering the main research question through the three distinct perspectives. Specifically, a multi-perspective approach offers an understanding of the role of business and social structures in the servitization trajectory and insights into how managers within servitizing firms can deal with these structures and their resulting dynamics.

The first study of this dissertation aims to understand better the emergent nature of the servitization process, particularly the role of dyadic inter-organizational tensions

as a driving force of emergence. Given the necessity to acquire specialized resources and the subsequent reconfiguration of business relationships (Kowalkowski et al., 2013; Salonen & Jaakkola, 2015), the study emphasizes how active participation from network counterparts can result in tensions, adding complexity to the servitization process (Struyf et al., 2021). Past research has primarily explored causes and managerial responses to these tensions (Burton et al., 2016; Galvani & Bocconcelli, 2022; Tóth et al., 2022), leaving a gap in understanding their influence on the servitization process. To address this, the study integrates dialectic theory and the activity-resource-actor (ARA) framework from the Industrial Purchasing and Marketing tradition to develop a dialectic process model for servitization. This theoretical approach provides a more granular comprehension of tensions in business relationships and how they drive servitization. The following research question guides this study:

**SQ1: How do inter-organizational tensions in business relationships influence the servitization process?**

The second study in this dissertation adopts a service ecosystem perspective to investigate the role of institutional complexity in inter-organizational tensions. Institutional complexity refers to “the multiplicity of institutional arrangements in service ecosystems, which confronts actors with incompatible prescriptions for action” (Siltaloppi et al., 2016, p. 334). So far, existing literature has paid limited attention to the role of institutional arrangements in the emergence of inter-organizational tensions (Galvani & Bocconcelli, 2022; Tóth et al., 2022). However, these social structures may be necessary for understanding inter-organizational tensions, particularly because digital servitization requires industry-transcending collaboration with partners that bring diverging routines, expectations, and perceptions (Favoretto et al., 2022; Makkonen et al., 2022). By adopting a service ecosystem perspective and focusing on institutional complexity, this study extends the discourse on inter-organizational tensions in digital servitization. The main question this study seeks to answer is:

**SQ2: How does institutional complexity contribute to the emergence of inter-organizational tensions in digital servitization, and how do actors respond to these tensions?**

The third study of this dissertation focuses on agile co-creation in digital servitization. Building on the emerging literature about agile co-creation in digital servitization, this research investigates the impact of two separate agile co-creation approaches, namely, agile development and lean startup, on service business performance. By differentiating between agile development and lean startup, this study clarifies the different practices comprising agile co-creation, which has been treated as a unified innovation approach (Chirumalla et al., 2023a; Sjödin et al., 2020). Agile development, a project management framework originating from the software industry, emphasizes the importance of diverse and autonomous teams, short iterative cycles, and frequent internal and external communication (Sjödin et al., 2020). On the other hand, lean startup is an agile method for business model innovation that revolves around the build-measure-learn loop, equipping entrepreneurs to navigate dynamic and complex business environments

effectively (Ghezzi & Cavallo, 2020a; Ries, 2011). To better understand the distinct roles of agile innovation approaches in digital servitization, we divide service business performance into financial and non-financial service business performance (Oliva et al., 2012). Considering these dimensions provides the opportunity to differentiate between the impact of agile development and lean startup on the financial bottom line of the service business and their impact on value co-creation with customers. This differentiation is essential as many firms struggle to translate customer satisfaction with digital services into economic benefits (Neely, 2008; Sjödin et al., 2020). The question guiding this study is:

**SQ3: What is the impact of agile development and lean startup on financial and non-financial service business performance?**

The fourth study in this dissertation extends the discourse on agile innovation in the servitization literature beyond the customer-supplier dyad to the broader ecosystem (Sjödin et al., 2020). This investigation acknowledges that servitization demands input from various ecosystem partners, including suppliers, technology partners, and intermediaries (Sklyar et al., 2019; Story et al., 2017). Situated in the business ecosystem perspective, this study investigates the connection between agile innovation approaches and dynamic capabilities. Dynamic capabilities provide a conceptual framework for understanding how firms adapt to changing environments and are considered essential for aligning internal and external stakeholders with an envisioned digital servitization strategy (Chirumalla et al., 2023a; Lütjen et al., 2019; Teece, 2007). This study answers the following question:

**SQ4: How do agile innovation approaches facilitate dynamic capabilities in the digital servitization context?**

Table 1.1 provides an overview of the four studies that make up this dissertation, including their sub-questions, primary theoretical basis, key concepts, and methods.

**Table 1.1 Overview of studies and sub-questions in this dissertation**

| Study | Sub question                                                                                                                                                          | Theoretical basis                              | Method                                                                                                                      | Outcomes                                                                                                                            |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1     | How do inter-organizational tensions in business relationships influence the servitization process?                                                                   | Dialectic theory; Industrial network approach  | Qualitative – Single case study of WinCo interacting with network counterparts to implement Façade-as-a-Service proposition | Dialectic process model for understanding the emergent nature of servitization in inter-organizational contexts.                    |
| 2     | How does institutional complexity contribute to the emergence of inter-organizational tensions in digital servitization, and how do actors respond to these tensions? | Service ecosystem                              | Qualitative – Single case study about MetalNet, a network organization developing a Spareparts-as-a-Service proposition.    | Process model for understanding the role of institutional complexity in inter-organizational tensions and organizational responses. |
| 3     | What is the efficacy of agile development and lean startup for servitization?                                                                                         | Servitization; Agile development; Lean startup | Quantitative – Survey research (N=114), analyzed through Partial Least Squares (PLS).                                       | Generalizable insights on the effectiveness of agile innovation approaches for servitization.                                       |
| 4     | How do agile innovation approaches facilitate dynamic capabilities in the digital servitization context?                                                              | Business ecosystem; Dynamic capabilities       | Qualitative – Interview-based study (N=24).                                                                                 | A nuanced understanding of the implications of agile innovation approaches for business ecosystem transformation and alignment.     |

## 1.6 Research design

This dissertation adopts a pragmatist philosophy of science. At the heart of pragmatism is an orientation towards problem-solving and a deep concern with the practical consequences and effects of concepts and theories, valuing action and application over mere abstract theorizing (D. L. Morgan, 2014). Inspired by a Darwinian evolutionary view of the world, the pragmatist ontology assumes an indeterminate world and becoming instead of being (Lorino, 2018; Simpson, 2018). Drawing from Darwin's observation that all species, including humans, are part of a fluid progression of life, pragmatists recognize that humans continuously cope with an evolving and uncertain environment (Lorino, 2018). While accepting a realist ontology, pragmatism emphasizes that the utility of actions in the real world is intertwined with objective facts and our socially constructed interpretations of these facts (Cherryholmes, 1992). For example, a pragmatist perspective on digital servitization as a growth strategy would consider objective observable technological capabilities and innovation processes and the social interpretations of these capabilities and processes that may influence their development and implementation. In this sense, pragmatism does not adhere strictly to the strong ontological assumptions characteristic of positivistic and constructivist traditions (Cresswell & Cresswell, 2017).

Pragmatism adopts an epistemological stance that emphasizes the provisional and evolving nature of knowledge. Pragmatism is concerned with the practical and achievable instead of the theoretical or ideal (Lorino, 2018). As such, pragmatist epistemology is not about revealing an objective or socially constructed reality. Instead, it considers knowledge as a tool, a means to navigate, adapt to, and intervene in a continuously changing world. Knowledge serves to address immediate problems and mitigate uncertainty (Simpson, 2018). Practical inventions like forks, knives, and microchips exemplify this approach as they are not immutable truths but solutions refined over time that help us cope with specific challenges in our uncertain environment (Simpson, 2018). This epistemological stance fits well with the focus of this dissertation on inter-organizational constellations. These constellations are constantly changing, presenting new challenges requiring innovative solutions.

This dissertation employs a mixed-method research approach to align with the pragmatic philosophy adopted. This approach reflects the methodological pluralism of pragmatism and its assertion that the research objective should shape methodology. Quantitative methods are suited when the aim is prediction and control, as is the case when investigating action and consequence. Conversely, when the objective is to understand human experiences and morality, non-naturalistic approaches such as case studies or ethnography become more suitable (Baert, 2004). Thus, pragmatism enables the design of a single study that explores the nature of a phenomenon from a participant's point of view while examining the relationships between measurable variables (Williams, 2007). This fits well with the aims of this dissertation, which strives to comprehend the emergent nature of servitization in an inter-organizational context (studies 1 and 2) and identify effective management approaches for the servitization process, such as agile development (studies 3 and 4). Notably, the methodological

pluralism in pragmatism does not imply that epistemological and methodological considerations become irrelevant. Instead, a clear articulation of why a specific approach is appropriate for the research objective and context is crucial (Denzin, 2012). Thus, this dissertation consciously oscillates between methodological foundations to comprehensively explore the complex dynamics involved in servitization.

The first study employed an abductive research approach, integrating theory building with data collection for a Dutch SME transitioning towards a servitization strategy. A qualitative research design was chosen to understand the complexities of business relationships and their role in the emergent nature of the servitization process, with the case study methodology as the basis (Yin, 2009). A single case study design was preferred because it provides a holistic view of a system and its structures and relations (Halinen & Törnroos, 2005), providing the necessary depth of analysis. Data was collected primarily from semi-structured interviews conducted in three rounds from December 2019 to December 2022, supplemented by a workshop and secondary data like industrial white papers and research reports. The data analysis process was composed of four phases, involving transcription and close reading of interviews, coding of data, deepening of analysis using the activity-resource-actor (ARA) framework (Håkansson & Snehota, 1995), and comparison of business relationship structures to uncover contradictions and tensions. This multi-dimensional approach, blending rich primary and secondary data sources with an iterative analytical framework, allowed for a comprehensive exploration of the studied phenomenon.

The second study used a single case study research design to gain an insider perspective on an inter-organizational collaboration called “MetalNet,” which aimed to develop a Spare-parts-as-Service proposition. This research design was used to study complex interaction dynamics within business relationships. It enabled the researcher to immerse themselves in the case setting, understanding the tacit knowledge and underlying social structures that guided the actions and perceptions of the group members. Based on a service ecosystem perspective, this study was particularly interested in interactions, activities, artifacts, and symbols as carriers of unobservable institutional arrangements (Koskela-Huotari & Vargo, 2016). Hence, data were gathered through ethnographic observations, semi-structured interviews, and internal documents. Thematic analysis was used to identify institutional arrangements, complexity, and coping practices to deal with institutional complexity.

In the third study, an international survey was conducted to investigate the performance implications of agile co-creation for digital servitization within manufacturing (NAICS 31–33) and industrial systems integration companies (NAICS 5415). The sample was diverse, comprising mainly small to large-sized enterprises, with a substantial share of their revenue generated from services. To ensure accuracy, proven scales were used for measuring service business performance, agile development, and lean startup constructs (Harms & Schwery, 2020; Malik et al., 2021; Oliva et al., 2012). Controls were implemented for variables like firm size, customer count, and the type of physical product offered. A non-response bias check compared early and late respondents’ scores. The data was analyzed using partial least squares (PLS) path modeling due to its

ability to handle complex models and small sample sizes (Hair et al., 2022; Henseler et al., 2016).

The fourth study employed an interview-based qualitative research design because of the explorative nature of the research into agile innovation approaches and dynamic capabilities. Furthermore, a qualitative approach was considered appropriate as dynamic capabilities cannot be observed directly but can only be studied by tapping into the experience of the individuals involved with them (Qiu et al., 2022). The study population consisted of managers from servitizing firms within the construction, equipment manufacturing, IT, and utility sectors. A two-round semi-structured interview approach was used for data collection. The first round focused on understanding the dynamics and outcomes of agile innovation practices in digital servitization. The second round delved deeper into ecosystem transformation for servitization, dynamic capabilities, and the impact of agile practices on these capabilities. The data analysis included three phases: transcription of interviews, open coding content analysis followed by thematic analysis, and then coding data that addressed the firm's dynamic capabilities and their role in ecosystem alignment and transformation. This approach provided a nuanced understanding of the role of agile innovation approaches in facilitating dynamic capabilities for ecosystem transformation and the necessary contingencies that affected this relationship.





# 2

## **INTER-ORGANIZATIONAL TENSIONS IN SERVITIZATION: A DIALECTIC PROCESS MODEL**

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# Abstract

This study investigates inter-organizational tensions in servitization and develops a dialectic process model to better understand the emergent nature of servitization. Inter-organizational tensions have gained increasing attention in servitization literature recently, but their role in the emergent nature of servitization requires further research. By combining dialectic theory and the ARA framework from the IMP tradition, we develop a dialectic process model that identifies inter-organizational tensions as generative forces in servitization. We present a single case study about the implementation of a circular service proposition in the construction industry called “Façade-as-a-Service”. Our results reveal multiple tensions in a servitization context and how they emerged as a result of contradictory relationship structures in the existing and envisioned situations. We found that the tensions may be either anticipated as a result from direct clash between the status quo and an envisioned situation or unanticipated because they are indirectly as a result of interdependent ARA dimension. Our findings contribute to the existing literature by (i) extending knowledge on servitization tensions to a circular economy context, (ii) portraying servitization as a dialectic process that requires integration of competing interests, and (iii) opening the black box of inter-organizational tensions by adopting the ARA framework.

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## 2.1 Introduction

*“We don’t offer Façade-as-a-Service because we think we need a new business model or want to earn more money, no, servitization is a means to reach our goal to become more circular” - (Manager, WinCo).*

Servitization refers to the integration of product and service offerings by manufacturing firms (Baines et al., 2009; Vandermerwe & Rada, 1988). It is a widely used response to product commoditization and can lead to enhanced financial and non-financial performance (Matthyssens & Vandenbempt, 2008; Wang et al., 2018). Offering services is associated with higher profits, more stable revenues, enhanced customer proximity, and increased offering differentiation (Baines et al., 2009). Servitization might enable and incentivize manufacturers to (i) design for longevity, (ii) optimize quality during operations, and (iii) increase the reusability of materials after disposal (Yang & Evans, 2019a). As indicated in the quote above, it is an important means for firms to operate in the circular economy (Centobelli et al., 2020). In this sense, it has a role to play in our global quest to reduce the industrial ecological footprint (Tukker, 2015).

Servitization was initially considered a unidirectional and planned process in which firms gradually increase the importance of service offerings and move from add-on services to relationship-based and process-centered services (Mathieu, 2001; Oliva & Kallenberg, 2003). More recently, this view was challenged by conceptualizing servitization as an emergent and unstructured process that requires an adaptive approach (Kowalkowski et al., 2012; Matthyssens & Vandenbempt, 2008). Because servitization requires highly specialized resources that are often distant from the core manufacturing knowledge (Paiola et al., 2021; Ulaga & Reinartz, 2011), firms tend to draw these resources from external actors (Salonen & Jaakkola, 2015). The resulting reconfigurations in business relationships require effort and investments from servitizing firms as well as from their network counterparts (Chakkol et al., 2014; Kowalkowski et al., 2013). In other words, network counterparts have to actively participate in developing and deploying the resources required for servitization (Raddats et al., 2017; Story et al., 2017). Therefore, co-evolutionary forces are important drivers of servitization (Martinez et al., 2017).

Actively involving network counterparts can lead to inter-organizational tensions which increase the complexity of servitization (Struyf et al., 2021). Network counterparts may respond defensively to a servitization initiative when it threatens their network position or does not fit with their goals and priorities (Öberg & Shih, 2014; Story et al., 2017). For example, servitizing firms may experience resistance from network counterparts due to the lack of service capabilities or vision, or unfavorable local conditions at intermediaries (Reim et al., 2019). Inter-organizational tensions arise when the envisioned servitization strategy contradicts and collides with the external environment in which servitizing firms operate (Galvani & Bocconcelli, 2022). For example, tensions arise when servitizing firms challenge the expertise, role, or professional identity of network counterparts through their product-service offering (Burton et al., 2016; Tóth et al., 2022). Servitizing firms need to respond adequately to these tensions when they emerge and prevent their potential negative effect on servitization outcomes (Galvani &

Bocconcelli, 2022; Tóth et al., 2022).

Research on inter-organizational tensions in servitization has been limited so far and emerged only recently. Most studies that have investigated them focus on tension categories, their causes, and how managers can adequately respond to them (Burton et al., 2016; Galvani & Bocconcelli, 2022; Tóth et al., 2022). The effect of tensions on servitization outcomes requires further research to better understand their role in the emergent and unstructured nature of servitization (Martinez et al., 2017; Tóth et al., 2022). Specifically, the impact of business relationships as an active force that influences servitization outcomes should be investigated more in-depth (Galvani & Bocconcelli, 2022). To respond to these calls, this paper aims to better understand the emergent nature of servitization by investigating how tensions in dyadic inter-organizational business relations drive this transformation. These insights extend our understanding of business relationships as a source of complexity in servitization and how this complexity drives servitization outcomes.

To study the effects of dyadic inter-organizational tensions on servitization outcomes, we integrated dialectic theory and the activity-resource-actor (ARA) framework from the Industrial Purchasing and Marketing (IMP) tradition (Van de Ven & Poole, 1995). Dialectic theory asserts that contradictory values, interests, and goals of different actors lead to tensions, confrontation, and change through thesis, antithesis, and synthesis (Van de Ven & Poole, 1995). Dialectic theory is particularly well-suited for studying process dynamics in relation to tensions (Putnam et al., 2016). To conceptualize dialectic processes in business relationships, we adopted the ARA framework. This framework is based on the premise that the business landscape consists of activities, resources, and actors that become mutually adapted and result in observable business relationship structures through ongoing interaction (Håkansson et al., 2009). Because ARA dimensions are interdependent and represent the basic micro-layer of business relationships (Håkansson et al., 2009; Kaartemo et al., 2020), they provide a more detailed understanding of the dialectic process in business relationships.

Based on a single case study of a servitization initiative to contribute to the circular economy in the construction industry, we make three contributions. First, we found several inter-organizational tensions that are related to servitization in general as well as specifically to servitization as a strategy to contribute to the circular economy. Second, we increased our understanding of the emergent and unpredictable nature of servitization by portraying it as a dialectic process. Our findings indicate that servitization involves inter-organizational tensions which emerge from contradictions between the thesis and an envisioned antithesis and drive servitization through synthesis. As such, it appears that dialectics is a generative force that drives the course of servitization and requires an iterative and agile management approach. Third, we portray the ARA framework as a micro-foundation of the dialectic process. This means that the ARA dimensions explain the emergence of dialectic tensions on a lower abstraction level (Felin et al., 2015a). Involving network counterparts in a servitization strategy leads to contradictions between existing and envisioned activity links, resource ties, and actor bonds. These dimensions are also interrelated, and contradictions within a single dimension may

lead to contradictions in other dimensions.

In the rest of this paper, we discuss the contemporary literature on servitization and develop our conceptual model by drawing on dialectic theory and the IMP tradition. We then elaborate on our research approach, case setting, and procedures for data collection and analysis. Finally, we present our case study results as well as our discussion and conclusions.

## **2.2 Conceptual background**

In this section, we elaborate on the theoretical underpinnings of our research. We start by describing the servitization phenomenon, its network-embedded character, and the role of inter-organizational tensions. This is followed by the development of our conceptual model, for which we drew on dialectic theory (Hegel, 1969; Van de Ven & Poole, 1995) and the Industrial Marketing and Purchasing (IMP) tradition (Håkansson & Snehota, 1995).

### **2.2.1 Servitization**

Servitization refers to the integration of products and services to offer customers better solutions through more complete market packages (Vandermerwe & Rada, 1988). It involves changes in operational processes, resources, and the organizational culture (Gebauer et al., 2005; Ulaga & Reinartz, 2011; Vendrell-Herrero et al., 2017). This process has long been considered unidirectional and linear, with manufacturers gradually shifting from basic product-oriented services to more sophisticated process-oriented services (Lütjen et al., 2017; Oliva & Kallenberg, 2003). However, this linear thinking is problematic as the process appears to be illogical, organic, and unpredictable (Kowalkowski et al., 2015; Martinez et al., 2017). So instead of considering servitization a unidirectional transition taken in a few large steps, the process should be envisaged as incremental and without a clear, predefined direction (Kowalkowski et al., 2012). One reason for the unpredictability of servitization is the need to involve business network partners, which reduces control over the process and therefore increases complexity (Kohtamäki, Parida, et al., 2019; Struyf et al., 2021; Visnjic et al., 2018).

Setting up new value constellations to deliver integrated product-service offerings requires firms to establish new relationships and reconfigure existing ones (Kowalkowski et al., 2013; Spring & Araujo, 2013). Such a business network reconfiguration requires effort from the servitizing firm as well as its network counterparts. Downstream actors such as agents and distributors need to develop service delivery and development capabilities, including identification of local needs, well-trained technicians, and competencies for data analysis (Reim et al., 2019; Story et al., 2017). Upstream actors such as suppliers need to integrate and tailor their resources with those of the servitizing firm (Chakkol et al., 2014). However, such network reconfigurations cannot be expected to always go smoothly. For example, business networks tend to maintain an equilibrium, which implies that the servitization of one firm requires the deservitization of another firm in the business network (Forkmann et al., 2017; Halinen et al., 1999). Thus,

servitization might threaten the network position of other network actors and trigger defensive attitudes from network counterparts (Story et al., 2017).

The need to involve network counterparts who also pursue their own agendas and priorities increases the complexity of servitization because this can result in inter-organizational tensions (Burton et al., 2016; Struyf et al., 2021). Tensions emerge when the implementation of a servitization strategy collides with the context in which servitizing firms operate (Galvani & Bocconcelli, 2022). The context can be economic, technological, or legal, or the perceived benefits of service offerings or an actor's existing identity (Reim et al., 2019). For example, digital services may collide with local regulation or with professional identities when a service offering interferes with operator tasks (Reim et al., 2019; Tóth et al., 2022). Different tensions emerge over time, depending on the servitization stage the firm is in (Dmitrijeva et al., 2022; Galvani & Bocconcelli, 2022). Even though tensions in servitization can be categorized in terms of organizing, learning, belonging, and performing (Dmitrijeva et al., 2022; Tóth et al., 2022), their specific form remains unpredictable, and dealing with tensions requires constant contextual evaluation and adaptation (Galvani & Bocconcelli, 2022).

Servitization scholars have mostly studied tensions to explain the challenging and complex nature of servitization. This means that the challenging nature of servitization stems from tensions that are generated by the transition and which require managerial actions to ensure desired servitization outcomes (Dmitrijeva et al., 2022; Kohtamäki, Einola, et al., 2020). However, we suspect that the relation between servitization and tensions is recurrent, that is, tensions also drive and shape servitization. Hints of such a recurrent relation can be found in the works of Galvani and Bocconcelli (2022) and Tóth et al. (2022). They found that managers respond to tensions by adapting their servitization strategy in terms of service design and service implementation. However, they did not explicitly focus on tensions as a generative force that affects the servitization path. Further investigating the role of tensions as a generative force in servitization increases our understanding of the emergent and unstructured nature of the servitization process (Kowalkowski et al., 2015; Martinez et al., 2017).

### **2.2.2 Towards a dialectic process model**

So far, inter-organizational tensions in servitization have mostly been studied through a paradox lens (e.g., Galvani & Bocconcelli, 2022; Tóth et al., 2022). The paradox lens focuses on contradictions between opposite yet interrelated elements that exist simultaneously and persist over time (Smith & Lewis, 2011). This focus enables us to study the unity of contradictory elements that are often perceived by actors separately (Lewis, 2000). Scholars and practitioners can make sense of the paradox in its totality and address tensions by balancing opposite elements instead of choosing one element over the other (Lüscher & Lewis, 2008). For example, servitizing firms cannot choose between customization or standardization but need to achieve both (Kohtamäki, Einola, et al., 2020). The paradox lens assumes that actors accept the co-presence of contradictory elements and engage in the act of balancing them (Hargrave & Van de Ven, 2017). In a network context, this may not always be the case as tensions can also arise

from a clash between actors who actively pursue their own goals (Corsaro & Snehota, 2011; Öberg & Shih, 2014).

To study tensions stemming from clashing perspectives, agendas, and priorities between network actors, we draw on dialectic theory (Hegel, 1969). Dialectic theory is rooted in Hegelian and Marxist philosophy and asserts that economic actors are situated in a pluralistic world consisting of actors who have contradictory values, interests, and goals which lead to tensions, confrontation, and change (Van de Ven & Poole, 1995). Because contradictions are inherent to organizational arrangements, clashes between existing and alternatively envisioned arrangements are inevitable (Benson, 1977; Seo & Creed, 2002). This clash between thesis (i.e., status quo) and antithesis (i.e., alternative social arrangement) triggers political struggle between proponents of both sides, which leads to transformation or synthesis that cannot be predicted a priori (Langley & Sloan, 2012; Van de Ven & Poole, 1995). Next to synthesis, which refers to change through integrating thesis and antithesis, contradictions can also be absorbed into existing arrangements or be suppressed by subscribers to the status quo (Farjoun & Fiss, 2022; Hargrave & Van de Ven, 2017). Through the process of confrontation between thesis and antithesis, dialectic theory assumes that innovation cannot be considered in isolation from what previously existed and that innovation outcomes reflect the integration of the old and the new (Bledow et al., 2009).

The focus of dialectic theory on tensions that drive organizational changes (Hargrave & Van de Ven, 2017). While process is an important concept in paradox studies (Farjoun, 2010), it has not been its central focus (Putnam et al., 2016). In contrast, dialectic theory focuses on change processes by studying actors whose interests are not being served and therefore engage in a political struggle to change the status quo (Hargrave & Van de Ven, 2017). The focus of dialectic theory on contradictions between interests among actors who have the agency to bring about change through political conflict makes it an approach well-suited to studying strategic change processes (Farjoun & Fiss, 2022)

Furthermore, dialectic theory adopts a wider view of underlying tensions by considering both dualisms and dualities (Smith & Lewis, 2011). This means that the dialectic process entails tensions that can be resolved through integration (i.e., dualisms) as well as ones that are persistent over time (i.e., dualities). To understand tensions in inter-organizational relationships and capture the dialectic process in business relationships, further conceptualization is needed. So far, dialectics have mostly been studied on the organizational level (Putnam et al., 2016), and a detailed framework is lacking to study tensions in complex and dynamic business relationships.

To conceptualize dialectics in business relationships, we draw on the IMP tradition. IMP has studied business relationships and networks for more than four decades and provides a strong theoretical base for understanding inter-organizational dynamics (Håkansson & Gadde, 2018). IMP asserts that business is conducted in long-term business relationships and puts inter-organizational interaction at the center of attention (Håkansson & Snehota, 1989; Waluszewski et al., 2019). Over time, interaction results in complex network structures that favor stability over change (Håkansson et al., 2009).

Because innovation involves the reconfiguration of existing social, technical, and economic relations between business network actors (Hoholm & Olsen, 2012), network tensions are inherently part of innovation (Chou & Zolkiewski, 2018; Vildåsen & Havenvid, 2018). The fierceness of these tensions depends on the combination of structure ‘heaviness’ and ‘variety’, which refers to the stickiness of the old and the extent to which the new deviates from the old (Håkansson & Waluszewski, 2002a). The focus on stability and long-term relationships stemming from interaction provides a more detailed understanding of how a thesis emerges in business relationships and how innovations could lead to tensions.

Tensions in business relationships become salient when an actor tries to mobilize other actors by actively developing or changing some aspects of existing relationships or network structures (Ford & Mouzas, 2013; Mouzas & Naudé, 2007). The mobilizing process consists of various management activities ranging from identifying relevant network counterparts to motivating them to commit resources to a change initiative and influencing how resources are integrated across actors (Aarikka-Stenroos et al., 2017). Network counterparts may not respond to a change initiative as the mobilizer anticipated because they pursue their priorities (Öberg & Shih, 2014). Consequently, mobilizers need to make compromises that direct innovation in different, unexpected ways (Hoholm & Olsen, 2012). Compromises may entail mutual adaptation within existing business relationships through constructive debate or dissolution of existing relationships to overcome value co-creation impediments (Chou & Zolkiewski, 2012; S. Fang et al., 2011; Tidström, 2014). This view on actor mobilization, including the need for compromising, enriches the understanding of the synthesis process in business relationships.

To capture the dialectic process in the context of business relationships, we adopt the ARA framework (Håkansson & Snehota, 1995), which is a widely used analytical framework to make sense of business relationships (Möller & Halinen, 2022). The ARA framework differentiates between three dimensions that make up business relationships: activity links, resource ties, and actor bonds. As these relationship dimensions represent the basic micro-elements of networks (Kaartemo et al., 2020), they provide a more detailed understanding of the dialectic process in business networks. In other words, thesis and antithesis in a dialectic process can be described in terms of activity links, resource ties, and actor bonds. More specifically, we consider the existing business relationship structure as the thesis situation and the mobilizer’s envisioned relationship structure as the antithesis. When these relationship dimensions clash in the dialectic process, inter-organizational tensions emerge and trigger the synthesis process, which is a political process that involves compromising (Hoholm & Olsen, 2012).

Activity links refer to a system of interdependent transformation (i.e., production) and transaction (i.e., exchange) activities carried out by different network actors (Dubois, 1998). Due to the interdependence between activities among network actors, a change in one activity configuration often requires other actors to change their activity configuration as well (Håkansson et al., 2009). However, changing activity configurations at one firm may negatively influence another firm’s efficiency and

elicit resistance (Dubois, 1998). Resource ties refer to the interfaces between tangible and intangible resources as a result of mutual adaptations and investments (Baraldi et al., 2012; Håkansson & Waluszewski, 2002b). While specialized resource interfaces can be valuable to firms, they may also be difficult to change due to their “heaviness” (Håkansson & Waluszewski, 2002b; Prenkert, 2016). Actor bonds refer to cognitive links between network counterparts including inter-personal relations, identity perspectives, norms, and ideas about how business relationships will evolve in the future. Network actors may have different and even contradictory views on relationship structures and how they evolve (Leek & Mason, 2009; Mattsson et al., 2015).

One of the key features of the ARA framework is the interdependence between the activity, resource, and actor dimensions (Håkansson et al., 2009). Activity links and resource ties are interdependent because resources enable activities (Bankvall, 2014). For example, production activities may require resources such as machinery and process knowledge. Vice versa, innovation activities affect resources as production activities may result in new products and innovation activities in new knowledge (Dubois, 1998; Möller, 2010). Activity links and actor bonds are interdependent because how actors perceive their own and each other’s identity and network role determines what behavior and activities are considered appropriate and affect what innovation opportunities are pursued (Tripsas, 2009). Resource ties and actor bonds are interdependent because resource structures can be expressed in activated structures and idea structures (Håkansson & Waluszewski, 2003). Activated structures refer to the existing resource structures whereas idea structures refer to the actor’s subjectively held interpretations of resource structures. While idea structures may be the source of resource tie development, they can also result in tensions due to contradictory idea structures among actors (Håkansson & Waluszewski, 2003).

Integrating dialectic theory and the ARA framework results in a dialectic process model to analyze and explain dialectics in business relationships (Figure 2.1). We use this model to study the dialectic process in the context of servitization. Our model asserts that servitizing firms operate in existing business relationships that are characterized by product-focused interaction (Reim et al., 2019; Sklyar et al., 2019). Because servitization involves network reconfigurations (Kowalkowski et al., 2013), implementing a servitization strategy requires firms to challenge the thesis by envisioning new relationship configurations in accordance with their servitization strategy (i.e., antithesis). Both thesis and antithesis can be described in terms of three interdependent elements, namely, activity links, resource ties, and actor bonds. However, the antithesis envisioned by the servitizing firm may be perceived as a threat by other network actors who will try to defend their network position through a defensive response (Story et al., 2017). Such defensive responses trigger the dialectic process manifested in inter-organizational tensions. Depending on how these tensions play out, they result in synthesis or dissolution.

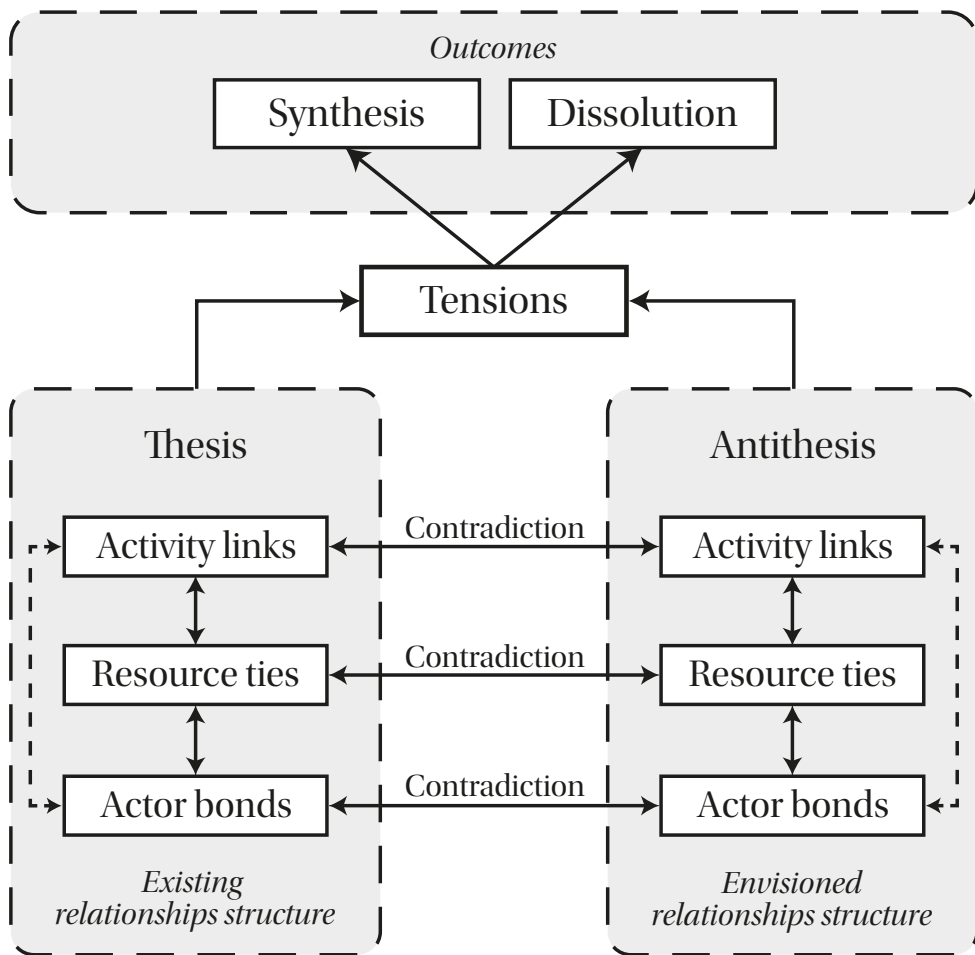


Figure 2.1 Dialectic process model

## 2.3 Method

This study presents a case of a Dutch SME in the construction industry that is implementing a servitization strategy to become more circular. We adopted an abductive research approach, which means that we switched between theory building and data collection during the research process. Specifically, we opted for a qualitative research design because it provided us with the required investigational flexibility (Yin, 2009). A qualitative research design allowed us to draw on multiple data sources and different informants and make adjustments in our data collection and analysis to deal with business relationship complexity. We chose to conduct a case study for two reasons. First, it helps us gain an understanding of the uniqueness and embeddedness of the studied phenomenon (Stake, 1995). And second, case studies give a holistic view of a

system, including its structure and relations (Halinen & Törnroos, 2005). To thoroughly understand how the clash between current and envisioned business relationship structures resulted in inter-organizational tensions, we preferred a single case study design over a multiple case study design. Given the available time and resources, a single case study provides us with the required depth of analysis, which is preferred over the breadth that can be obtained from multiple case studies (Yin, 2009).

We picked a case from the Dutch construction industry for two reasons. First, the circular economy is a high-priority topic in this industry since construction is responsible for 40% of total energy consumption, 38% of all waste, and 36% of CO<sub>2</sub> emissions in the EU (European Commission, 2019). Along with building design and choice of materials, service business models are expected to play an important role in making the construction industry more adherent to the principles of the circular economy. This makes servitization a highly relevant topic within this industrial context. Second, we expected that studying servitization in the construction industry would increase the quality of our data since the phenomenon of servitization is noticeable in this industry (Visnjic et al., 2018). As a result, we expected informants to be more aware of the topic under investigation, which might encourage them to provide more in-depth and higher-quality insights. To scope this case, we included network actors that play an essential role in the materialization of the service offering under study. This scope emerged during the study, and we relied on references from case informants to identify other relevant informants.

We collected empirical data from a Dutch façade builder (acronym: WinCo) and its surrounding network. A façade is a building's exterior, and its main function is to offer protection against wind, water, and noise while allowing daylight through. Traditionally, this is done by integrating an aluminum system and glass, but recently façades are being equipped with components such as active heating, ventilation, air-conditioning systems, artificial lighting, energy storage, or solar thermal collectors (Prieto et al., 2017). As a result, façades are becoming increasingly complex products that can be augmented with services. To anticipate this trend, WinCo, an SME with approximately 120 employees and yearly revenues between 20 and 30 million euros, is developing and implementing servitization through Façade-as-a-Service (FaaS). With this offering, WinCo targets utility buildings and large residential towers. Offering building components through a service offering is an increasingly observed trend in the construction industry to push the transition to the circular economy. By implementing service business models which involve long-term collaboration between project partners, the construction industry is expected to implement more energy-saving technologies and use sustainable materials (Azcárate-Aguerre et al., 2018). By implementing a servitization strategy, WinCo follows industry pioneers such as Mitsubishi (M-Use®) that have used servitization to adapt to the circular economy.

In FaaS, WinCo envisions integrating its capital goods with operation and maintenance services in a leasing service. So instead of selling the façade to the end-user, WinCo aims to remain the legal owner of the capital good and becomes responsible for maintaining and redeploying or recycling the capital good during and after its life cycle. This new

business model makes sense from both an economic and a circularity perspective. Regarding the economic aspect, WinCo traditionally created value by engineering and producing the capital good; now this is extended by the exploitation and afterlife phases in FaaS. By extending the scope of value creation, WinCo expects to capture some of this value and increase its bottom-line performance. From a circular economy perspective, WinCo becomes economically incentivized to design and maintain the capital good for longevity and circularity, resulting in higher resource utilization. We were mainly interested in the networked character of implementing FaaS and thus in the interaction between WinCo and counterparts. To this end, WinCo was a suitable case because many different network actors got involved with the implementation of FaaS, and many interactions could be studied.

When we got in touch with WinCo, they had already been working on the FaaS proposition for some years. But at this point, they were starting to incorporate FaaS into real projects, which led to multiple tensions in several dyadic relations becoming salient. During the data collection, WinCo continued developing and adapting the FaaS proposition, and they had multiple encounters with several network counterparts, including financiers, suppliers, and project developers. During these interactions, WinCo increasingly realized the importance of adopting an inter-organizational perspective on FaaS and how interdependence between them and network counterparts impacted the successful implementation of FaaS. When we ended our data collection in June 2021, FaaS was still in the implementation phase, but negotiations were taking place to implement FaaS in multiple projects. This indicates that WinCo made progress in shaping its FaaS proposition.

### **2.3.1 Data collection**

We collected data between December 2019 and December 2022 through semi-structured interviews and a workshop. We identified relevant informants who, according to other network actors, have played a significant role in the materialization of FaaS. The majority of our data was collected between December 2019 and June 2021. In this period, we first conducted 25 semi-structured interviews of approximately an hour each, divided into three rounds. The first interview round was executed by the lead author and assisted by a bachelor student who was working on their thesis. Through the interviews, we got acquainted with the case and WinCo's network. Interviewees were identified by asking each one to indicate any important network counterparts for the traditional and FaaS value propositions. We asked general questions about each network counterpart, their role in the network, their relations with network counterparts, and how they envisioned their role in FaaS. The first interview round resulted in a general overview of the actors involved in the network, their roles, and the interrelations between them. The interview guide can be found in appendix 2A.

In the second interview round, executed by the lead author, we gathered more detailed data on dyadic network structures in a FaaS context. We collected dyadic business relationship data on activity links, resource ties, and actor bonds between the interviewee's organization and WinCo as well as between their organization and other

network partners. By asking questions about resource streams, innovation activities, product sales, service delivery, and personal relations between organizations, we got a detailed overview of the business relationship structures. This data was collected for both the current context and the envisioned FaaS setting. As a result of the reflective nature of the questions, interviewees often shared concerns and tensions that emerged from the difference between the current business relationship structure and the envisioned FaaS structure. Also, respondents often reflected on firm-level and network-level factors that played a role in these tensions. Even though we focused on the dyadic level, we asked further questions about these different interaction levels when we considered them to be important context information for our research. The interview guide can be found in appendix 2B.

After the initial data analysis, the lead author and a master student conducted seven additional interviews to ask about specific developments within FaaS and the relevant network relations. Table 2.1 gives an overview and additional information about the interviewees in this third round.

**Table 2. 1 Overview of interview informants**

| #                 | Company description<br>(acronym) | Role of informant              | Date     | Duration       |
|-------------------|----------------------------------|--------------------------------|----------|----------------|
| Interview round 1 |                                  |                                |          |                |
| 1                 | Façade builder (WinCo)           | Manager                        | 09-12-19 | 01:17:23       |
| 2                 | Façade builder (WinCo)           | Manager                        | 03-02-20 | 01:08:04       |
| 3                 | Façade builder (WinCo)           | Technical director             | 03-02-20 | 00:25:19       |
| 4                 | Façade builder (WinCo)           | Executive director             | 03-02-20 | 00:36:18       |
| 5                 | Glass supplier (GlassCo)         | Product manager                | 13-02-20 | 00:54:06       |
| 6                 | System supplier 1 (SysCo)        | Product manager                | 18-03-20 | 00:45:00 (+/-) |
| 7                 | System integrator                | Engineer                       | 31-03-20 | 00:54:15       |
| 8                 | Sunscreen supplier               | Head of digital transformation | 17-04-20 | 00:56:15       |
| 9                 | Developer 1                      | Project developer              | 27-05-20 | 01:15:11       |
| 10                | Architect 1                      | Architect 1                    | 15-06-20 | 00:39:08       |
| 11                | Contractor                       | Project developer              | 25-06-20 | 00:48:38       |
| Interview round 2 |                                  |                                |          |                |
| 12                | System supplier 1                | Product manager                | 13-08-20 | 01:04:32       |
| 13                | Glass supplier (GlassCo)         | Product manager                | 20-08-20 | 00:54:41       |
| 14                | System integrator                | Engineer                       | 01-09-20 | 01:04:13       |
| 15                | Developer 1                      | Project developer              | 25-09-20 | 00:56:19       |
| 16                | Architect 2                      | Architect 2                    | 14-10-20 | 00:56:02       |
| 17                | Financier                        | Banker                         | 15-10-20 | 00:45:00 (+/-) |
| 18                | Façade builder (WinCo)           | Manager                        | 16-10-20 | 01:03:31       |
| Interview round 3 |                                  |                                |          |                |
| 19                | Façade builder 1 (WinCo)         | Manager                        | 07-05-21 | 00:35:39       |
| 20                | Façade builder 2                 | Technical director             | 19-05-21 | 00:47:30       |
| 21                | Façade builder 1 (WinCo)         | Manager                        | 21-05-21 | 00:41:56       |
| 22                | Glass supplier (GlassCo)         | Product manager                | 21-05-21 | 00:44:22       |
| 23                | Developer 2                      | Project developer              | 21-05-21 | 00:22:56       |
| 24                | Façade builder 1 (WinCo)         | Manager                        | 25-05-21 | 00:55:28       |
| 25                | System supplier 2                | Product manager                | 04-06-21 | 00:47:30       |

The interviews were complemented by a workshop with WinCo and some of its suppliers to identify potential systematic challenges or risks of FaaS. This workshop took place on 28 October 2020, lasted 3.5 hours, and was attended by two façade builders, GlassCo and SysCo, a blinds supplier, and a system integrator. The workshop identified general challenges and risks with FaaS network alignment as well as challenges and risks specifically for every participating firm. The outcome was a set of network-wide and company-specific challenges to successfully implement FaaS.

Finally, we collected secondary data, including industrial white papers, practice-oriented research reports, and relevant interviews in trade magazines to complement our primary database. These data sources supplemented the existing database but also allowed for data triangulation to increase the internal validity of our results. In December 2022, we conducted a final interview to collect data about the most recent status concerning the implementation of FaaS.

### **2.3.2 Data analysis**

The process of data analysis consisted of four phases. First, we started by transcribing all interview recordings verbatim and reading them closely to get acquainted with the data. Whenever we came across potentially interesting chunks of data, we highlighted them and made comments. In the second phase, we coded all chunks of data that reflected the current state of the façade network (i.e., thesis), WinCo's envisioned FaaS network (i.e., antithesis), or results from clashes between both (i.e., tensions). In the third phase, we deepened the analysis by structuring the thesis and antithesis separately in terms of the activity-resource-actor (ARA) framework (Håkansson & Snehota, 1995). Specifically, we categorized all codes which were associated with the business relationship structure in the thesis or antithesis situation to one of the ARA dimensions. In the fourth phase, we compared business relationship structures in the thesis and antithesis to find any contradictions and looked at how these contradictions connected to the observed tensions. We investigated why certain tensions emerged and if and how these tensions related to observed contradictions between the thesis and antithesis. While one of the researchers was actively involved with the case, collecting and coding data, the other authors critically reviewed the emerging codes, categories, and interrelations.

## **2.4 Results**

In this section, we present our findings on the business relationships between WinCo, financier, architect, SysCo, and GlassCo. These business relationships were chosen because we found the most notable tensions there that affected the development of the FaaS offering. Table 2.2 summarizes our findings. We identified several tensions that resulted from envisioned activity links, resource ties, and actor bonds which contradicted the status quo. We also found that tensions resulted from first- or second-order contradictions. First-order contradictions refer to a clash between WinCo's purposefully envisioned antithesis and the thesis. Second-order contradictions are unintended ones that result from the interdependence among ARA dimensions. Finally, we found that tensions drove WinCo's servitization process and the development of its

FaaS proposition. The remainder of this section discusses each tension and its effect on the FaaS proposition in depth. We also provide visual representations of the dialectic processes through our conceptual dialectic process model.

Table 2. 2 Summary of inter-organizational tensions

| Figure | Counterpart | Observed tension                     | Contradiction (ARA) | Contradiction (first-order/second-order) | Tension outcome(s) |
|--------|-------------|--------------------------------------|---------------------|------------------------------------------|--------------------|
| 2.2    | Financier   | Lack of trust                        | Actor bonds         | Second-order                             | Synthesis          |
| 2.3    | Financier   | Pricing process rigidity             | Activity links      | Second-order                             | Synthesis          |
| 2.3    | Financier   | Regulative barriers                  | Activity links      | Second-order                             | Synthesis          |
| 2.4    | SysCo       | Reactive attitude                    | Activity links      | Second-order                             | Dissolution        |
| 2.5    | SysCo       | Organizational rigidity              | Actor bonds         | Second-order                             | Dissolution        |
| 2.6    | Architect   | Professional identity threat         | Actor bonds         | Second-order                             | Synthesis          |
| 2.7    | Architect   | Limited control over façade measures | Resource ties       | First-order                              | Synthesis          |
| 2.8    | Architect   | Greenwashing concerns                | Actor bonds         | First-order                              | Synthesis          |
| 2.9    | GlassCo     | Risk distribution                    | Actor bonds         | Second-order                             | Synthesis          |

2.4.1 WinCo - Financier

When offering FaaS, WinCo needs to pre-finance the physical façade, which is earned back through cash flows from the service contract. Because WinCo does not have the financial resources to pre-finance the physical façade themselves, they need to draw funds from external sources. When engaging financiers for FaaS, two tensions appeared in the relationship between WinCo and the financier. These tensions are a lack of trust (Figure 2.2) and pricing process rigidity (Figure 2.3).

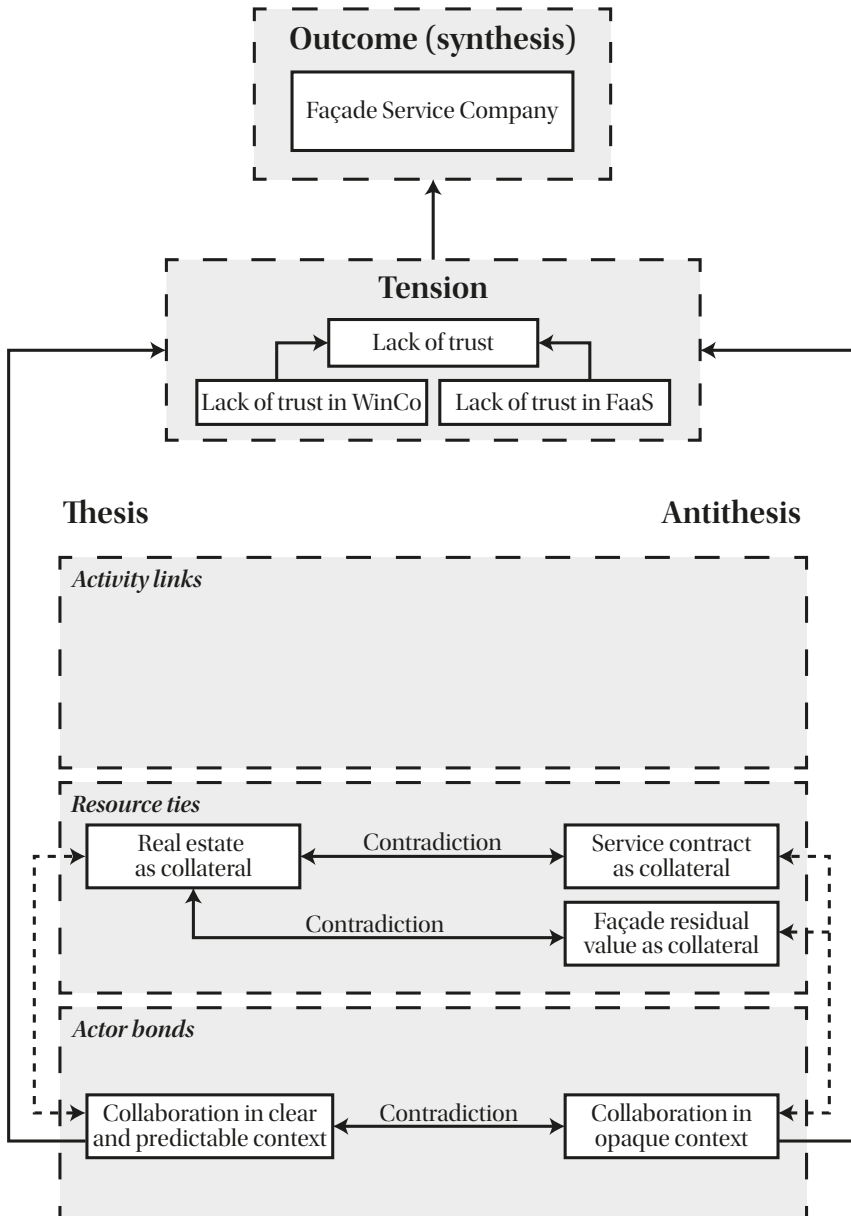
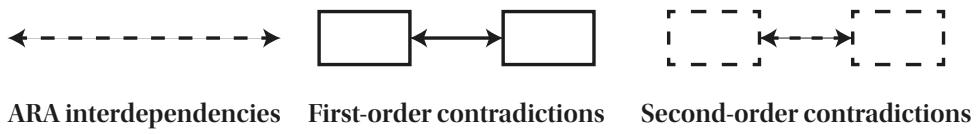


Figure 2.2 Trust tensions WinCo – Financier dyad

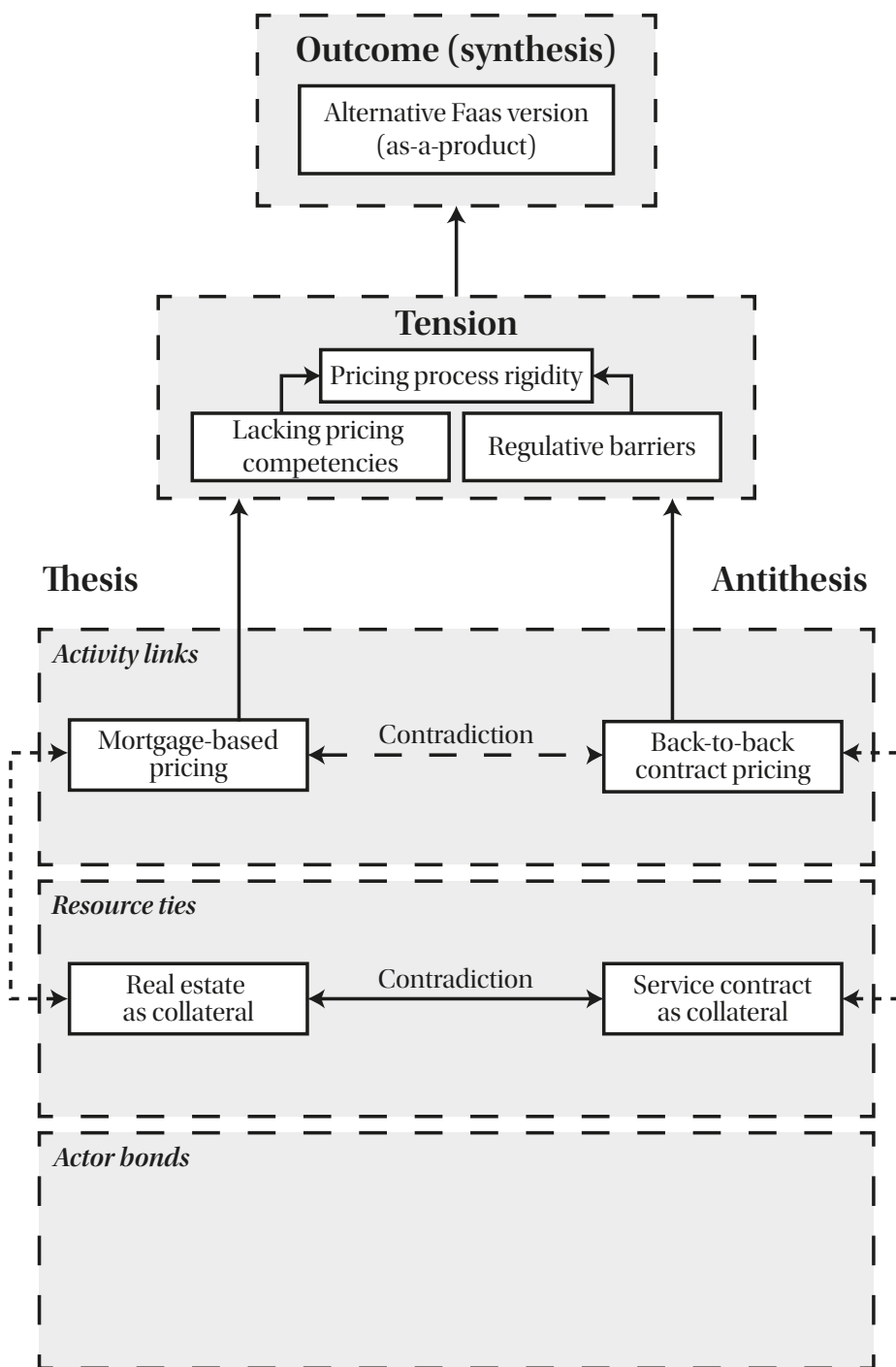


Figure 2.3 Process rigidity tensions WinCo – Financier dyad

#### 2.4.1.1 Lack of trust (Figure 2.2)

To finance FaaS, WinCo looked for novel financing structures and thereby directly contradicted existing resource ties and indirectly, actor bonds. Resource ties in the thesis situation are characterized by mortgage-based loans. Mortgage-based loans stem from the financier's perception of real estate as a trustworthy security that can be seized and resold in case of debt default. Because mortgage-based loans enable a predictable interaction context between WinCo and the financier, resource ties and actor bonds are interrelated in this case. WinCo's envisioned FaaS antithesis directly challenged resource ties and actor bonds indirectly in the thesis situation. Instead of using real estate as collateral, WinCo envisioned that the service contract and façade residual value would serve as collateral. WinCo envisioned this antithetical resource tie because the façade has no intrinsic value in FaaS since façade ownership is separated from the rest of the building and cannot be easily seized in case of debt default. Therefore, mortgage-based loans were not an option. WinCo considered contract-based financing a suitable alternative because cash flows stemming from service contracts are predictable and quantifiable. However, even though WinCo's envisioned resource ties are used by financiers in different contexts (e.g., automotive), they normally provide such financing based on transparent and predictable market conditions, as illustrated in the following quotes:

*"Car leasing companies know exactly what a car will be worth in five years with 250,000 km on the counter" (Banker).*

*"Companies like Mitsubishi [M-Use<sup>®</sup>] or Phillips [Light-as-a-Service] can pre-finance some projects from their own resources [...] and at some point when the market sees "okay, this works" (Banker).*

In contrast to using the envisioned financing structures as mentioned above, using contract-based financing for FaaS is associated with many unknowns such as customer readiness, façade residual value, or WinCo's long-term survivability. In other words, WinCo's envisioned antithesis implies collaboration with financiers in an opaque market situation. Therefore, WinCo's envisioned antithesis led not only to first-order contradictions in the resource layer but to second-order contradictions in the actor layer as well.

Second-order contradictions in the actor layer resulted in trust-related tensions. One part of this tension was related to the ability of WinCo to secure the collateral and generate long-term cash-flow from the service contract. Financiers were concerned with cash-flow instability stemming from volatile resource prices and risks associated with WinCo's long-term ability to meet service-level agreements. The latter could be jeopardized by bankruptcy, for example. Another part of this tension was related to uncertainty regarding the FaaS concept itself. Because the adoption rate of FaaS and façade residual value development were highly uncertain, financiers were not able to calculate risks. This uncertainty resulted in a hesitant attitude from financiers to provide financing as envisioned by WinCo. WinCo responded to these tensions by establishing

‘Façade Service Company’, a joint venture with another façade builder and one of its main suppliers. Because WinCo made multiple agreements with joint venture partners that reduced uncertainties associated with cash-flow stability and façade residual value, financiers were more willing to provide WinCo’s envisioned financing. This situation illustrates how second-order contradictions resulted in trust-related issues that influenced WinCo’s servitization in the form of establishing a joint venture to offer FaaS.

#### 2.4.1.2 Pricing process rigidity (Figure 2.3)

The first-order contradiction between existing and envisioned resource ties also led to second-order contradictions in the activity layer. In a traditional product setting, activity links in which financiers determine appropriate interest rates through sophisticated pricing models are based on real estate as collateral (i.e., resource ties). WinCo’s vision to use service contracts as collateral indirectly contradicted activity links. Instead of using mortgage-based pricing models, financiers must calculate the risks of back-to-back contract structures. These are contractual forms in which agreements between the financier and the Façade Service Company also apply to subcontracting firms participating in this joint venture. By using such legal constructions, financiers reduce risks of cost overruns and loan charges by assigning some financial liability to the firms, including WinCo, that back the joint venture.

Second-order contradictions in the activity layer led to pricing process rigidity-related tensions. As seen in the following quote, financiers have trouble effectively assessing the risks associated with such contracts:

*“To obtain financing [based on back-to-back contracts], colleagues from the risk department must also give their approval. This is a bottleneck because the models used for calculations are now based on mortgage rights. [...] In addition, the question is which return and risk model this product should be placed in.”* (Financier whitepaper)

In addition to lacking pricing competencies, the banking sector is also heavily regulated. This regulation prevents them from freely experimenting with new financing products that suit FaaS. This is reflected in the following quote:

*“One can value them [service cash-flows] in the same way [as bonds], by discounting future cash flows. [...]. Only the banking sector has its own rules, which are agreed upon internationally”* (Banker)

Tensions related to the lack of pricing competencies and industry regulation have prevented financiers so far from providing WinCo with a suitable financing product. This situation illustrates how a contradiction in resource ties can indirectly result in tensions due to the interdependence between resource ties and activity links. As a result of these tensions, financiers could only provide unsecured loans. Because these loans include high-interest rates, they drastically increase WinCo’s cost of capital and jeopardize the FaaS business case. As a response to this tension, WinCo decided to also offer a FaaS

version in which they would sell the façade with a separate service contract. In this case, ownership is still transferred to the end-user but WinCo remained in touch with their products through a long-term service contract.

#### **2.4.2 WinCo – SysCo**

SysCo is a multinational company that sells high-quality aluminum profile systems at a premium price point. SysCo's product is the most important component of WinCo's façades. Because WinCo has worked with SysCo for a long time and has had good experiences with their products and services, they wanted to involve them in the development of FaaS. However, when interacting with SysCo about FaaS, WinCo experienced a reactive attitude (Figure 2.4) and organizational rigidity as tensions with SysCo (Figure 2.5).

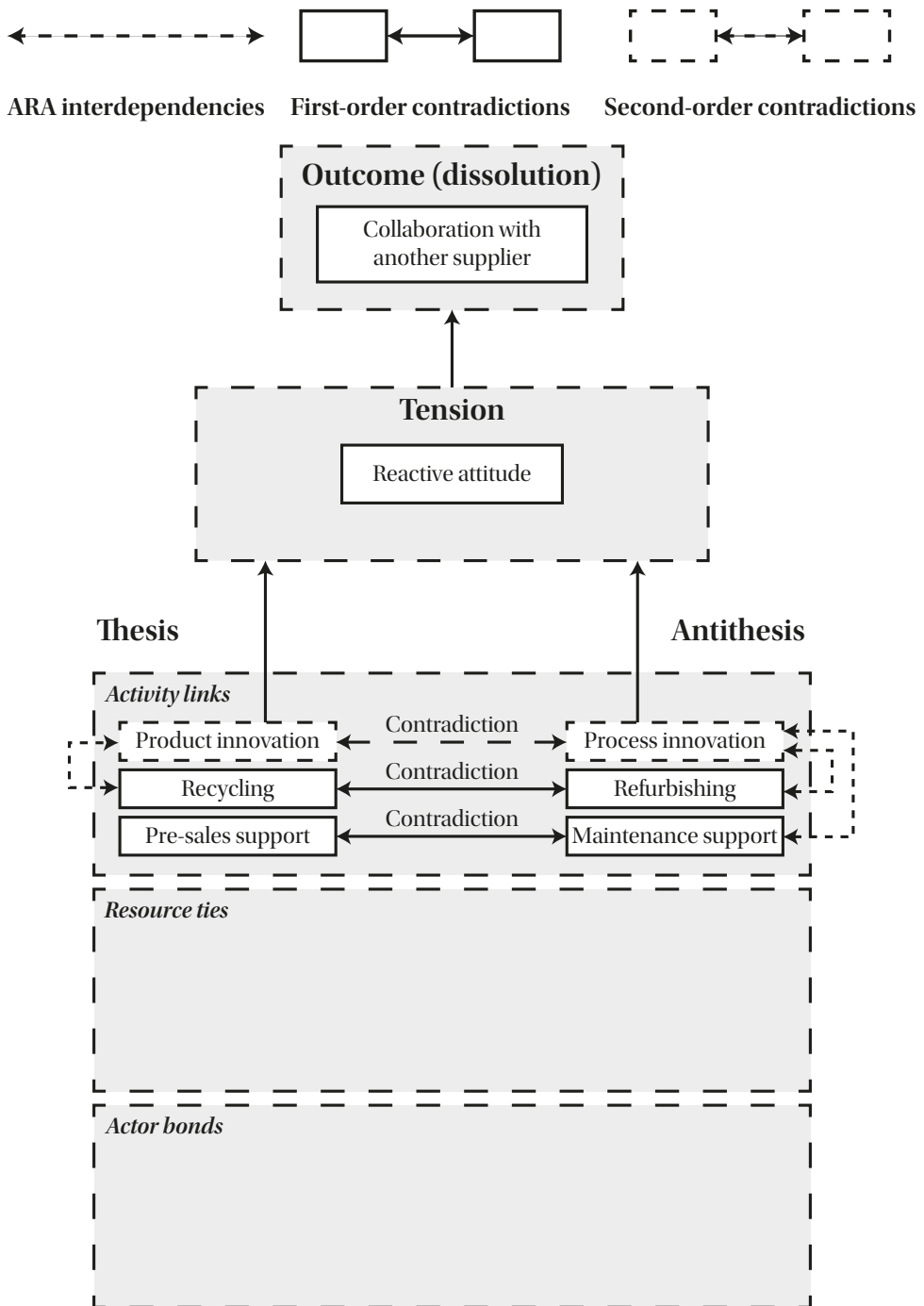


Figure 2. 4 Reactive attitude tensions WinCo - SysCo dyad

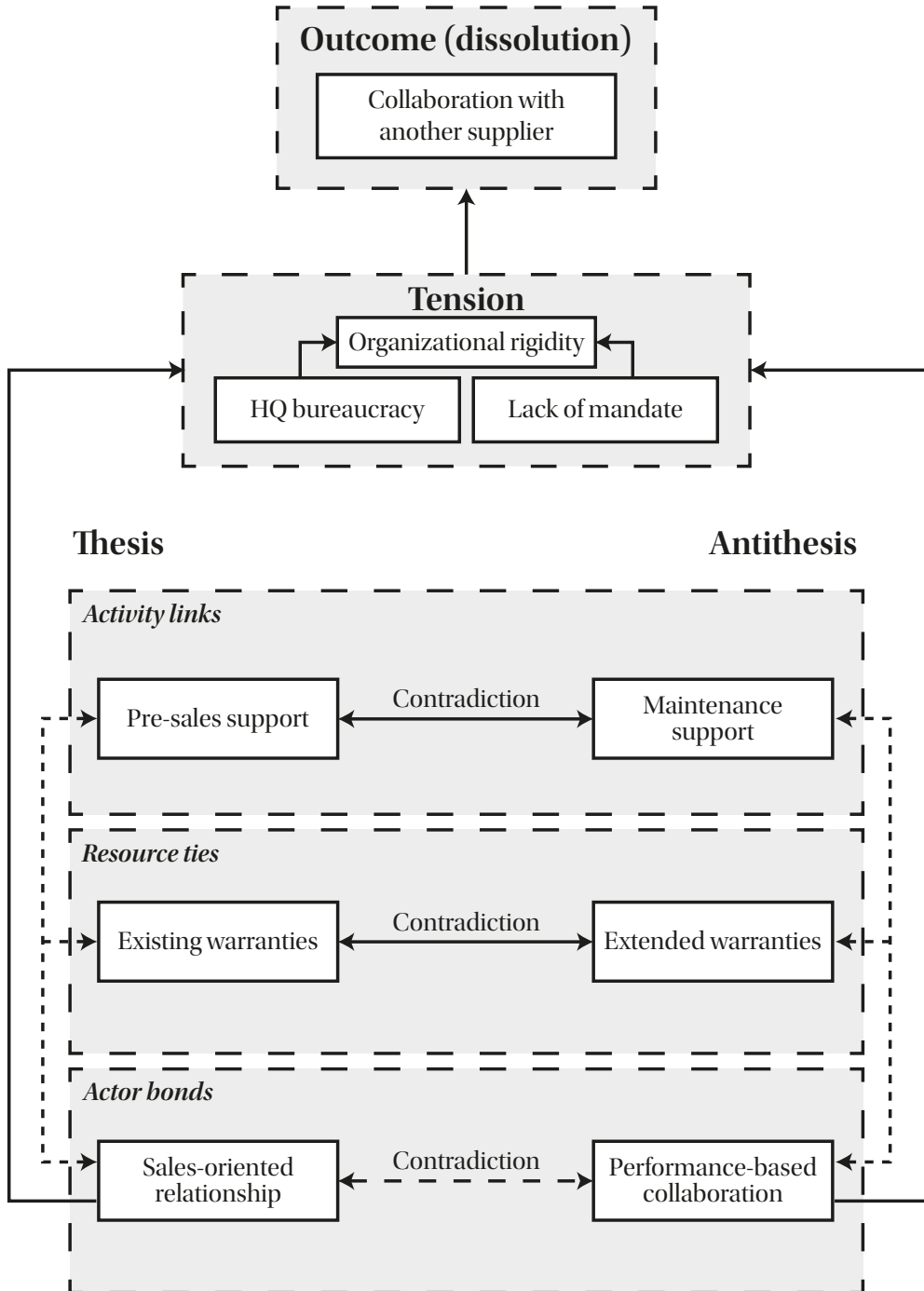


Figure 2. 5 Rigidity tensions WinCo - SysCo dyad

#### 2.4.2.1 Reactive attitude (Figure 2.4)

In the thesis situation, SysCo strongly focuses on pre-sales support in the tendering and engineering phases to justify its premium prices. Furthermore, SysCo aims to maintain its brand as a premium profile supplier by offering products that contain state-of-the-art technology that fits with industry requirements. Therefore, SysCo significantly invests in product innovation to improve the thermal performance and energy consumption of its products. To respond to industry trends regarding ecological sustainability, SysCo aims to improve product recyclability through product innovation. Because both recycling and pre-sales support activities are driven by product innovation, these activity links are interdependent. By envisioning activity links in which SysCo provides maintenance support during the façade exploitation phase and engages in façade refurbishing when the façade returns, WinCo directly contradicts existing activity links. Also, these first-order contradictions imply second-order contradictions because maintenance support and refurbishing require process innovation on the part of SysCo while they are traditionally focused on product innovation.

This second-order contradiction led to tension between WinCo and SysCo, which was characterized by a reactive attitude from SysCo towards FaaS. This tension is illustrated in the following quotes:

*“SysCo has existing business processes and a certain strategy, which is why it is difficult to roll out a different business concept. We try to support WinCo’s servitization within our existing processes”*  
(Product manager, SysCo)

*“We thought [SysCo] innovated a lot, but it appeared that it was not the innovation we needed”* (Manager, WinCo).

It appeared that SysCo lacked the capacity and knowledge to implement WinCo’s requirements. This situation illustrates how contradictions within the activity layer can be interdependent as transactional activities (i.e., maintenance involvement and refurbishing) were associated with transformational activities (i.e., process innovation activities). SysCo’s reactive attitude ultimately led to the dissolution of the business relationship. After interacting for more than a year, WinCo realized that SysCo would not implement its requests and dissolved the relation with SysCo in the FaaS context. Instead, WinCo partnered with one of SysCo’s competitors to support FaaS. This new supplier was included in the joint venture, and they have also collaboratively developed an innovative and modular façade. In this instance, second-order contradictions within the activity layer and SysCo’s reactive attitude affected WinCo’s servitization in terms of network configuration. So instead of partnering with its main supplier, the tensions made WinCo approach a different supplier to offer FaaS.

#### 2.4.2.2 Organizational rigidity (Figure 2.5)

In addition to extensive pre-sales support, SysCo's products came with ten years of warranty on the profile system and one year on peripherals such as locks, hinges, and electronic components. Pre-sales support and existing warranties were based on a sales-oriented relationship in which SysCo was mainly focused on optimizing product sales. WinCo directly contradicted existing activity links and resource ties by envisioning an antithesis in which SysCo would get involved in maintenance activities and provide extended warranties of fifteen years on all components. These first-order contradictions in the activity and resource layers also resulted in second-order contradictions in the actor layer. By requiring SysCo to get involved with maintenance activities and provide extended warranties, WinCo assumed a performance-based collaboration with SysCo. This performance-based collaboration also assumed a new role for SysCo to become a "resource financier" instead of a company that sells kilograms of aluminum. Such a collaboration contradicted the sales-oriented actor bond.

This second-order contradiction in the actor layer led to tensions in terms of a lacking mandate and HQ bureaucracy. Because WinCo was traditionally in contact with a sales representative, this person lacked the mandate to implement WinCo's requirements. Consequently, this person had to steer FaaS requirements through HQ, but this was complicated, as illustrated in the following quote:

*"But you have to deal with the [national] mentality,  
and that can be difficult at times. I now have a person,  
[SysCo manager], who shifts up a bit faster there.  
But gosh, that's all so bureaucratic, awful." (Manager, WinCo).*

So, despite WinCo's contact person being enthusiastic about FaaS and willing to implement WinCo's envisioned changes, he lacked the mandate to implement those innovations himself and failed to steer the initiative through the organization. In this instance, the envisioned resource ties and activity links also involved a newly envisioned actor bond. The contradiction between actor bonds in the thesis and antithesis resulted in organizational rigidity to adapt to FaaS. This tension exacerbated the reactive attitude identified earlier and contributed to the demise of the business relationship between WinCo and SysCo in the FaaS setting.

#### 2.4.3 WinCo - Architect

Architects have an important role in the early stages of a construction project. Because they are involved in design and engineering activities and are responsible for the visual appearance of buildings, they greatly influence the adoption of FaaS. Therefore, they are important counterparts to be taken into account by WinCo. When WinCo engaged architects in FaaS, tensions regarding professional identity threat (Figure 2.6), lack of control over façade components (Figure 2.7), and greenwashing concerns emerged (Figure 2.8).

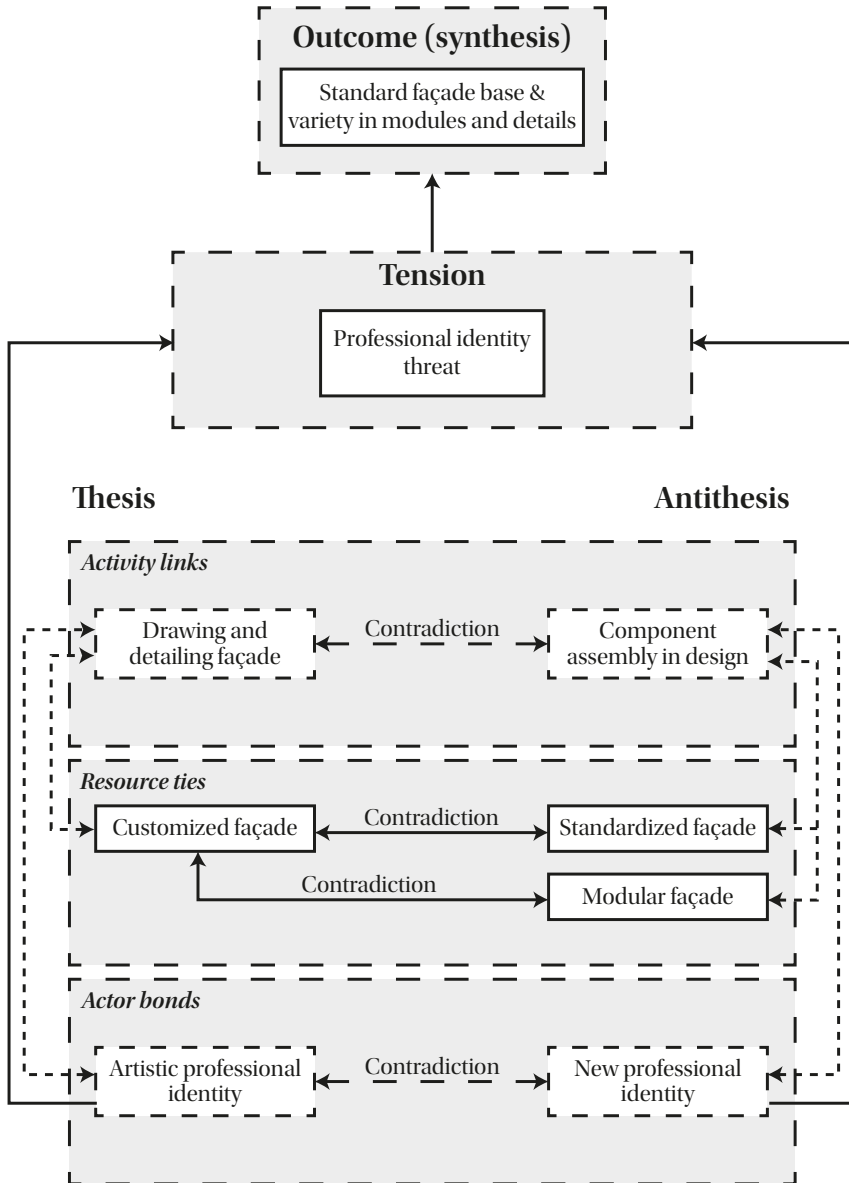
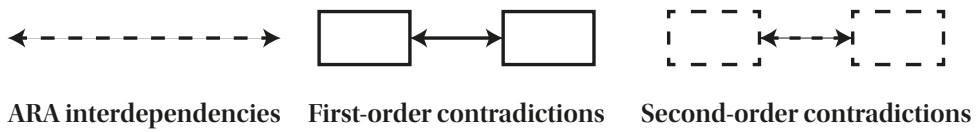


Figure 2.6 Identity tensions WinCo – Architect dyad

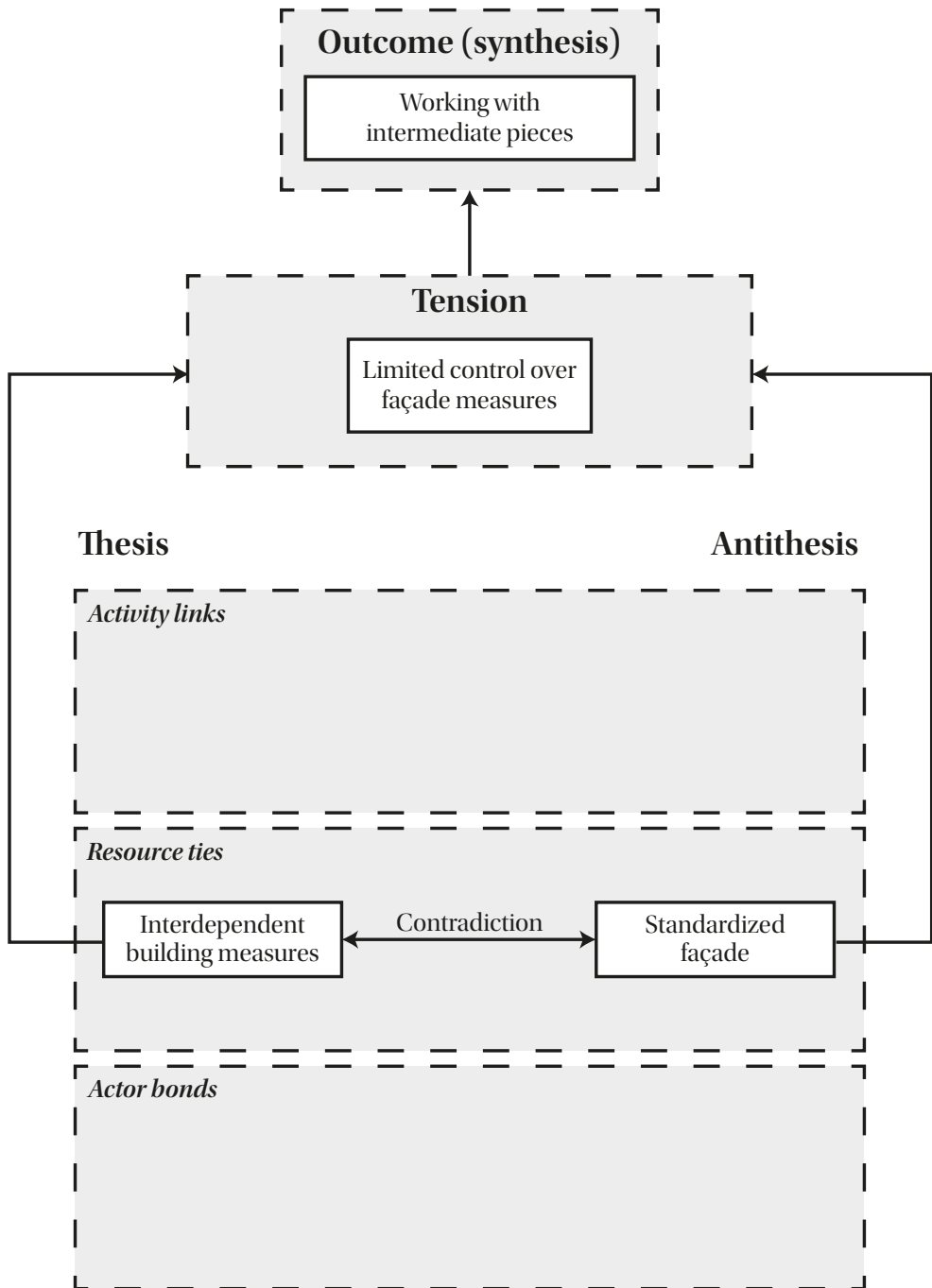


Figure 2. 7 Control tensions WinCo - Architect dyad

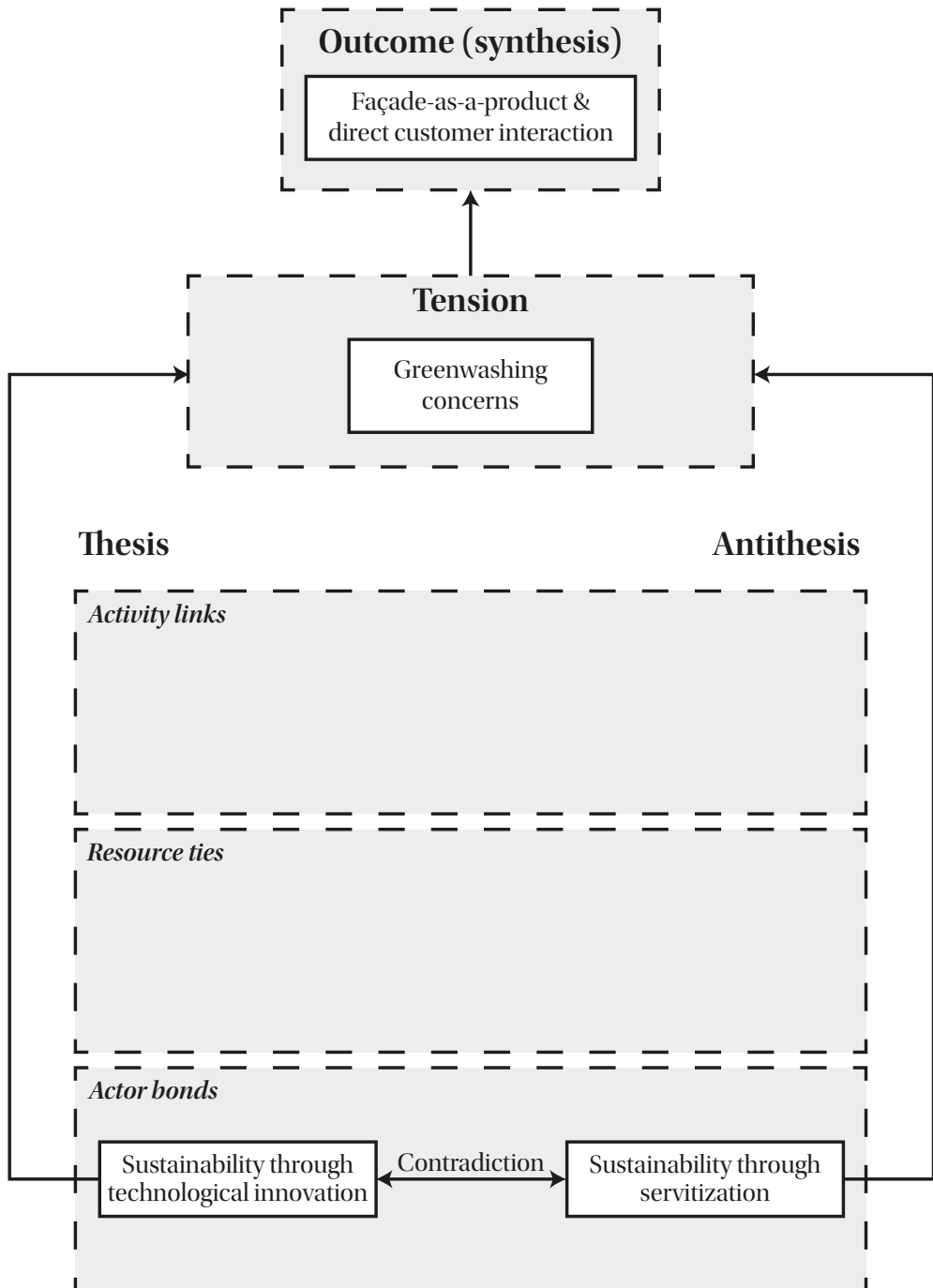


Figure 2. 8 Greenwashing tensions WinCo - Architect dyad

#### 2.4.3.1 Professional identity threat (Figure 2.6)

By increasing the standardization and modularity of the façade, WinCo directly contradicted existing resource ties and indirectly, existing activity links and actor bonds. In the thesis situation, actor bonds are characterized by architects considering themselves artists because they have been educated as such. A good architect is considered one who can express artistic freedom within the given cultural and geographical context of the assignment. This actor bond is interdependent with activity links because architects used to draw and detail façades themselves to express their artistic identity. Because the façade has a great influence on the visual appearance of a building, architects relied on customized façades to express their artistic freedom (i.e., resource ties).

WinCo directly challenges these resource ties by envisioning standardized and modular façade design in FaaS. Standardized and modular façades enable WinCo to minimize maintenance costs and maximize façade reusability. The first-order contradiction in terms of resource ties created second-order contradictions in the activity and actor layers. Enhanced façade standardization and modularity required architects to adapt their design activities to component assembly-based design. Because drawing activities are strongly related to an architect's professional identity as an artist, these alternative activity links had consequences for actor bonds as well. By contradicting existing resource ties, WinCo indirectly contradicted the professional identity of architects as well.

This second-order contraction resulted in tensions related to professional identity threats by architects as illustrated in the following quote:

*"That whole view [of the architect profession] is changing considerably because some form of component assembly has crept into our profession [...]. We work with a catalog of façade systems. [...], it has become a different kind of profession but many architects do not yet accept that" (Architect 1).*

This situation illustrates how a contradiction in the resource layer (i.e., standardized façades) resulted in other contradictions in the activity layer (i.e., component assembly design) and actor layers (i.e., different professional identity). As a response to this tension, WinCo decided to use standardized base systems with equal measures but still allow a high variety in detail such as color, style, and peripherals. This enabled WinCo to standardize maintenance activities and easily redeploy façades while reducing the impact of FaaS on the artistic identity of architects by providing room for customization. They realized this balance between standardization and customization by developing a modular façade concept and a digital "module library" that architects could use in their engineering processes. This library consisted of modules that fitted in the standardized base of the façade but enabled many different combinations so that architects could still design a custom façade. Thus, the second-order contradictions in the actor layer and the resulting identity threat tensions drove FaaS development to balance standardization and modularization.

#### 2.4.3.2 Limited control over façade measures (Figure 2.7)

WinCo's envisioned façade standardization also contradicted thesis resource ties in terms of various building components. For example, the engineering of the building skeleton, interior layout, and façade measures are interdependent. However, these elements have their own requirements such as safety norms, customer requirements, and aesthetical requirements, which may conflict at times. So, in the thesis situation, architects must balance these competing requirements in their design activities and adapt façade measures to other building components. This interdependence between building components and the need to adjust them accordingly contradicted WinCo's envisioned standardized façade measures. Because of component interdependence, architects could not always adopt the standardized measure envisioned by WinCo in the FaaS antithesis.

This first-order contradiction in the resource layer led to tensions between WinCo and architects. Even if architects were willing to implement FaaS and the standard measure, resources from other actors sometimes prevented them from doing so:

*"It is not a rejection but there are other factors that determine the sizing of the façade. At [project] it was also removed; we first had a nice modular size but then for some reason [architect] deviated from that pattern"* (Manager, WinCo).

This case illustrates how tensions emerged from the first-order contradiction between envisioned and existing resource ties. By using fitting pieces, WinCo was still able to implement standardized façades and mitigate tensions with architects. However, WinCo indicated that this is a suboptimal outcome as these fitting pieces cannot be reused and will end up as landfill after the façade is deconstructed. This instance shows that WinCo's servitization was driven by first-order contradictions in the resource layer and the resulting tensions. The tension required WinCo to adapt the technical design of the façade by using fitting pieces to work around the tension.

#### 2.4.3.3 Greenwashing concerns (Figure 2.8)

The third contradiction between the thesis and WinCo's envisioned antithesis was observed in the actor layer. Like many actors in the construction industry, architects also actively focused on implementing a circular strategy and adapting their business accordingly. In the thesis situation, architects envisioned that a circular construction industry is achieved through enhanced building flexibility and detachability. Ideas of future developments regarding the circular economy were focused on technological developments. These actor bonds in the thesis partially contradicted WinCo's envisioned antithesis of circularity through service contracts. While improving flexibility and detachability through technological innovation is inherently part of FaaS, WinCo also considered new contractual forms to be important. According to WinCo, these contractual forms are essential to creating proper incentives for a circular supply chain in the construction industry. Thus, WinCo's view on how to implement a circular economy in the construction industry partially contradicted those of architects.

This clash between thesis and antithesis resulted in greenwashing concerns on the part of architects. Architects were skeptical about the effectiveness of the new contract forms to improve circularity. The following quote illustrates an architect's perception of the circularity potential of FaaS:

*"The incentives to adopt service concepts are mainly financially driven as far as I am concerned [...], they are too often framed as sustainable and circular and I just have my doubts about that" (Architect 2).*

The architect is exhibiting a lack of trust in WinCo's incentives. Also, architects indicated that they doubted whether WinCo would be able to ensure the circular process given the long-term character of its service contract. As the industry and the entire world will change significantly in fifteen years (i.e., duration of the service contract), they were skeptical about a detailed refurbishing strategy in FaaS. Tensions emerged because of contradictory actor bonds in the thesis and antithesis. WinCo responded to this tension by offering their circular façade as a product and directly contacting end-users after the circular façade was integrated into the building. The option to sell the façade as a product and negotiate a service contract afterward was enabled by the design principles of WinCo's circular façade because they were of high quality and based on principles of modularity. As such, the product design enabled WinCo to circumvent architects when they did not want to adopt façade-as-a-service.

#### **2.4.4 WinCo - GlassCo**

GlassCo provides WinCo with several types of glass, which is another important component of WinCo's façades. Because GlassCo was one of WinCo's main glass suppliers and in close geographical proximity, WinCo decided to involve GlassCo in the development of FaaS. We identified one tension related to risk distribution in the relation between WinCo and GlassCo (Figure 2.9).

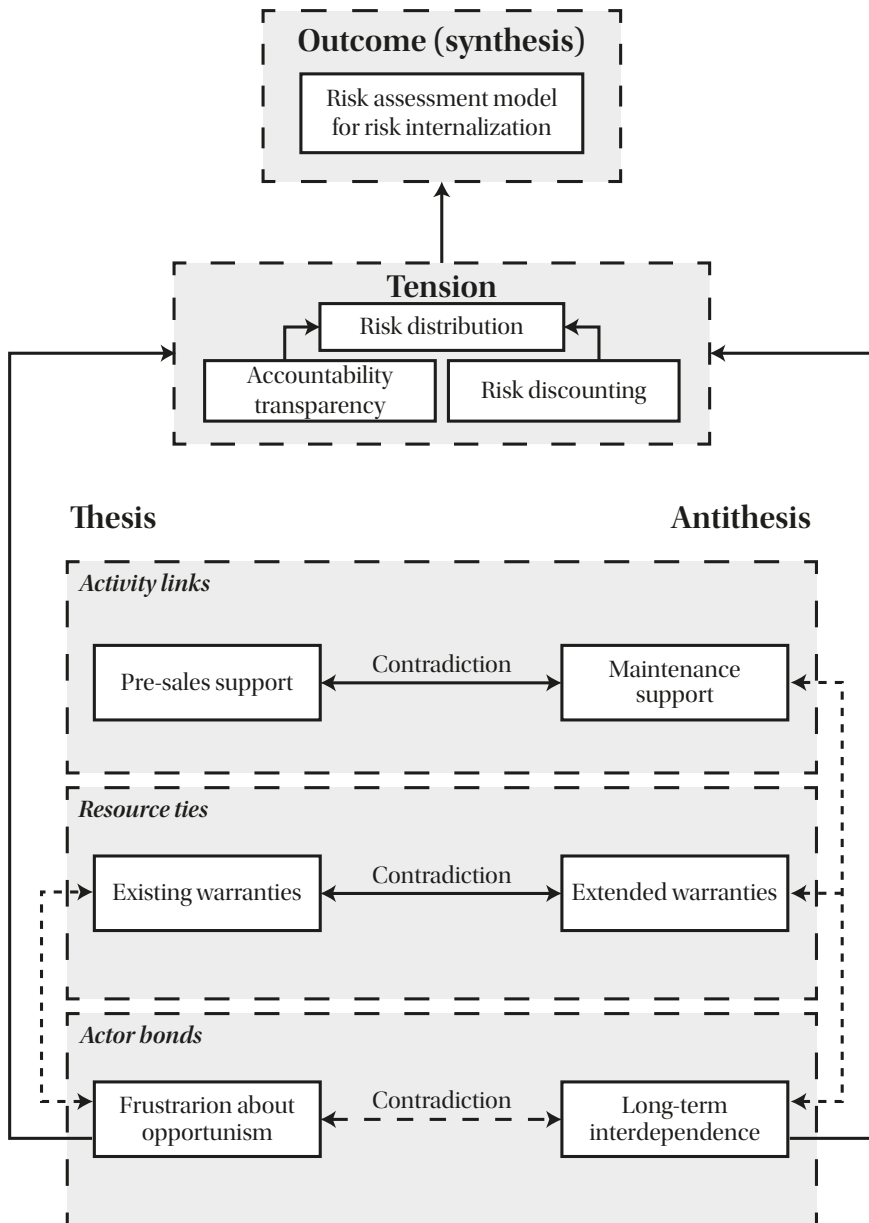
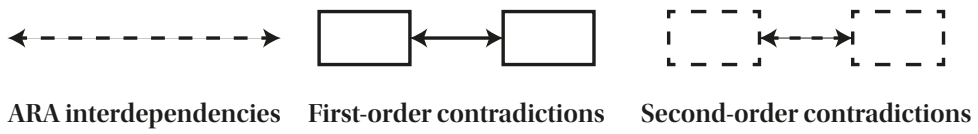


Figure 2.9 Risk distribution tensions WinCo – GlassCo dyad

#### 2.4.4.1 Risk distribution (Figure 2.9)

In the thesis situation, GlassCo strongly focuses on extending its pre-sales support services by offering e-learning modules or tracking services. Also, GlassCo provides its glass products with ten years of warranty on the glass and two years on peripheral products. However, without specifically referring to WinCo, GlassCo was frustrated about opportunistic behavior concerning these warranties. GlassCo indicated that many malfunctions, for which they were held accountable, were not caused by GlassCo's wrongdoings but by other actors instead. For example, electrical components may not work because wires are accidentally cut in the production process of the façade builders or windows are not installed according to the prescribed standards. For FaaS, WinCo envisions an antithesis in which GlassCo gets involved in the maintenance activities and provides extended warranties. This antithesis involves first-order contradictions in the activity and resource layers by contradicting existing warranties and pre-sales support activities. It also created a second-order contradiction in the actor layer. By requiring maintenance support and extended warranties, WinCo implicitly envisioned actor bonds to become characterized by long-term interdependence beyond the sales phase. Such long-term relationships contradicted existing actor bonds characterized by opportunistic behavior because long-term interdependence would make GlassCo more vulnerable to opportunism.

The second-order contradiction in the actor layer resulted in risk distribution tensions, despite GlassCo's insistence that they trusted WinCo not to take advantage of them. The first part of this tension was related to accountability transparency. GlassCo indicated that they would only provide extended warranties if the transparency regarding malfunction accountability was improved. In other words, the responsible actor can be pointed out and held accountable in case of a malfunction. Along with enhanced accountability transparency, another tension related to risk discounting is illustrated in the following quote:

*"If they want to implement their products, the business case has to be right. The benefits must be very clear, including the financials. And they do think along with it, but they still discount risks and that makes it expensive and less interesting for my customers"* (Manager, WinCo).

The second part of this tension was related to risk discounting. To mitigate the risks associated with extended warranties, GlassCo made very conservative estimates about the potential breakdowns of their products. Consequently, their offer was unattractive for WinCo concerning their FaaS business case. This situation illustrates how first-order contradictions in the activity and resource layers can indirectly lead to tensions through second-order contradictions in the actor layer. Because it did not seem that GlassCo would internalize the operational risks of their products in FaaS, WinCo opted for a compromise. The outcome of this tension is that WinCo continued working with GlassCo's existing support activities and warranties but asked them for an elaborate risk model of their product. This model enabled WinCo to better manage risks associated with GlassCo's products. So instead of internalizing the risks, WinCo required GlassCo

to actively support them in managing risks.

## 2.5 Conclusion and discussion

This paper sets out to better understand the emergent nature of servitization by investigating how tensions in inter-organizational business relations drive this transformation. By integrating dialectic theory and the ARA framework from the IMP tradition, we developed a conceptual model to study the emergence of inter-organizational tensions. Drawing on a single case study in the construction industry, we found that servitization in a circular economy setting is associated with multiple inter-organizational tensions. These tensions had a driving effect on the course of servitization as they influenced the content and implementation of the strategy. Tensions resulted from the clash between contradictory activity links, resource ties, and actor bonds in the thesis and antithesis. Furthermore, we identified first-order and second-order contradictions. These contradictions differentiate anticipated outcomes (from the clash between the status quo and the envisioned situation) from unintended outcomes (stemming from tensions that are indirectly triggered through the interdependence of ARA dimensions). This finding implies that firms cannot fully predict upfront what inter-organizational tensions will arise and how these tensions will play out in terms of their effect on the envisioned servitization strategy. As such, inter-organizational tensions are an important mechanism to understand the emergent and unstructured nature of servitization.

Our findings have multiple implications for the servitization literature. Our first contribution is that we studied servitization-related network tensions in a circular economy context and found that they partly overlap with servitization-related network tensions in other settings. Tensions have previously been studied in a servitization context (Reim et al., 2019) and a technological and digital innovation context (Burton et al., 2016; Galvani & Bocconcelli, 2022; Tóth et al., 2022). Our findings overlap with these works because we also found inter-organizational tensions associated with an entrenched organizational culture, identity threats, and diverging priorities among network actors. Also, we found that implementing a servitization strategy to adapt to the circular economy can be hindered by regulatory frameworks, lack of service provision capabilities, and contact persons with limited influence within their organization. These findings specifically overlap with the tensions found by Burton et al. (2016) and Reim et al. (2019). As such, these inter-organizational tensions may be generalized to a setting in which servitization is applied for enhanced circularity.

However, we also identified two inter-organizational tensions that have not been observed before and may become more salient when servitization is implemented for circular economy purposes. One of them relates to a lack of confidence in the economic feasibility of a service proposition and little trust in the circularity incentives driving the implementation of a servitization strategy. The former is especially troublesome for smaller firms that strongly rely on external financiers for prefinancing the service business. In contrast to large firms, smaller firms do not have the financial means to cover risks associated with new service concepts that have not been proven, and they

need financiers to internalize these risks. The latter reflects the mixed results about the effect of servitization on ecological sustainability. On the one hand, servitization could incentivize firms to design more durable and reusable products (Yang & Evans, 2019). On the other, it may create a rebound effect with increased and more reckless use of resources (Herring & Sorrel, 2009; Tukker, 2004). The ambiguity regarding the effect of servitization on circular performance creates suspicion about the “real” incentives of firms that predicate implementing servitization because of its circular character.

The other inter-organizational tension we found relates to risk distribution among the servitizing firm and its counterparts. By retaining ownership of the product, servitizing firms internalize more operational risks related to the proper functioning of the product and its disposal. Our findings showed that firms may need to externalize some of these operational risks to suppliers to balance the increased internalized risks and decreased control over their underlying activities due to reliance on external actors (Visnjic et al., 2018). However, when suppliers are not yet organized to maintain their components or take them back after their lifecycle, they may refuse to take on these risks. Alternatively, suppliers estimate these risks very conservatively and make offers to internalize these risks that are economically infeasible for the service business case. Consequently, servitizing firms may be overexposed to operational risks they cannot control or influence. Failing to effectively deal with these tensions may endanger the servitization initiative.

Our second contribution is the development of a dialectic process model to explain the emergent nature of servitization. We extend the works of Tóth et al. (2022) and Galvani & Bocconcetti (2022) by explicitly linking inter-organizational tensions with servitization progress. Tensions can indeed be a source of complexity, but this complexity also influences the course of servitization by affecting the content and implementation of the servitization strategy. We found that some tensions can be overcome by synthesizing the underlying contradictions while others resulted in the dissolution of existing business relationships and establishment of new ones. Thus, we portray servitization as a political process that involves negotiation, integration, and confrontation to work through or around inter-organizational tensions (Mørk et al., 2018).

This view on servitization implies that servitizing firms must match their offering with the goals and preferences not only of the customer but also of counterparts in the wider network (Mikhailova & Olsen, 2018). The iterative dialectic process of thesis, antithesis, and synthesis opposes the view that servitization is a linear and rather predictable process (Oliva & Kallenberg, 2003). Instead, servitization is driven by clashes between different and opposing agendas in the networked environment. Because interdependencies between agendas and priorities are too complex for firms to fully comprehend (Henneberg et al., 2006; Mouzas et al., 2008), servitization appears as an unstructured and incremental process that should be approached with an agile mindset (Kowalkowski et al., 2012; Martinez et al., 2017; Sjödin et al., 2020).

Our third contribution is opening the black box of inter-organizational tensions by adopting the activity-resource-actor (ARA) framework. Because the ARA dimensions

constitute the basic micro-elements of business relationships (Kaartemo et al., 2020), they operate at a lower aggregation level than the dialectic process and constitute the thesis and antithesis. We consider the ARA dimensions as micro-foundations of the dialectic process (Felin et al., 2015a). Our findings illustrate that the implementation of a servitization strategy can lead to inter-organizational tensions through contradictions in various business relationship dimensions. The ARA dimensions correspond with the micro-foundation categories of Felin et al. (2012), which include individuals, processes, and structure. Based on our findings, we state that inter-organizational tensions can result from contradictions in terms of (i) value creation activities (i.e., activity links – processes), (ii) technical and offering structures (i.e., resource ties – structures), and (iii) cognitions and personal relations (i.e., actor bonds – individuals). Using the ARA framework as a micro-foundation for inter-organizational tensions may pave the way for a better understanding of why some tensions are functional while others lead to business relationship dissolution (S. Fang et al., 2011; Mele, 2011; Vaaland & Håkansson, 2003).

Also, the interdependence between ARA dimensions enabled us to increase our understanding of the unpredictable and emergent nature of servitization. In some instances, the interdependence between ARA dimensions led to second-order contradictions. That is, by contradicting one ARA dimension that is interdependent with another dimension, a servitizing firm may indirectly contradict multiple relationship dimensions by directly contradicting a single one. For example, a change in product design directly contradicts existing resource ties but may indirectly challenge existing professional identities (i.e., actor bonds) because actor bonds and resource ties are interdependent. This finding implies that multiple micro-foundations of inter-organizational tensions (i.e., ARA dimensions) are interdependent, as described by (Felin et al., 2012). The interdependence among ARA dimensions explains why firms encounter unforeseen problems when innovating in a networked context (Mørk et al., 2018). Because these problems are often not recognized until actors engage with network counterparts, developing network insight through heedful interaction with counterparts is essential in the synthesis process (Mouzas et al., 2008).

## 2.6 Practical implications

Our findings have multiple implications for practitioners. First, when managers implement a servitization strategy to adapt to the circular economy and need to mobilize external actors, they must anticipate several tensions. Some tensions are associated with servitization in a general sense, such as established product culture, identity threats, risk distribution, and conflicting priorities among network actors. When managers use servitization as a means to become circular, they must anticipate potential skepticism regarding their incentives and the economic feasibility of the initiative by being clear and transparent about how servitization will improve circularity KPIs.

Second, managers should consider servitization as a dialectic process that involves the synthesis of the existing business relationship structure and an envisioned structure in a servitization context. This iterative process implies that managers must

continuously adapt their offering not only to customer needs (Sjödin et al., 2020) but also to those of other counterparts. Managers can use our dialectic process model to map existing network configurations and their envisioned network configurations to identify contradictions and spot potential inter-organizational tensions. However, because networks are complex and understanding of one's business network is in a constant state of evolution (Mouzas et al., 2008), managers should adopt an agile approach towards servitization. Our dialectic process model can aid them in explicitly formulating their underlying assumptions concerning servitization and engaging in focused experimentation accordingly.

Third, managers should understand that business relationships with network counterparts have multiple dimensions and that these dimensions are interdependent. Consequently, they should be aware that challenging a single relationship dimension (e.g., product design) may indirectly contradict other relationship dimensions and result in inter-organizational tensions. They should engage in multiple episodes of heedful and fact-based interaction with counterparts to understand these network complexities and be able to effectively counter them in the synthesizing process.

## **2.7 Limitations and future research**

Our research has some limitations that pose opportunities for further research. First, generalizability is one of the main limitations of case studies, and future research could investigate network tensions in other contexts. We have studied servitization in the Dutch construction industry because circularity and servitization are important topics in that context. However, the construction industry is characterized by project-based collaboration and may therefore yield different results compared to manufacturing machinery or transportation settings.

Second, the abductive and exploratory research design limited the formulation of specific and focused interview questions before engaging with the case study. This limitation affected the strength of our conclusions, which require validation through a deductive research design. For example, future research can assess the validity of our process model in different case contexts or adopt our process model to shape servitization in an action research design.

Third, future research may adopt a network perspective when studying dialectic tensions in servitization. While we found some indications that firm-level and network-level dimensions played a role in the emergence of tensions, we have not systematically included these analysis levels in our research. (Öberg et al., 2020) demonstrated that resolved tensions within a dyad may resurface somewhere else in the network. Adopting a longer timeframe in combination with a network perspective might shed light on the interrelation of dialectic tensions across network relations. Also, a longer timeframe might increase our understanding of dialectics as a cyclical process in which the outcome of one dialectic cycle may be the start of another one (Farjoun & Fiss, 2022; Raisch et al., 2018).

# Appendix 2A – Interview outline round 1

## Introduction

1. Can you briefly tell us about your organization in terms of product/service, annual turnover, number of employees?
2. Can you briefly tell us about your position within the organization?

## Value proposition

3. Can you describe how a façade supports the core business of the customer? Be as specific as possible.
4. What are the requirements (most frequently mentioned) that customers place on a façade for carrying out their core business and achieving KPIs?

## Network structure

5. What is your role in the process of producing and/or installing a façade?
6. What are other roles in the network that are important to realize a façade?
7. Who are the main actors performing these roles?
8. What does the flow of input and output look like in the network? Who supplies what to whom? Think of components, information, knowledge, etc.

## Relationships and collaboration

9. Can you describe your relationship with [WinCo]?
10. Can you describe your relationship with other organizations involved in the production and installation of a façade?
11. Can you say something about the relationship with the most important actors in your network in terms of:
  - a. Power relations
  - b. Commitment
  - c. Trust
  - d. Mutual understanding
  - e. History of the collaboration
  - f. Future expectations

### **Façade-as-a-service**

12. To what extent are you already involved in [WinCo's] FaaS idea?

13. What implications would FaaS have for your role or position in the network?

14. What implications would FaaS have for the network and the collaboration between parties?

15. What implications would FaaS have for the customer's role throughout the process

## Appendix 2B – Interview outline round 2

### Introduction

In the first interview round, we got a global picture of the network around the “façade-as-a-service” proposition. This interview round aimed to systematically inventory all possible relationships within this network using the activities, resources, and actors framework.

Based on the previous interview round, we identified the following actors.

- |                                                                        |                                           |                                   |
|------------------------------------------------------------------------|-------------------------------------------|-----------------------------------|
| • Façade builder                                                       | • System integrator                       | • Bank/Financier                  |
| • Glass supplier                                                       | • Installer                               | • Government/legislator           |
| • Profile system supplier                                              | • Architect                               | • Urban mining organization       |
| • Sunscreen supplier                                                   | • Project developer                       | • End-user/owner/facility manager |
| • Manufacturer of other systems (e.g., ventilation or heating systems) | • Service supplier - property maintenance |                                   |

1. Of the above actors, can you indicate which actors you have a relationship with?
2. Can you describe (and/or validate existing data) per direct relation and in the context of Façade-as-a-Service:

A. The relationship in general

B. Resource ties

- I. What does the value stream look like, for example, the exchange of products, services, information, knowledge, and/or money?
- II. Total volume purchased/sold in relation to the total.

C. Activity links

- I. To what extent is the way you sell products or services influenced by the other party, and/or vice versa?
- II. To what extent is the way you produce products influenced by the other party, and/or vice versa?
- III. To what extent is the way you perform services influenced by the other party, and/or vice versa?
- IV. To what extent are you involved in the other party's innovation and to what extent is the other party involved with yours?

D. Actor bonds

- I. What does the typical interaction look like (e.g., purely business, informal, what kind of things are discussed, under what circumstances, etc.)?
- II. How do you see the identity of this actor?
- III. Can you say something about trust and commitment in the relationship?





# 3

## **INSTITUTIONAL COMPLEXITY IN DIGITAL SERVITIZATION: A SERVICE ECOSYSTEM PERSPECTIVE**

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## Abstract

This study investigates multi-actor collaborations for digital servitization through a service ecosystem lens. Previous studies highlighted the need for industry transcending collaboration when implementing a digital servitization strategy and the complexity of diverging goals and perceptions, resulting in inter-organizational tensions. By adopting a service ecosystem perspective, we study the role of institutional complexity in the emergence of inter-organizational tensions and how firms handle these challenges. Based on a single case study of a multi-actor collaboration, this study highlights how diverging institutional backgrounds across collaborating actors culminate in inter-organizational tensions. Furthermore, the results yield insights into how actors can respond to inter-organizational tensions and facilitate value co-creation in service ecosystems. This study contributes to the existing literature by (i) providing a process model for understanding inter-organizational tensions more in-depth, (ii) emphasizing the multi-level character of institutional complexity, and (iii) offering novel perspectives on ecosystem orchestration, explicitly highlighting the critical role of structuring and reconciliation practices to respond to inter-organizational tensions

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### 3.1 Introduction

Manufacturing firms are increasingly embracing a digital servitization strategy. Defined by Sjödin et al. (2020) as “the transformation in processes, capabilities, and offerings within industrial firms and their associated ecosystem to progressively create, deliver, and capture increased service value arising from a broad range of enabling digital technologies,” this strategy is often used as a response to the growing prevalence of digital technologies and data. Manufacturers can enhance customer responsiveness by adopting digital servitization, tapping into novel revenue streams, and leveraging digital technologies (Favoretto et al., 2022). Shifting to a business model based on digital services is inherently inter-organizational, demanding extensive integration and collaboration across organizations. For example, Cenamor et al. (2017) illustrate how digital services rely on technological infrastructures that integrate various actors and enable them to share data. While the potential benefits of digital servitization for firm performance are recognized (Faramarzi et al., 2023; W. Wang et al., 2018), many firms encounter challenges in its successful implementation (Kohtamäki, Einola, et al., 2020; Sjödin et al., 2020).

A significant challenge of digital servitization is managing its complexity, which partly stems from inter-organizational tensions (Stegehuis et al., 2023; Tóth et al., 2022). Recent studies have advocated a multi-level perspective to understand better the complicated nature of digital servitization (Struyf et al., 2021). This approach recognizes that collaboration for digital servitization often transcends traditional industry boundaries, bringing together diverse actors such as technology partners, software developers, and governmental agencies (Favoretto et al., 2022; Makkonen et al., 2022). The heterogeneity of these groups, as highlighted by Tóth et al. (2022) and Galvani and Bocconcelli (2022), often leads to divergent goals and perceptions, contributing to intra and inter-organizational tensions. Understanding these tensions, especially in an interconnected organizational landscape, is essential to harnessing the potential of digital servitization. Such understanding necessitates further exploration into how external contexts influence strategic complexity and how firms can navigate these challenges effectively (Galvani & Bocconcelli, 2022; Kohtamäki, Rabetino, et al., 2022; Struyf et al., 2021). Additionally, examining the nuances of inter-organizational tensions through various theoretical lenses will provide deeper insights into the phenomenon (Tóth et al., 2022).

We adopt a service ecosystem perspective to understand better the complexity stemming from inter-organizational heterogeneity. Rooted in the Service-Dominant Logic, service ecosystems are defined as “relatively self-contained, self-adjusting systems of resource-integrating actors linked by shared institutional arrangements and mutual value creation through service exchange” (Vargo & Lusch, 2016, pp. 10-11). Emphasizing the coordinating role of institutional arrangements in value co-creation, service ecosystems provide a theoretical lens to study the role of institutional complexity in inter-organizational tensions. Institutional complexity refers to the challenges stemming from the multiplicity of institutional arrangements and diverging perceptions and behavior among resource-integrating actors (Sajtos et al., 2018; Siltaloppi et al., 2016).

Existing literature has paid limited attention to the role of institutional arrangements in shaping diverging behavior and perceptions among actors and giving rise to paradoxical and dialectical tensions (Galvani & Bocconcelli, 2022; Stegehuis et al., 2023; Tóth et al., 2022). Adopting a service ecosystem perspective and exploring institutional complexity in digital servitization addresses this gap by illuminating the role of conflicting social structures in inter-organizational collaboration for digital servitization. Therefore, this paper aims to uncover how institutional complexity gives rise to inter-organizational tension and how firms respond to mitigate the adverse effects of these tensions. The following research question guides this study: *How does institutional complexity contribute to the emergence of inter-organizational tensions in digital servitization, and how do actors respond to these tensions?*

We address this research question through a single case study of a collaboration between multiple Dutch SMEs to develop a novel service offering based on various digital technologies. Our results illustrate how institutional complexity gives rise to inter-organizational tensions. Furthermore, our results shed light on the practices firms used to respond to tensions and mitigate their disrupting effect on value co-creation. As such, our results make several contributions to the digital servitization literature. First, extend the literature on inter-organizational tensions by developing a process model that highlights the role of institutional complexity and provides more analytical depth (Galvani & Bocconcelli, 2022; Stegehuis et al., 2023; Tóth et al., 2022). Second, we extend the multi-level perspective on digital servitization by illustrating that inter-organizational tensions originate from industry, organizational, and individual-level institutions (Struyf et al., 2021). Third, we enrich the ecosystem transformation and alignment literature, identifying multiple response practices to deal with inter-organizational tensions (Kolagar et al., 2022; Makkonen et al., 2022). Fourth, we extend the SDL literature by highlighting that actors can respond to institutional complexity without reconciling diverging institutional arrangements (Chandler et al., 2019; C. H. Lee & Hsieh, 2023).

## 3.2 Theory

### 3.2.1 Digital servitization

Digital servitization represents a fundamental organizational transformation involving developing digital and service capabilities (Paschou et al., 2020; Ulaga & Reinartz, 2011). This shift also demands a deep-rooted cultural change to accommodate the new service-oriented business model (Fliess & Lexutt, 2019). As organizations adopt new resources and cultural attributes, they can either complement or conflict with pre-existing ones (Vendrell-Herrero et al., 2014). Thus, digital servitization is a process of business model reconfiguration facilitated or hindered by existing business (Forkmann et al., 2017; Paiola & Gebauer, 2020; W. Zhang & Banerji, 2017). Significantly, the effects of this reconfiguration are not confined within the limits of a single firm. By their very nature, business models often relate to those of external entities such as suppliers, intermediaries, and customers (Reim et al., 2019; Story et al., 2017; Zott & Amit, 2010). Reflecting this widespread impact, recent research on digital servitization has shifted

towards a multi-actor perspective, emphasizing collaboration among ecosystem partners (Kohtamäki, Parida, et al., 2019; Struyf et al., 2021; J. Wang et al., 2023).

Digital servitization increases the number and complexity of required capabilities (Ulaga & Reinartz, 2011). Consequently, firms engage in multi-actor collaboration to access, exchange, and combine complementary knowledge, capabilities, and resources (Fliess & Lexutt, 2019; J. Wang et al., 2023). Complementarity, in this sense, refers to the alignment of pre-existing capabilities across actors and includes newly developed capabilities by integrating knowledge and resources (Raddats et al., 2017). Previous research on inter-organizational collaboration emphasizes the importance of shared innovation goals and a shared understanding of the potential reciprocal value proposition of a digital servitization initiative (Polova & Thomas, 2020; Tronvoll et al., 2020). So, collaborating partners must be on the same page about what they are trying to achieve and how they will achieve it so that their efforts and contributions align with each other and result in an integrated product-service solution. However, such inter-organizational alignment is not a given. It requires proactive alignment efforts to establish a shared vision, incentivizing external actors to contribute to the digital servitization initiative, and aligned value co-creation (Kolagar et al., 2022; Sjödin et al., 2022).

Aligning multiple actors is challenging due to the varied objectives and routines of external counterparts (Makkonen et al., 2022; Story et al., 2017). Digital servitization extends beyond the involvement of traditional ecosystem partners like suppliers and intermediaries and also encompasses distant actors such as research institutions, standard-setting bodies, and governmental agencies (Makkonen et al., 2022; Tóth et al., 2022). Involving such a diverse set of actors adds to the complexity of digital servitization by giving rise to multiple inter-organizational tensions (Galvani & Bocconcelli, 2022; Tóth et al., 2022). These tensions often stem from divergent objectives and perspectives towards digital servitization initiatives (Stegehuis et al., 2023). While some of these diverging objectives and views emerge from internal factors like existing business models, strategies, or professional identities (Stegehuis et al., 2023; Tóth et al., 2022), external elements, such as technological and legal landscapes also shape diverging attitudes of partners (Reim et al., 2019; Zolkiewski et al., 2023). These findings underscore the importance of considering the distinct contextual backgrounds of ecosystem partners. Considering this variety of contextual backgrounds fosters our understanding of inter-organizational dynamics and alignment challenges in digital servitization.

### **3.2.2 Service ecosystem perspective**

Service ecosystems provide a theoretical lens to study the multi-actor nature of digital servitization by emphasizing the contextuality of value co-creation (Chandler & Vargo, 2011; Sklyar et al., 2019). Service ecosystems emerged as the Service-Dominant Logic shifted from a dyadic perspective on value co-creation to service ecosystems, an encompassing actor-to-actor view (Akaka & Vargo, 2015; Vargo & Lusch, 2016). This shift occurred because value co-creation – service exchange by integrating operant resources – involves multiple actors (Lusch & Nambisan, 2015). For example, to experience value

from a smartphone by contacting other people, the user not only needs to integrate their knowledge of using a smartphone with the smartphone itself. Additionally, reaching out to others via a smartphone demands resources from various actors: the mobile network that facilitates connectivity and applications designed to streamline communication. Integrating multiple resources depends on their availability and the social structures that guide human behavior.

Service ecosystems provide a conceptual lens to understand the role of social structures in value co-creation by emphasizing the importance of institutional arrangements (Vargo & Lusch, 2016). Institutional arrangements are assemblages of cognitive, normative, and legal institutions that shape the behavior and perception of resource-integrating actors through a set of material practices, symbols, and organizing principles (Edvardsson et al., 2014; Koskela-Huotari & Vargo, 2016; Vink et al., 2019). These arrangements are vital to value co-creation by providing rules and guidelines for behaving in social situations (Greenwood et al., 2011; Vargo & Lusch, 2016). Also, institutional arrangements determine how actors frame their environment, thereby acting as a reference base for assessing the behavior of other actors and value outcomes (Edvardsson et al., 2014; Koskela-Huotari & Vargo, 2016). Building on the smartphone example, consider how communication between devices relies on telecom industry standards and how socio-cultural and legal developments regarding data privacy influence users' perception of online services. For actors to efficiently collaborate and co-create value, they require overlapping mental schema, norms, and rules (Lusch & Vargo, 2014).

Institutional arrangements coordinate value co-creation in service ecosystems (Vargo & Lusch, 2016). However, the extent to which actors share institutional arrangements can differ considerably (Koskela-Huotari & Vargo, 2016; Sajtos et al., 2018). Institutional complexity characterizes service ecosystems as actors operate in multiple contexts that shape their beliefs and expectations (Akaka et al., 2013). Institutional complexity refers to “the multiplicity of institutional arrangements in service ecosystems, which confronts actors with incompatible prescriptions for action” (Sitaloppi et al., 2016, p. 334). While institutional complexity can spark creativity and drive innovation in service ecosystems (Chandler et al., 2019; Koskela-Huotari & Vargo, 2016), it may also complicate or disrupt value co-creation. Two or more actors may have different belief systems, resulting in conflicting perceptions about appropriate goals and means, which prevent collaboration (Sitaloppi et al., 2016). Diverging and incompatible institutional arrangements can result in inter-organizational tensions related to (i) the group task, (ii) how to accomplish group tasks, (iii) the distribution of responsibilities among actors, (iv) interpersonal compatibility, and (v) values (Mele et al., 2018).

Institutional complexity can have transformative or disrupting implications for value co-creation (Bonamigo et al., 2020; Sitaloppi et al., 2016). How inter-organizational tensions affect value co-creation depends on how actors respond to them. As actors notice structural conflict during interaction with other actors, they can build reflexivity and become aware of institutional complexity (C. H. Lee & Hsieh, 2023; Vink & Koskela-Huotari, 2022). Once actors discover inconsistent institutional arrangements, they can engage in institutional reconciliation to converge ideas and align competing institutions

(Chandler et al., 2019). For example, actors can acquire new operant resources or reconstruct relationships with other actors to overcome institutional complexity (C. H. Lee & Hsieh, 2023). However, institutional reconciliation is a dialectic process that may not always result in a novel synthesized institutional arrangement (Chandler et al., 2019; Hargrave & Van de Ven, 2017). When actors cannot overcome institutional complexity through integration, they may address it through structural separation (Greenwood et al., 2011). Such structuring responses mitigate the disrupting effect of institutional complexity without directly confronting it. For example, actors may decide to work in localized subunits that embody different institutional logics. Institutional reconciliation and separation can overcome conflicting institutional arrangements among actors and facilitate value co-creation in service ecosystems.

### **3.2.3 Institutional complexity in digital servitization**

We developed a conceptual process model based on the service ecosystem perspective to understand inter-organizational tensions and ecosystem transformation better (Figure 3.1). The model differentiates between three stages of value co-creation in multi-actor constellations. When firms collaborate with actors from different institutional backgrounds, they face institutional complexity. For example, when manufacturers implement a servitization strategy, they might collaborate with software developers (Favoretto et al., 2022), which adopt agile organizing principles that differ from traditional plan-based principles in the manufacturing sector (Campanelli & Parreiras, 2015; Huikkola, Kohtamäki, & Ylimäki, 2022). Servitizing firms experience institutional complexity through inter-organizational tensions, which are task-related, process-related, role-related, affective, and values-related (Mele et al., 2018). Once firms experience these tensions, they get into a state of reflexivity that allows them to recognize and address conflicting institutional arrangements (C. H. Lee & Hsieh, 2023; Vink & Koskela-Huotari, 2022). Firms can respond to institutional complexity by reconciling or separating incompatible institutional arrangements (Chandler et al., 2019; Greenwood et al., 2011). Addressing institutional complexity and overcoming inter-organizational tensions facilitates a stabilized service ecosystem and value co-creation between actors. We assert that servitization involves institutional complexity, which manifests as inter-organizational tensions. By effectively responding to these tensions, servitizing firms can leverage tensions to shape a service ecosystem that facilitates value co-creation.

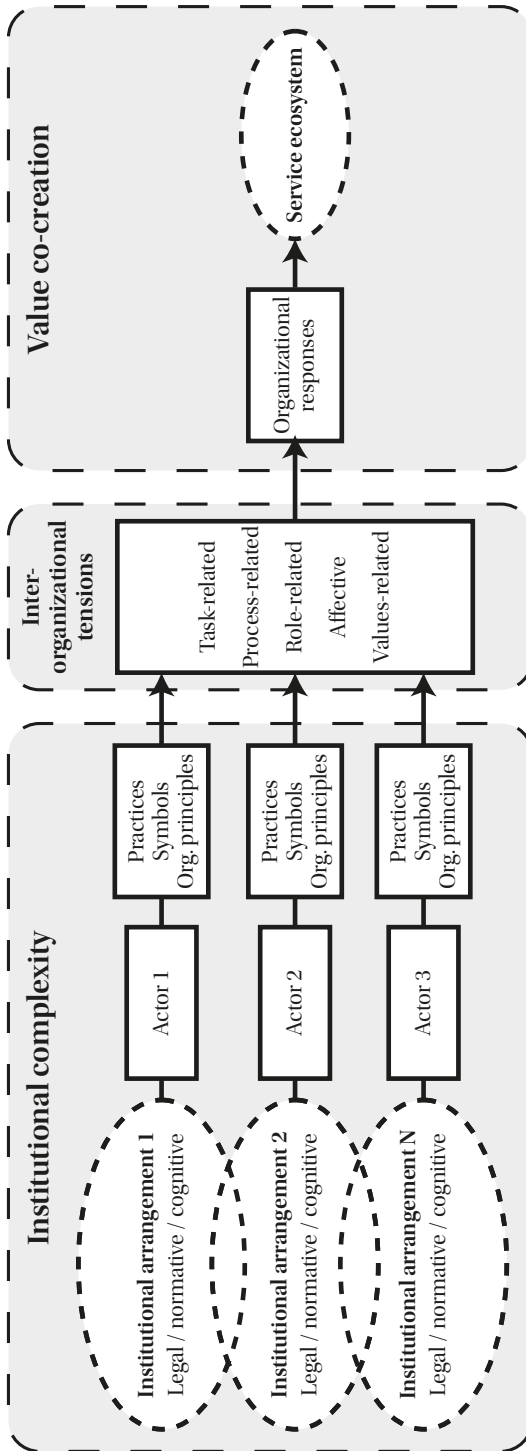


Figure 3.1 Institutional complexity driving tensions in service ecosystems

## 3.3 Method

This study employs a single case study research design. We chose this approach because it facilitates a rich and holistic understanding of complex theoretical and empirical phenomena, such as inter-organizational constellations (Halinen & Törnroos, 2005). It is instrumental in exploring tacit institutional arrangements manifested through practices, symbols, and organizing principles. Because these arrangements are often implicit and taken for granted, they require an insider perspective for a nuanced understanding (Kohtamäki, Whittington et al., 2022). To gain this insider perspective and enhance the study's analytical depth, the lead author actively interacted with the case context, regularly attending meetings and engaging in reflective discussions. This immersive involvement allowed a nuanced exploration of the institutional arrangements and practices under study. Meanwhile, the other authors maintained a distance, ensuring objectivity and safeguarding against potential biases in the research process.

### 3.3.1 Case description

This study presents a case about an inter-organizational innovation project named “MetalNet.” MetalNet aims to develop a novel Spare parts-as-a-Service (SpaaS) proposition for the Dutch railway sector to respond to price pressures and leverage technological developments. SpaaS, a subscription-based service, leverages digital twinning and 3D printing technologies for high-mix, low-volume production. Recognizing the need for enhanced competencies and external collaboration, one of the MetalNet partners brought together diverse actors to foster innovation and develop essential capabilities such as 3D scanning, additive manufacturing, and predictive maintenance. Once the required partners were mobilized and MetalNet was formed in September 2019, it comprised ten organizations with heterogeneous and complementary resources, ranging from manufacturers and IT companies to research institutes, as detailed in Tables 3.1 and 3.2.

MetalNet's goal was defined as “developing (IT) technologies to cost-efficiently (re) manufacture spare parts in high-mix low-volume (i.e., volumes up to 100 pieces) within a short time frame”. To achieve this goal, MetalNet partners conducted research based on concrete use cases from the market. These so-called “assets” are obsolete spare parts that fail unpredictably and must be replaced quickly and cost-efficiently. MetalNet also aimed to investigate service business models to commercialize these technologies.

The case reached its conclusion in July 2023, marked by the drafting and submitting the final report. The realization of multiple significant outcomes crucial to SpaaS characterized this pivotal moment. Key results included the development of 3D scanning and rapid casting capabilities, establishing a digital warehouse, and creating a price estimation tool that automates the quotation process, facilitating decisions even with limited customer technical knowledge. Furthermore, MetalNet partners developed a new IoT device to enable predictive maintenance in the Rail-Infra context. Central to achieving these outcomes was the emergence of a shared ecosystem that facilitated the seamless sharing and integration of knowledge and capabilities from various MetalNet partners.

Table 3.1 Overview of MetalNet partners

| Organization (acronym) | Type       | Offering                            | Institutional context (sector)   | Founded | Size (FTE) | Core competencies                                 | MetalNet stake                                                                                                                                                    | MetalNet role                                                                    |
|------------------------|------------|-------------------------------------|----------------------------------|---------|------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Alpha                  | For-profit | Casting products                    | Rail-infra                       | 1953    | 11-50      | Non-ferro casting, engineering                    | Developing SpaaS business model                                                                                                                                   | Formative manufacturing, business model innovation, project management           |
| Beta                   | For-profit | IoT safety systems                  | Rail-infra                       | 2008    | 51-250     | Hardware, software development                    | Extend offering portfolio with a digital warehouse                                                                                                                | Developing digital library and predictive maintenance tech                       |
| Gamma                  | For-profit | Product development and prototyping | Automotive                       | 2002    | 11-50      | Subtractive manufacturing, (reverse) engineering  | Expanding customer base in new markets, improving 3D scanning tech                                                                                                | Reverse engineering, subtractive manufacturing, 3D scanning                      |
| Delta                  | For-profit | 3D printers, filament               | Variety of B2B and B2C customers | 2012    | 11-50      | Filament development & manufacturing              | Expanding expertise in 3D printing through the lost wax method, developing and testing new filaments.                                                             | Additive manufacturing of spare parts and casting molds.                         |
| Iota                   | Non-profit | Research, education                 | Chemistry                        | 2012    | 11-50      | Additive manufacturing, composite materials       | Developing and testing new composite materials, 3D-scan technologies, additive manufacturing technology, and knowledge valorization among students and companies. | Investigating and developing composite materials for spare part reproduction.    |
| Mu                     | For-profit | Software development                | Logistics                        | 2018    | 1-10       | Software development                              | Developing expertise in artificial intelligence, big data analytics, and expanding customer portfolio                                                             | Developing software platforms, developing rule-based decision system             |
| Pi                     | Non-profit | Research, education                 | Education                        | 1996    | 1000+      | Additive manufacturing, business model innovation | Research in additive manufacturing and servitization, knowledge valorization among students and companies.                                                        | Reverse engineering, additive manufacturing, service business model development. |
| Customer 1             | For-profit | Transportation                      | Rail-infra                       | 1938    | 1000+      | Logistics                                         | Cost-reduction of obsolete spare part sourcing                                                                                                                    | Steering group                                                                   |
| Customer 2             | For-profit | Contracting                         | Rail-infra                       | 1921    | 1000+      | Management and maintenance                        | Increasing efficiency of management and maintenance of rail tracks                                                                                                | Steering group                                                                   |
| Customer 3             | For-profit | Railway safety systems              | Rail-infra                       | 1879    | 51-200     | Product design, engineering                       | Adapting to changing demands from customers in rail-infra                                                                                                         | Steering group                                                                   |

**Table 3.2**                      **Overview individuals participating in MetalNet**

| Organization | Organizational role       | Age   | Education background | Previous sector(s)     | Previous role(s)                              |
|--------------|---------------------------|-------|----------------------|------------------------|-----------------------------------------------|
| Alpha        | CEO                       | 40-60 | Political sciences   | Entertainment          | Commercial director, managing director        |
| Alpha        | Technical director        | 40-60 | Engineering          | Manufacturing          | Sales engineer, R&D employee                  |
| Beta         | Managing director         | 30-40 | Computer science     | Telecom                | Product owner, functional manager             |
| Gamma        | Technical director        | 40-60 | Model making         | Manufacturing          | Model maker, line manager, functional manager |
| Gamma        | Engineer                  | 20-30 | Engineering          | Manufacturing          | Designer, engineer                            |
| Delta        | Financial director        | 40-60 | Finance              | Variety of industries  | Financial controller, financial manager       |
| Delta        | Strategic account manager | 30-40 | Engineering          | Manufacturing          | Engineer, sales engineer                      |
| Iota         | Researcher                | 40-60 | Engineering          | Education, research    | Lecturer                                      |
| Mu           | Co-owner                  | 30-40 | Business             | Digital innovation     | Marketeer, account manager                    |
| Pi           | Researcher (1)            | 40-60 | Engineering          | Additive manufacturing | Business owner, functional manager            |
| Pi           | Researcher (2)            | 30-40 | Business             | Education              | Lecturer                                      |

### 3.3.2 Data collection

Our research focused on the individual actors and their interactions in the MetalNet ecosystem as the unit of analysis. The study employed a data collection strategy consisting of ethnographic observations, semi-structured interviews, and internal documents. Ethnographic observations were our primary data source as the lead author participated in numerous formal and informal network meetings, engaging with various partners to gain an insider perspective. Formal sessions were recorded, coded, and stored in ATLAS.ti, and additional observational data were captured in a research journal.

Two rounds of semi-structured interviews were conducted to complement the ethnographic observations (see Table 3.3). The first interview round was conducted when the lead author entered the case context and focused on partner roles, goals, and views on the current collaboration (Appendix 3A). Insights from the first interview round served as the starting point for the second interview round. During this round, problems and potential solutions mentioned in the first interview round were anonymously presented to other MetalNet partners, and they were asked to reflect on and respond to them (Appendix 3B). These questions led MetalNet partners to actively reflect on the problems and potential solutions brought up by other partners. The reflections resulted in an extended and consolidated understanding of collaboration processes within MetalNet. During both interview rounds, we only interviewed MetalNet partners actively participating in MetalNet and were available for an interview as they understood the inter-organizational collaboration between partners. Customer organizations only had supervisory roles and were not available for interviews.

We complemented observational and interview data with secondary data. These include publicly available reports, content, and internal documents such as email threads containing communication between MetalNet actors. During the research, MetalNet partners consented to collecting and utilizing data for research purposes. Their anonymity was ensured at all times.

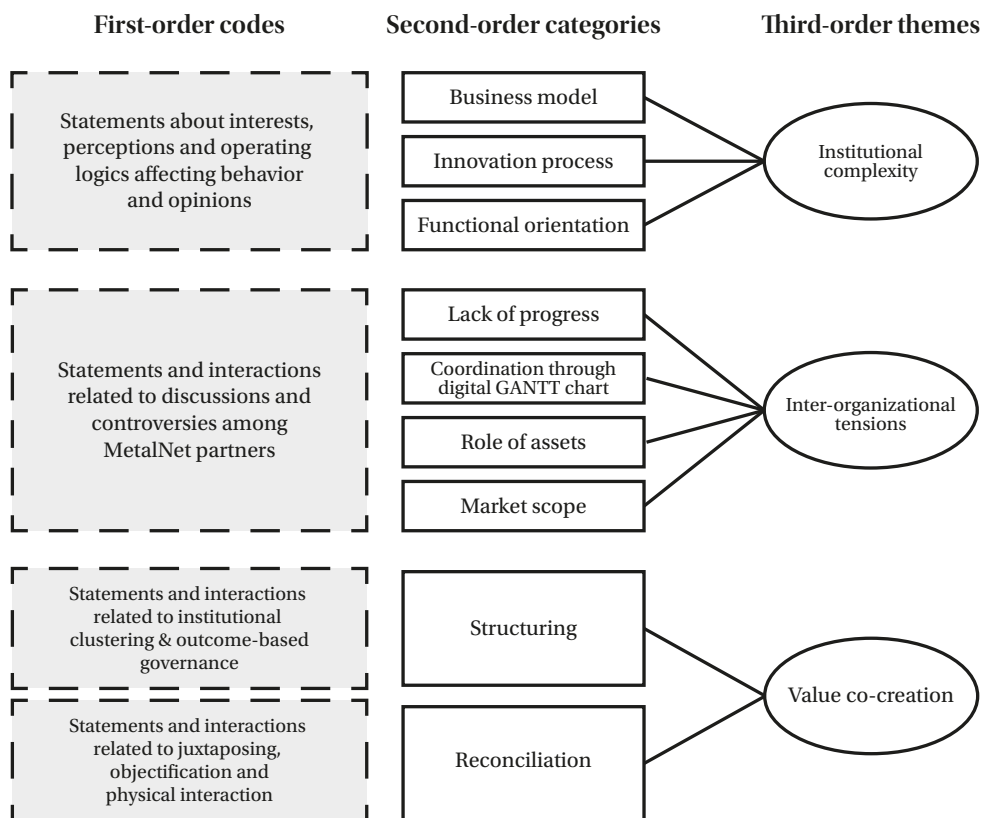
**Table 3.3**                    **Overview of interview informants**

| #                        | Company (role)        | Role of informant              | Date       | Duration |
|--------------------------|-----------------------|--------------------------------|------------|----------|
| <b>Interview round 1</b> |                       |                                |            |          |
| 1                        | Gamma (Manufacturing) | Technical director             | 12-01-2021 | 00:46:50 |
| 2                        | Beta (IT)             | Managing director              | 14-01-2021 | 00:52:02 |
| 3                        | Alpha (Manufacturing) | Technical director             | 15-01-2021 | 00:55:29 |
| 4                        | Delta (Manufacturing) | Financial director             | 15-01-2021 | 00:44:12 |
| 5                        | Iota (Research)       | Business Developer             | 18-01-2021 | 00:58:40 |
| 6                        | Mu (IT)               | Managing director              | 19-01-2021 | 00:52:27 |
| 7                        | Alpha (Manufacturing) | Managing director              | 20-01-2021 | 00:57:56 |
| 8                        | Pi (Research)         | Researcher 1                   | 22-01-2021 | 00:54:04 |
| 9                        | Delta (Manufacturing) | Strategic account manager      | 26-01-2021 | 00:28:02 |
| <b>Interview round 2</b> |                       |                                |            |          |
| 10                       | Delta (Manufacturing) | Strategic account manager      | 16-03-2021 | 00:58:37 |
| 11                       | Alpha(Manufacturing)  | Managing director              | 16-03-2021 | 00:58:38 |
| 12                       | Delta (Manufacturing) | Operational/financial director | 17-03-2021 | 00:42:47 |
| 13                       | Gamma (Manufacturing) | Technical director             | 18-03-2021 | 00:46:01 |
| 14                       | Gamma (Manufacturing) | Engineer director              | 18-03-2021 | 00:50:56 |
| 15                       | Iota (Research)       | Business Developer             | 18-03-2021 | 00:42:34 |
| 16                       | Alpha (Manufacturing) | Technical director             | 18-03-2021 | 00:24:29 |
| 17                       | Pi (Research)         | Researcher 2                   | 19-03-2021 | 01:18:07 |
| 18                       | Mu (IT)               | Managing director              | 23-03-2021 | 00:40:11 |
| 19                       | Beta (IT)             | Managing director              | 23-03-2021 | 00:47:27 |
| 20                       | Pi (Research)         | Researcher 1                   | 26-03-2021 | 00:50:37 |

### 3.3.3 Data analysis

The data analysis process involved thematic analysis of qualitative data tailored to unravel the intricate dynamics within MetalNet's collaborative context. We followed Saldaña (2015) by labeling relevant data excerpts with first-order codes and then clustering all data in second-order categories and third-order themes. During the first-order coding process, we looked for data excerpts related to symbols, artifacts, activities, and interactions because these are carriers of institutional arrangements in service ecosystems (Scott, 2014; Vink et al., 2021). We inferred the underlying institutional arrangements by triangulating different types of data (i.e., ethnographic observations, interviews, and secondary data) from various sources. For example, we inferred that the business model of Pi was an essential driver of the preference of Pi's researcher 1 to work on non-commercial assets by combining statements from Pi's researcher 1, Beta's managing director, and observations during monthly MetalNet meetings. Combining these statements with observations revealed that Pi's researcher 1 prioritized finishing a subsidy project because the organization allocated hours based on expected subsidy incomes.

We divided the thematic analysis into three stages. To begin, we systematically identified statements about actions, artifacts, and symbols that shed light on the interests, perceptions, and operational logic of the individual MetalNet partners. This process offered valuable insights into institutional complexity in MetalNet. Subsequently, we delved into discussions and controversies among the partners to discern inter-organizational tensions in MetalNet. Finally, we investigated the practices used to respond to the tensions arising from institutional complexity. This comprehensive analysis illuminated the nuanced interplay of institutional elements and their impact on MetalNet's collaborative endeavors, providing valuable contributions to understanding inter-organizational dynamics in complex ecosystems. The coding process resulted in the data structure presented in Figure 3.2



**Table 3.2      Data structure**

## 3.4 Results

### 3.4.1 Institutional complexity

MetalNet consisted of individuals from different organizations and different contextual backgrounds. These differences can be categorized as complexities stemming from different business models, innovation processes, and functional orientations

#### 3.4.1.1 Business model

The first institutional complexity involved different organizational logic and priorities among MetalNet partners, rooted in the various business models among partners. This diversity separated individuals from commercial organizations with a profit incentive from those working at state-funded research institutes. Individuals from commercial entities such as Alpha, Beta, Delta, Gamma, and Mu perceived MetalNet as a catalyst for fostering innovation and achieving long-term organizational goals. Individuals from these organizations, driven by profit motives and customer demands, utilized MetalNet as an instrumental tool to execute their organizational strategies and innovate business models for sustainable growth and competitiveness. These commercial actors pursued long-term economic goals by participating in MetalNet, including accessing new markets (Gamma), developing a new product (Beta), and changing the existing business model (Alpha).

Contrastingly, actors from research institutes like Pi and Iota embodied a different institutional logic. Fundamentally driven by research and knowledge dissemination, these state-funded institutes leveraged MetalNet to augment their research capabilities and educational outreach. Projects like MetalNet were an integrated component of its core business strategy. Actors from research institutes considered MetalNet key in facilitating academic and practical research endeavors. Beta's managing director also recognized this alternative approach: *"I think that in terms of companies, we are all moving in the direction of 'we have to make a commercial success,' but with [Pi], you see a different approach of 'we have to complete a research project within three years'"* (Managing Director, Beta) The institutional complexity regarding business models within MetalNet was marked by divergent organizational logic and objectives. Actors from commercial organizations were future-oriented, aligning their participation with long-term economic ambitions and adaptability to market demands. In contrast, actors from research institutes prioritized immediate scholarly contributions and educational enhancement, viewing MetalNet as a vital resource for fulfilling these institutional responsibilities rather than a means for long-term strategic transformation.

#### 3.4.1.2 Innovation process

In the diverse innovation landscape of MetalNet, a segment of actors came from organizations that predominantly manufactured physical products, such as Alpha, Delta, and Gamma. These actors navigated innovation through a plan-based approach. The following quote illustrates the cognitive and normative origins of such plan-based

approaches: *“Everyone approaches a project in different ways. We are structured much more based on the automotive industry. So you have to meet dates and requirements to be allowed to deliver”* (Technical Director, Gamma). Rooted in precision and driven by stringent deadlines, MetalNet actors operating in a manufacturing context approached innovation through meticulous upfront planning. This plan-based approach was reinforced by high upfront investments in tooling and machinery, necessitating a well-orchestrated innovation pathway to mitigate risks and avoid costly adjustments.

Similarly, actors from research institutes like Iota and Pi structured their innovation processes through a plan-based approach but more rigidly aligned with their core educational businesses. As these research institutes involved students in conducting part of the research in MetalNet, actors from these organizations had to structure and plan their innovation efforts in line with education curricula: *“We have a research assignment for 20 weeks within the CFD with three students and an expert, and we are letting it run. So you are dependent and must deal with these differences”* (Researcher, Iota).

In contrast, actors from Beta and Mu, originating from the IT industry, embraced agility as the cornerstone of their innovation process, as Mu’s managing director indicated: *“We work according to Scrum. So, we define a scope in advance and determine several sprints. And then, we establish a backlog with all the software functionalities we will develop. We share this backlog with the customer and then meet every two weeks to discuss the sprint’s outcomes and what we will do in the next sprint.”* (Managing Director, Mu). Adopting a Scrum framework, their approach emphasized flexibility, enabling responsive navigation through the dynamic landscapes of software development and customer needs. This iterative model accommodated unforeseen variables and emerging customer needs, allowing for adaptive realignment of innovation trajectories. It is important to note that while Beta adopted Scrum for innovation purposes, it combined this approach with a stage-gate approach. This combination implies that their innovation approach was rooted in plan-based and agile practices.

Collectively, the complexity of innovation processes within MetalNet unveiled various approaches, each reflecting the unique operational contexts, priorities, and imperatives of the involved actors. This diversity ranged from rigid, plan-centric models driven by high upfront investment and precision to adaptive, agile frameworks that promote flexibility and continuous improvement.

#### **3.4.1.3 Functional orientation**

The institutional landscape of MetalNet could be further differentiated between actors with various functional orientations. This complexity operated at the level of individual actors as individuals from the same organization could be embedded in different institutional logics. Multiple actors within MetalNet, exemplified by leadership figures such as CEOs and managing directors from Alpha, Beta, Delta, and Mu, and researcher 2 from Pi, gravitated towards a business-centric functional orientation. This orientation entailed a sensitivity to commercial business opportunities and an interest in strategic

issues like formulating value propositions, designing revenue models, and engaging in customer interaction. In other words, a business orientation entailed a strategic and commercial mindset with a sensitivity to customer problems and potential innovative solutions.

In contrast, multiple other actors within MetalNet, notably technical directors, engineers from Alpha and Gamma, and researchers from Iota and Pi, embodied a technical functional orientation. This orientation manifested in sensitivity and fascination with manufacturing capabilities and technological innovation. Central to this cognition was professional pride rooted in technical expertise and craftsmanship, reflecting an appreciation for the artistry of technological creations and advancements. An illustrative moment capturing this essence was witnessed during a technical team’s visit to Gamma, where the technical director passionately shared his professional manufacturing experiences. An exemplary quote from Alpha’s technical director further crystallizes the nature of this technical cognition: *“You sit together as technicians, and of course, you do not want to be embarrassed there. [...] You can see how today everyone stood around that 3D printer as if it were a holy grail”* (Technical Director, Alpha). This quote reflects the interaction between like-minded individuals who share a deep interest in manufacturing and technology and take pride in their technical expertise.

3.4.1.4 Synthesizing institutional complexity

Table 3.4 presents an integrated overview of the institutional embeddedness of MetalNet actors along the three institutional complexity categories discussed previously. Integrating all institutional dimensions provides a more comprehensive picture of institutional complexity in MetalNet. As an actor was simultaneously embedded in the three institutional categories, and because these categories were not necessarily interdependent, many different combinations of institutions were possible. As such, an integrated view reveals the wide variety of institutional arrangements that shaped the behavior and perceptions of actors in MetalNet.

Table 3.4 Institutional complexity in MetalNet

| Organization | Organizational role       | Business model     | Innovation process | Cognitive orientation |
|--------------|---------------------------|--------------------|--------------------|-----------------------|
| Alpha        | CEO                       | Commercial         | Plan-based         | Business-oriented     |
| Alpha        | Technical director        | Commercial         | Plan-based         | Technical-oriented    |
| Beta         | Managing director         | Commercial         | Agile / Plan-based | Business-oriented     |
| Gamma        | Technical director        | Commercial         | Plan-based         | Technical-oriented    |
| Gamma        | Engineer                  | Commercial         | Plan-based         | Technical-oriented    |
| Delta        | Financial director        | Commercial         | Plan-based         | Technical-oriented    |
| Delta        | Strategic account manager | Commercial         | Plan-based         | Technical-oriented    |
| Iota         | Business Developer        | Research Institute | Plan-based         | Technical-oriented    |
| Mu           | Co-owner                  | Commercial         | Agile              | Business-oriented     |
| Pi           | Researcher                | Research Institute | Plan-based         | Technical-oriented    |
| Pi           | Researcher/lecturer       | Research Institute | Plan-based         | Business-oriented     |

### 3.4.2 Inter-organizational tensions

In the MetalNet collaboration, controversies arose among partners, highlighting the inter-organizational tensions in MetalNet. Each actor's unique institutional arrangement shaped their viewpoints and interactions, creating diverse opinions and approaches towards specific issues such as progress in MetalNet, utilizing digital GANT charts for coordination, the role of assets within MetalNet, and defining MetalNet's market scope. The observable interactions, practices, and symbols in the disputes exemplify how institutional complexity influenced inter-organizational tensions within the collaboration.

#### 3.4.2.1 Lack of progress

The initial point of discussion within MetalNet revolved around the perceived lack of progress and output relative to the expectations outlined in the original subsidy application. This issue became particularly critical concerning asset reproduction, triggered by the withdrawal of a key customer steering partner from MetalNet. The incident catalyzed discussions among MetalNet partners about how to collaborate. During these discussions, provoking debates emerged around defining and evaluating "progress" within the collaboration.

Discussions about the role of the subsidy application revealed divergent perceptions of progress. Actors from research institutes aligned progress with the completion of predefined working packages essential for securing subsidy income. Pi researcher 1 emphasized the significance of adhering to the submitted research proposals to ensure the receipt of subsidy funds: *I see [CEO, Alpha] commenting in his last email that 'we are now going to focus purely on the commercial objectives,' but MetalNet is about realizing research as we submitted it, at least if we want to obtain that subsidy. [...] If we don't get the subsidy because the subsidy provider says "you haven't done what you promised, "that is a bad ding because we budget hours for that"* (Researcher 1, Pi). On the contrary, actors from commercial partners like Alpha, Beta, and Delta envisioned progress through a commercial lens, prioritizing long-term business success. They regarded subsidies as a means to offset investment costs rather than ends. The Managing Director of Beta underscored this difference in perspective: *"I can start building all sorts of software and spend 100k, but if no one is going to use it and we cannot sell it, it makes no sense to build it. And the approach of [Researcher 1, Pi] is 'but we have a commitment, and we have to complete a project,' that is a completely different mindset"* (Managing Director, Beta). This quote delineates a practical, market-oriented viewpoint, highlighting a fundamental dichotomy in interpreting progress within MetalNet.

The diverse institutional arrangements within MetalNet underscore the institutional complexity at the heart of debates concerning progress. Research institutes within MetalNet, significantly reliant on subsidies as a crucial revenue stream, structure their business models and objectives around securing and maximizing subsidy income. Consequently, actors from these organizations equated progress primarily with completing work packages that ensure the flow of these funds, which is essential for

covering personnel expenses and sustaining operations. Contrastingly, actors from the commercial entities within MetalNet, driven by long-term commercial imperatives, adopted an instrumental approach toward participation. For them, progress was intrinsically linked to the return on investment from their activities within the network. They viewed subsidies as tools to alleviate investment costs, but the ultimate justification for participation lay in the potential for revenue generation. This exploration reveals that the diversity of business models within MetalNet emerged as tensions related to progress in MetalNet. These tensions were reflected in the varied interpretations and priorities attached to the concept of progress.

#### **3.4.2.2 Coordination through digital GANTT chart**

To address the lack of progress within MetalNet, the CEO of Alpha introduced a digital GANTT chart tool. The tool was designed to consolidate the two-year planning of all MetalNet partners, requiring them to update and report their progress monthly. He elaborated, *“We have a GANTT chart, which shows at the start of the match ‘what is the tactic’ and drawing on the board; that is what we have done”* (CEO, Alpha). This initiative, aimed at fostering a clear and coordinated approach, ignited discussions and varied responses among the MetalNet partners about the appropriateness of a GANTT chart for coordinating MetalNet activities.

A majority supported the digital GANTT chart, referring to enhanced transparency and clarity in responsibilities and progress as their argument. For instance, Researcher 1 from Pi noted, *“I am emphasizing the digital GANTT chart because everything becomes visible to everyone, what needs to be done, where we stand, and so on,”* indicating a positive outlook towards the tool’s potential to foster visibility and coordination. However, there were reservations, too. For instance, the Managing Director from Mu exhibited skepticism: *“I vowed never to look at a GANTT chart again [...]. It is just a classic waterfall method, and when developing software, you have absolutely no idea where we will be in two years, so why are we already planning where we will be in two years?”* (Managing Director, Mu). This sentiment underscores a reluctance to embrace the tool due to its inherent rigidity and the unpredictable nature of software development. Instead, he advocated for a more flexible approach to accommodate uncertainties and evolving needs in project execution.

The divergent viewpoints regarding implementing the digital GANTT chart underscore how institutional complexity related to innovation processes developed in tensions. Partners favoring a plan-based approach appreciated the comprehensive overview and enhanced transparency the digital GANTT chart offered. Conversely, actors from organizations that adopt an agile innovation approach, characterized by flexibility, iterative development, and less rigid upfront planning, found the tool restrictive. Thus, the discussions about adopting the GANTT chart echo a broader conflict between structured, plan-oriented strategies and dynamic, agile approaches to innovation. In this context, institutional complexity complicated setting up governance structures and improving MetalNet productivity.

### 3.4.2.3 Role of assets in MetalNet

After productivity improved within MetalNet and partners began to progress, a new focal point emerged in their discussions: the role and availability of assets. This discussion arose when experimentation was hindered due to a lack of suitable technical assets to facilitate the work. Until this point, Alpha's CEO sourced most assets and maintained the most extensive customer contacts. However, this approach led to dissatisfaction among some MetalNet partners, who questioned the suitability and variety of the assets introduced into the collaboration. This discontent spurred discussions, challenging whether the focus should be on commercial or non-commercial assets and scrutinizing the impact of such choices on MetalNet's objectives and workflow.

Opinions varied significantly regarding the utilization of commercial versus non-commercial assets. Alpha's CEO advocated for the prioritization of real-life commercial assets, which is reflected in the following quote: *"[Pi researcher 1] wants to do research, but I'd like to call it commercial research where we push those assets through the pipeline and get results to answer our research questions. [...] For me, the success of MetalNet is very dependent on how we conclude those assets for customers. If we successfully conclude a few assets, we will create enthusiasm among customers and gain traction in the industry. (CEO, Alpha)"* His perspective underscored a strategic alignment toward commercial viability and industry traction, viewing the handling of assets as instrumental to the commercial success of MetalNet.

In contrast, other actors, such as those from Gamma, Iota, and Pi, championed a focus on non-commercial assets. They argued, as illustrated by a quote from Pi's researcher that an over-reliance on commercial assets could jeopardize the project's completion: *"If we make ourselves too dependent on commercial assets from the market, we become too dependent and that the realization of the work packages might become problematic."* (Researcher 1, Pi). This perspective underscores the potential vulnerabilities in finishing the MetalNet project associated with a strong commercial focus on assets. Similarly, the Technical Director from Gamma pointed out the limited technical capabilities within MetalNet. He cautioned against overcommitting to customer expectations: *"But this asset, for example, is too complex for MetalNet. And [CEO, Alpha] starts to push a bit, saying 'guys, we have to get results,' but I think we should not give the customer the illusion that we will sort this out because that is not the case"* (Technical director, Gamma). This quote reflects his viewpoint and preference for working on non-commercial assets due to their less intense pressure and expectations while allowing for a focus on enhancing MetalNet's technical capabilities.

Different viewpoints on the role of assets among MetalNet partners are rooted in diverse institutional arrangements. For instance, Alpha's CEO, driven by a business-centric functional orientation and a for-profit business model, emphasizes the importance of focusing on commercial assets. This approach aligns with a vision of leveraging assets for industrial traction and commercial success. Contrastingly, research institutes like Iota and Pi, relying on subsidy income, expressed reservations against too much dependence on commercial assets. Their focus leaned toward non-commercial assets

to secure their primary objective of completing the MetalNet project. This perspective was further reinforced by a technical orientation prevalent among some actors. Because of their technical orientation, some actors prioritized technical enhancement over immediate commercial pursuits. Tensions in this context were driven by institutional complexity stemming from different business models and functional orientations. These institutional differences shaped divergent viewpoints on asset utilization and influenced mutual perceptions among partners. For example, the researcher from Iota perceived Alpha's CEO actions to be rushed towards commercialization, possibly at the expense of thorough research and technical refinement. In contrast, Alpha's CEO considered the focus on the research of some of the MetalNet partners to be a barrier to realizing commercial success. Overall, institutional complexity fostered an environment marked by heightened discussions, misunderstandings, and tensions, posing significant challenges to the collaboration and future trajectory of MetalNet.

#### **3.4.2.4 MetalNet market scope**

In response to the lack of assets in MetalNet, Beta's managing director introduced an asset from the Railway sector. This asset was a new product developed by Beta, but since it was not an obsolete spare part, it fell outside the initially defined scope of MetalNet. The introduction of this asset ignited a series of discussions among partners regarding the flexibility and adaptability of MetalNet's market scope.

MetalNet partners exhibited a spectrum of opinions regarding Beta's asset and the consequent expansion of market scope beyond obsolete spare parts. These positions ranged from a dynamic commercial outlook, open to leveraging emergent opportunities, to a more conservative, technically anchored perspective emphasizing adherence to original project parameters and a realistic appraisal of MetalNet's capabilities. Beta's managing director and Alpha's CEO welcomed the new asset, seeing it as a conduit of commercial opportunity. They adopted a dynamic perspective to MetalNet's market scope, viewing the scarcity of assets as an opportunity to broaden MetalNet's horizon beyond its initial focus on spare part obsolescence. Alpha's CEO stated: *"Beta develops a new product of which they need 50 or 100. With MetalNet, we will first look at product development and make a prototype. Then they will put it into production, scale it up, and need a thousand over the next thirty years"* (CEO, Alpha). This quote projects a flexible, business-oriented approach.

Contrastingly, various technical actors from multiple organizations, including Alpha, Gamma, Delta, Iota, and Pi, leaned towards a more conservative stance. Their viewpoint resonated with adherence to the original objectives outlined in the subsidy grant application. For instance, a researcher from Pi emphasized the project's core focus: *"The project is about spare parts, about replacing worn parts, and then this is really not part of it"* (Researcher 1, Pi). His perspective aligned with the initial boundaries set for MetalNet's scope. Furthermore, certain technically oriented actors presented a pragmatic viewpoint, questioning MetalNet's technical ability to navigate new product development landscapes. Their skepticism was echoed in practical concerns regarding MetalNet's distinctive competence, questioning whether it harbored a competitive edge

in new product development realms. One exemplary reflection came from a strategic account manager from Delta: *“Why would MetalNet be better at this than another company? [...] we are not set up to do product development”* (Strategic Account Manager, Delta), underlining a perceived mismatch between MetalNet’s existing capabilities and the demands of new product development.

The differing perspectives regarding MetalNet’s scope provide insights about the role of institutional complexity, primarily fueled by diverging functional orientations, in the emergence of tensions. Alpha’s CEO and Beta’s Managing Director experienced misunderstandings and resistance when proposing an extension of scope beyond obsolete spare parts. This experience was vividly captured in the CEO’s words: *“It was not explained to me, and [Managing Director, Beta] found it astonishing [...]. It became clear that technicians think differently than people much more involved in business development and commerce”* (CEO, Alpha). This statement underscores a critical divide, segregating the technically oriented from the business-oriented individuals in MetalNet. Differing functional orientations translated into varied strategic decision-making approaches. Business-oriented actors engaged in strategic decision-making focused on exploring new opportunities and envisioning the requisite capabilities for potential commercial success. Conversely, technically oriented actors made strategic decisions by anchoring their deliberations in existing capabilities and looking for the best avenues to deploy them. Consequently, institutional complexity related to diverging functional orientations complicated decision-making in the MetalNet partnership.

### **3.4.3 Value co-creation**

In the MetalNet partnership, actors employed diverse responses to address inter-organizational tensions and foster value co-creation. These responses fall into two distinct categories. The first, structuring, involves responses that alleviate tensions and promote collaborative alignment without directly addressing underlying institutional complexities. The second category, reconciliation, concerns responses that actively bridge conflicting institutional norms and routines. This approach harmonizes collaboration by directly addressing institutional complexities underlying tensions, paving the way for a unified institutional arrangement.

#### **3.4.3.1 Structuring**

Structuring alleviates inter-organizational tensions without addressing the underlying institutional complexity and entails institutional clustering and outcome-based governance. Institutional clustering is a structuring approach in which actors with a shared institutional background are grouped. Institutional clustering appeared in MetalNet through working in separate committees. Confronted with challenges such as a lack of progress and clear responsibility delineation in asset reproduction, institutional clustering aimed at fostering focused and efficient collaboration. The formation of an “asset committee” marked the introduction of institutional clustering within MetalNet. Comprising technically oriented individuals, such as technical directors from Alpha and Gamma and researchers from Pi and Iota, this committee created a space for specialized

technical discussions. It allowed participants to engage deeply, leveraging their technical expertise and fostering a collective sense of purpose and involvement. Subsequently, this clustering practice was expanded to encompass various focal areas, such as IT and revenue model development, facilitating clusters of actors sharing similar institutional backgrounds. For instance, IT-related committees included actors with commercially oriented cognitive dispositions who adopted agile innovation approaches.

Institutional clustering cultivated an environment of harmonious interaction among partners, grounded in mutual understanding and shared language and practices. For instance, IT-related committees, including business-oriented actors, had a mutual understanding of software development through agile innovation practices, streamlining the process. As such, institutional clustering alleviated tensions arising from divergent innovation processes and functional orientations. However, while enhancing operational efficiency and mutual understanding, institutional clustering intensified discussions and tensions between groups of actors with a shared institutional background. As groups of actors with a shared opinion perceived to have more power than an individual actor, it increased institutional complexity. This dynamic became apparent during the discussions about the MetalNet market scope, which characterized tensions between actors from the asset committee and the IT and revenue model committees. In conclusion, institutional clustering became a critical way to respond to tensions by fostering aligned, efficient, and harmonized collaborative environments.

Outcome-based governance was the second structuring response and entailed frequently reporting on progress. A lack of accountability characterized the initial interactions among MetalNet partners, attributed partly to a prevailing social distance. Partners often found it easy to evade responsibilities, as noted by Beta's managing director: "It is too easy just to listen reactively, looking at what others have done and say 'great job guys, see you next time'" (Managing Director, Beta). Addressing this, MetalNet partners engaged in outcome-based governance by implementing weekly planning and monthly progress reporting routines. This approach leveraged the commitment of all MetalNet partners during the formation phase and created normative pressure to deliver output. While partners held differing opinions on process structure and choice of artifacts, particularly evident in discussions surrounding the digital GANTT chart, outcome-based governance routines created social pressure that reminded them of their shared commitments. Pi's Researcher 2 pointed out, "Regularly demonstrating results to MetalNet partners creates pressure because you have to show something, and you do not want to show that you have done nothing" (Researcher 2, Pi). Outcome-based governance alleviated tensions stemming from business model and innovation process-related complexity. By prioritizing outcomes over processes, this response allowed all partners the freedom to shape innovation processes in alignment with their institutional arrangements while upholding the initial commitment of the entire MetalNet partnership.

### **3.4.3.2 Reconciliation**

Reconciliation addresses institutional complexity by bridging and reducing

contradictions between various institutional arrangements. Reconciliation encompasses three practices: juxtaposing, objectification, and physical interaction. Juxtaposing involves explicitly mentioning different perceptions or goals among partners to spark constructive discussions that reduce institutional differences. An illustrative example was when Alpha's CEO juxtaposed working on commercial assets with conducting research. Explicitly highlighting and questioning the contradiction between assets and research created a platform for fruitful discussion among MetalNet partners aimed at finding commonalities and complementarities between contradictory goals and priorities. This practice facilitated addressing institutional complexities by encouraging MetalNet partners to explore and appreciate each other's viewpoints. Through juxtaposing, MetalNet partners addressed complexities related to business models, unveiling the complementarities between research and commercial activities and fostering alignment and a shared perception of individual and collective goals

Objectification involves adopting a systematic and data-driven approach to decision-making. This routine made MetalNet partners reflect on their intuitions and heuristics rooted in their institutional arrangements. As such, objectification provided an engagement platform to integrate various institutions into a shared institutional arrangement. In MetalNet, objectification was exemplified in the formulation of asset selection criteria. This selection criteria became a pivotal artifact, facilitating the harmonization of diverse goals among partners, particularly concerning the role of assets and the market scope of MetalNet. For instance, Alpha's CEO championed using well-defined criteria for accepting or rejecting assets from potential customers. By collaboratively defining selection criteria, all partners needed to articulate their specific needs and goals regarding the assets. As a result, they developed an asset selection criteria matrix, symbolizing an integration of the varied needs and objectives of the MetalNet partners that were rooted in their institutional arrangements. This artifact served as a collective lens through which partners could uniformly assess and approach market opportunities in asset acquisition. By doing so, objectification assisted in mitigating institutional complexity rooted in different business models and functional orientations, enabling the partners to overcome and align different institutional arrangements.

The critical role of physical interaction in institutional hybridization became apparent during the COVID-19 pandemic. MetalNet partners transitioned to digital communication platforms like MS Teams during this event. Although these digital tools enabled continuity in operations, they inherently lacked the richness and depth fostered by physical interactions. MetalNet partners preferred in-person meetings, as they facilitated meaningful exchanges and mutual understanding, essential for institutional reconciliation. Beta's managing director encapsulated this sentiment by highlighting that physical sessions included casual social interactions over coffee, fostering a deeper insight into each partner's perspectives, operational choices, and priorities. Such insights promoted more cohesive and empathetic relationships among MetalNet partners. These face-to-face encounters were instrumental in alleviating tensions by increasing mutual understanding of diverse institutional backgrounds and orientations among the MetalNet partners. This understanding ultimately reduced institutional

complexity and fostered value co-creation between actors.

## **3.5 Conclusion and discussion**

This paper set out to investigate the role of institutional complexity in inter-organizational tensions in digital servitization through a service ecosystem perspective. Our findings emphasize that inter-organizational tensions are rooted in diverging institutional arrangements among collaborating actors. As digital servitization requires collaboration with actors from different institutional backgrounds (Favoretto et al., 2022; Makkonen et al., 2022), diverging institutional arrangements give rise to inter-organizational tensions through incompatible perceptions of goals and prescriptions for behavior. We found that institutional complexity related to diverging business models, innovation processes, and functional orientations created tensions and complicated inter-organizational collaboration. Furthermore, our findings illuminate different practices actors adopt to respond to and mitigate inter-organizational tensions. These responses can be categorized as practices that address inter-organizational tensions without confronting the underlying institutional complexity and responses that reconcile institutional differences and align institutional arrangements. Both types of responses can mitigate the disrupting effect of inter-organizational tensions and facilitate value co-creation for digital servitization. Our study indicates that while inter-organizational tensions complicate digital servitization, they are essential cues of institutional complexity that can be leveraged to achieve alignment between heterogeneous actors.

### **3.5.1 Theoretical contributions**

Our findings contribute to the digital servitization literature in various ways. First, our results contribute to the ongoing discourse about inter-organizational tensions in digital servitization. Adopting a service ecosystem perspective provides a new theoretical angle to study inter-organizational tensions. Compared to the paradox lens and the industrial network approach adopted by previous studies (Galvani & Bocconcelli, 2022; Stegehuis et al., 2023; Tóth et al., 2022), provides a richer understanding of the sources of inter-organizational tensions. Focusing on institutional arrangements and complexity reveals that tensions originate in a combination of cognitive, normative, and legal prescriptions for action, making inter-organizational tensions multifaceted. For example, an actor's response or perception towards digital servitization may be simultaneously shaped by the organization's business model and by an actor's functional orientation. This insight adds depth to our understanding of inter-organizational dynamics, suggesting that the interplay among diverse institutional forces yields various perspectives, opinions, and behaviors in digital servitization environments. Overall, a service ecosystem lens accommodates studying paradoxical and dialectical tensions, providing an integrated lens to explore inter-organizational tensions in digital servitization (Hargrave & Van de Ven, 2017).

Second, our findings extend the multi-level perspective on digital servitization outlined by Struyf et al. (2021). While institutional complexity has predominantly been examined at the intra-organizational and inter-organizational levels (Galvani & Bocconcelli,

2022; Lenka et al., 2018b; Tóth et al., 2022), our research reveals that such complexity also emerges from contradictions at the individual and industry levels. Behavior and perceptions of business actors are not only shaped by the organizational or inter-organizational context but also through higher and lower-level forces such as industry best practices or individual functional orientations. For instance, we identified that differing experiences among individual actors, or disparate industry best practices, can result in inter-organizational tensions and complicate digital servitization. Our study suggests that a comprehensive understanding of complexity in digital servitization requires expanding the analytical scope beyond the organizational and inter-organizational levels by including individual and contextual dynamics.

Third, our research enriches our understanding of ecosystem transformation and orchestration in digital servitization (Makkonen et al., 2022; Sjödin et al., 2022). We reveal that ecosystem orchestration extends beyond setting standardized “rules of the game,” as previously suggested by Kolagar et al. (2022), and involves navigating the diverse “rules” that individuals bring to the collaboration. These existing rules originate from the unique and disparate institutional arrangements in which actors are embedded before engaging with digital servitization. This diversity, indicative of institutional complexity, complicates orchestration efforts. For example, methods such as GANTT charts or sprint-based workflows might be met with resistance from actors whose institutional norms conflict with such practices. Our findings indicate that firms can respond to institutional complexity through structuring or reconciliation practices. As such, this study highlights practices to facilitate alignment of institutional arrangements (Makkonen et al., 2022). Actors can respond by grouping actors with similar institutional backgrounds or engage in institutional reconciliation to bridge and integrate diverging institutional backgrounds. These responses enable firms to mitigate the disrupting force of inter-organizational tensions and facilitate value co-creation in multi-actor collaborations. By highlighting practices to mitigate institutional complexity and align institutional arrangements (Makkonen et al., 2022), our study responds to the call of Kohtamäki et al. (2020) to further investigate practices to cope with challenges in digital servitization.

Fourth, by applying the service ecosystem perspective to the empirical context of digital servitization, our study contributes to the SDL literature. So far, the literature has focused on institutional reconciliation as the driving force of innovation in service ecosystems (Chandler et al., 2019; C. H. Lee & Hsieh, 2023; Siltaloppi & Vargo, 2014). However, our findings indicate that institutional complexity does not necessarily require reconciliation to facilitate innovation. Alternatively, actors can mitigate institutional complexity without addressing institutional divergence between resource-integrating actors (Greenwood et al., 2011). Our findings identify structuring practices that mitigate the disruptive consequences of institutional complexity for value co-creation without directly confronting institutional complexity itself. These findings are particularly relevant in the face of paradoxical tensions that involve irreconcilable complexities that persist over time (Smith & Lewis, 2011).

### **3.5.2 Practical implications**

Our findings provide various implications for practitioners. First, managers should be aware of the diverse institutional backgrounds among actors when engaging in industry-transcending collaboration. Because of this diversity, collaboration partners may have different expectations, interpretations, and opinions about an envisioned digital servitization strategy and how such strategy should be implemented. Second, managers should be attentive to inter-organizational tensions and appreciate the transformative opportunities of these tensions. While tensions complicate and slow down decision-making and achieving predefined goals, they can facilitate reflexivity among the involved actors. Reflexivity enables transformation as institutional arrangements, often implemented and reenacted unconsciously, become visible. Such visibility allows actors to address the underlying institutional complexity, work towards shared institutional arrangements, and build a service ecosystem that facilitates resource integration and value co-creation. Third, managers can use different approaches and tools to respond to inter-organizational tensions and unlock their transformative potential. They can use separation practices to smoothen decision-making or engage in institutional reconciliation to bridge different institutional arrangements and move towards a shared institutional arrangement. We advise managers to use both to balance short-term progress and the long-term viability of the service ecosystem.

### **3.5.3 Limitations and future research**

While our research contributes to theory and practice, it also presents limitations that open avenues for further exploration. Firstly, our study focuses on a single case study involving inter-organizational collaboration initiated by a manufacturer. However, the impetus for digital servitization can also originate from other actors, such as customers, governmental entities, or suppliers. Future studies could explore the impact of institutional complexity and response practices in these varied contexts to enhance the generalizability of our findings. Secondly, our application of a service ecosystem perspective focuses on a specific, self-contained collaboration and examines institutional complexity within this collaboration. To gain a more comprehensive understanding of the role of the institutional context in which digital servitization occurs, future research could apply a service ecosystem perspective on a larger scale. For example, exploring how firms navigate challenges like existing contracting routines, compliance with data-sharing legislation, or addressing customer reluctance to shift from ownership to usage models could provide valuable insights. Investigating these aspects would enrich our comprehension of the complexities of successfully implementing a digital servitization strategy, offering a more comprehensive picture of the institutional contexts shaping these initiatives. Third, future research could further investigate the conceptual model presented in this research. Specifically, future research should investigate the relationship between tension types and organizational responses and how various structuring and reconciliation practices facilitate value co-creation. Such research would increase our understanding of institutional complexity and shape service ecosystems to enable value co-creation (Vink et al., 2021; Vink & Koskela-Huotari, 2022).

## **Appendix 3A - Interview outline round 1**

### **Introduction**

1. Can you briefly tell me about your organization and your role in MetalNet?
2. Why did you decide to participate in MetalNet?
3. What goals do you hope to achieve by participating in MetalNet?

### **Collaboration MetalNet**

4. What is your opinion on collaboration regarding the assets?
5. Which assets have been addressed in MetalNet so far?
6. Can you describe the process and duration for each asset?
7. To what extent do these processes correspond to the normative flowchart?
8. In your opinion, why is the current asset lead time so far above the target time?
9. What needs to happen/be coordinated to set up the process to achieve a lead time of 10 days?
10. What is your opinion on the complementarity between MetalNet partners?
  - a. (Complementarity = the extent to which resources and knowledge of different actors reinforce each other to create (new) valuable products or services).
11. What is your opinion on the compatibility between MetalNet partners?
  - a. (Compatibility = the degree of “fit” between the different organizations’ cultures, interests, and work styles).
12. What is your opinion on the commitment of your organization and the other MetalNet partners to contribute to MetalNet?
13. Can you describe your trust level in the participating organizations within MetalNet?
14. Can you describe your trust level in the participating individuals within MetalNet?

### **Ecosystem**

15. To what extent is the following value proposition correct and complete: “Manufacture of high-mix, low-volume spare parts within ten working days”?
16. How is the production method determined for a spare part during an intake, and who is responsible?
17. What does the production process look like after the intake? Which parties are involved, and what does this involvement look like?
18. Are other essential actors in value creation not currently involved in MetalNet (e.g., your suppliers or partners)?

### **Revenue model/value capture**

19. What financial goals do you want to achieve by participating in MetalNet?
20. What is the added value of the association in MetalNet for the customer compared to offering the various services/products separately?
  - a. How does the customer benefit from the association of different companies in a network organization compared to the existing situation?
21. What requirements should the distribution of MetalNet profits meet, in your opinion?
22. How should MetalNet deal with the tension between the individual interests of the MetalNet partners and the collective interests of MetalNet?

## **Appendix 3B - Interview outline round 2**

### **Problem Analysis**

1. Separate the red problem cards that apply to MetalNet. Explain for each card what the problem means and why it is a problem. In other words, which goal does this problem hinder?
2. If you think there are still problems not represented on the red cards, write them on a blank red card and place them with the cards from step one.
  - a. Please explain what the problem means (including an example) and why it is a problem.
3. If applicable, can you describe how problems are interconnected?

### **Prioritizing Problems**

4. Prioritize the cards that you separated in step 1, i.e., which problems are the biggest obstacles to the success of MetalNet? Place these cards at the top.
5. Please explain your ranking.

### **Matching Solutions to Problems**

6. Drag the green solution cards currently being used to the corresponding red problem card (a green card can be associated with multiple problem cards).
  - a. Explain why the green card belongs to the problem card and how the solution can impact the problem.
7. Drag the green solution cards that are not yet used but could help the corresponding red problem card (again, a green card can be associated with multiple problem cards).
  - a. Explain why the green card belongs to the problem card and how the solution can impact the problem.
8. If there are missing current or potential solutions on the green cards, add them to an empty green card and drag them to the appropriate location.





# 4

## **AGILE CO-CREATION IN DIGITAL SERVITIZATION: INVESTIGATING THE IMPACT OF AGILE DEVELOPMENT AND LEAN STARTUP ON SERVICE BUSINESS PERFORMANCE**

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# Abstract

This study examines the role of agile co-creation in digital servitization. Despite the significance of agile innovation structures and processes in digital servitization, most investigations to date have been qualitative, necessitating further validation. Therefore, we quantitatively analyze the interplay between agile development, lean startup, and service business performance to address this gap. Using PLS-SEM based on 114 respondents, our findings indicate that agile development and lean startup relate positively to non-financial service business performance. However, only agile development shows a positive relationship with financial performance, a link partially mediated by non-financial performance. Notably, our results reveal a positive interaction between agile development and lean startup in relation to non-financial performance. Our findings extend the literature by (i) increasing the conceptual clarity of agile co-creation, (ii) providing empirical evidence of its performance implications in digital servitization, and (iii) enhancing the understanding of the synergy between agile development and lean startup. In conclusion, while implementing agile development and lean startup in tandem benefits servitizing firms in the initial stages, agile development, and its focus on micro-services, becomes more significant as firms advance in their servitization journey, especially when scaling a service business.

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## 4.1 Introduction

The servitization literature has increasingly focused on the role of digital technologies, such as the internet of things, big data analytics, and cloud computing (Paschou et al., 2020). This trend resulted in the emergence of digital servitization (Vendrell-Herrero et al., 2017), which is defined as “the transformation in processes, capabilities, and offering within industrial firms and their associate ecosystem to progressively create, deliver, and capture increased service value arising from a broad range of enabling digital technologies” (Sjödin et al., 2020, p. 478). While digitalization enhances service delivery (Paschou et al., 2020), it complicates servitization. Firms are facing challenges such as increased offering complexity, rising service delivery costs, and rapid technological developments (Porter & Heppelmann, 2014; Sjödin et al., 2020). To mitigate these challenges, servitizing firms must modify their organizational structure and business processes related to value creation and service delivery (Favoretto et al., 2022).

As the shift to digital services is considered emergent and associated with continuous innovation and uncertainty (Baines et al., 2020; Martinez et al., 2017), firms are believed to benefit from agile organizational forms (Chirumalla et al., 2023a; Favoretto et al., 2022). Specifically, the digital servitization literature has recently turned toward the role of agile co-creation, which involves a fast-paced, incremental approach to innovation characterized by small, focused investments in “micro-services,” iterative development cycles, and a “learning-by-doing” mentality (Sjödin et al., 2020). By employing short, time-boxed development cycles, known as “sprints,” companies can reduce the time to market and adapt quickly to evolving technological and customer landscapes (Huikkola, Kohtamäki, Rabetino, et al., 2022; Tronvoll et al., 2020). This agility creates a modular portfolio of micro-services, offering scalability and flexibility that are crucial to maintaining profitability in a service-based business model (Linde et al., 2023; Sjödin et al., 2020).

Two important gaps remain despite the increasing focus on agile co-creation to structure digital servitization. First, previous research investigating agile co-creation has relied on qualitative case-based studies, which provide a process view on agile co-creation in a digital servitization context and an elaborate account of its implementation (e.g., Linde et al., 2023; Sjödin et al., 2020). However, quantitative research could provide further insights into the relationship between agile co-creation and digital servitization outcomes. Second, agile co-creation has been considered a unified approach to innovation, even though it incorporates practices from multiple agile innovation approaches. Agile co-creation has been rooted in practices from agile development and lean startup (Huikkola, Kohtamäki, Rabetino, et al., 2022; Sjödin et al., 2020). It is essential to distinguish between the two approaches because they originate from different literature streams, consist of other practices, and may play different roles in digital servitization.

Agile development is a project management framework initially designed to structure software development processes by emphasizing short iterations and autonomous teams, facilitating rapid adjustments (Bianchi et al., 2020; Malik et al., 2021). On the

other hand, lean startup stems from the entrepreneurship domain and is tailored to business model innovation in high-uncertainty contexts through cycles of formulating hypotheses, conducting experiments, and learning through validation (Ghezzi & Cavallo, 2020a; Zellweger & Zenger, 2023). Agile development and lean startup practices play different roles in digital servitization as firms move from small-scale experiments to market growth through development and implementation (Linde et al., 2023). These different roles imply that they may have other performance implications and must be investigated separately.

To address these gaps and develop our understanding of agile co-creation in digital servitization, this paper aims to investigate the performance implications of agile development and lean startup. We follow Oliva et al. (2012) by differentiating between financial and non-financial service business performance. While financial service business performance relates to the profit and loss performance of a service business unit (SBU), non-financial performance relates to value co-creation reflected by customer satisfaction and loyalty (Oliva et al., 2012). Differentiating between financial and non-financial performance has two significant benefits. First, the non-linear relationship between servitization and financial performance indicates a considerable time lag between servitization efforts and financial pay-offs (E. Fang et al., 2008; Visnjic Kastalli & Van Looy, 2013). Customer satisfaction and loyalty are often considered antecedents of financial performance (Macdonald et al., 2016); differentiating between financial and non-financial performance accounts for this time lag (Oliva et al., 2012). Second, as well as tapping into new revenue pools, firms implement digital servitization to strengthen the traditional product business through a differentiation strategy (Salonen et al., 2017; Zimmer et al., 2020). As this strategic imperative implies that customer loyalty and satisfaction are more important than the financial performance of the SBU, differentiating between financial and non-financial performance enables us to capture different strategic motives.

To sum up, investigating the performance implications of agile development and lean startup through financial and non-financial service business performance provides a detailed and nuanced understanding of the role of agile development and lean startup in digital servitization. Hence, we answer the following research question: *What is the impact of agile development and lean startup on financial and non-financial service business performance?*

Using PLS-SEM to analyze survey data from 114 manufacturers, we advance the discussion on agile co-creation in digital servitization by providing novel insights into the separate performance implications of agile development and lean startup. Specifically, we report on (i) the direct consequences of agile development and lean startup for service business performance, (ii) the mediating role of non-financial service business performance in the relationships between agile development, lean startup, and financial service business performance, and (iii) the interaction effect between agile development and lean startup. These findings contribute to the literature about agile co-creation in digital servitization in multiple ways. First, we provide conceptual clarity by demonstrating that agile co-creation can be empirically observed through

agile development and lean startup practices. Second, we provide a more detailed understanding of agile co-creation in digital servitization by emphasizing the disparate roles of agile practices in digital servitization (Huikkola, Kohtamäki, Rabetino, et al., 2022; Sjödin et al., 2020). Our expectation is that, initially, agile development and lean startup jointly enhance value co-creation, but agile development becomes crucial as firms aim for scale.

## 4.2 Theory

### 4.2.1 Digital servitization

The literature has discussed servitization over three decades, emphasizing the shift from products to integrated product-service offerings (Baines et al., 2009; Vandermerwe & Rada, 1988). While digital technology has been part of the servitization narrative from its inception, the scope and accessibility of digital technology have increased significantly (Porter & Heppelmann, 2014). For example, servitizing firms increasingly rely on digital technologies such as the internet of things (IoT), big data analytics, and artificial intelligence (Paschou et al., 2020). Consequently, the literature on servitization has increasingly focused on digital servitization, highlighting the mutual relationships between servitization and digitalization. On the one hand, digital technologies are key enablers for offering integrated product-service packages (Abou-foul et al., 2020). On the other hand, servitization enables firms to recoup investments from digitalization efforts.

Servitization can offer firms economic, marketing, and strategic benefits (Baines et al., 2009). Economically, it promises direct financial advantages, such as stable revenue and profits, thanks to the revenue potential from servicing the installed base, higher service profit margins, and the relative resilience of service sales against economic downturns compared to product offerings (Kowalkowski, Gebauer, Kamp, et al., 2017; Oliva & Kallenberg, 2003). However, achieving profitability can be challenging because of disproportionately rising costs compared to revenue (Neely, 2008; Sjödin et al., 2020). In marketing, services can foster stronger customer relationships, enabling firms to understand their customers better and refine service and product offerings based on customer insights (Faramarzi et al., 2023; Salonen et al., 2017). Ultimately, improved customer relations enhance customer loyalty and satisfaction (Oliva et al., 2012). Strategically, services can facilitate differentiation from competitors. As manufacturers transition to services, they capitalize on their product knowledge, resulting in spillover benefits (E. Fang et al., 2008; Reinartz & Ulaga, 2008). Furthermore, given the intangible and labor-intensive nature of services, they are harder to replicate (Gebauer et al., 2005; Oliva & Kallenberg, 2003). Additionally, moving into the solution business signals quality to the market, enabling firms to differentiate themselves from their competitors (Zimmer et al., 2020). The multifaceted nature of these potential advantages should also be reflected in the conceptualization of digital servitization performance outcomes.

To fully understand the varied benefits of digital servitization, we conceptualize its outcomes as financial and non-financial service business performance. The financial

performance revolves around the notion that the service business has a separate profit and loss responsibility and involves generating revenue and profits from services (Antioco et al., 2008). Conversely, non-financial performance emphasizes intangible dimensions such as customer loyalty and satisfaction with the services provided (Homburg et al., 2003; Oliva et al., 2012). This performance dimension is intrinsically linked to value co-creation in digital servitization (Kamalaldin et al., 2020; Sjödin et al., 2016), a process defined by the collaborative melding of resources steered by value propositions predicting potential outcomes (Vargo, 2020; Vargo & Lusch, 2008). In other words, non-financial performance reflects the procedural and content-related consequences of value co-creation.

Differentiating between financial and non-financial performance dimensions is essential for several reasons. First, not all benefits of digital servitization translate directly into financial gains at the service business level (Salonen et al., 2017; W. Wang et al., 2018). For example, the market quality signals of solution offerings may enhance both service and traditional product businesses (Zimmer et al., 2020), with some benefits potentially not appearing in the service business's profit and loss statement but emerging at the broader firm level. Second, distinguishing between financial and non-financial performance aids in capturing the temporal nuances associated with transforming to digital services. A service-centric business model shift may require time to manifest financial returns, emphasized by the non-linear relationship between servitization and financial performance (E. Fang et al., 2008; Visnjic Kastalli & Van Looy, 2013). Since non-financial metrics such as customer loyalty and satisfaction are considered important precursors to financial outcomes (Macdonald et al., 2016; Tuli et al., 2007), they become vital indicators during the early stages of digital servitization.

#### **4.2.2 Agile co-creation in digital servitization**

The digital servitization literature increasingly discusses the role of agile co-creation (Chirumalla et al., 2023a; Ferreira & Lind, 2023). The focus on agile co-creation emerged from specific challenges associated with digital servitization. These include overly customized offerings, low willingness by the customer to pay for a technologically complex solution, and unexpectedly high digital transformation costs (Sjödin et al., 2020). Agile co-creation refers to an incremental innovation approach that involves short development cycles, frequent small investments, and learning by doing (Linde et al., 2023; Sjödin et al., 2020). Agile innovation processes and organizational structures are considered particularly important for dealing with the complex, non-linear, and fast-paced character of value co-creation associated with digital servitization (Favoretto et al., 2022; Huikkola, Kohtamäki, Rabetino, et al., 2022; Sjödin et al., 2020).

Agile innovation approaches originate from the software development context. They are defined in the agile manifesto as people-centered and responsive approaches focused on delivering working software in close collaboration with customers (Beck et al., 2001). Over time, this general understanding of agile development has resulted in many different methods and practices for software development, such as Scrum, Extreme Programming, and Kanban (Campanelli & Parreiras, 2015). Agile development

approaches are project management frameworks for software development stemming from the information systems literature (Bambauer-Sachse & Helbling, 2021; G. Lee & Xia, 2010). Inspired by agile software development approaches, a specific approach for business model innovation emerged in the entrepreneurship domain, namely, lean startup (Ghezzi & Cavallo, 2020a; Ries, 2011). In contrast to agile development practices, lean startup approaches are geared toward business model validation and originate from the entrepreneurship literature (Ghezzi & Cavallo, 2020a; Zellweger & Zenger, 2023). Despite the closeness and partial overlap of agile development and lean startup practices (Sandstø & Reme-Ness, 2021), they can be considered separate concepts.

Agile development is a software development approach built around self-organizing multidisciplinary teams that work in time-boxed sprints and active customer participation throughout the process (Dingsøyr et al., 2012). Working in autonomous and multidisciplinary teams enables firms to rely on the expertise of teams by giving them decision-making authority over their working processes, work prioritization, and planning (Cockburn & Highsmith, 2001; Shastri et al., 2021). Furthermore, agile teams break tasks down into smaller pieces and work in time-boxed iterations of two to four weeks (Bianchi et al., 2020). During these sprints, teams work on a limited number of predefined tasks to remain focused and prevent distractions (Pikkarainen et al., 2008). Every sprint concludes with a deliverable that can be presented to the customer to collect customer feedback (Edwards et al., 2019). The agile development process entails regular informal face-to-face communication, internally and with external stakeholders, to improve trust and reduce documentation (Bambauer-Sachse & Helbling, 2021; Serrador & Pinto, 2015). Informal communication occurs through various practices such as short daily standup meetings among team members, sessions in which deliverables are demonstrated to customers, and retrospectives in which teams discuss the collaboration process and how this process can be improved (Schwaber & Sutherland, 2020).

While agile development focuses on team dynamics and iterative processes, its principles have implications for the architecture of software and service offerings. Another essential element of agile development that emerged more recently is the concept of micro-services. In software development, micro-services refer to a modular software architecture existing of discrete and loosely coupled applications (Jamshidi et al., 2018). Firms using a micro-service approach design software systems as separate subsystems, each with a single function, operating independently. In this sense, micro-services can be contrasted with so-called “monoliths,” which refer to building and deploying software systems with multiple functionalities in one application (Ponce Mella et al., 2019). In digital servitization, micro-services refer to digital services that deliver a specific functionality addressing a particular customer need (Sjödín et al., 2020). For example, instead of developing an elaborate fleet management solution that addresses multiple needs, firms develop small functionalities that address one need at a time.

Lean startup is an agile innovation approach tailored to business model innovation (Ghezzi & Cavallo, 2020a). It is based on a pragmatist approach to entrepreneurship, which posits that entrepreneurs create value in uncertain environments through

inferential action, forming beliefs, testing these beliefs, and responding to test results (Zellweger & Zenger, 2023). The lean startup approach poses that business model innovation in uncertain environments requires falsifiable hypotheses to be tested through systematic experimentation instead of drafting detailed business plans (Blank & Dorf, 2012; Ries, 2011). Lean startup is based on the “build–measure–learn” loop, which is planned in reverse (Allen, 2022). So, firms start by formulating hypotheses that underpin their business ideas and then develop well-designed experiments and metrics to test these hypotheses (Ries, 2011). Based on the outcomes of these experiments, firms decide whether they will pursue the idea and test more assumptions or pivot in another direction (Mansoori, 2017). By adopting a methodologically rigorous and objective approach to business model innovation, firms can mitigate decision-making biases such as confirmation and optimism bias (Eisenmann et al., 2011). Also, the experiment-based approach of lean startup offers firms the means to understand, quantify, and reduce the risks associated with business model innovation, such as the cost structure, the customer’s willingness to pay, and interdependencies with external actors (Ganguly & Euchner, 2018; Teece, 2018).

While the digital servitization literature has explored various aspects of agile co-creation, it has not recognized the nuanced conceptual differences between agile development and lean startup. Sjödin et al. (2020) introduced an agile co-creation approach that integrates agile development and lean startup elements, such as sprint-based working and experimentation, into a unified innovation approach. However, it is essential to note that firms might not consistently integrate agile development and lean startup practices (Zorzetti et al., 2022). Additionally, agile development and lean startup practices play differing roles at various stages of the digital servitization process. The early stages, characterized by significant uncertainty, benefit from lean startup practices such as hypothesis-testing and business model validation, whereas iterative agile development practices become more important as firms progress and shift toward service implementation and optimization (Linde et al., 2023). This differentiation suggests that agile development and lean startup might relate differently to digital servitization outcomes.

### **4.2.3 Hypothesis development**

#### **4.2.3.1 Direct relationship between agile development and service business performance**

Agile development provides opportunities to deal with the increasing importance of the experiential nature of value in servitization. Digital servitization involves a shift from selling products based on specifications to selling value-in-use (Keränen et al., 2021; Ulaga & Loveland, 2014). This shift to value-in-use implies an experienced, contextual, and processual nature of value (Chandler & Vargo, 2011; Macdonald et al., 2016). To discover and monitor value-in-use, firms must stay in constant dialogue with customers and build trust-based relationships so that customers share essential information, which may put them in a vulnerable position (Raddats et al., 2019). As agile development is associated with increasing trust, transparency, and expectation

alignment with customers (Bambauer-Sachse & Helbling, 2021; Dybå & Dingsøyr, 2008), it can support servitizing firms in dealing with the experiential nature of value. Engaging in sprint-based working, constantly releasing small value increments, and high customer responsiveness build trust with the customer and enhance customer-supplier alignment (Sjodin et al., 2021). As such, agile development enables servitizing firms to understand the customer's goals and context, thus discovering opportunities to facilitate value-in-use through product-service offerings.

Next, to transition to a value-in-use perspective, servitizing firms must be able to exploit customer insights by feeding them back into the service development and sales process (Prohl & Kleinaltenkamp, 2020). Bundling products and services into packaged solutions requires the integration of tacit knowledge from various functions and disciplines (Storbacka, 2011). Particularly in digital servitization, firms face the challenge of integrating manufacturing, service, and digital capabilities (Favoretto et al., 2022; Ulaga & Reinartz, 2011). Integrating these disparate capability bases could benefit from the collaboration practices promoted by agile development focused on transferring tacit knowledge (Beck et al., 2001; Serrador & Pinto, 2015). By working in diverse and autonomous teams that adopt communication practices such as daily standups, demo sessions, and retrospectives, agile development provides a platform for information exchange by bringing together team members and various stakeholders (Pikkarainen et al., 2008). The frequent and informal nature of communication enables team members from various backgrounds and departments to develop working relations, reduce sociocultural distance, and increase intra-team information flow (Herbsleb & Mockus, 2003; Holmström et al., 2006). Based on these dynamics, agile development enables firms to integrate various capabilities to develop digital product-service offerings.

Overly customized service offerings often complicate the shift to a contextual and experiential nature of value, and servitizing firms often struggle to set up a scalable service business because of the resource-intensive nature of customized services (Johnson & Selnes, 2004; Kowalkowski et al., 2015; Nordin et al., 2011). Agile development facilitates balancing service customization and standardization by promoting incremental innovation and developing small increments that provide customer value. Agile development utilizes a "product backlog," a repository for outputs and tasks (Dingsøyr et al., 2018). This approach encourages large deliverables to be broken down into smaller components, which entail discrete value increments that can be developed and released within one or more sprints. Over time, releasing these discrete value increments results in a portfolio of micro-services, which can be easily combined and adjusted across customers as they act as standalone functionalities (Sjodin et al., 2020). Also, adjustments can be made in individual micro-services without needing to test the entire system (Jamshidi et al., 2018). This modular approach enables servitizing firms to balance customization and standardization and prevent service development and delivery costs from disproportionately increasing compared to revenue (Cenamor et al., 2017; Hasselblatt et al., 2018; Sjodin et al., 2020).

In summary, agile development gives servitizing firms the tools to navigate the experiential and processual nature of value in digital services, to integrate

multidisciplinary knowledge, and to maintain a balance between customization and standardization. As such, we hypothesize that:

*H1a: Agile development relates positively to financial service business performance.*

*H1b: Agile development relates positively to non-financial service business performance.*

#### **4.2.3.2 Direct relationship between lean startup and service business performance**

As an approach to opportunity exploration, lean startup principles offer a framework for fostering a learning orientation, particularly when implementing a digital servitization strategy (Huikkola, Kohtamäki, Rabetino, et al., 2022; Tronvoll et al., 2020). This orientation is vital for addressing complex challenges in digital servitization, such as emerging customer demands, new regulatory landscapes, and fast-paced technological innovation (Benedettini et al., 2015; Favoretto et al., 2022). The learning process, rooted in hands-on experience in offering services (Eggert et al., 2011, 2014), is enhanced by lean startup practices. Hypothesis-testing, MVP experimentation, and systematic data analysis reduce development time and resource wastage (Allen, 2022; Felin & Zenger, 2017; Lizarelli et al., 2022). Objective MVP performance evaluations allow managers to sharpen their decision-making (Dobrigkeit & de Paula, 2017), while the flexible nature of lean startup techniques helps firms to pivot away from unfeasible business ideas, preserving valuable resources (Lizarelli et al., 2022). Ultimately, these methods allow servitizing firms to build their service portfolio and quickly eliminate non-viable service concepts, paving the way to scaling their service business profitably, often within a significantly reduced time frame (Huikkola, Kohtamäki, Rabetino, et al., 2022).

A specific focus for servitizing firms is understanding customer needs. As previously noted, servitization focuses on facilitating value-in-use, a concept encompassing explicit and latent customer needs (Mahr et al., 2014). Customers may be unaware of their business needs or struggle to articulate them effectively (Aarikka-Stenroos & Jaakkola, 2012). This complexity is exacerbated by the diversity of needs among various stakeholders within a customer organization, each requiring tailored value propositions from the service provider (Homburg & Kuehn, 2014; Payne et al., 2017). The experimental approach of lean startup enables firms to navigate this complexity. Unlike traditional market research methods, which may capture customer feedback in artificial contexts such as surveys, lean startup experimentation aligns with a proactive market orientation. This orientation allows firms to explore and satisfy underlying customer needs. In particular, the intangibility, perishability, and customization of services make them challenging to test within artificial boundaries, reinforcing the importance of experimentation (Schultz et al., 2019; Thomke, 2003). Hence, experimentation with MVPs in the customer context provides genuine and valid insights into a customer's business needs, setting it apart from traditional techniques (Blank & Dorf, 2012; Ries, 2011). Constant experimentation in the customer context offers servitizing firms essential insights that cannot be collected in an R&D lab (Sjödin et al., 2020; Thomke, 2003).

Shifting to a value-in-use logic requires a shift from a technology-oriented to a customer-centered organizational identity, and from a manufacturing to a digital technology firm (Favoretto et al., 2022; Huikkola, Kohtamäki, Rabetino, et al., 2022). This shift is supported by establishing a vision for digital servitization, legitimizing it through managerial narratives, and exposing key stakeholders to new digital technologies and service offerings (Kohtamäki et al., 2021; Tronvoll et al., 2020). Iterative experimentation facilitates stakeholder dialogue, internal learning, and justifying resource allocation to service propositions (Bocken, Schuit, et al., 2018; Huikkola, Kohtamäki, Rabetino, et al., 2022). By defining hypotheses related to value creation, lean startup helps managers to recognize biases and include diverse insights in innovation (Mansoori, 2017). Systematic experiments and predefined metrics improve communication and build support for new service businesses (Ganguly & Euchner, 2018; Mansoori, 2017). By grounding service concepts in real-life experiments and predefined KPIs, servitizing firms can overcome resistance rooted in the traditional product-oriented logic of manufacturing firms (Bocken, Schuit, et al., 2018; Lütjen et al., 2017).

In summary, the lean startup approach equips servitizing firms with empirically driven practices aligned with a proactive market orientation. These practices foster a learning orientation and enable firms to better understand complex and latent customer needs. They facilitate organizational transformations toward customer-centric and digital technology orientations while optimizing resource allocation. Consequently, lean startup principles present a robust framework for improving the financial and non-financial aspects of service business performance. Therefore, we hypothesize that:

*H2a: Lean startup relates positively to financial service business performance.*

*H2b: Lean startup relates positively to non-financial service business performance.*

#### **4.2.3.3 Indirect relationship between agile development, lean startup, and financial service business performance**

So far, we have hypothesized that agile development and lean startup contribute directly to the financial bottom line of the service business and value co-creation reflected in non-financial performance. Investigating the mediating role of non-financial business performance provides a nuanced understanding of the relationship between agile development, lean startup, and service business performance.

Customer satisfaction and loyalty are well-documented crucial antecedents of financial performance (Macdonald et al., 2016; Tuli et al., 2007). However, value creation does not guarantee value capture because many features that customers value are “nice to have” for which they are unwilling to pay (Amit & Zott, 2001; Linde et al., 2023). Concerning the distinction between value creation and value capture (Amit & Zott, 2001), the servitization literature reveals challenges in capitalizing on services that customers value, with many firms struggling to charge fees for previously free services (Witell & Löfgren, 2013). This struggle is rooted in a manufacturing-oriented worldview of the sales force and a lack of capability to develop and implement new revenue models

for product-service offerings (Linde et al., 2023; Mustak et al., 2023). These challenges highlight the complexity of translating non-financial performance into financial returns, implying that value co-creation outcomes cannot always be fully reflected in economic gains.

Furthermore, service development and delivery costs drive the profit dimension of financial performance. Servitizing firms often struggle with controlling the delivery costs of digital services, as they are more labor-intensive, require more working capital, and are often less controllable in performance-related service contracts than product offerings (Coreynen et al., 2017; Neely, 2008). Also, digital services involve significant maintenance costs, as service providers must regularly update software and sensors in line with rapid technological changes, further increasing delivery costs (Sjödin et al., 2020). These costs are a critical element of the economic outcomes of digital servitization. However, non-financial service business performance does not fully reflect this cost dimension, which means that non-financial service business performance only partially predicts financial performance. Our expectation is that agile development and lean startup benefit non-financial service business performance. However, they do not fully translate into economic gains because of the imperfect relationship between value creation and capture. Additionally, costs linked to service development and delivery significantly impact financial performance. These costs are not fully reflected in non-financial service business performance. Hence, we hypothesize that:

*H3: Non-financial service business performance partially mediates the relationship between agile development and financial service business performance.*

*H4: Non-financial service business performance partially mediates the relationship between lean startup and financial service business performance.*

#### **4.2.3.4 Interaction effect between agile development and lean startup**

So far, we have discussed separately the expected impact of agile development and lean startup on service business performance. However, the two approaches seem complementary, and we expect their joint adoption to have specific benefits (Cano et al., 2021; Silva et al., 2020).

Agile development and lean startup practices complement each other in developing offerings that meet customer needs (Cano et al., 2021). As a project management framework in a service context, agile development improves customer satisfaction in terms of the process but not the output (Bambauer-Sachse & Helbling, 2021). This finding might be related to a lack of focus of agile development on understanding problems and finding solutions, as it assumes that the development teams are aware of customer needs (Dobrigkeit & de Paula, 2017; Ghezzi & Cavallo, 2020a). This limitation can be addressed by combining agile development with lean startup practices. By integrating agile development and lean startup, every sprint revolves around a build-measure-learn cycle in which validated learning becomes the primary objective rather

than the delivery of an increment (Cano et al., 2021). So, development teams dedicate sprints to running experiments and gathering data to test a limited set of hypotheses (Blank, 2013). Insights from these experiments feed back into the product backlog and form the basis of the next sprint. As such, integrating lean startup and agile development increases the market focus of agile development teams and offers them the means to translate customer feedback into actionable learning points (Ximenes et al., 2015).

Moreover, combining agile development and lean startup gives teams the required structure and processes for successful experimentation. Lean startup is mainly adopted in young and nascent firms, which is a substantially different context than established firms in terms of organizational structure, culture, and organizational legacy (Chesbrough & Tucci, 2020). Successful experimentation requires teams to have autonomy and a “license to fail” (Signoretti et al., 2019), which is not a given in established firms that tend to promote hierarchical control and not tolerate failure (Euchner, 2019). Also, successful experimentation through lean startup practices requires a cross-functional team (Edison et al., 2018). When firms implement agile development structures alongside lean startup processes, teams get the autonomy and flexibility to experiment with new service concepts and act upon emerging insights. So, agile development informs how to structure and position teams experimenting with new service concepts through the lean startup approach.

Besides dealing with the experiential nature of value in servitization, combining agile development and lean startup may also lead to efficiency gains, enabling servitizing firms to better use their scarce resources. Both agile development and lean startup adopt an incremental approach to releasing output to the customer, and the roles of these releases differ in the two approaches (Cano et al., 2021). Agile development focuses on frequent and consistent releases of product increments enabled by sprint-based working (Schwaber & Sutherland, 2020). However, because of the technical orientation of agile development, teams risk becoming trapped in their product vision, which may result in a suboptimal product-market fit (Griffin, 2021). Because lean startup considers incremental releases (i.e., MVPs) to be a means to validate learning (Ries, 2011), lean startup increases the market focus of agile development teams and the quality of their decision-making (Dobrigkeit & de Paula, 2017; Ximenes et al., 2015). So, combining agile development and lean startup increases the efficiency and effectiveness of experimentation, resulting in higher resource efficiency in innovation.

Complementarity between agile development and lean startup is also evident in how the two approaches facilitate knowledge management. As discussed earlier, lean startup facilitates knowledge management by increasing openness to others’ opinions and views and enabling support-building among stakeholders through a fact-based approach to innovation (Ganguly & Euchner, 2018; Mansoori, 2017). In addition, agile development facilitates tacit knowledge transfer through events that stimulate frequent informal face-to-face interaction (Holmström et al., 2006; Serrador & Pinto, 2015). So, agile development provides a platform for bringing together the relevant stakeholders (Pikkarainen et al., 2008), and lean startup provides fact-based content that facilitates

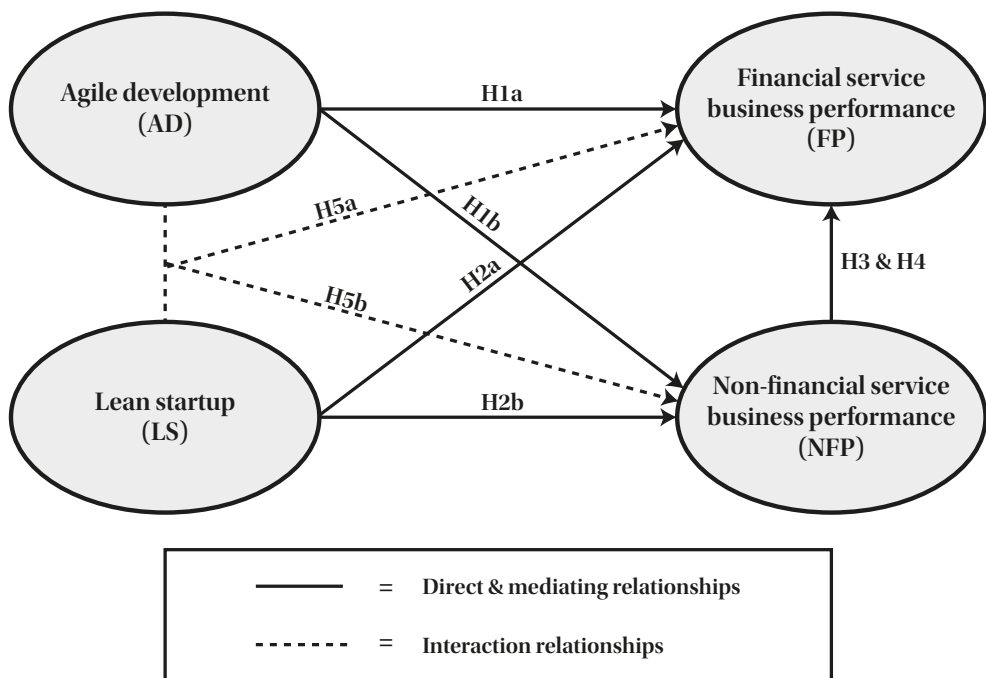
productive dialogue between these stakeholders (Bocken, Schuit, et al., 2018).

In sum, we expect agile development and lean startup to complement each other when implementing a digital servitization strategy. By combining the two approaches, servitizing firms can better deal with the complex and experiential nature of value in servitization, better use scarce resources, and integrate knowledge from internal and external stakeholders. This results in both financial and non-financial benefits. Hence, we hypothesize that:

*H5a: Agile development and lean startup positively interact in relation to financial service business performance.*

*H5b: Agile development and lean startup positively interact in relation to non-financial service business performance.*

In summary, Figure 4.1 (below) provides our theoretical model and illustrates the developed hypotheses.



**Figure 4. 1** Theoretical model, including hypotheses

## 4.3 Method

We used partial least squares (PLS) path modeling to test our theoretical model. PLS is a variance-based structural equation modeling (SEM) technique that explains variance by estimating model parameters based on proxies stemming from linear combinations of observed variables (Henseler et al., 2016). We chose PLS as our analysis technique for three reasons. First, because PLS uses an iterative estimation process and estimates latent variable proxy scores instead of using a sample covariance matrix, it can handle complex models and relatively small sample sizes (Hair et al., 2022; Henseler et al., 2016). Second, PLS is a limited information estimation technique and hence less likely to be affected by model misspecification in some parts of the model (Antonakis et al., 2010). As our measurement instruments are relatively new and, in some cases, require further development (Harms & Schwery, 2020), PLS provides more robust results to potential model misspecification. Third, PLS is the method of choice when a theoretical model contains composite factors (Henseler et al., 2016), which is the case in our theoretical model. To conduct the PLS analyses, we used SmartPLS 4 (Ringle et al., 2022).

### 4.3.1 Data and sample

To study the performance implications of agile development and lean startup in a digital servitization context, we collected survey data from an international set of companies that either manufactured products (NAICS 31–33) or engaged in systems integration of industrial products (NAICS 5415). We collected the data between May 2021 and May 2023. Respondents were approached through various means. First, we contacted 419 companies from a database comprising Dutch manufacturing companies. We also contacted eligible companies through network organizations and consultancies with an extensive network in manufacturing industries. Finally, we sampled respondents from the authors' and colleagues' personal networks through LinkedIn. We contacted companies by phone to introduce them to the study and ensure that we found the right person within the firm to complete the survey. Specifically, we looked for individuals who were closely involved with developing new services; we then sent the survey to the right person within the firm. By approaching respondents through various channels and assessing whether these companies actively offered services, our sample was strategically curated, mitigating sampling bias and enhancing the robustness and generalizability of our findings.

After excluding incomplete survey responses, our sample comprised 114 observations. Four responses did not provide input on the company size or country; however, we included these observations because they provided input on the constructs of interest and contributed to our analysis. The positions of informants included executive managers (33.6%), business unit directors (23.9%), project managers (7.1%), production managers (16.8%), and business developers (12.4%). Other roles included consultant, engineer, and developer (6.2%). Respondents' involvement with new service development was, on average, 58% of a whole working week, ranging from 5% to 100%. Hence, we assumed that our respondents could provide valid answers to our survey questions about structure, processes, and performance concerning new service development. Table 4.1 provides details of our sample and shows that most of our sample comprised small and

medium enterprises. On average, the sample firms got 30% of their revenue from add-on services and integrated product-service contracts compared to physical goods. Most sample firms (65.8%) were at or above the critical service ratio value at which they could expect positive financial results (E. Fang et al., 2008).

**Table 4. 1**                      **Sample descriptive statistics (N = 114)**

|                                       | Frequency | Percentage |
|---------------------------------------|-----------|------------|
| <b>Size</b>                           |           |            |
| Small-sized enterprises (<50 FTE)     | 28        | 24.6%      |
| Middle-sized enterprises (50-249 FTE) | 36        | 31.6%      |
| Large-sized enterprises (> 250 FTE)   | 46        | 40.4%      |
| Missing                               | 4         | 3.5%       |
| <b>Share revenues from services</b>   |           |            |
| <20%                                  | 39        | 34.2%      |
| 20-30%                                | 32        | 28.1%      |
| >30 %                                 | 43        | 37.7%      |
| <b>Country</b>                        |           |            |
| The Netherlands                       | 78        | 68.4%      |
| Others                                | 30        | 26.3%      |
| Missing                               | 6         | 5.3%       |

### 4.3.2 Variables and operationalization

We used previously developed and tested scales to measure the service business performance, agile development, and lean startup constructs. Items were translated from English to Dutch, and back from Dutch to English, to ensure translation equivalence (Brislin, 1970). We used the instrument of Oliva et al. (2012) to measure service business performance. This measurement instrument differentiates between financial and non-financial service business performance, which aligns with the frequently reported benefits of servitization (Faramarzi et al., 2023; W. Wang et al., 2018). All performance measures were based on a 7-point Likert scale. We used the validated measurement instrument of Malik et al. (2021) to measure agile development. They differentiate between four dimensions of agile development, including team autonomy, team diversity, iterative development, and agile communication. Again, all measures were based on a 7-point Likert scale. Finally, we used the work of Harms and Schwery (2020) to measure lean startup. They differentiate between five dimensions: customer insight, hypothesis-testing, iterative experimentation, validation, and learning. While their instrument shows acceptable measurement validity, it could benefit from additional items. Hence, we added other questions to measure the dimensions of learning, hypothesis-testing, and validation.

During the data analysis, we controlled for the impact of various variables on service business performance. First, we controlled for the effect of firm size because larger firms tend to engage more in servitization and might benefit from higher resource slack more than smaller firms (E. Fang et al., 2008; Neely, 2008). We measured firm size in terms of

full-time employees (FTE). Second, we controlled the number of customers because we expected firms with fewer customers to have closer customer relations. Such relations enable firms to offer services that meet customer requirements and are profitable (T. Morgan et al., 2019). Third, we controlled for the type of physical product offered (i.e., machinery or non-machinery). While providing additional services to an installed machinery base is more natural, adding services to raw materials or component offerings requires the entire mental model of companies to be changed (Storbacka et al., 2013). After running the analysis, we found that only the type of product being offered affected one of the dependent variables, so we omitted the other control variables from our model.

#### **4.3.3 Non-response bias**

Following Armstrong and Overton (1977), we checked our data for non-response bias by comparing the indicator scores related to our dependent and independent variables of early and late respondents. We divided respondents into two groups: one who filled in the survey within the first 12 months and another who filled it in during the last 12 months. Because our data on dependent and independent variables were not normally distributed, we conducted a non-parametric test to compare the means of both groups. After performing a Mann-Whitney U Test, we found that late respondents scored significantly higher on 16 of the 48 indicators than early respondents.

While this pattern raised the possibility of non-response bias, it is essential to consider the distribution of these differences. Specifically, two significantly different indicators were related to financial service business performance. In contrast, none of the indicators associated with non-financial performance differed between early and late respondents. Concerning our independent variables, it is worth noting that all indicators related to the variable “validation” showed significantly higher scores among late respondents. Notably, the “validation” variable is a subcomponent of the broader composite construct of lean startup, suggesting that the observed non-response bias in this variable is unlikely to significantly affect the overarching findings of our study. The remaining differences among indicators were scattered across multiple independent variables, indicating no further concentration of non-response bias in any particular area of inquiry. The scattered nature of the differing indicator scores across multiple variables mitigates the risk that non-response bias systematically affected our overall conclusions. In other words, while there may be a specific risk of bias in the validation variable, the broader array of indicators showing differences suggests that our study may still offer a broadly applicable perspective.

#### **4.3.4 Common method bias**

We conducted a full collinearity test to assess whether common method bias might have affected our results (Kock, 2015). We first added a new column to our dataset, with random values ranging from 0 to 1. We then created a latent variable representing the random dummy variable and drew a path model in which all first-order reflective factors pointed to the new random dummy variable. After running the PLS regression

algorithm and inspecting the VIFs for scores above 3.3, we found that the VIF values of “agile communication,” and our product type control variable, indicated minor common method bias ( $VIF_{\text{agile\_communication}} = 3.494$ ,  $VIF_{\text{prodtype\_machines}} = 3.683$ ). However, because only two factors suffered from common method bias and the values were slightly above the 3.3 threshold, we concluded that our results were not severely affected by common method bias and proceeded with the analysis.

## 4.4 Results

### 4.4.1 Overall model fit

First, we evaluated potential misspecification issues for the measurement and structural models. We conducted three distinct goodness-of-fit tests: the standardized root mean square residual (SRMR), the unweighted least squares discrepancy (dULS), and the geodesic discrepancy (dG). To determine an acceptable model fit, the test scores from the estimated model were compared with specified bootstrap quantiles at both the 95% and 99% confidence levels. These criteria serve as general guidelines. A lower score indicates a better fit, but the test scores should not exceed the bootstrap quantiles at the 95% and 99% confidence levels for a model to be deemed acceptable.

**Table 4.2**                      **Model fit indices (estimated model results)**

| Criterion | Value | HI95  | HI99  |
|-----------|-------|-------|-------|
| SRMR      | 0.077 | 0.070 | 0.084 |
| dULS      | 0.465 | 0.380 | 0.549 |
| dG        | 0.235 | 0.237 | 0.290 |

The test results from the direct path model (Table 4.2) showed an acceptable fit. Specifically, the SRMR and dULS scores slightly exceeded the 95% confidence level but remained within the acceptable range of the 99% confidence level. On the other hand, the dG score fell below the 95% confidence level, signifying a good fit between the model and the observed data. These results suggest that the model has acceptable robustness, providing a reliable framework for interpreting the data.

### 4.4.2 Measurement model

We adopted a disjoint two-stage approach for modeling agile development and lean startup, following the guidelines of Sarstedt et al. (2019). We modeled agile development and lean startup as reflective–formative higher-order constructs. We first modeled the first-order factors. The latent factor scores derived from this step were then utilized as formative indicators to estimate our formative higher-order construct of the agile innovation approach.

Following Hair et al. (2022), we assessed the first-order reflective measurement model through convergent validity, reliability, and discriminant validity. Our initial analysis failed to establish discriminant validity between agile communication and

iterative working. The heterotrait-monotrait ratio of correlation (HTMT) for these two constructs was 0.915 and thus exceeded the recommended threshold of 0.9 (Henseler et al., 2015). To address this issue, we merged the two constructs into a single reflective factor called “sprint-based working.” This merger was theoretically justifiable, as agile communication and iterative working are core aspects of how agile teams organize their work during sprints (Schwaber & Sutherland, 2020).

Tables 4.3 and 4.4 show the reliability and validity scores for the first-order reflective constructs, including the merged sprint-based working factor. We reassessed measurement model quality based on Cronbach’s alpha and PA for reliability (Dijkstra & Henseler, 2015), the average variance extracted (AVE) score for convergent validity, and the HTMT ratio for discriminant validity (Henseler et al., 2016). Table 4.3 shows that all factor measures were reliable ( $\alpha > 0.7$ ,  $PA > 0.7$ ) and that convergent validity was established ( $AVE > 0.5$ ). Table 4.4 shows that all first-order factors showed discriminant validity ( $HTMT < 0.9$ ). Hence, we concluded that our first-order factor measurements were valid and reliable.

**Table 4.3 First-order factor assessment and results**

| Factor construct (incl. indicators)                                                                                               | Code     | Standardized factor loading | Cronbach's alpha ( $\geq 0.7$ ) | PA ( $\geq 0.7$ ) | AVE ( $\geq 0.5$ ) |
|-----------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------|---------------------------------|-------------------|--------------------|
| <b>Service business performance (Oliva et al., 2012)</b>                                                                          |          |                             | 0.812                           | 0.902             | 0.731              |
| Financial service business performance (i.e., the financial performance of the service business)                                  |          |                             |                                 |                   |                    |
| The services we provide are profitable.                                                                                           | Servfp1  | 0.648                       |                                 |                   |                    |
| A significant portion of our total profit is generated by our service business.                                                   | Servfp2  | 0.946                       |                                 |                   |                    |
| A significant fraction of our total revenue is generated by our service business.                                                 | Servfp3  | 0.937                       |                                 |                   |                    |
| <b>Non-financial service business performance (i.e., the perceived ability to satisfy customers through the service business)</b> |          |                             | 0.894                           | 0.935             | 0.755              |
| Customers are very satisfied with the quality of our services.                                                                    | Servnfp1 | 0.881                       |                                 |                   |                    |
| The collaboration with our customers is smooth.                                                                                   | Servnfp2 | 0.903                       |                                 |                   |                    |
| We regularly satisfy our customers.                                                                                               | Servnfp3 | 0.926                       |                                 |                   |                    |
| We are able to retain the majority of our customers.                                                                              | Servnfp4 | 0.755                       |                                 |                   |                    |
| <b>Agile development (Malik et al., 2021)</b>                                                                                     |          |                             | 0.915                           | 0.943             | 0.797              |
| Team autonomy (i.e., degree of self-management in service innovation)                                                             |          |                             |                                 |                   |                    |
| The service project team is allowed to freely choose tools and technologies.                                                      | Ta1      | 0.932                       |                                 |                   |                    |
| The service project team has control over what they are supposed to accomplish.                                                   | Ta2      | 0.916                       |                                 |                   |                    |
| The service project team is granted autonomy on how to handle user requirement changes.                                           | Ta3      | 0.917                       |                                 |                   |                    |
| The service project team is free to assign personnel to the project.                                                              | Ta4      | 0.799                       |                                 |                   |                    |
| <b>Team diversity (i.e., varied backgrounds, skills, and perspectives of team members)</b>                                        |          |                             | 0.932                           | 0.941             | 0.831              |
| The members of the service project team are from different areas of expertise.                                                    | Td1      | 0.896                       |                                 |                   |                    |
| The members of the service project team have skills that complement one another.                                                  | Td2      | 0.931                       |                                 |                   |                    |
| The members of the service project team have a variety of different experiences.                                                  | Td3      | 0.913                       |                                 |                   |                    |
| The members of the service project team vary in functional backgrounds.                                                           | Td4      | 0.905                       |                                 |                   |                    |

**Table 4.3 (Continued) First-order factor assessment and results**

| Factor construct (incl. indicators)                                                                                                                       | Code | Standardized factor loading | Cronbach's alpha ( $\geq 0.7$ ) | PA ( $\geq 0.7$ ) | AVE ( $\geq 0.5$ ) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------|---------------------------------|-------------------|--------------------|
| <b>Sprint-based working</b> (i.e., structured, iterative cycles of planning, execution, review, and adaptation)                                           |      |                             | 0.925                           | 0.929             | 0.655              |
| The service project team usually works in short iterations or cycles to carry out work on their project.                                                  | It1  | 0.837                       |                                 |                   |                    |
| The amount of work that is performed on each project phase or cycle is strictly defined.                                                                  | It2  | 0.844                       |                                 |                   |                    |
| Major tasks or product/service features are typically split into smaller components to fit with the length of a development phase or cycle.               | It3  | 0.773                       |                                 |                   |                    |
| Service project team members have constant access (either visually in the workplace or through a software system) to information such as project status.  | Ac1  | 0.798                       |                                 |                   |                    |
| The customer/user (or their representative) is available and helps to answer questions regarding service requirements.                                    | Ac2  | 0.814                       |                                 |                   |                    |
| The service project teams organize daily team meetings to discuss the progress and adjust future work if necessary.                                       | Ac3  | 0.823                       |                                 |                   |                    |
| A meeting is conducted after an iteration in which the team presents its goal/scope, the work completed, key decisions, and a demo of the completed work. | Ac4  | 0.816                       |                                 |                   |                    |
| The service project team emphasizes face-to-face communication for conveying information within the team.                                                 | Ac5  | 0.768                       |                                 |                   |                    |

**Table 4.3 (Continued) First-order factor assessment and results**

| Factor construct (incl. indicators)                                                                                                                                         | Code | Standardized factor loading | Cronbach's alpha ( $\geq 0.7$ ) | PA ( $\geq 0.7$ ) | AVE ( $\geq 0.5$ ) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------|---------------------------------|-------------------|--------------------|
| <b>Hypothesis-testing</b> (i.e., translating vision into falsifiable assumptions)                                                                                           |      |                             | 0.937                           | 0.941             | 0.841              |
| The service project team formulated a series of testable and falsifiable assumptions about the market needs and how to best implement this analysis.                        | Ht1  | 0.903                       |                                 |                   |                    |
| The service project team translates the vision of the service and its value proposition into falsifiable assumptions.                                                       | Ht2  | 0.918                       |                                 |                   |                    |
| The service project team explicitly formulates short and concise sentences reflecting expectations about the market needs and how best to deliver them.*                    | Ht3  | 0.923                       |                                 |                   |                    |
| The service project team formulates a series of assumptions that represent the expectations about relations between elements of market needs and how best to deliver them.* | Ht4  | 0.923                       |                                 |                   |                    |
| <b>Customer insight</b> (i.e., deep customer understanding)                                                                                                                 |      |                             | 0.909                           | 0.923             | 0.784              |
| It is important to gain deep market insight (= talking directly to customers) to better understand the customer's challenge.                                                | Ci1  | 0.890                       |                                 |                   |                    |
| When developing the solution, the service project team always keeps the customer in mind.                                                                                   | Ci2  | 0.920                       |                                 |                   |                    |
| The service project team invested significant effort into understanding the challenge and learning about the user and the social context.                                   | Ci3  | 0.841                       |                                 |                   |                    |
| It is important to gain deep market insight into how the solution resolves the customer's challenge.                                                                        | Ci4  | 0.890                       |                                 |                   |                    |

**Table 4.3 (Continued) First-order factor assessment and results**

| Factor construct (incl. indicators)                                                                                                                            | Code | Standardized factor loading | Cronbach's alpha ( $\geq 0.7$ ) | PA ( $\geq 0.7$ ) | AVE ( $\geq 0.5$ ) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------|---------------------------------|-------------------|--------------------|
| <b>Iterative experimentation (i.e., continuous experimentation to test business model elements)</b>                                                            |      |                             | 0.898                           | 0.949             | 0.657              |
| The service project team viewed new service development as cycles of experiments, learning, and additional experiments.                                        | Ie1  | 0.815                       |                                 |                   |                    |
| The service project team tried many different service solutions before the right one was identified.                                                           | Ie2  | 0.851                       |                                 |                   |                    |
| The service project team engaged in many trial-and-error processes in service development before we had a complete understanding of the market and technology. | Ie3  | 0.832                       |                                 |                   |                    |
| The service project team repeated the process of testing until all key business model assumptions had been validated.                                          | Ie4  | 0.802                       |                                 |                   |                    |
| The service project team takes an experimental approach that relies on frequent trial and error to identify the right product/service solution.                | Ie5  | 0.789                       |                                 |                   |                    |
| The service project team frequently designs and conducts experiments on elements of the business model.                                                        | Ie6  | 0.770                       |                                 |                   |                    |
| <b>Validation (i.e., adopting an experimental and data-driven approach to decision-making and outcome monitoring)</b>                                          |      |                             | 0.938                           | 0.950             | 0.801              |
| The service project team used metrics to measure the impact of service improvements on customer behavior.                                                      | Val1 | 0.843                       |                                 |                   |                    |
| The service project team uses data-driven tests to improve human judgment in the decision-making process.                                                      | Val2 | 0.910                       |                                 |                   |                    |
| The service project team has metrics available to test the service acceptance by customers and sales performance.                                              | Val3 | 0.907                       |                                 |                   |                    |
| The service project team systematically analyzes its metrics.*                                                                                                 | Val4 | 0.922                       |                                 |                   |                    |
| The service project team relies on data to assess and interpret metrics.*                                                                                      | Val5 | 0.889                       |                                 |                   |                    |

| Factor construct (incl. indicators)                                                             | Code | Standardized factor loading | Cronbach's alpha ( $\geq 0.7$ ) | PA ( $\geq 0.7$ ) | AVE ( $\geq 0.5$ ) |
|-------------------------------------------------------------------------------------------------|------|-----------------------------|---------------------------------|-------------------|--------------------|
| <b>Learning</b> (i.e., using new information to update business ideas and beliefs)              |      |                             | 0.920                           | 0.936             | 0.719              |
| The organization's ability to learn is considered to be key to our competitive advantage.       | Le1  | 0.900                       |                                 |                   |                    |
| The basic values of our organization include learning as a key to improvement.                  | Le2  | 0.918                       |                                 |                   |                    |
| Venture learning is an investment, not an expense.                                              | Le3  | 0.876                       |                                 |                   |                    |
| Learning in our organization is a key commodity necessary to guarantee organizational survival. | Le4  | 0.866                       |                                 |                   |                    |
| The service project team is able to translate failure and mistakes into subsequent actions.*    | Le5  | 0.814                       |                                 |                   |                    |
| Actions and decisions of the service project team are based on past experiences and results.*   | Le6  | 0.694                       |                                 |                   |                    |

*\*New measurement items added to the instrument of Harms and Schwery (2020)*

|                           | Customer insight | Iterative experimentation | Financial performance | Hypothesis-testing | Learning | Non-financial performance | Sprint-based working | Team autonomy | Team diversity |
|---------------------------|------------------|---------------------------|-----------------------|--------------------|----------|---------------------------|----------------------|---------------|----------------|
| Customer insight          |                  |                           |                       |                    |          |                           |                      |               |                |
| Iterative experimentation | 0.616            |                           |                       |                    |          |                           |                      |               |                |
| Financial performance     | 0.426            | 0.309                     |                       |                    |          |                           |                      |               |                |
| Hypothesis-testing        | 0.629            | 0.778                     | 0.348                 |                    |          |                           |                      |               |                |
| Learning                  | 0.818            | 0.541                     | 0.368                 | 0.601              |          |                           |                      |               |                |
| Non-financial performance | 0.452            | 0.232                     | 0.468                 | 0.295              | 0.501    |                           |                      |               |                |
| Sprint-based working      | 0.620            | 0.786                     | 0.495                 | 0.646              | 0.559    | 0.409                     |                      |               |                |
| Team autonomy             | 0.679            | 0.629                     | 0.409                 | 0.519              | 0.576    | 0.491                     | 0.755                |               |                |
| Team diversity            | 0.686            | 0.565                     | 0.489                 | 0.516              | 0.599    | 0.471                     | 0.801                | 0.787         |                |
| Validation                | 0.416            | 0.645                     | 0.418                 | 0.693              | 0.420    | 0.193                     | 0.621                | 0.398         | 0.404          |

**Table 4.4**      Heterotrait-monotrait ratio of correlation (HTMT)

Next, we modeled the second-order composite factors: agile development and lean startup. To model these second-order composite factors, we preserved the factor scores of the first-order latent variables and used them as indicators. We examined the quality of the composite measurement model by checking for collinearity between indicators through the variance inflation factor (VIF). This step revealed no collinearity issues, with all scores falling below the 3.3 threshold (Hair et al., 2022). Subsequently, we evaluated the outer weights and p-values ( $\alpha = 0.10$ ) of all composite factors to ascertain whether all first-order factors significantly contributed to their respective second-order factors (Table 4.5).

**Table 4.5      Second-order composite assessment and results**

| Composite construct | Indicator                  | Weight | T-values | P-values |
|---------------------|----------------------------|--------|----------|----------|
| Agile development   | Team autonomy              | 0.318  | 1.361    | 0.087    |
|                     | Team diversity             | 0.445  | 1.850    | 0.032    |
|                     | Sprint-based working       | 0.340  | 1.445    | 0.074    |
| Lean startup        | Customer insight           | 0.451  | 2.055    | 0.020    |
|                     | Hypothesis-testing*        | NA     | NA       | NA       |
|                     | Iterative experimentation* | NA     | NA       | NA       |
|                     | Validation                 | 0.284  | 1.492    | 0.068    |
|                     | Learning                   | 0.454  | 1.876    | 0.030    |

*\*Indicator eliminated because of non-significant outer weight*

Our analysis showed that two indicators held non-significant outer weights on the lean startup construct, namely, “hypothesis-testing” (t-value = 0.285,  $p = 0.775$ ) and “iterative experimentation” (t-value = 0.855,  $p = 0.393$ ). Although one should be careful when eliminating indicators from a composite factor measurement model (Henseler et al., 2009), we decided to exclude both indicators because of their lack of contribution to the composite factor.

We also decided to omit the iterative experimentation indicator because of its unexpected outer weight of -0.143. This unexpected finding implied that the measurement scale for the iterative experiment measured other constructs in a manufacturing context compared to its original context of software startups. Specifically, we assumed that our questionnaire captured practices closely related to experimentation, such as prototyping and piloting. Like experimentation, these practices entail testing through tangible artifacts, and trial-and-error are prevalent practices for product innovation in a manufacturing context (Geissdoerfer et al., 2022). Therefore, we argue that iterative experimentation, as measured through our instrument, does not function as a defining factor for the lean startup construct within a manufacturing context.

Consequently, we removed this indicator from the lean startup measurement model. Even though hypothesis-testing and iterative experimentation are defining elements of lean startup, dropping these dimensions from the measurement model still enabled us to measure the core principles of lean startup. A combination of customer insight, metric-based validation, and learning still resembled the “get-out-the-building” mentality and validated the learning principles of lean startup.

#### 4.4.3 Structural model and hypothesis-testing

Figure 4.2 presents the structural path model to evaluate the direct relationships between agile development, lean startup, and the dependent variables of financial and non-financial service business performance. The critical metric for assessing the utility of this structural model is based on  $R^2$  values associated with the dependent variables. In our analysis, the  $R^2$  values for financial and non-financial service business performance were 0.302 and 0.367, respectively. These values fell below the commonly accepted threshold of 0.5, indicating low to moderate variance explained by the model. This suggests that other factors not accounted for in the current model also significantly influence business performance in financial and non-financial service sectors. Despite this limitation, the explained variance attributed to agile development and lean startup practices is not negligible and offers valuable insights into their impact on service business performance.

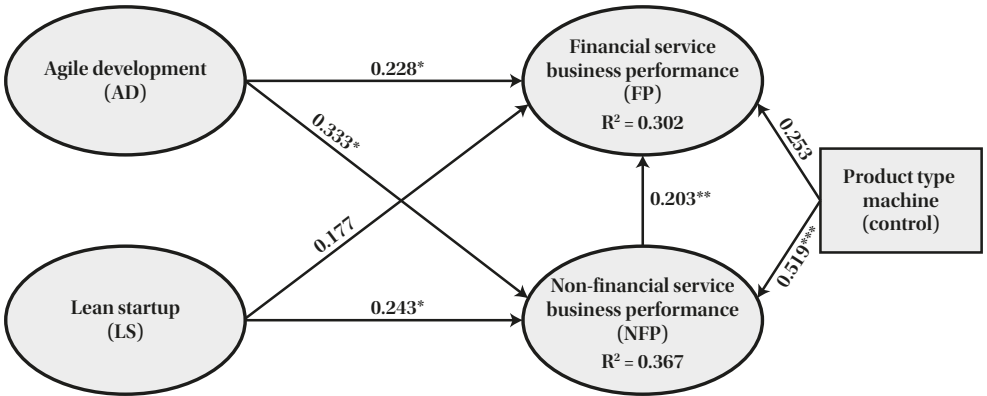


Figure 4. 2 Direct path model

(\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ )

<sup>1</sup> We used  $f^2$  to assess relationship strength and considered values above 0.02, 0.15, and 0.32 to be small, moderate, and strong, respectively (Henseler et al., 2016).

The analysis is structured into three key stages: (i) examining the direct relationships between agile development and service business performance, and between lean startup and service business performance (H1a,b and H2a,b); (ii) assessing the mediating role of non-financial service business performance in these relationships (H3, H4); and (iii) investigating the interaction effects between agile development and lean startup (H5a,b).

4.4.3.1 Testing direct effects (H1a,b and H2a,b)

As indicated in Table 4.6, agile development has a small1 positive effect on financial service business performance (path coefficient = 0.228, p = 0,036, f2 = 0.049) and a small-to-moderate positive effect on non-financial service business performance (path coefficient = 0.333, p = 0,004, f2 = 0.107). In contrast, lean startup has a small positive effect on non-financial performance (path coefficient = 0.243, p = 0,038, f2 = 0.068). At the same time, we observed no such relationship for financial performance (path coefficient = 0.177, p = 0,124, f2 = 0.034). Consequently, our findings support hypotheses H1a and H1b, as well as H2b, while refuting H2a. Moreover, these positive relationships remain consistent in more complex path models, as evidenced by Table 4.8. Taken together, our results support H1a, H1b, and H2b but not H2a.

Table 4. 6      Direct effects between constructs

| Hypothesis | Path     | 95% CI         | Significance (p-value) | Hypothesis supported |
|------------|----------|----------------|------------------------|----------------------|
| H1a        | AD → FP  | 0.005 – 0.416  | 0.036                  | Yes                  |
| H1b        | AD → NFP | 0.130 – 0.526  | 0.004                  | Yes                  |
| H2a        | LS → FP  | -0.028 – 0.407 | 0.124                  | No                   |
| H2b        | LS → NFP | 0.018 – 0.451  | 0.038                  | Yes                  |

#### 4.4.3.2 Testing indirect effects (H3 and H4)

Table 4.7 reveals the outcomes of indirect path modeling, which assesses the mediating role of non-financial service business performance in the relationship between agile development and lean startup toward financial service business performance. Our analysis indicates a mediating effect of non-financial performance in the relationship between agile development and financial performance (path coefficient = 0.067,  $p = 0,044$ ). Notably, this mediation is partial, as we also observed a direct effect between agile development and financial service business performance, thereby confirming H3. In contrast, for lean startup, no such mediating effect was found (path coefficient = 0.050,  $p = 0,099$ ), leading to a lack of support for H4.

**Table 4. 7      Mediating effect of non-financial service business performance**

| Hypothesis | Path          | 95% CI        | Significance (p-value) | Hypothesis supported |
|------------|---------------|---------------|------------------------|----------------------|
| H3         | AD → NFP → FP | 0.011 – 0.138 | 0.044                  | Yes                  |
| H4         | LS → NFP → FP | 0.000 – 0.123 | 0.099                  | No                   |

#### 4.4.3.3 Testing interaction effects (H5a,b)

Figure 4.3 and Table 4.8 present findings exploring the interaction effects between agile development and lean startup on financial and non-financial service business performance. Our data do not indicate an interaction effect between agile development and lean startup regarding financial service business performance (path coefficient = 0.068,  $p = 0.133$ ,  $f^2 = 0.049$ ), thereby rejecting H5a. Conversely, we observed a significant and moderate-to-strong positive interaction effect between agile development and lean startup on non-financial service business performance (path coefficient = 0.233,  $p = 0.000$ ,  $f^2 = 0.200$ ). This result supports H5b and suggests that agile development and lean startup practices have a synergistic relation toward non-financial service business performance.

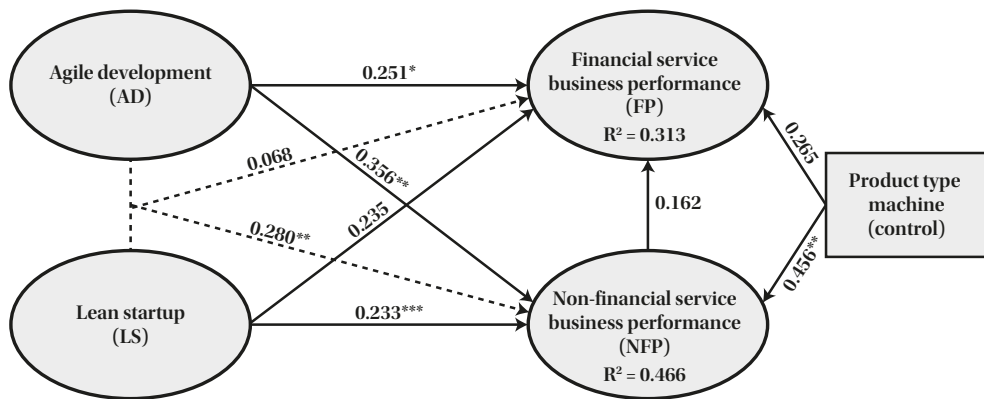


Figure 4.3 Interaction analysis AD\*LS

(\* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ )

Table 4.8 Interaction effects between agile development and lean startup

| Hypothesis | Path        | 95% CI<br>Direct path model | 95% CI<br>Moderation path model | Hypothesis supported |
|------------|-------------|-----------------------------|---------------------------------|----------------------|
| H1a        | AD → FP     | 0.005–0.416                 | 0.026–0.441                     | Yes                  |
| H2a        | LS → FP     | -0.028–0.407                | -0.011–0.471                    | No                   |
| H5a        | AD*LS → FP  |                             | -0.043–0.176                    | No                   |
| H1b        | AD → NFP    | 0.130–0.526                 | 0.136–0.591                     | Yes                  |
| H2b        | LS → NFP    | 0.018–0.451                 | 0.154–0.672                     | Yes                  |
| H5b        | AD*LS → NFP |                             | 0.122–0.339                     | Yes                  |

## 4.5 Discussion

Our findings have multiple implications for understanding agile co-creation in digital servitization. First, our study provides conceptual clarity on agile co-creation in digital servitization, suggesting that it encompasses practices from two distinct agile innovation approaches: agile development and lean startup. We achieved discriminant validity between first-order factors representing practices from both agile development and lean startup. This empirical evidence supports our argument that these approaches are separate conceptual entities. Such findings resonate with the work of Zorzetti et al. (2022) and Ghezzi and Cavallo (2020), who indicated that firms might utilize these approaches individually, catering to specific purposes. As such, we refine the work of Sjödin et al. (2020), who presented a unified agile co-creation process. Instead,

our results suggest that servitizing firms perceive agile co-creation as a repertoire of practices, choosing to implement them as standalone processes tailored to their unique needs. Consequently, instead of treating agile co-creation as an aggregate approach to digital servitization (e.g., Chirumalla et al., 2023; Sjödin et al., 2022), a more detailed view that includes different agile innovation approaches such as agile development and lean startup provides a better understanding for structuring the shift to digital services.

Second, our results uncover distinct, positive relationships between agile development, lean startup, and non-financial service business performance (H1b and H2b), emphasizing their unique roles in enhancing non-financial outcomes. This distinction becomes evident through the value co-creation lens in digital servitization, which refers to the process of resource integration guided by value propositions that entail its anticipated outcomes (Vargo, 2020; Vargo & Lusch, 2008). As a project management framework, agile development aligns primarily with the processual dimension of value co-creation, enhancing customer satisfaction in cooperative processes, as Bambauer-Sachse and Helbling (2021) noted. In contrast, lean startup resonates more with the content dimension because it focuses on discovering and addressing latent customer needs through compelling value propositions (Bortolini et al., 2018; Sala et al., 2022). Our findings offer empirical evidence for the distinct, yet complementary, roles of agile development and lean startup in value co-creation.

The synergy between agile development and lean startup in the value co-creation process is further highlighted by our discovery of an interaction effect between them in relation to non-financial service business performance (H5b). This synergy can be attributed to their distinct focuses and practices. Agile development centers on converting customer needs into tangible and intangible outputs, leveraging sprint-based working and agile communication practices (Malik et al., 2021; Schwaber & Sutherland, 2020). This approach presupposes an understanding of customer needs, from the perspective of the customer or the service provider. Yet, as firms transform to digital services, articulating these needs is often challenging for service providers and customers (Huikkola, Kohtamäki, Rabetino, et al., 2022; Tronvoll et al., 2020).

Our findings suggest that lean startup practices fill this gap, equipping organizations with tools and processes to unearth latent customer needs. Specifically, practices related to experimentation and validation strengthen a firm's market orientation and offer the means to interpret and act upon customer feedback in a more informed way. On the other hand, while lean startup is grounded in the idea of testing business model assumptions and learning from all outcomes, both positive and negative (Ries, 2011), this experimental mindset, based on a "license to fail," might not be a natural fit within established firms (Chesbrough & Tucci, 2020; Signoretti et al., 2019). Our findings suggest that when firms integrate the structured processes of agile development alongside the adaptive practices of lean startup, they create the required environment for deriving maximum value from experimentation and validation. Our findings emphasize the compounded benefits of deploying agile development and lean startup within the digital servitization co-creation process.

Third, the differential relationships of agile development and lean startup with financial service business performance hint at their unique roles in the digital servitization process. While the positive relationship between agile development and financial service business performance matched our expectations (H1a), the non-relationship for lean startup (H2a) was not anticipated. This absence of a positive relationship between lean startup and financial service business performance underscores the complexity of converting customer insights into a profitable service business model. A possible explanation is the challenge that servitizing firms face when attempting to generalize insights on potential value-in-use beyond specific contexts of launching customers. Firms might inadvertently tailor their service business models too closely to the particular circumstances of initial customers. This is a result of aligning their assumptions and value propositions too precisely with these initial contexts (Zellweger & Zenger, 2023). Such overfitting can hinder the adaptability of these models beyond these customer segments, obstructing the scaling of service business models. Furthermore, excessive customization of the service business model based on the experimentation context might result in revenue gains that do not outweigh service delivery costs when applied beyond the original environment (Cenamor et al., 2017; Neely, 2008). In light of these challenges, we assert that while lean startup practices are effective for generating customer-specific insights, other approaches are needed to convert these insights into scalable and profitable service business models.

Furthermore, our findings highlight a direct positive relationship between agile development and financial service business performance (H1a). The partial mediation of this relationship by non-financial performance (H3) suggests that the contributions of agile development to economic outcomes are multifaceted. In addition to facilitating value co-creation, we posit that the direct financial benefits of agile development also stem from its emphasis on micro-services. Adopting a modular approach to digital servitization enhances the scalability and maintainability of the digital infrastructure that is essential for the service business (Ghofrani & Lübke, 2018; Jamshidi et al., 2018), enabling firms to expand their service business cost-efficiently. These insights align with the existing literature, which underscores the significance of striking a balance between customization and standardization (Cenamor et al., 2019; Hasselblatt et al., 2018), as well as managing the costs associated with developing and delivering digital services (Coreynen et al., 2017; Sjödin et al., 2020).

Altogether, our results provide insights into agile development and lean startup roles throughout the digital servitization process. At the outset of the digital servitization trajectory, both agile development and lean startup practices are instrumental, highlighting the emphasis on identifying customer needs through value co-creation (Baines et al., 2020). However, as the journey progresses and firms aim to scale their service business, which is reflected in financial performance, the contribution of agile development becomes more important. These findings align with the dynamic view of Linde et al. (2023), suggesting that servitizing firms initially prioritize development through experimentation and gradually shift toward implementation and scaling the service business. Therefore, while the combination of agile development and lean startup benefits firms initially, agile development becomes more important over time,

with its practices helping to scale the service business. Our findings underscore the importance of adopting and adapting different agile co-creation practices in line with the evolving needs of the digital servitization trajectory.

## 4.6 Conclusion

This paper investigates the impacts of agile development and lean startup on service business performance in digital servitization. Our findings indicate that agile development and lean startup contribute to value co-creation, reflected in non-financial service business performance. Additionally, our results reveal a positive interaction effect between agile development and lean startup in relation to non-financial service business performance, indicating a synergistic benefit from their combined implementation. In contrast to non-financial performance, our findings reveal that only agile development contributes to financial service business performance. As this relationship is partially mediated by non-financial service business performance, agile development benefits financial performance by contributing to value co-creation and other mechanisms, such as service delivery cost development.

Based on these findings, our research contributes to the developing research about agile co-creation in digital servitization. By adopting a quantitative research design, we provide further empirical evidence of the importance of adopting an agile co-creation approach to digital servitization (Favoretto et al., 2022; Huikkola, Kohtamäki, Rabetino, et al., 2022; Sjödin et al., 2020). Our findings indicate that agile co-creation exists in practices from different agile innovation approaches, including agile development and lean startup. Thereby, we increase the conceptual clarity of agile co-creation and assert that agile co-creation should be considered a set of innovation practices instead of a unified innovation approach, as implied in the existing literature (e.g., Chirumalla et al., 2023; Sjödin et al., 2022). This assertion is supported by the different roles of agile development and lean startup practices throughout the digital servitization trajectory. As our findings indicate that a combination of agile development and lean startup practices benefit value co-creation in the initial stages of digital servitization, agile development becomes more important as firms progress and try to scale their service business. As such, our research contributes to the dynamic understanding of agile co-creation throughout the digital servitization trajectory and provides empirical support in line with the work of Linde et al. (2023).

Besides the theoretical implications, our study offers crucial insights for managers venturing into digital servitization. Both agile development and lean startup practices have distinct advantages. When managers aim to enhance customer satisfaction and loyalty by developing service offerings that fit customers' needs, incorporating elements from both these approaches is beneficial. Such integration can be operationalized by setting up autonomous teams comprising members from varied skills and domains that engage in iterative customer-centric experimentation. Whether these are specialized task forces or separate business units depends on the firm's resources. Once established, these autonomous teams work iteratively according to the build-measure-learn cycle, which is concluded with a value increment. In a manufacturing context,

a value increment should not be restricted to minimum viable products but can be anything testable that yields validated learning by enabling teams to test predefined hypotheses regarding the service business model. Such value increments include wire prototypes, competitor assessments, and personas. Setting up diverse teams committed to constant customer-driven experimentation yields deep customer insights, enables swift responses to feedback, and minimizes decision biases.

As services evolve and managers seek growth beyond the initial customer base, the importance of agile development, specifically the micro-services approach, is amplified. This approach, which emphasizes addressing distinct customer needs one at a time, helps firms to avoid overstretching their resources and keeps service development costs manageable. Additionally, the modularity of micro-services ensures that customization and maintenance expenses remain controlled. Given that individual micro-services can be adjusted independently without impacting the broader system, it provides firms with flexibility and adaptability. In essence, the micro-services approach allows businesses to scale their service offerings and digital infrastructure cost-efficiently, helping to mitigate some of the financial pitfalls commonly encountered in digital servitization.

While our study contributes to theory and practice, it has certain limitations that present opportunities for future research. First, external factors such as mood or general optimism might have influenced our subjective performance measures. Future research could consider using objective performance indicators for validation. Second, our sample size restricted us from examining more complex theoretical models. As a next step, researchers might explore the implications of our findings in a broader context, examining potential boundary conditions or underlying mechanisms of the agile co-creation process in digital servitization. For example, future research might expand the ongoing work on ecosystem transformation and orchestration through the role of agile development and lean startup practices (Makkonen et al., 2022; Sjödin et al., 2022). Third, the cross-sectional design of our study does not account for potential delayed effects or evolutions in the implementation of agile development and lean startup practices over time. A longitudinal design in future research could capture these nuances. Lastly, while we focused on agile development and lean startup, future research could investigate other innovation approaches, such as design thinking (Lichtenthaler, 2020) or combining agile co-creation with stage-gate models (Cooper & Sommer, 2016b).

Furthermore, we acknowledge certain limitations regarding our application of Partial Least Squares Structural Equation Modeling (PLS-SEM). A critical reflection, prompted by scholarly exchange, revealed that our reliance on the two-stage approach of Sarstedt et al. (2019) to model reflective-formative hierarchical constructs may yield inconsistent estimates (Schuberth, 2021). This limitation is rooted in the controversy in the literature about the applicability of PLS-SEM in estimating reflective measurement models (Schuberth 2021; Hair et al., 2022). Even though a two-stage approach is often considered an acceptable way of estimating reflective measurement models, it is prone to sensitivity issues, thereby failing to disconfirm invalid measurement models (Schuberth, 2021). In other words, the two-stage approach adopted in this study raises potential type II errors in the measurement models of agile development and

lean startup. To increase validity and reliability of the findings, re-estimation of the reflective-formative hierarchical constructs through a three-stage approach (Van Riel et al., 2017) is recommendable in further research.





# 5

## **AGILE CO-CREATION IN DIGITAL SERVITIZATION: A DYNAMIC CAPABILITIES PERSPECTIVE**

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## Abstract

While digital servitization literature recognizes the role of agile co-creation in dynamic capabilities, particularly in seizing, its potential in facilitating sensing and reconfiguration remains underexplored. Therefore, our study systematically explores how agile innovation approaches enable dynamic capabilities across sensing, seizing, and reconfiguration. Based on 31 interviews with 24 respondents, we identified multiple characteristics and outcomes of agile co-creation in digital servitization that align with sensing, seizing, and reconfiguration dimensions. Additionally, we reveal governance and cultural boundary conditions impacting this relationship. Our findings contribute to the ongoing digital servitization literature by (i) broadening the inquiry scope beyond seizing capabilities and clarifying the link between agile co-creation and dynamic capabilities, (ii) illuminating the underlying mechanisms through which agile co-creation fosters these dynamic capabilities, deepening our comprehension of their interrelationship, (iii) increasing the scope of agile innovation beyond the customer-supplier dyad and to an ecosystem context, and (iv) providing a nuanced understanding of the link between agile co-creation and dynamic capabilities by identifying boundary conditions .

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## 5.1 Introduction

Manufacturers are nowadays navigating a challenging business landscape marked by intense global competition and rapid technological advancements. This landscape often requires them to rethink their product-centric business models (Porter & Heppelmann, 2014; Ulaga & Reinartz, 2011). Consequently, manufacturers increasingly adopt a digital servitization strategy (Favoretto et al., 2022), which refers to *“the transformation in processes, capabilities, and offerings within industrial firms and their associate ecosystem to progressively create, deliver, and capture increased service value arising from a broad range of enabling digital technologies”* (Sjödin et al., 2020, p. 478). This service transformation is beneficial for manufacturers, offering higher profit margins, greater market differentiation, and strengthened customer relationships, enhancing competitive advantage (Baines et al., 2009; Faramarzi et al., 2023).

While the benefits of digital servitization are clear, many firms struggle to successfully implement a digital servitization strategy and experience positive financial outcomes (Benedettini et al., 2015; Neely, 2008; Sjödin et al., 2020). This struggle is rooted in the need to rethink organizational processes, structures, and culture (Favoretto et al., 2022; Tronvoll et al., 2020; Ziaee Bigdeli et al., 2021). For example, manufacturers are often not used to innovating and implementing digital technologies and software, which may cause them to overestimate revenue streams or underestimate delivery costs (Huikkola, Kohtamäki, Rabetino, et al., 2022; Sjödin et al., 2020). Also, the radical nature of the service transformation may cause a loss in strategic focus and the effective allocation of scarce resources to various transformation and development initiatives (Castka et al., 2024; Fang et al., 2008).

In the face of the profound intra and inter-organizational changes required for digital servitization (Huikkola, Kohtamäki, & Ylimäki, 2022; Makkonen et al., 2022; Tóth et al., 2022), firms find themselves operating in an environment best described as volatile, uncertain, complex, and ambiguous (VUCA). To effectively operate in this context, recent servitization literature emphasizes the critical role of dynamic capabilities (Chirumalla et al., 2023). Dynamic capabilities refer to “the firm’s ability to integrate, build, and reconfigure internal and external competences” (Teece et al., 1997, p. 516) and are divided into sensing, seizing, and reconfiguration capabilities (Teece, 2007). In the context of digital servitization, dynamic capabilities are vital for tasks like developing new services, innovating, managing the interplay of products and services, and coordinating with external partners (Castka et al., 2024; Kanninen et al., 2017; Lütjen et al., 2019). Furthermore, the development and deployment of these dynamic capabilities are underpinned by various micro-foundational processes and structures (Chirumalla et al., 2023; Saul & Gebauer, 2018).

Agile co-creation is considered a crucial micro-foundation of dynamic capabilities for digital servitization (Chirumalla et al., 2023). Agile co-creation entails an iterative sprint-based approach based on constant experimentation, quick customer feedback loops, and learning through trial and error (Linde et al., 2023; Sjödin et al., 2020). Firms have adopted agile innovation approaches to deal with the complex and dynamic character

of digital servitization (Bustinza et al., 2018; Huikkola, Kohtamäki, Rabetino, et al., 2022). When engaging in agile co-creation, servitizing firms initially focus on identifying latent customer needs and their willingness to pay for solutions through experimentation. They gradually emphasize developing and implementing service offerings (Linde et al., 2023; Sjödin et al., 2020). By adopting agile innovation approaches, servitizing firms increase their organizational flexibility and the overall clock speed of service innovation (Huikkola, Kohtamäki, Rabetino, et al., 2022; Sjödin et al., 2020).

Despite the increasing attention to agile co-creation as an essential micro-foundation of dynamic capabilities, we identified two important literature gaps. First, as a micro-foundation of seizing capability, agile co-creation is only considered to benefit opportunity exploitation through strategic decision-making and securing commitment (Chirumalla et al., 2023). However, agile innovation approaches can also benefit opportunity discovery and routine transformation by promoting experiential learning, sprint-based working, and close collaboration with key stakeholders (Sjödin et al., 2020; Thomson et al., 2023). Because these benefits align with sensing and reconfiguration capabilities, we argue that further research is needed to explore how agile innovation approaches facilitate sensing and reconfiguring capabilities. Second, agile co-creation has mainly been investigated in the customer-supplier context (Sjödin et al., 2020). However, digital servitization requires coordination and transformation among various ecosystem actors (Kohtamäki et al., 2019; Makkonen et al., 2022). Because managing in business ecosystems requires specific dynamic capabilities (Foss et al., 2023; Lütjen et al., 2019), further research is needed into how agile innovation approaches facilitate dynamic capabilities in an ecosystem context (Sjödin et al., 2020).

To address the two literature gaps mentioned above, this study aims to expand the understanding regarding the role of agile co-creation in dynamic capabilities for digital servitization. Specifically, we employ Teece's (2007) sensing, seizing, and reconfiguration framework to systematically investigate how agile innovation approaches facilitate dynamic capabilities. As such, we explore the role of agile co-creation beyond seizing capability (Chirumalla et al., 2023). Furthermore, we explore the role of agile co-creation in dynamic capabilities concerning both intra and inter-organizational dynamics. In other words, we investigate the role of agile co-creation in facilitating internal transformation processes and shaping an ecosystem strategy, that is, aligning partners and securing an envisioned role in a competitive ecosystem (Adner, 2017, p. 47). This dual focus on intra and inter-organizational dynamics provides novel insights into the role of agile co-creation beyond the customer-supplier context (Sjödin et al., 2020). The guiding research question for this study is: *How do agile innovation approaches facilitate dynamic capabilities in the digital servitization context?*

We make three contributions to the digital servitization literature using a qualitative interview-based research design. First, we broaden and deepen our understanding of the role of agile innovation approaches in dynamic capabilities for digital servitization. Our results emphasize their micro-foundational role beyond seizing and providing the underlying mechanisms through which agile co-creation facilitates dynamic capabilities (Chirumalla et al., 2023). Second, we contribute to the discourse on agile

innovation by extending its application beyond customer-supplier relationships to a broader ecosystem context, thereby filling an existing research gap and contributing to the multi-level perspective on digital servitization (Sjödin et al., 2020; Struyf et al., 2021). Third, we provide a nuanced understanding of the role of agile innovation approaches in digital servitization by identifying important governance and culture-related boundary conditions that affect the outcomes of agile innovation approaches regarding a firm's ability to implement digital servitization. Jointly, our work provides a comprehensive lens through which firms can better understand the complexities and preconditions associated with successfully leveraging agile innovation approaches in digital servitization.

## 5.2 Theory

### 5.2.1 Dynamic capabilities

Dynamic capabilities refer to a firm's ability to maintain evolutionary fitness by sensing opportunities and threats, seizing them through timely decision-making, and reconfiguring the business by realizing change (Teece, 2007). As dynamic capabilities involve integrating, building, and reconfiguring internal and external structures and competencies in line with external development (Teece et al., 1997), they are inherently related to operating in business ecosystems (Adner, 2017; Teece, 2007). Specifically, dynamic capabilities are crucial for ecosystem leaders to initiate and orchestrate business ecosystems through resource mobilization and standard setting (Foss et al., 2023; Teece, 2018). Furthermore, dynamic capabilities are important for complementors because they enable them to identify and pursue potential bottleneck positions within the ecosystem and respond to ecosystem dynamics (Qiu et al., 2022; Teece, 2018). To combine the role of dynamic capabilities for ecosystem leaders and complementors, we assert that dynamic capabilities enable firms to devise and implement an ecosystem strategy, which refers to *"the way in which a focal firm approaches the alignment of partners and secures its role in a competitive ecosystem"* (Adner, 2017, p. 47). So, firms draw on dynamic capabilities to identify developments in their business ecosystem and beyond, exploit these developments through business models and envisioned ecosystem structures, and reconfigure internal and external structures aligned with their envisioned strategy.

Dynamic capabilities are essential in facilitating the radical intra and inter-organizational transformation associated with digital servitization (Chirumalla et al., 2023). Within the firm, dynamic capabilities enable firms to develop new services and digital capabilities that fit customer needs (Kanninen et al., 2017; Saul & Gebauer, 2018). Also, dynamic capabilities facilitate shifting from the existing product-oriented culture to a service-oriented one (Huikkola, Kohtamäki, & Ylimäki, 2022; Tronvoll et al., 2020). Externally, digital servitization requires deep involvement and adaptations from a wide variety of ecosystem actors (Raddats et al., 2017; Story et al., 2017), which calls for alignment between a firm's digital servitization strategy and the ecosystem (Kindström et al., 2013; Martinez et al., 2017). In this context, dynamic capabilities facilitate digital servitization by identifying and accessing external capabilities and influencing the business

ecosystem actors to facilitate co-development in line with the envisioned servitization strategy (Chirumalla et al., 2023; Lütjen et al., 2019).

Dynamic capabilities are often conceptualized as sensing, seizing, and reconfiguring capabilities (Teece, 2007). *Sensing* refers to the firm's ability to identify opportunities and threats through continuous scanning, filtering, and market exploration (Helfat & Peteraf, 2015; Teece, 2007). In the digital servitization context, sensing involves observing potential services emerging technologies, and identifying suitable partners (Chirumalla et al., 2023; den Hertog et al., 2010). Essentially, sensing is about being exposed to environmental signals and interpreting them effectively (Kohtamäki et al., 2020; Li et al., 2017; Lütjen et al., 2019). This capability extends beyond internal processes, encompassing the involvement of external actors to augment opportunity identification and leverage external sensing capabilities, as Qiu et al. (2022) discussed. However, sensing is not without its challenges. Identifying viable service opportunities and understanding the nuances of their contextual constraints can be complex due to their often latent and complex nature (Mahr et al., 2014; Stegehuis et al., 2023).

*Seizing* focuses on exploiting identified opportunities by crafting and selecting appropriate business models and securing commitment to mobilize resources (Teece, 2007). In digital servitization, seizing involves the development of service business models, the establishment of suitable technological infrastructures, and the envisioning of ecosystem structures and strategies to access external capabilities (Castka et al., 2024; Huikkola, Kohtamäki, & Ylimäki, 2022; Linde et al., 2023). Furthermore, seizing in digital servitization involves fostering commitment from internal and external stakeholders. Therefore, servitizing firms require capabilities to involve them in the co-creation process and the ability to design incentives that align with the goals and strategies of stakeholders within the organization and of potential ecosystem partners (Chirumalla et al., 2023; Kolagar et al., 2022; Lütjen et al., 2019). Incentive structures, such as those based on data access and revenue sharing, motivate ecosystem actors to collaborate and invest in the envisioned service business model (Parida & Jovanovic, 2021; Sjödin et al., 2022).

*Reconfiguration* entails transforming resources and organizational structures (Teece, 2007). In digital servitization, reconfiguration refers to the capabilities to implement the envisioned service business model, realizing the necessary innovations, and transforming intra and inter-organizational structures (Huikkola, Kohtamäki, & Ylimäki, 2022; Lütjen et al., 2019). Structural transformation includes technological and institutional structures (Chirumalla et al., 2023). Technological structures are the resource and capability interfaces between firms. For example, servitization requires new, aligned data-sharing interfaces and analytics capabilities (Favoretto et al., 2022). Institutional structures are the routinized processes and thinking patterns associated with product-oriented business models (Thomson et al., 2023). For example, servitization requires an identity shift from a product orientation based on efficiency and cost minimization to a service orientation based on value maximization and total cost of ownership (Gebauer et al., 2012; Töytäri et al., 2018). Reconfiguration capabilities regarding intentional narratives or change management facilitate this shift (Huikkola,

Kohtamäki, & Ylimäki, 2022; Kohtamäki et al., 2021). Moreover, reconfiguration involves capabilities to address external challenges, such as adapting to the demands and resistance of various stakeholders. (Galvani & Bocconcelli, 2022; Qiu et al., 2022; Tóth et al., 2022). For example, stakeholders may consider servitization a threat to their existing business and try to preserve the status quo or promote an alternative innovation direction (Burton et al., 2016; Stegehuis et al., 2023; Story et al., 2017).

As the understanding of dynamic capabilities in digital servitization evolves, the literature has increasingly focused on their micro-foundations (Chirumalla et al., 2023; Saul & Gebauer, 2018). DC micro-foundations are the skills, processes, procedures, organizational structures, decision rules, and disciplines that underpin and give rise to sensing, seizing, and reconfiguration capabilities (Teece, 2007). Micro-foundations represent the intricate interplay of lower-level processes, structural elements, and individuals that drive and enable a firm's dynamic capabilities (Felin et al., 2015; Schilke et al., 2018). Examining these micro-foundations provides a more detailed understanding of organizing principles and individual competencies needed to develop and leverage dynamic capabilities (Foss & Pedersen, 2016). In the context of digital servitization, there has been a growing emphasis on agile co-creation as a critical DC micro-foundation (Chirumalla et al., 2023; Sjödin et al., 2020) (Chirumalla et al., 2023; Sjödin et al., 2020).

### **5.2.2 Agile innovation approaches**

The growing emphasis on agile innovation approaches in digital servitization is driven by recognizing its inherently emergent, unpredictable, and iterative nature (Huikkola, Kohtamäki, & Ylimäki, 2022; Martinez et al., 2017). Unlike traditional methods that often rely on predefined plans or centralized strategies, digital servitization benefits from a more exploratory and improvisational approach, adapting to unforeseen developments both within and outside of the firm (Bustinza et al., 2018; Huikkola, Kohtamäki, Rabetino, et al., 2022; Kowalkowski et al., 2012). Consequently, servitizing firms have increasingly adopted agile innovation practices (Linde et al., 2023; Sjödin et al., 2020; Thomson et al., 2023). Agile innovation approaches, such as agile development and lean startup methods, are characterized by flexible, iterative, and learning-oriented innovation processes (Ghezzi & Cavallo, 2020). Originating in the software industry, these methods promote a sprint-based development process leveraging self-organizing, multidisciplinary teams collaborating closely with customers (Dingsøyr et al., 2012). In the context of digital servitization, this agile approach is marked by rapid and incremental service development through short sprints, continuous experimentation, and quick customer feedback loops, culminating in iterative learning cycles (Linde et al., 2023; Sjödin et al., 2020; Thomson et al., 2023).

So far, agile innovation approaches have been primarily linked to enhancing the seizing capability due to their impact on strategic decision-making and business model innovation (Chirumalla et al., 2023). For example, agile co-creation, characterized by sprint-based working, enables firms to swiftly adapt their service offerings to evolving customer needs (Huikkola, Kohtamäki, Rabetino, et al., 2022; Thomson et al., 2023).

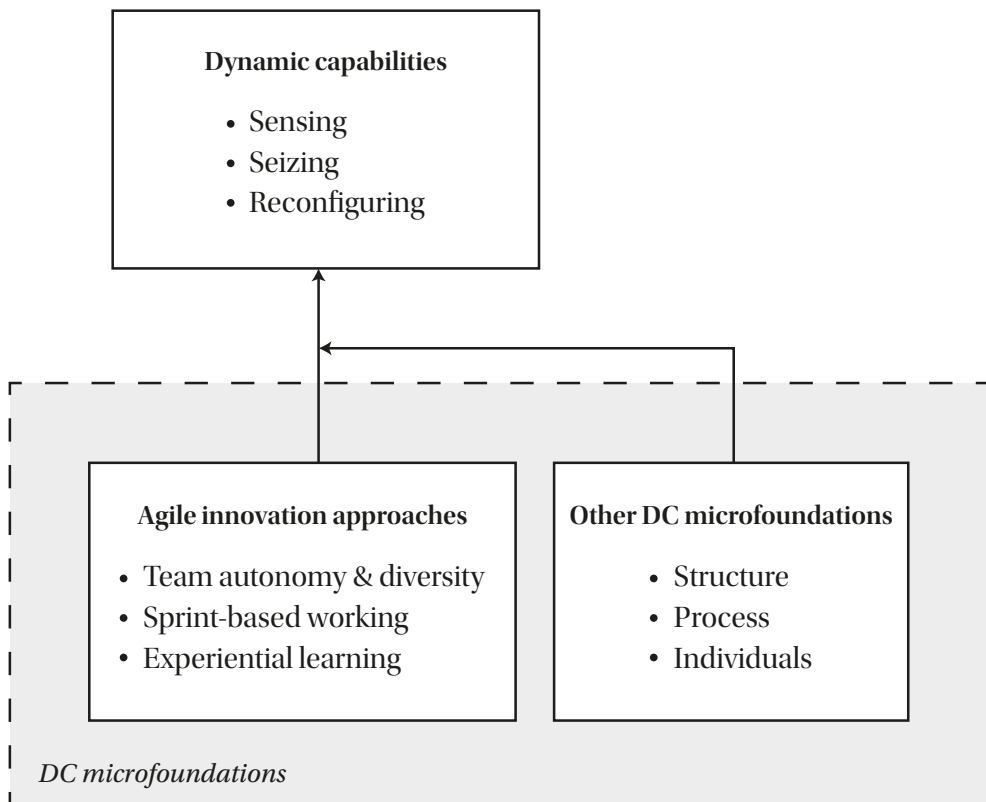
Additionally, the agile approach aids in mitigating decision-making biases by fostering a sense of psychological empowerment among team members (Eisenmann et al., 2011; Holmström et al., 2006; Malik et al., 2021). Furthermore, a key feature of agile innovation is its facilitation of a modular portfolio of micro-services (Jamshidi et al., 2018), allowing services to operate independently or in combination with others. This modularity is particularly beneficial for developing seizing capabilities because it helps firms balance customization with standardization (Sjödin et al., 2020). Lastly, implementing autonomous, multidisciplinary teams in agile innovation processes enhances knowledge management. This approach effectively breaks down organizational knowledge silos and integrates diverse expertise, which is crucial for the development and delivery of product-service offerings (Antioco et al., 2008; Malik et al., 2021; Martinez et al., 2010).

While the literature recognized the role of agile co-creation in opportunity exploitation, it is also associated with multiple dynamics and outcomes that align with sensing capabilities. First, the reliance on short, iterative cycles and close collaboration with customers and ecosystem partners establishes a continuous feedback loop. This loop is instrumental in uncovering latent customer needs and keeping track of technological developments (Eggert et al., 2018; Sjödin et al., 2020, 2022). Second, experiential learning through systematic hypothesis testing and validation aids firms in interpreting opportunity signals. This learning process, where firms continually test, validate, and update their beliefs, enhances their ability to make sense of various opportunity signals (Linde et al., 2023; Thomson et al., 2023; Zellweger & Zenger, 2023). Third, the emphasis on autonomous and diverse teams in agile innovation approaches further aligns with sensing capabilities. Such structures expand the cognitive frameworks teams use to interpret signals, enabling a more nuanced understanding and quicker response to opportunities without the loss of information often experienced as data moves through an organization (Teece, 2007).

Agile co-creation may also play a vital role in supporting reconfiguration capabilities. Specifically, sprint-based working and agile communication practices facilitate the transfer of tacit knowledge. This knowledge transfer is crucial for developing reconfiguration capabilities because it enables the integration of expertise from within and outside the firm (Pikkarainen et al., 2008; Serrador & Pinto, 2015). Furthermore, the involvement of external stakeholders in the agile co-creation process is particularly valuable for managing co-innovation because it increases the coordination of resource integration across functional and organizational boundaries (Sjödin et al., 2022). Finally, sprint-based working and a micro-service approach can notably reduce the disruptiveness often associated with novel service business models (Sjödin et al., 2020; Tóth et al., 2022). This approach allows firms to implement change incrementally, mitigating the anxiety and resistance typically encountered during significant transformation efforts (Teece, 2007).

As mentioned, existing literature predominantly associates agile co-creation with the seizing capabilities in digital servitization (Chirumalla et al., 2023). However, our review of studies on agile innovation approaches reveals their potential micro-foundational role in sensing and reconfiguration capabilities as well. To explore these relationships, we have developed a conceptual model (Figure. 5.1). This model conceptualizes agile

innovation approaches, exiting of structural and processual dimensions, as a micro-foundation for sensing, seizing, and reconfiguration capabilities. As such, this model provides a broader scope of the role of agile innovation approaches in dynamic capabilities for digital servitization (Chirumalla et al., 2023). It is essential to note that DC micro-foundations do not function in isolation. Instead, they shape the development of dynamic capabilities through their dynamic interplay (Chen et al., 2023; Schilke et al., 2018). For instance, the effectiveness of organizational processes like agile development relies on the skills and competencies of the individuals implementing them (Helfat & Peteraf, 2015; Jarzabkowski et al., 2016). Therefore, it becomes essential to consider the moderating effect of various DC micro-foundations when analyzing the role of agile innovation approaches in shaping dynamic capabilities.



**Figure 5.1** The micro-foundational role of agile innovation approaches in dynamic capabilities.

## 5.3 Method

Considering the explorative nature of our research, the complex and dynamic character of ecosystems, and the multi-faceted nature of dynamic capabilities (Lütjen et al., 2019), this study draws on an interview-based qualitative research design to address the research question. An interview-based research design is appropriate for research into dynamic capabilities because it is challenging to observe interactive dynamic capabilities directly and thus requires tapping into the experiences of the individuals involved (Qiu et al., 2022).

This study identified potential interviewees as managers from servitizing firms in various sectors. We specifically targeted the construction, equipment manufacturing, IT, and utility sectors, as servitization is prevalent (Visnjic et al., 2018). We reason that individuals in these sectors are considered knowledgeable about servitization and can provide high-quality input. Within these sectors, we identified companies that were implementing a servitization strategy. We identified potential interviewees through the professional networks of the involved researchers and based on the importance of servitization in the company. To be eligible to participate in this study, interviewees had to be involved in and knowledgeable about practices and processes used for servitization and how this affected interaction with ecosystem partners. As a result, we focused on executives or senior managers, depending on the firm's size. An overview of our sample can be found in Table 5.1.

**Table 5.1      Sample overview**

|    | Acronym | Country | Sector                  | Size (FTE) | Interview #1 duration | Interview #2 duration |
|----|---------|---------|-------------------------|------------|-----------------------|-----------------------|
| 1  | Alpha   | DE      | Equipment manufacturing | 50 - 250   | 01:27:02              | -                     |
| 2  | Beta    | NL      | IT                      | 5-250      | 00:47:31              | 00:31:36              |
| 3  | Gamma   | NL      | Utility                 | 1 - 10     | 01:06:01              | -                     |
| 4  | Delta   | NL      | Construction            | 1000+      | 00:58:13              | -                     |
| 5  | Epsilon | JPN     | Equipment manufacturing | 1000+      | 01:03:12              | 00:57:54              |
| 6  | Zeta    | NL      | Construction            | 1000+      | 01:12:14              | 00:56:45              |
| 7  | Eta     | DE      | IT                      | 50-250     | 00:58:04              | -                     |
| 8  | Theta   | NL      | Construction            | 10-50      | 00:40:40              | 00:55:50              |
| 9  | Iota    | NL      | Equipment manufacturing | 250-500    | 00:49:52              | 00:59:11              |
| 10 | Kappa   | CH      | Equipment manufacturing | 500-1000   | 01:02:47              | 01:08:17              |
| 11 | Lambda  | NL      | Equipment manufacturing | 50-250     | 00:40:52              | -                     |
| 12 | Mu      | NL      | Equipment manufacturing | 500-1000   | 01:12:27              | -                     |
| 13 | Nu      | NL      | Equipment manufacturing | 50-250     | 01:01:46              | -                     |
| 14 | Xi      | NL      | Equipment manufacturing | 50-250     | 00:57:23              | -                     |
| 15 | Omicron | NL      | Equipment manufacturing | 50-250     | 00:57:25              | -                     |
| 16 | Pi      | NL      | Equipment manufacturing | 50-250     | 01:09:57              | -                     |
| 17 | Rho     | NL      | IT                      | 500 - 1000 | 01:02:11              | -                     |
| 18 | Sigma   | NL      | Equipment manufacturing | 1-10       | n.a.                  | -                     |
| 19 | Tau     | NL      | Construction            | 1-10       | 00:58:14              | -                     |
| 20 | Upsilon | NL      | Equipment manufacturing | 50-250     | 00:52:09              | -                     |
| 21 | Phi     | NL      | Equipment manufacturing | 50-250     | 00:57:16              | -                     |
| 22 | Chi     | NL      | Equipment manufacturing | 10-50      | 00:41:00              | -                     |
| 23 | Psi     | NL      | Equipment manufacturing | 10-50      | 00:27:01              | 00:59:22              |
| 24 | Omega   | NL      | Equipment manufacturing | 10-50      | 00:41:48              | -                     |

### **5.3.1 Data collection**

For the data collection, we conducted two rounds of semi-structured interviews. The first round of interviews focused on understanding the practices and methods used for servitization within the firm. We specifically focused on how agile development and lean startup practices were used and implemented and what outcomes were associated with these practices. To mitigate biased answers, we first asked open-ended questions about the processes or methods used for servitization before asking for specific agile development and lean startup practices. Additionally, we guided respondents to reflect on objective events as examples of how agile innovation manifested in the organization (Davis & Eisenhardt, 2011). To identify appropriate prompts for specific agile development and lean startup, we drew on the operationalizations of Malik et al. (2021) and Harms and Schwery (2020). During the first round of interviews, we also asked how firms engaged with external stakeholders and how they realized changes in the ecosystem required for servitization. The outline for the first interview round can be found in Appendix 5A.

During the second interview round, we focused on the role of agile innovation practices in how a firm approaches and interacts with stakeholders within and outside the organization. We asked respondents how the firm identified service opportunities, engaged in decision-making, and implemented these decisions through internal and external transformation. These questions provided insights on the dynamic capabilities of the firm. We also asked respondents how the outcomes of agile innovation approaches, identified in the first interview round, related to the firm's ability to engage and interact with key stakeholders. The outline for the second interview round can be found in Appendix 5B.

### **5.3.2 Data analysis**

For the data analysis of the interview data, we adopted a procedure consisting of two stages. The first stage adopted an inductive thematic coding approach following (Saldaña, 2015). So, we transcribed all interviews verbatim and familiarized ourselves with the data by closely reading the transcripts. Then, we started with open coding content analysis followed by thematic analysis, an appropriate approach when the topic under scrutiny is not well established and lacks comprehensive theoretical understanding (Braun & Clarke, 2006). During the first coding stage, we identified and coded statements about the agile co-creation practices used to shape digital servitization, their characteristics, and what outcomes were associated with these practices. These codes were then organized into first-order categories and second-order themes and yielded mechanisms through which agile co-creation practices benefited firms during their servitization journey.

In the second coding stage, we adopted a deductive approach. We allocated the seven second-order themes related to mechanisms through which servitizing firms benefitted from agile co-creation practices to the three dynamic capabilities, i.e., sensing, seizing, and reconfiguration. So, we assessed whether the benefits that encompassed the second-order themes related to identifying opportunities or threats (i.e., sensing), decision-

making or resource mobilization (i.e., seizing), or transforming existing structures and capabilities (i.e., reconfiguration).

During the analysis, categories and themes emerged representing boundary conditions essential to benefit from agile innovation approaches. When respondents talked about the outcomes of the outcomes of agile innovation approaches during the interviews, they sometimes also talked about the lack of results and their underlying reasons. While this topic was not included in the interview outline, this input provided a more nuanced understanding of the outcomes of agile innovation approaches and was therefore considered relevant to include in this research. During the analysis phase, data excerpts related to lack of agile innovation approaches outcomes and their underlying reasons were labeled in separate categories and themes and emerged as the aggregated dimension of “agile co-creation boundary conditions.” The first-order categories and second-order themes constituting the agile co-creation boundary conditions dimension represent the role of other dynamic capability micro-foundations in the relationship between agile innovation approaches and dynamic capabilities. The second coding round resulted in the final data structure, as illustrated in Figure 5.2.

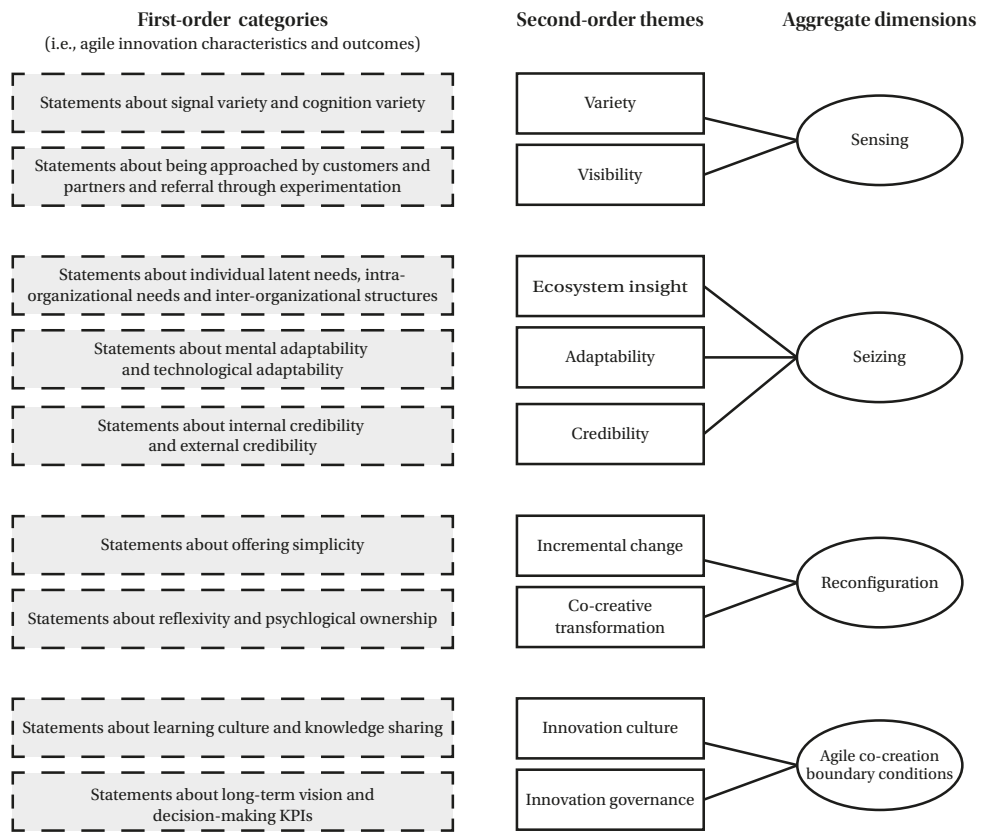


Figure 5.2 Data structure

## 5.4 Results

### 5.4.1 Sensing

Agile innovation approaches enhance a firm's sensing capability through two dimensions: variety and visibility. The concept of variety emphasizes the importance of diverse teams and engaging with external stakeholders to increase exposure to opportunity signals. Visibility centers on the strategic and open interaction with external actors through iterative experimentation, opening pathways to identify potential opportunities and partners for servitization.

#### 5.4.1.1 Variety

By promoting diverse teams that frequently interact with external stakeholders, agile innovation approaches facilitate the sensing capability through signal variety, which enhances a firm's exposure to opportunity signals. Diversity within the firm encompasses including individuals with different functional backgrounds and interests in the agile co-creation process. For example, one respondent highlighted the importance of variety in functional experiences: *"Servitization is fueled by input from the field. [...] We have many technicians on the shop floor who provide valuable insights. Additionally, we receive signals from sales about product performance, and we gather information from various product lines. This enables us to identify potential opportunities from several sources"* (Iota, interview 1). This quote highlights the importance of working in diverse teams with members from different functional backgrounds. So, while individuals from the sales department comprehensively understand customer needs through their ongoing interactions with customers, support and maintenance personnel, who engage with customers at the operational level, have insights into how customers interact with the supplier's offerings and the associated challenges. Furthermore, R&D and purchasing personnel focus more on technological development and potential technology partners. This finding indicates that working in varied teams with people from various functional backgrounds exposes a team to complementary signals because different functions are oriented towards other parts of the environment.

Besides functional orientation, team diversity also drives signal variety through different interests among team members. For example, one respondent indicated that people from different functional backgrounds are also exposed to other opportunity signals due to their interests: *"For example, the engineers who are really into PLC technology and using electricity efficiently delve into it, while others delve into other topics"* (Chi, interview 2). So, involving individuals from different functional backgrounds also implies involving people with various interests. Because these interests influence an individual's information-seeking behavior, they increase the variety of potential signals to which a team is exposed. So, the diverse interests of team members lead them to consult different information sources, such as papers and websites, further enhancing the variety of opportunity signals available to the firm.

In addition to diversity within the firm, involving external actors in the agile co-creation processes exposes firms to various opportunity signals. For example, one respondent highlighted the importance of engaging with channel partners to identify market trends: *"We have partners from whom we can use two hours of their time every month to talk about the business model and what they see happening in the market"* (Rho, interview 1). Regular consultations with channel partners enable firms to gather valuable information about market developments identified by these partners. Similarly, suppliers are instrumental in signaling technological advancements, as indicated in this quote: *"The role of our suppliers is to keep us informed about the latest developments in the technology they provide, ensuring we stay up to date"* (Upsilon, interview 1). Through frequent interactions with suppliers, servitizing firms position themselves to receive signals about emerging technologies and innovative solutions that address a customer's need.

Team diversity also facilitates the sensing capability through cognition variety, which revolves around the distinct sensemaking schemata among members of an agile development team. By incorporating individuals from various functional backgrounds and with different personality traits, the team benefits from heterogeneity in attention, perspectives, and interpretations when exposed to opportunity signals. For example, one respondent indicated that various team members look at different aspects of a potential opportunity: *"Some mainly focus on whether it is commercially or technically feasible, but I also consider the legal implications"* (Chi, interview 2). This quote illustrates that different interests among team members may lead them to pay attention to various elements of an opportunity signal. Furthermore, diverse skills and knowledge among team members may also contribute to sensemaking, as illustrated in the following quote: *"We often hold a so-called 'pizza session,' where we sit down with a few technicians to closely examine an opportunity that appears commercially viable and determine whether it also makes technical sense."* (Omega, interview 1). This quote illustrates that people from different functional backgrounds bring various knowledge to the table through which they assess an opportunity signal. As such, variety in interests and functional experiences enables teams to evaluate and make sense of opportunity signals more comprehensively.

#### **5.4.1.2 Visibility**

Another element of agile innovation is interacting with external actors through iterative experimentation. Respondents indicated that this experiment-based approach to innovation facilitated their sensing capability in two ways.

First, sensing was facilitated by *referral through experimentation*, which entails the identification of potential partners through active engagement in the marketplace. One respondent indicated that customers referred them to a technology partner in experimentation: *"We began experimenting with a self-developed IoT device. At one point, our customer suggested, 'Hey, you should also talk to [technology partner], as I think they have developed something similar.' That's how we initiated contact with [technology partner]"* (Epsilon, interview 2). By openly interacting with external actors about an intended servitization strategy and actively experimenting in the marketplace,

firms can tap into the network of counterparts. Such referrals provide opportunities to develop further or refine an envisioned offering or strategy, as indicated in the following quote: *“The moment we engage in activities X or Y with a certain partner using the lean startup approach, it sometimes occurs that a partner might suggest, for instance, ‘If you’re interested in biobased material, you should talk to [supplier], as they are interested in doing a pilot with a hemp fiber plate’”* (Theta, interview 2). These examples demonstrate that the willingness to experiment with external actors openly creates visibility of the envisioned strategy in the ecosystem, which results in opportunities to access potential partners with complementary resources. In other words, agile co-creation enabled servitizing firms to tap into the extended networks of their counterparts.

Second, an agile approach and visibly presenting experiment outcomes and learnings in the market resulted in servitizing firms ***being approached by customers and partners***. One respondent indicated that openly sharing experiment artifacts such as prototypes and experiment outcomes attracted external actors and exposed them to opportunity signals: *“By frequently experimenting, you naturally gain increased visibility. [...] showcasing prototypes at trade fairs, along with proofs of technologies or concepts, attracts people’s attention”* (Beta, interview 2). This quote illustrates how the firm shared the tangible output of the experimentation process and how this facilitated market visibility and exposure to opportunity signals. This is further exemplified in the following quote: *“We are often invited as keynote speakers at trade fairs, which establishes us as a kind of authority in [the market]. Furthermore, because of our brand’s reputation, parties seeking innovation are particularly interested in the topics we are currently discussing”* (Tau, interview 1). So, openness and proactive engagement in the marketplace increase a firm’s visibility and attract customer and partner interest in pursuing collaborative opportunities. The findings indicate that through experimentation and working with Minimum Viable Products (MVPs), firms experienced heightened visibility in the market and gained recognition as innovative organizations tackling significant problems.

### 5.4.2 Seizing

Agile innovation approaches strengthen seizing capability through ecosystem insight, adaptability, and credibility. Ecosystem insight underscores the importance of understanding the intricate network of relationships and interdependencies within an ecosystem, providing a framework for more effective engagement and collaboration. Adaptability entails mental and technological aspects, emphasizing the need for responsiveness to deal with the dynamic nature of innovation. Credibility relates to building trust internally and externally, facilitating the alignment and mobilization of resources essential for servitization.

#### 5.4.2.1 Ecosystem insight

Results showed that an agile innovation approach provided firms with ecosystem insight on three levels, namely, on the individual, intra-organizational, and inter-organizational levels. These insights enabled firms to develop their service business model and seize the opportunity for servitization.

First, the results indicated that an agile innovation approach facilitated discovering **individual latent needs**. Understanding customer needs and the potential value-in-use of product-service offerings is essential for business model development. However, customers often have trouble expressing their needs, as indicated in the following quote: *"We assess whether there is demand from the customer, and sometimes we nurture that demand because some customers have not yet realized the available possibilities"* (Tau, interview 1). So, latent needs derive from the fact that customers are not always up-to-date about the current state of technology. However, when customers have an initial understanding of their needs, they might not be fully able to express them, as indicated in the following quote: *"When we present a demo to customers after three weeks, they often have insights we had not considered, like 'But what I had not thought about was...' These insights can significantly impact our software architecture"* (Beta, interview 2). Like other respondents, the above quote shows that frequently releasing product increments (i.e., MVPs) to customers facilitates discovering these latent needs. Also, writing so-called user stories can reveal latent customer needs: *"When we develop user stories within the team, we present them as detailed narratives for the customer to review. Often, the customer responds with feedback, pointing out missing elements, like 'Yes, but then I miss this, or do I miss that'"* (Beta, interview 2). These findings illustrate the importance of tangible output to put customers in a state of reflection and enable them to express their needs that would otherwise remain latent.

Next to the individual level, engaging in agile co-creation through iterative experimentation and frequent product releases reveals **intra-organizational needs**. Servitization and product-service offerings affect multiple stakeholders within the customer's organization with different needs, which become visible when experimentation with the customer to validate business model hypotheses: *"By speaking with a variety of people, you begin to gain a deeper understanding of the customer. Moreover, this interaction clarifies how decision-making processes work within the customer's organization"* (Theta, interview 1). This quote illustrates that engaging with customers reveals the internal structures and the various stakeholders involved. Furthermore, such engagement also shows the goals and needs of multiple stakeholders, as indicated in the following quote: *"Often with innovations, there is a disconnect between the wishes of the C-level executives and their translation into operational actions. For instance, the purchasing department might think, 'I have my goals; I need to negotiate discounts. I prefer sticking with the usual contracts, and this [new approach] is inconvenient'"* (Tau, interview 1). So, servitizing firms may face different and contradictory goals among stakeholders within the customer organization. By actively engaging with customers through agile co-creation practices, these structures and the goals of various stakeholders within the customer organization can be revealed.

More specifically, the results indicate that actively involving stakeholders throughout the customer organization in the experimentation process provides an understanding of how an envisioned offering affects each stakeholder. The following quote exemplifies this insight: *"The management of the customer was enthusiastic; they liked the idea. However, when they began testing, they quickly concluded that the people on the shop floor would not use it because the VR headset caused motion sickness"* (Omicron,

interview 1). Individual stakeholders within the customer organization do not always understand the needs of other stakeholders or can oversee how an envisioned offering affects other stakeholders. So, releasing tangible output in the form of an MVP, a functional prototype, or a mere visual representation within the customer organization provides valuable insights into the effect of an envisioned offering or business model on the relevant stakeholders. Such interaction, enabled by tangible output, offers helpful information for refining value propositions and designing customer-centric solutions.

On the highest level, agile co-creation facilitates seizing capability by activating and providing insight into *inter-organizational structures* and the embeddedness of an envisioned service business model in the ecosystem. The results show that through experimentation with MVPs and pilots, firms could understand which capabilities should be kept in-house and which should be outsourced. This process allowed firms to assess the value and applicability of specific partner offerings and technologies, enabling them to make informed choices. For instance, the findings revealed that in one case, testing an external IoT platform with customers demonstrated that the benefits did not outweigh the costs, leading the firm to develop its system: *“We tested [external IoT platform] with two customers and received similar feedback from both. They commented, ‘that is nice but too expensive.’ Thus, while there was a benefit, it did not justify the costs”* (Alpha, interview 1). So, experimentation provides valuable insights into how external capabilities relate to customer needs, which is instrumental in the decision-making process about firm boundaries and capability sourcing from the ecosystem.

Next to delineating firm boundaries and deciding on potential partners for capability access, firms must also understand how their envisioned service business model affects existing ecosystem structures. Multiple respondents indicated that aligning an envisioned service proposition with the goals and needs of ecosystem partners such as suppliers and intermediaries was essential for securing commitment but also challenging. Involving ecosystem partners in the agile co-creation and experimentation process triggers responses from ecosystem partners that provide valuable insights into their goals, as is illustrated in the following quote: *“After we introduced a prototype to the market, some channel partners expressed concerns, saying, ‘you’re destroying my market; this is not acceptable.’ However, such reactions indicated to us that there was potential in what we were doing, and it was important to further explore this. From there, we could then develop a model that satisfies both parties”* (Rho, interview 1). So, presenting tangible manifestations of an envisioned business model (e.g., through MVPs) to ecosystem partners triggers responses that reveal important cues about their goals and sensemaking processes. These cues facilitate the development of the envisioned business model that includes incentives aligning with the ecosystem partner’s goals and priorities.

Altogether, agile innovation approaches are instrumental in seizing capability for servitizing firms. By embracing a responsive and iterative process, firms can uncover latent needs, navigate complex stakeholder landscapes, and make informed decisions about partnerships and firm boundaries. These findings underscore the versatility and strategic importance of agile practices in the contemporary business ecosystem.

#### 5.4.2.2 Adaptability

The second facilitating factor of agile innovation approaches for seizing capability is two forms of adaptability. First, *mental adaptability* was an essential dimension of agile innovation approaches in promoting seizing capability. Mental adaptability refers to the readiness to pivot away from unviable ideas and the openness to new ideas and decisions. By focusing on hypothesis formulation and testing, an agile innovation approach creates a sense of fallibilism towards business ideas, which is the belief of possible doubt regarding the truthfulness of a thought. The following quote demonstrates this dynamic: *"We start by filling out a business model canvas, then proceed to test our most significant assumptions. One of the first assumptions we tested revealed something interesting. After just three conversations, customers consistently responded, 'well, we don't feel like that at all'"* (Theta, interview 1). By stressing that all ideas contain assumptions that should be tested, individual actors become less emotionally attached to their ideas and will more proactively challenge their beliefs. Ultimately, this makes it easier to pivot away from unviable ideas.

Emotional attachment is further mitigated by working in short sprints and reducing the time spent working on an idea, as indicated in the following quote: *"Every month, after further exploration, we release a new version. We didn't predetermine the outcomes, but whenever new issues were identified, we would discuss them, and then they were addressed. Moreover, with each new insight, something new was created"* (Rho, interview 1). Sprint-based working reduces mental attachment to released products as teams dedicate little time, effort, and resources to developing MVPs. Consequently, individuals and teams remain open to iterating and improving their ideas. This is particularly the case if there is a formal go/no-go moment during which the assumptions are assessed, as indicated in the following quote: *"We formulate hypotheses and describe them in detail. We have established clear planning and milestones. Furthermore, if our assumptions are not positively validated, the project is halted; we pull the plug"* (Delta, interview 1). So, by engaging in experimentation guided by predefined assumptions that should be falsified, firms have clear points during which a project is deliberately continued or abandoned.

Furthermore, agile innovation increases mental adaptability by frequently releasing MVPs. The following quote highlights the importance of having an existing product running successfully in the market, which reduces the pressure to release something new immediately: *"[...] so you already have a successful product in the market, and you might need to discard three-quarters of it to incorporate an additional parameter. However, this is acceptable because you are starting with something that functions effectively"* (Mu, interview 1). This quote indicates that working in sprints and frequently releasing product increments reduces the risk of radical decisions and changing directions because there is already a working offering or technology. So, frequent releases to the market enhance mental adaptability, as releasing Minimum Viable Products (MVPs) fosters a culture of continuous improvement and learning. This culture promotes bold decision-making and continuously improving existing offerings in response to ecosystem dynamics and evolving customer needs.

Another finding was that an agile innovation approach enables **technological adaptability** during development. As indicated, such adaptability was necessary to deal with technological complexity and a priori unknown technological interdependencies within the ecosystem. The results suggest that an agile innovation approach supports firms in dealing with technological uncertainty: *“In my opinion, agile development is effective in dealing with uncertainty. The key question then becomes: How can you make adjustments as quickly as possible without necessitating excessive redoing?”* (Beta, interview 2). *“With agile methodology, the use of small increments allows for continuous adjustments. In contrast, with the waterfall approach, you might build something only to later discover that a specific technology should have been used. In such cases, you have to dismantle the existing technology and then modify it”* (Zeta, interview 2). Organizations can integrate new insights and technologies more effectively during development by employing an incremental sprint-based approach. The agile innovation approach allows the technological layout to emerge over time. This incremental approach to technological development supports flexibility in dealing with technical complexity and facilitates the timely adoption of emerging technologies throughout the process.

#### 5.4.2.3 Credibility

The ability of experimentation to showcase concrete and actual results and adopting a learning approach to innovation is instrumental in building credibility within and outside the organization. Credibility is essential to ensure trust and commitment and mobilize resources for servitization. By demonstrating practical success and profitability, experimentation facilitates **internal credibility** to ensure buy-in from key organizational stakeholders. One respondent provided an illustrative example where the profitability of the service business was showcased, prompting the company to allocate significant resources towards structuring and formalizing it: *“At a certain point, they realized that these services represent an ongoing business opportunity and that a viable business case could be made out of it. Consequently, last year, they hired a service director to fully manage this business”* (Xi, interview 1). The quote emphasized that showcasing concrete outcomes resulted in broader adoption of the service business throughout the organization. Specifically, compelling evidence from experimentation facilitated the creation of buy-in and mobilized essential resources for scaling up the service business.

Another respondent highlighted the power of demonstrating returns on investment over time: *“Some entities were very enthusiastic about our services, while others were less so. We initially collaborated with the enthusiastic ones, and after a year, we demonstrated a positive return from these efforts. This success led to a gradual change in attitude among the less enthusiastic entities, who began to express interest”* (Iota, interview 1). By initially involving departments that expressed enthusiasm and willingness and showcasing positive outcomes to less enthusiastic entities, they gradually changed their stance, expressing a desire to become more involved and gain knowledge about the new service initiative. These examples demonstrate the role of agile innovation in building internal credibility. Through experimentation and the demonstration of tangible results, initially skeptical stakeholders can be engaged and aligned with the organization's shift toward service transformation.

Furthermore, approaching external actors with a learning mindset facilitates *external credibility* and the creation of trust, as indicated in the following quote: “*One important lesson we’ve learned is to listen closely to customers to identify where the real demand lies and understand what is needed to address this need. This approach also helps to alleviate some of the customers’ suspicions. As a result, we receive much more valuable input, which is instrumental in shaping our services*” (Iota, interview 2). This quote underscores the importance of an empathetic and attentive approach toward customers enabled by agile innovation. Organizations build a foundation of trust and credibility by engaging customers not merely as consumers but as collaborators in the learning process and by actively seeking their input rather than pushing for immediate sales. The practice of actively listening to customers’ needs and concerns aligns with the principles of agile innovation, where continuous feedback and adaptability are essential. In this way, agile innovation approaches facilitate the development of products and services that more accurately meet customer demands and nurture customer relationships based on trust.

### 5.4.3 Reconfiguration

Agile innovation enhances reconfiguration capability through two distinct but complementary dimensions: incremental change and co-creative transformation. The aspect of incremental change highlights the sprint-based approach of agile innovation to craft service offerings in small increments. Co-creative transformation emphasizes the collaborative and reflective processes inherent in agile innovation. It engages internal and external stakeholders to critically evaluate their roles and cultivate a sense of psychological ownership.

#### 5.4.3.1 Incremental change

The results illustrate that agile innovation approaches are instrumental in shaping markets by embracing *offering simplicity*. This dimension is particularly salient in addressing the challenges of technological complexity and the unfamiliarity associated with product-as-a-service propositions and service business models. One respondent illustrates this by stating, “*You have to remove all obstacles because in this segment, the phenomenon where you don’t buy something physical, but rather the output, is completely new.*” (Pi, interview 1). This quote highlights the disruptive nature of servitization and the imperative to navigate it thoughtfully. Sprint-based working, marked by frequent releases of small product increments, is a practical approach to address the unfamiliarity barrier, as it enables servitizing firms to engage critical stakeholders in the change processes associated with servitization. Another interview reflected this insight, “*For me, agile is developing something small and receiving continuous feedback. As a result, you can take the customers along much more easily.*” (Psi, interview 1). By crafting service offerings in small increments and maintaining their simplicity and relatability, firms ensure they resonate with external stakeholders. Simplicity and relatability are critical because stakeholders tend to juxtapose new service offerings with existing ones. Overly complex service offerings result in customers sticking with familiar options and resisting the changes associated with servitization. Thus, sprint-based working and frequently releasing small offering increments foster familiarity and acceptance among

stakeholders, nurturing their receptivity to change.

#### **5.4.3.2 Co-creative transformation**

By actively involving internal and external stakeholders in agile co-creation activities like experimentation, organizations are prompting them to reflect on their roles and functions within the ecosystem critically. One respondent illustrated this reflective process: *“Our partners are not familiar with the nuances of as-a-service conversations. So we introduce them to the end customer, helping them envision the potential new roles they can play”* (Rho, interview 1). Reflexivity encourages stakeholders to reevaluate their positioning, catalyzing potential business changes. In essence, involving ecosystem partners in experimentation fosters the transformation of existing ecosystem structures and paves the way for aligning business practices and capabilities with an envisioned digital servitization strategy.

Involving stakeholders in agile co-creation processes also fosters a sense of psychological ownership, where stakeholders feel an emotional connection to the project and perceive it as “theirs.” Such ownership leads to increased engagement, responsibility, and reduced resistance to change. As one respondent indicated, asking stakeholders inside the organization for input increases involvement in the transformation: *“For example, in the workshops where we occasionally demonstrate our work, we ask for input, which results in significant involvement”* (Delta, interview 1). In other words, involving internal stakeholders in the experimentation process and asking for their input primes them for the service transformation as they feel heard and become part of the process. Another respondent indicated this dynamic applies to stakeholders outside the organization: *“First, we assembled a pilot group consisting of people from various departments at the customer’s organization. They provided feedback, which we actively utilized. [...] I believe that this approach also ensured they felt their input was taken seriously. It’s possible that because of this, we encountered much less resistance than anticipated.”* (Beta, interview 1). Thus, involving internal and external stakeholders in the experimentation process creates an emotional connection to the change process. Through such psychological ownership, agile innovation approaches foster reconfiguration capabilities.

#### **5.4.4 Agile co-creation boundary conditions**

Besides the role of agile co-creation in facilitating dynamic capabilities, we also identified two boundary conditions that affect the relationship between agile innovation practices and dynamic capabilities. Notably, we found the importance of innovation culture and governance.

##### **5.4.4.1 Innovation culture**

Innovation culture refers to an organizational belief that reflects the importance of continuous learning and knowledge sharing and a culture that accepts uncertainty and iterative progress in servitization. Results reveal that a *learning culture* within an organization is an essential precondition to harness the potential of agile innovation

practices. One respondent reflected on the challenges faced by conventional manufacturing companies, noting: *"Again, we see the struggle of a traditional manufacturing company; they are very accustomed to working with a kind of 100% plan. [...] However, what people in this company still struggle with is developing something akin to an MVP. [...] This shift in mindset and culture represents a significant barrier to efforts towards servitization."* (Epsilon, interview 2). Another observation highlights the contrast between manufacturing and service businesses and their effect on innovation approaches: *"We don't adhere to a fail-fast principle here; instead, it's more about getting it 180% right on paper. [...] Service innovation demands rapid prototyping and making small, incremental investments. In contrast, traditional product development eventually requires setting up machines, which represents a substantial investment."* (Epsilon, interview 2). These insights underscore that the prevailing organizational culture of manufacturing firms places a premium on certainty and predictability due to the substantial investments traditionally required in manufacturing.

In contrast to the manufacturing-centric mindset towards innovation, several respondents highlighted a learning culture within their organizations that endorsed and facilitated the agile co-creation process. While immediate objectives can be defined, the broader, long-term goals often unfold over time. One respondent provided insight into the organizational mindset, underscoring how senior management's tolerance for uncertainty and failure can be instrumental to iterative-based innovation: *"Our financial director never fixated on immediate sales targets. Instead, he emphasized progress, viewing even failure as a form of it. To him, failure signaled that while a particular approach might not have aligned with market needs or technology, it still provided invaluable learning."* (Zeta, interview 1). Hence, an authentic learning culture should not only be confined to the operational teams involved with servitization but should also be embedded in executive management, ensuring that a commitment to learning and adaptation shapes strategic decisions.

Another tenet of the required innovation culture is a culture of **knowledge sharing**, particularly in facilitating hypothesis-driven agile co-creation. The quality of hypotheses and consequent experiments greatly depend on the collective knowledge and expertise within the organization. However, much of this knowledge resides within the minds of individuals throughout various departments. To fully harness the potential of collective wisdom in the hypothesis development stage, a proactive attitude to sharing knowledge and experiences by employees is essential. A respondent encapsulated this challenge, observing, *"Lean startup presents a promising path forward. However, a significant challenge is motivating individuals within the company to share their knowledge. [...] The crucial task is not just about discipline, but about fostering a specific mindset that encourages and makes individuals feel comfortable in articulating their assumptions"* (Alpha, interview 1). In essence, fostering a culture that promotes inter-departmental knowledge exchange is paramount for hypothesis generation. When insights from varied functional backgrounds converge, the resulting hypotheses are invariably richer and multifaceted. Such collaboration improves the quality of the hypotheses and enhances the novelty and efficacy of subsequent experiments.

Beyond the internal boundaries of an organization, the outcomes of agile co-creation are influenced by external knowledge sharing with various stakeholders. Addressing technological complexities and leveraging inter-organizational synergies requires knowledge and data to flow across organizational borders. Illustratively, one respondent pinpointed the reluctance of their organization to share knowledge externally as a significant impediment to the servitization journey: *“We deliver the end product, so we own the data, but this approach has its limitations. For instance, when the idea of creating a data pool arose, we immediately insisted on complete control, which ultimately hindered its development. Thus, it wasn’t just technological limitations that impeded progress, but also our reluctance to move beyond a one-sided business model focused solely on our ownership. This mindset reflects a lack of consideration for ecosystems and co-development”* (Epsilon, interview 2). This narrative underscores the importance of organizational receptiveness and an ingrained knowledge-sharing culture with external entities. Embracing such a culture is pivotal to engaging a broad spectrum of external stakeholders, including technological partners and end-users, in the agile co-creation process. A closed innovation culture might deter these external entities from participating in the experimentation phase, consequently undermining the quality and scope of the co-creation process.

#### **5.4.4.2 Innovation governance**

Innovation governance refers to the processes, rules, and metrics top management uses to guide an organization’s innovation efforts. In the context of servitization, particular governance elements such as a long-term vision and learning-oriented KPIs were considered important factors influencing the outcomes of agile co-creation.

The first governance element is a **long-term vision** that supports innovation efforts with a long pay-off horizon. As indicated by one respondent, building a digital backbone for complex services requires a long-term investment horizon: *“We are very consciously building up the backbone of our system, guided by a forward-thinking philosophy. We are inputting data into the system that will be useful to us in five to ten years. This is crucial because, in a few years, we will need a system capable of making service predictions. It will allow us to monitor how customers’ lines are running and record this information. This data will be instrumental in shaping the next step of our service provision.”* (Iota, interview 2). So, building the digital infrastructure for the firm’s service offerings in the future requires a long-term vision.

However, firms must overcome the challenge of being overly focused on short-term financial goals, as indicated in the following quote: *“These developments are propelled by the existing organization, which is focused on a very short-term goal: making a profit this year. If [servitization] were placed in an incubator-like environment, with finances structured to accelerate its growth, I believe the results would be significantly better. However, since this is not the current approach, we tend to prematurely abandon promising initiatives, effectively ‘killing our darlings’ with potential”* (Epsilon, interview 2). When firms focus too much on short-term financial gains, they may assign too few resources to facilitate experimenting with potentially profitable service offerings.

Ultimately, allocating too few resources to experimentation might result in type II errors, which refer to false negatives by rejecting beneficial service concepts to the organization. Ensuring a balance between a long-term vision and immediate financial objectives is critical to effective innovation governance and agile co-creation.

The second governance element important for fostering agile co-creation processes relates to the *decision-making KPIs* adopted by top management. While agile co-creation processes enable firms to work through uncertainty and deal with subjectivity in servitization, such techniques require decision-making criteria that differ from traditional manufacturing. When top management primarily adopts decision-making criteria oriented toward short-term financial gains and efficiency, they might inadvertently starve experimentation of essential resources, as indicated in the following quote: *“Many of these [servitization] initiatives falter because management demands a solid business case at an early stage. However, it is often difficult to accurately determine the benefits and costs during the initial stages of development.”* (Epsilon, interview 1). Not adopting learning-oriented decision-making criteria suffocates experimentation as experimenting teams focus too much on financial details too early instead of uncovering value-in-use of envisioned service business models. Hence, learning-oriented KPIs such as the experimentation rate, feedback loop efficiency, and knowledge retention rate facilitate teams to optimize learning and leverage agile co-creation processes.

Another dimension to consider in assessing agile co-creation processes is recognizing the firm-wide benefits of servitization. These benefits often transcend direct financial gains, encompassing improved customer relationships and a unique market position that can boost traditional product business. As one respondent put it: *“If we assist customers with programming, it will lead to their need for more machines. [...] While our Platform-as-a-Service proposition may not be profitable by itself, it contributes to the bigger picture by driving increased machine sales, from which we profit”* (Omicron, interview 1). This highlights the importance of a holistic perspective, where top management evaluates the impact of servitization on the overall business. It is crucial to consider how servitization experiments and outcomes interplay with other business segments to ensure optimal resource allocation to agile co-creation processes. Adopting learning-oriented KPIs and a holistic business-wide perspective is pivotal in harnessing the full potential of agile co-creation processes in the servitization journey.

## 5.5 Conclusion and discussion

This paper investigates the role of agile innovation approaches in digital servitization, focusing specifically on their ability to facilitate dynamic capabilities across sensing, seizing, and reconfiguration dimensions. Our research underscores the instrumental role of agile innovation practices in augmenting these capabilities. For instance, by fostering diverse teams and open engagement with external stakeholders, agile innovation enhances the sensing capability by broadening the firm's exposure to opportunity signals and potential partners and enriching its cognitive base for interpreting these signals. Likewise, agile innovation facilitates seizing capabilities through iterative experimentation and sprint-based working. These practices foster ecosystem insight

by activating latent needs and ecosystem structures. Furthermore, reducing time spent on development and adopting a learning mindset to innovation increases mental and technological flexibility and internal and external credibility. Finally, an agile innovation approach advances reconfiguration capabilities by leveraging participative co-creation and sprint-based working, including various stakeholders in service transformation, triggering reflexivity, and cultivating a sense of psychological ownership among involved parties. Our study offers valuable insights into how agile innovation can catalyze equipping firms with the dynamic capabilities necessary for successful digital servitization.

Furthermore, our study reveals various boundary conditions that influence the facilitating role of agile innovation approaches to dynamic capabilities. These boundary conditions relate to the governance and culture of innovation within the firm. A supportive organizational culture that values learning, experimentation, and knowledge sharing appears to be essential to benefit from agile co-creation. Without an organizational culture that is receptive to learning through failure and venturing into unknown terrains, the core tenets of agile innovation, like iterative development and experimentation, may be stifled. A learning-oriented, transparent innovation culture is critical to garnering trust and involving external stakeholders in co-creation. The role of top management further amplifies these dynamics; effective agile co-creation necessitates a long-term vision articulated by the leadership and decision-making that balances short-term financial gains with learning outcomes for sustained success. These governance practices set the stage for the iterative, trial-and-error learning intrinsic to agile methods. Our findings suggest that the relationship between agile innovation approaches and dynamic capabilities depends on other structural, processual, and individual dynamic capability micro-foundations.

### **5.5.1 Theoretical contributions**

Our findings contribute to the existing literature in multiple ways. First, we extend the understanding of the role of agile innovation approaches in developing dynamic capabilities for digital servitization. While previous research predominantly positions agile co-creation as a micro-foundation for seizing capabilities (Chirumalla et al., 2023), our findings reveal its broader impact in facilitating sensing and reconfiguration capabilities. This extension enriches the debate around dynamic capabilities and deepens our understanding of how agile co-creation practices influence dynamic capabilities. By zooming in on the role of agile innovation approaches, we identified seven mechanisms that underpin their relationship with dynamic capabilities. Consequently, our research enriches the multi-level perspective on dynamic capabilities, offering a more nuanced and comprehensive understanding of their micro-foundations in digital servitization (Struyf et al., 2021).

Second, our study contributes to the emerging literature on agile co-creation in digital servitization by illuminating its role in facilitating ecosystem alignment (Sjödin et al., 2020). Previous research has predominantly focused on how agile innovation enables the co-creation of digital solutions within specific ecosystem constellations (Sjödin

et al., 2022). However, it has remained relatively silent on partner identification and involvement and the challenge of ecosystem transformation. We address this gap by demonstrating that agile innovation practices facilitate the identification of suitable ecosystem partners through an open innovation approach. This openness allows external actors to approach the servitizing firm directly or refer potentially relevant collaborators, thereby widening the pool of innovation partners. Also, we emphasize the significance of adopting a learning mindset when engaging with these partners, which enhances their active participation in co-creative endeavors. Finally, our study unveils that agile innovation triggers reflexivity among stakeholders about existing and prospective ecosystem structures, elements that are vital drivers of ecosystem transformation (Vink & Koskela-Huotari, 2022). Together, these insights expand the discourse on agile co-creation beyond the customer-supplier dyad, extending it to a broader business ecosystem context.

Third, our research contributes to the literature on organizational readiness for digital servitization. Prior work in this area highlighted the importance of technological competencies and service orientation as critical preconditions for successful servitization (Baines et al., 2020; Coreynen et al., 2018). However, the boundary conditions that influence the relationship between agile co-creation practices and dynamic capabilities bring forth additional dimensions of organizational readiness that are particularly pertinent to digital servitization. Besides technical competencies and service orientation, we argue that a learning culture and learning-oriented governance practices are essential attributes that determine an organization's readiness for digital servitization. These dimensions become especially critical as digital servitization involves and requires agile co-creation (Linde et al., 2023; Sjödin et al., 2020). Therefore, we suggest that any assessment of organizational readiness for digital servitization must also incorporate a firm's preparedness for agile co-creation in terms of organizational culture and governance.

Fourth, we contribute to the digital servitization literature by illustrating how agile innovation approaches enable firms to deal with the emergent nature of this transformation (Baines et al., 2020; Martinez et al., 2017). Digital servitization is emergent because customer needs are latent, contextual, and dynamic (Akaka et al., 2013; Biggemann et al., 2013). Our results indicate that experimentation enables firms to deal with this challenge by increasing visibility in the market and activating latent customer needs. Furthermore, the emergent nature of digital servitization is driven by its multi-actor nature and diverging goals and priorities among stakeholders (Stegehuis et al., 2023; Tóth et al., 2022). Our findings illustrate that agile innovation approaches enable firms to deal with this challenge as experimentation provides valuable insights into the underlying structures, thereby providing a helpful understanding of ecosystem structures that must be navigated. Also, an iterative and incremental innovation approach facilitates ecosystem transformation by striking the right balance between disrupting and maintaining existing structures (Koskela-Huotari et al., 2016). Finally, the emergent nature of digital servitization depends on rapid technological developments (Parida et al., 2019; Porter & Heppelmann, 2014). Working in autonomous and diverse teams increases firms' ability to identify and make sense of these developments. Moreover,

an iterative and sprint-based innovation approach provides the flexibility needed to respond to technological developments and adjust the technical infrastructure of a service offering.

### **5.5.2 Practical implications**

This research presents several critical insights for practitioners. First, it suggests that managers in servitizing companies should adopt agile innovation methods to align their internal and external stakeholders with an envisioned digital servitization strategy. By forming independent, varied teams, embracing iterative processes, and constantly experimenting, companies can increase their market presence and more effectively recognize and meet stakeholders' evolving needs and desires internally and externally. This approach also fosters support from crucial stakeholders for digital servitization, facilitating the transformation of the company's internal and external environments. Second, the study advises managers to engage critical stakeholders in the agile co-creation process. Involving key stakeholders broadens the perspective for spotting potential risks and opportunities, thus improving an organization's ability to identify service opportunities, find ecosystem partners, and discover new technologies. Stakeholder involvement also aids in the transformation of the ecosystem through increased reflexivity and a sense of ownership. Third, the research indicates that firms should evaluate their existing governance structures and corporate culture to see how well they accommodate agile innovation practices. Top management should define a clear vision for digital servitization, adopt decision-making criteria prioritizing long-term learning over immediate financial returns, and encourage a culture of knowledge sharing and learning. Without such governance and cultural foundations, the integration of agile co-creation might be hindered, potentially leading to the isolation of digital servitization efforts from the organization's core operations.

### **5.5.3 Limitations and future Research**

Our research design and scope are associated with multiple limitations that provide opportunities for future research. First, our respondents were mainly executives and senior managers within servitizing firms, leaving out the perspectives of middle managers and frontline employees, which also play an essential role in the agile co-creation process (Sjödin et al., 2020). Future research could consider these perspectives to shed further light on the role of agile innovation approaches in digital servitization. Second, the cross-sectional nature of our study limits the ability to observe changes over time. While respondents were able to retrospectively reflect on agile co-creation processes, a longitudinal research design would provide more in-depth insights into the transformative power of agile co-creation. Third, even though our study focused on the role of agile innovation approaches in reconfiguration, we did not take into account the adoption of agile innovation approaches as an important reconfiguration dimension of digital servitization. Future research may study agile transformation as another dimension of digital servitization and shed further light on how firms develop agile competencies required for digital servitization and the risks associated with combining digital servitization with an agile transformation. Fourth, we only focused on the role

of agile innovation approaches in dynamic capabilities without considering other potential innovation approaches. Future research could compare multiple innovation approaches, such as stage-gate or stage-gate agile hybrids, to shape digital servitization (Cooper & Sommer, 2016).

# Appendix 5A - Interview outline round 1

## Introduction

- 1) Can you briefly introduce your firm and your role?
- 2) How strong is the relevance of service innovation or servitization in your organization?
- 3) What service types does your firm offer and/or develop?

## B - Agile / Lean startup & servitization

- 4) Can you briefly describe the service development process and (best) practices/methods that the firm uses?

Next, we want to know how you have implemented (if at all) agile development and lean startup practices.

- 5) How does the firm implement an agile organizational structure (i.e., working in autonomous and diverse teams) in service innovation/servitization?
- 6) How does the firm implement sprint-based working (i.e., working in short iterative cycles) in service innovation/servitization?
- 7) How does the firm implement agile communication practices in service innovation/servitization (e.g., customer participation, demo sessions with customers, regular scrum meetings, Kanban board, retrospective meetings)?
- 8) How does the firm implement experimentation (i.e., hypothesis formulation, developing and deploying minimum viable products) in service innovation/servitization?
- 9) How does the firm implement validated learning (i.e., experimental data and systematic analysis) in service innovation/servitization?

## Networks & servitization

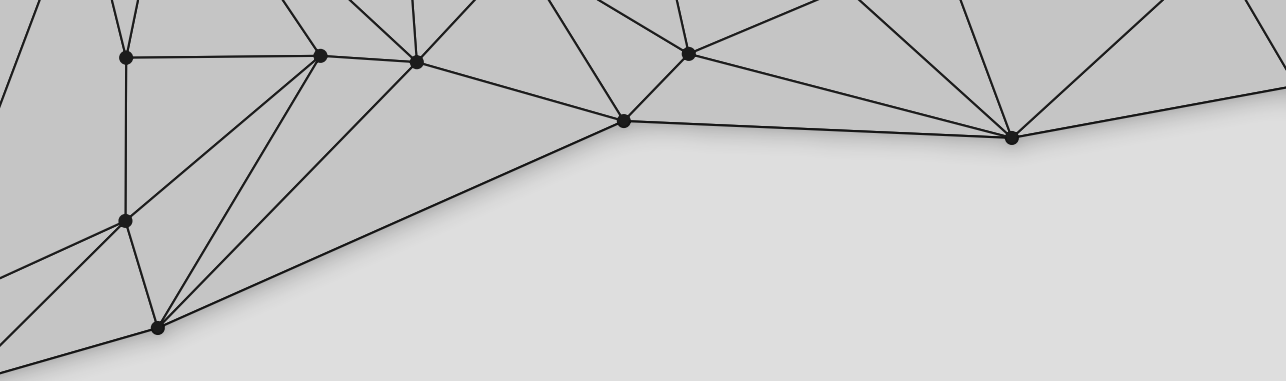
- 10) Can you briefly describe the role of external stakeholders when servitizing the firm, providing some examples of different types of stakeholders that may be involved?
- 11) What changes to the ecosystem were needed when servitizing the firm? How were new partnerships established and existing relationships reconfigured regarding what is being exchanged and how?

## Appendix 5B - Interview outline round 2

Thank you for participating in this interview. Our goal is to understand how your company navigated the process of servitization—transitioning from pure manufacturing to providing services. We are particularly interested in how organizational structures and processes affected this transition. Your experiences and insights will be invaluable in exploring these elements and their implications for the servitization journey.

- 1) Can you describe how the firm identifies technological developments, service opportunities, and potential partners for servitization in the external environment?
  - a. Can you describe how the firm's organizational structure and processes have influenced exposure to opportunities?
- 2) Can you describe the role of different cognitive styles and understanding of the world among team members in interpreting and making sense of opportunity signals?
  - a. Please describe how employees' different ways of thinking in servitization impacted the interpretation and understanding of opportunities.
- 3) Can you describe the firm's process to identify potential suppliers, technology providers, or channel partners in the servitization journey?
  - a. How did team structure or processes facilitate identifying potential partners?
- 4) Have you noticed any patterns in how customers and partners approach the firm with ideas or opportunities? Can you share an example of when such an approach led to a new opportunity?
  - a. Please reflect on how the firm's way of working (or doesn't) facilitates such proactive encounters between stakeholders and your firm.
- 5) How did the firm gain insights about unexpressed or hidden customer needs regarding (product-)service offerings? Can you recall a time when you successfully identified such customer needs?
  - a. Please describe how the firm's organizational structure and processes contributed to understanding unexpressed or hidden customer needs.
- 6) What was the role of gaining insights about network structures and the goals of critical stakeholders in the servitization process?
  - a. Please describe how the firm's organizational structure and processes helped gain insight into network structures and the goals of key stakeholders.
- 7) How did the firm decide which capabilities to keep in-house and which to outsource to external actors?
  - a. Please describe the role of the firm's organizational structure and processes in understanding the capabilities of external actors and envisioning the firm's ecosystem strategy.

- 8) What was the role of the flexibility of thinking in the firm's servitization process?
  - a. How did the firm's organizational structure and processes facilitate thinking flexibility? So, how did structures and processes enable the firm to keep an open mindset to new ideas and abandon faulty ones?
- 9) What was the role of technological flexibility in the firm's servitization process? So what was the role of the ability to adapt technology in servitization?
  - a. How did the firm's organizational structure and processes facilitate technological flexibility?
- 10) How has credibility played a role in garnering support and resources for your service offerings? Can you provide an example?
  - a. What was the role of the organizational structure and processes in creating credibility within and outside the organization?
- 11) How did the firm deal with (and overcome) internal and external adoption barriers and resistance to servitization?
- 12) How did the firm change and align invisible structures (e.g., culture or operating logic) within and across the firm?
  - a. What was the role of organizational structure and processes in changing invisible structures?
- 13) How did the firm change and align tangible structures (e.g., technological interfaces) within and across the firm?
- 14) What was the role of organizational structure and processes in changing tangible structures and coordinating co-innovation among stakeholders?



# 6

## DISCUSSION

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## 6. Discussion

This dissertation embarked on a journey to explore the growing field of servitization in manufacturing sectors. This trend has drawn significant attention from academics, practitioners, and policymakers in light of digitalization and climate change. While the positive financial and non-financial outcomes of servitization are well-established (Faramarzi et al., 2023; W. Wang et al., 2018), many firms struggle to realize these benefits (Benedettini et al., 2015; Xing et al., 2023). The network-embedded character of servitization and the uncertain and unpredictable nature of this transformation have been pinpointed as reasons behind this struggle (Bustinza et al., 2018; Struyf et al., 2021). While these challenges have been acknowledged and discussed separately, an integrated view was missing. Specifically, the existing literature provided a limited understanding of the multi-level forces that drive emergence in servitization and how agile co-creation facilitates firms in navigating these forces (Galvani & Bocconcelli, 2022; Sjödin et al., 2020). Hence, this dissertation sought to bridge the gap between digital servitization in inter-organizational constellations and agile co-creation by investigating the following central research question: ***“How does servitization unfold in inter-organizational constellations, and what is the role of agile innovation approaches in facilitating this transformation?”*** This research question was divided into four sub-questions, focusing on different facets of this overarching research question.

This chapter weaves together the distinct threads of individual research by presenting an integrated overview of the findings and their contributions to the ongoing servitization literature. While previous chapters discussed the contributions of each study, the current chapter emphasizes how the aggregated results of all chapters jointly enrich our understanding of servitization. This chapter is structured first to delineate the key findings of each study, addressing each sub-question separately, thereby providing a comprehensive overview of the findings. After examining each sub-question, synthesizing all conclusions will lead to answering the main research question. Next, the findings are discussed in the context of the existing literature, clarifying how this dissertation extends current knowledge of servitization. After this theoretical reflection, the chapter turns its attention to real-world applicability, delving into the practical implications of the research, with a particular focus on the ramifications for firms and individual managers engaged with servitization.

### 6.1 Key findings

#### 6.1.1 Study 1: Inter-organizational tensions in servitization: A dialectic process model

The first study focused on better understanding the emergent nature of servitization, specifically by exploring the influence of inter-organizational tensions on the servitization process. The following question guided the research: “How do inter-organizational tensions in business relationships influence the servitization process?” The research design entailed a single case study focusing on developing Façade-as-a-Service (FaaS), a novel circular service proposition in the construction sector. Given the

intricate web of interdependent stakeholders in the construction sector, it offered fertile ground for examining inter-organizational dynamics emerging during the execution of a servitization strategy. The focal firm, WinCo, and its interaction with various network counterparts were studied over three years. The methodological approach encompassed multiple phases of interviews, the facilitation of a workshop, and the in-depth analysis of secondary data, including industrial white papers, research reports, and publicly available interviews.

The first key finding of this study resides in conceptualizing a dialectic process model grounded in dialectic theory and integrated with IMP's ARA framework. This model posits inter-organizational tensions as an essential driver shaping the course and outcomes of a servitization trajectory. It asserts that the divergent goals and agendas of multiple actors within business networks manifest in contradictions, giving rise to tensions that affect the outcomes of deliberate attempts to modify network structures in line with an envisaged service proposition. In this context, a dialectical interplay between actors supporting the status quo, represented as the thesis, and those advocating alternative network configurations, expressed as the antithesis, drive servitization outcomes (Langley & Sloan, 2012; Van de Ven & Poole, 1995). The clash between thesis and antithesis stems from the embeddedness of servitizing firms in existing network structures, consisting of activity links, resource ties, and actor bonds (Håkansson & Snehota, 1995), and the need to operate in alternative network configurations when moving from products to product-service offerings (Kowalkowski et al., 2013; Reim et al., 2019). The resulting struggle between network counterparts and achieving synthesis requires compromising (Hoholm & Olsen, 2012), which results in deviations from the initially conceived service proposition and network structure.

The second key finding of this study revolves around the interdependence of the activity, resource, and actor business relationship dimensions. This interdependence resulted in the emergence of second-order tension, which refer to indirect contradictions that drive inter-organizational tensions. This complexity became evident in WinCo's shift from customized to standardized façades, which directly challenged existing product designs (i.e., resource ties). However, this shift also indirectly challenged the architect's professional identity (i.e., actor bond), which fuelled inter-organizational tensions. This situation illustrates how a contradiction within a single business relationship dimension can ripple through other dimensions. These interconnected dimensions often conceal the root cause of tensions, making them challenging to identify and address. This complexity points to the inherent unpredictability in the servitization process. It underscores the need for a nuanced and deep understanding of business relationships and their dynamics because they influence both the trajectory and the eventual outcome of transitioning to a service-oriented business model.

To answer the first sub-question, *inter-organizational tensions influence the servitization process by triggering a dialectic process of synthesis, and the interdependent nature of relationship elements makes this process complex and unpredictable, thereby contributing to the emergent nature of servitization.*

### **6.1.2 Study 2: Institutional complexity in digital servitization: A service ecosystem perspective**

The second study focused on the role of divergent social structures in inter-organizational tensions among collaborating partners. A service ecosystem perspective guided this research because it provided a suitable lens to study social structures by emphasizing institutional arrangements. This research revolved around the emergence and development of institutional complexity within inter-organizational collaboration in digital servitization, guided by the question: “How does institutional complexity contribute to the emergence of inter-organizational tensions in digital servitization, and how do actors respond to these tensions?” The chapter presented a detailed analysis of the “MetalNet” case study to address this question. This case encapsulated an inter-organizational partnership to develop a new service proposition and foster the necessary capabilities. Because the collaborating partners came from diverse institutional backgrounds, it provided a suitable context to study institutional complexity and inter-organizational tensions. Data was collected through ethnographic observations, semi-structured interviews, and secondary data collection. The data was analyzed through thematic coding. During the study, multiple controversies and tensions emerged, offering unique insights into institutional complexity in servitization and how actors responded to these tensions.

This study yields two major insights regarding institutional complexity in digital servitization. First, the study presents a process model that delineates how institutional complexity gives rise to inter-organizational tensions. In servitization, where industry transcending collaboration is essential for developing novel capabilities (Makkonen et al., 2022; Raddats et al., 2017), partnerships often involve actors with diverse institutional backgrounds. Such institutional complexity, wherein collaborating partners bring varying perspectives, expectations, and practices to the collaboration, may develop into inter-organizational tensions. Tensions emerge when the different perceptions, expectations, and best practices among partners clash. In this case, institutional complexity becomes a challenge, impeding aligned decision-making, hampering smooth collaboration, and obstructing the formation of trust-based relationships. As such, the study illuminates that complexity in servitization originates from the diverse goals, perceptions, and prescribed behaviors that stem from the different institutional backgrounds of involved partners. In other words, institutional complexity manifests in servitization through inter-organizational tensions and emerges from the need to collaborate with partners from different institutional environments.

The second key finding of this study concerns organizational responses that managers can employ to navigate institutional complexity and their resulting tensions. The results revealed two sets of practices. Firstly, structuring practices allow managers to mitigate the disrupting effects of inter-organizational tensions without addressing the underlying institutional complexity. This approach involves institutional clustering and outcome-based governance practices. These practices mitigated tensions by grouping actors with similar institutional backgrounds, smoothing the collaboration, and creating normative pressure to provide output that partners committed to. Secondly, managers can use

reconciliation practices to overcome inter-organizational tensions by directly addressing the underlying institutional complexity. Reconciliation practices included juxtaposing institutional contradictions, objectification, and physical interaction to engage in constructive discussion that facilitate mutual understanding and more cohesive and empathic relationships. Structuring and reconciliation practices provide an extensive conceptual toolbox to respond to and deal with institutional complexity, overcome inter-organizational tensions, and build a cohesive service ecosystem enabling value co-creation.

To address the second sub-question, *institutional complexity in digital servitization arises from collaborations across functional departments (e.g., engineering, sales, and executive management), different industries (e.g., manufacturing, software development, and technology development), and organizational forms (e.g., for-profit, educational institutes, and governmental bodies), frequently resulting in inter-organizational tensions. In response to these tensions, actors engage in structuring and reconciliation practices. These practices are instrumental in fostering a nascent service ecosystem characterized by aligned institutional arrangements, which, in turn, facilitate the co-creation of value.*

### **6.1.3 Study 3: Agile co-creation in digital servitization: Investigating the impact of agile development and lean startup on service business performance**

In the third study, the focus shifted to agile co-creation in digital servitization, a concept gaining increased traction in the literature (Chirumalla et al., 2023; Sjödin et al., 2020). Despite its growing importance, previous discussions on agile co-creation have often treated it as a unified innovation approach, predominantly explored through qualitative research methods, thereby limiting the granularity and generalizability of our understanding. To address this gap, the study investigated the impact of two agile innovation approaches, agile development and lean startup, on both financial and non-financial aspects of service business performance. Guided by the research question, “What is the impact of agile development and lean startup on financial and non-financial service business performance?” the study used Partial Least Squares Structural Equation Modeling (PLS-SEM) to analyze survey data from 114 respondents, providing novel insights into the role of agile co-creation in digital servitization.

The key finding of this study underscores the distinct impacts of agile development and lean startup on service business performance. Agile development positively influenced financial and non-financial performance metrics, whereas lean startup showed a positive relationship only with non-financial performance. This finding delineates the unique contributions of each approach in digital servitization and their synergetic effect: lean startup facilitates identifying latent customer needs and shaping business models and service propositions accordingly, while agile development, as a project management framework, enhances customer satisfaction by optimizing the co-creation process. As reflected in the observed interaction effect, agile development and lean startup complement each other. Notably, while both approaches benefit non-financial performance, only agile development positively relates to financial outcomes.

This underscores the importance of agile development in scaling service businesses, a key driver of economic success in servitization (E. Fang et al., 2008). Agile development facilitates this scaling through its micro-service approach, characterized by incremental and modular development that balances customization and standardization (Jamshidi et al., 2018; Sjödin et al., 2020). These findings suggest that while agile development and lean startup are valuable and complementary in the initial phases of digital servitization, the importance of agile development grows as firms advance in their digital service transformation.

To answer the third sub-question, *this study finds that while agile development and lean startup positively impact non-financial service business performance, only agile development shows a positive relationship with financial performance. This finding highlights their distinct roles in facilitating digital servitization, with agile development crucial for enhancing economic outcomes.*

#### **6.1.4 Study 4: Agile co-creation in digital servitization: A dynamic capabilities perspective**

The fourth study of this dissertation investigated the role of agile innovation approaches in the development of dynamic capabilities. As dynamic capabilities enable firms to respond to and shape their external environment (Foss et al., 2023; Teece, 2007), they are associated with operating in business ecosystems and devising and implementing an ecosystem strategy. Such strategy refers to “the way in which a focal firm approaches the alignment of partners and secures its role in a competitive ecosystem” (Adner, 2017, p. 47). Utilizing the sensing, seizing, reconfiguration framework of Teece (2007), the study investigated the role of agile co-creation in the capacity of servitizing firms to comprehend, respond to and shape their dynamic environment. This study centered on the research question: “How do agile innovation approaches facilitate dynamic capabilities in digital servitization?” A qualitative study was conducted to address this question, involving 31 interviews from 24 respondents involved with servitization, utilizing two rounds of semi-structured interviews for data collection and thematic coding for analysis.

This study yielded two key insights about the relationship between agile co-creation and dynamic capabilities in digital servitization. The first insight concerns how agile co-creation enhances a servitizing firm’s capacity to sense, seize, and reconfigure. These findings extend the breadth and depth of our understanding about the facilitating role of agile co-creation in a firm’s dynamic capabilities (Chirumalla et al., 2023). The results illustrate that agile team structures and experimentation increase a firm’s exposure to opportunity signals, including potential customer demands, partners, and technology developments. Also, the variety in agile teams enhanced a firm’s ability to interpret these signals effectively. Regarding seizing capability, an iterative approach to experimentation and development is instrumental in facilitating timely and appropriate decisions regarding service propositions, technological infrastructure, and organizational boundaries. This process also fosters internal and external credibility, which is critical for mobilizing the necessary resources aligned

with these strategic decisions. Lastly, agile co-creation supports reconfiguration by employing an incremental and collaborative approach to innovation. Such an approach effectively navigates the inertial forces within intra- and inter-organizational structures. Firms can deal with these challenges by keeping offerings simple and securing the involvement of key stakeholders by involving them in the co-creation process.

The second key finding of this study highlights the boundary conditions that influence the effectiveness of agile co-creation in enhancing dynamic capabilities. The results suggest that a supportive innovation culture and robust governance processes are crucial for realizing the potential benefits of agile co-creation. For instance, the diversity within agile teams is leveraged only in an organizational culture that fosters knowledge sharing. Similarly, the role of top management is pivotal; servitizing firms only benefit from agile co-creation when decision-making prioritizes long-term learning over short-term financial gains. In the absence of such governance practices, firms risk that experimentation is prematurely terminated, preventing the emergence of valuable learnings about novel service business models.

To answer the fourth sub-question, *agile innovation approaches enhance the dynamic capabilities of servitizing firms by leveraging diverse teams, involving key stakeholders in co-creation, and employing iterative experimentation. These practices are instrumental in broadening the firms' comprehension of their environment, facilitating informed and timely decision-making, and securing essential commitment from internal and external stakeholders. However, fully realizing these benefits is contingent upon a supportive innovation culture and robust governance principles.*

### **6.1.5 Synthesis of key findings**

This synthesis builds upon the comprehensive findings of previous studies to address the central research question of this dissertation: *“How does servitization unfold in inter-organizational constellations, and what is the role of agile innovation approaches in facilitating this transformation?”* Initially, the discussion will delve into the dynamics of servitization within inter-organizational constellations, providing insights into its complexities and nuances. Subsequently, the focus will explore how agile innovation approaches contribute to and shape this transformative process.

Servitization within inter-organizational constellations is marked by tensions arising from existing network structures. As firms engage with external actors to develop new resources and capabilities for servitization, they will experience diverging goals, expectations, priorities, routines, and opinions among (potential) partners. These differences stem from the unique embeddedness of each partner in existing social and technological network structures. For example, an existing supplier may resist an envisioned servitization strategy because the supplier tries to defend its profitable product-oriented business model instead of selling its products through service-level agreements. Alternatively, network counterparts may resist a proposed business model transformation due to their cognitive investments in a product-oriented one. These examples emphasize that two types of structures shape a counterpart's response to a

servitization strategy. These include tangible structures like resource and activity interfaces and social structures, which are often unconsciously enacted. So, this dissertation illustrates how the financial and mental investments in existing network structures and the resulting tensions affect the unfolding of servitization in inter-organizational constellations.

The inherent complexity and interdependence of network structures dictate that servitization unfolds in an emergent and iterative manner. Firms attempting to understand how these structures influence their counterparts' goals and priorities often confront a limited, partial view of their multifaceted network environment. This environment, encompassing cognitive, economic, and technological dimensions, presents challenges in perceiving and implementing servitization strategies. For instance, physical resource interfaces might be tightly interwoven with social structures like professional identities. As such interdependencies might not always be visible or consciously enacted, firms must base their initial servitization strategies on this incomplete information, adapting them iteratively through ongoing interaction and feedback. As such, this dissertation highlights the role of inter-organizational dynamics in the emergent and iterative nature of servitization.

To manage the emergent and iterative nature of servitization and navigate complexities stemming from inter-organizational dynamics, servitizing firms benefit from agile innovation approaches. Specifically, firms can benefit from implementing agile development and lean startup practices. For example, iterative experimentation allows firms to better comprehend their external environment in a shorter timeframe. Low-fidelity prototypes or MVPs will elicit latent network structures and enable servitizing firms to probe the responses of network counterparts to an envisioned servitization strategy, thereby increasing network insights. Another example is iterative development, consisting of sprint-based working and a micro-services approach, which enhances flexibility to deal with emerging insights and developments in the external environments. Breaking up tasks and service offerings into smaller discrete parts enables servitizing firms to adjust their offerings and the technological infrastructure to emerging customer needs and technologies. Also, an incremental innovation approach facilitates reconfiguring existing network structures as the proposed network changes are less disruptive than when releasing a fully developed service offering. Altogether, this dissertation illustrates how agile innovation approaches enable firms to navigate challenges in servitization. Specifically, agile innovation approaches facilitate a deeper understanding of the external environment and allow for more adaptable and responsive strategy development in the face of emergent network dynamics.

To answer the main research question, *this dissertation establishes that servitization in inter-organizational constellations unfolds in an emergent and iterative fashion, driven by structural diversity among network actors, which results in tensions. By promoting diverse and autonomous teams engaged in sprint-based working and iterative experimentation, agile innovation approaches enable servitizing firms to navigate emerging inter-organizational dynamics and complexities. This dissertation underscores the importance of adaptability and responsiveness in shaping servitization in inter-organizational constellations.*

## 6.2 Theoretical contributions

This dissertation identified three research gaps in the servitization literature: a limited understanding of inter-organizational collaborations, the interplay between strategic intent and emergence, and the role of agile co-creation approaches to servitization. The findings in this dissertation address these gaps and make multiple theoretical contributions to the servitization literature. Specifically, the results deepen our understanding of servitization in inter-organizational constellations, shedding new light on the interplay between strategic intent and emergence. Also, this dissertation extends the discourse on the role of agile co-creation in digital servitization. Finally, the dissertation brings forth a novel model to study transformations, like servitization, that involve various actors from multiple angles. The remainder of this section will discuss these contributions in more depth.

### 6.2.1 Servitization in inter-organizational constellations

Understanding its dynamics within inter-organizational constellations has emerged as a pivotal research area in servitization. This dissertation responds to calls for more profound insight, as highlighted by Raddats et al. (2019) and Story et al. (2017), into the transformative challenges and adjustments required at the inter-organizational level, a topic that has gained traction in recent literature (Galvani & Bocconcelli, 2022; Tóth et al., 2022). Building upon these foundational studies, this work contributes novel insights and perspectives on inter-organizational collaboration and tensions when implementing a servitization strategy.

First, this dissertation contributes to understanding the challenges inherent in reconfiguring and establishing inter-organizational constellations for servitization. Through the lenses of the ARA framework and institutional arrangements, this research highlights how pre-existing structures at the inter-organizational level introduce complexities, primarily through emergent tensions (Galvani & Bocconcelli, 2022; Tóth et al., 2022). These structures influence the goals, routines, and perceptions of various actors in the network (Håkansson et al., 2009; Koskela-Huotari & Vargo, 2016). Given the distinct network embeddedness of each participant (Edvardsson et al., 2014; Håkansson et al., 2009), the dissertation observes a diverse range of responses to proposed servitization strategies. This is exemplified in the first study, which presents a case where the required reconfigurations for reverse logistics in a supplier's operations clashed with its existing product-oriented business model. These findings underscore the complexity of servitization, highlighting how existing networked structures and the heterogeneity inherent in inter-organizational relationships complicate servitization (Huikkola, Kohtamäki, Rabetino, et al., 2022; W. Zhang & Banerji, 2017). This research thereby extends the understanding of servitization, demonstrating that while a strategy may appear coherent and sensible to a focal firm, it may not align with the perspectives or goals of network counterparts.

Second, this dissertation sheds light on the underlying forces that drive tensions across existing and new inter-organizational relationships. Drawing insights from study one it

reveals that pre-existing ties imposed constraints on strategic flexibility. Servitization requires firms to break with existing routines, resource interfaces, and actor identities (Galvani & Bocconcelli, 2022; Tóth et al., 2022). However, due to the the rigidity of existing business relationships, such changes often result in inter-organizational tensions. In contrast, study two illustrates that forming new partnerships presents a different challenge: institutional complexity. In this case, even though the substance of existing business relationships does not complicate interaction, they still bring beliefs and expectations to the collaboration. Mainly when servitization requires collaboration between actors from various industries or backgrounds (Favoretto et al., 2022; Makkonen et al., 2022). This dissertation underscores that servitization in inter-organizational constellations involves dealing with inter-organizational tensions, albeit by reconfiguring existing business relationships or building new ones.

Third, this dissertation enhances our conceptual understanding of servitization as an emergent and iterative process (Martinez et al., 2017). Studies one and two reveal how inter-organizational tensions often emerge unpredictably due to the interconnected and latent nature of organizational structures. The complex interplay of physical and social elements in these structures leads to unpredictability in how counterparts respond to servitization strategies or transformation efforts. Moreover, actors are often unaware of the structures governing their behavior and perceptions, further complicating upfront predictions of counterpart responses. This unpredictability results in cycles of action and reaction within inter-organizational constellations. Servitizing firms continuously adjust their offerings and business models based on the feedback and reactions they receive after revealing them to counterparts. For example, study one demonstrates how firms adapt their service business models in response to these emerging tensions. As such, this dissertation contributes to the literature on the interplay between strategic intent and emergence (Kohtamäki et al., 2021). This interplay between strategic intent and emergence highlights that only through engaging in and responding to network dynamics can firms genuinely grasp the implications of their servitization efforts, underscoring the need for agility in navigating these complex, network-level dynamics (Galvani & Bocconcelli, 2022).

### **6.2.2 Agile structures and processes for servitization**

Servitization involves value propositions, revenue models, and technological configurations unfamiliar to many manufacturers (Favoretto et al., 2022). Consequently, servitization is considered an evolutionary learning challenge that requires an agile innovation approach (Sandvik et al., 2022; Thomson et al., 2023). While agility has been considered in the servitization literature at the start of writing this dissertation (e.g., Bustinza et al., 2018), its prevalence has increased in recent years. Notably, the work of Sjödin et al. (2020) has sparked attention to agile innovation approaches by presenting an agile co-creation approach for digital servitization. Today, agile co-creation is considered an essential structural and processual prerequisite for successful servitization (Chirumalla et al., 2023; Huikkola, Kohtamäki, & Ylimäki, 2022). This dissertation contributes to this trend by providing further empirical evidence and a more nuanced understanding of agile innovation approaches.

While the existing servitization literature often views agile co-creation as a unified innovation approach (Chirumalla et al., 2023; Sjödin et al., 2020), this dissertation introduces a nuanced perspective by differentiating between agile development and lean startup. Aligning with recent scholarly discourse (Linde et al., 2023; Thomson et al., 2023), we emphasize the distinct yet complementary roles of these two methodologies in co-creating digital services. The empirical analyses in studies three and four substantiate this differentiation. As evidenced in study four, agile development enhances a firm's adaptability and responsiveness to technological shifts and market insights (Parida et al., 2019), primarily through sprint-based working that fosters emotional and technological agility. In a complementary fashion, the lean startup approach effectively counters intra and inter-organizational resistance (Galvani & Bocconcelli, 2022; Lenka et al., 2018a; Tóth et al., 2022), by bolstering internal and external credibility. This synergetic relationship between agile development and lean startup underscores their joint contribution to the servitization process, offering a comprehensive toolkit for firms navigating the complexities of servitization. These insights enrich the understanding of agile co-creation by delineating the roles and interplay of varied agile innovation approaches in servitization.

This dissertation extends the discourse on agile innovation in servitization by moving beyond the traditional customer-supplier dyad and exploring agile roles in ecosystem alignment (Sjödin et al., 2020, 2022). The empirical analysis from studies one and two highlights the challenges of involving external stakeholders and managing inter-organizational tensions, illustrating the dialectic nature of servitization within networked environments. Study four further reveals how agile development and lean startup practices enable firms to navigate these complexities, for example, by illustrating how these practices activate latent ecosystem structures and allow firms to adapt business models and technological infrastructures in response to inter-organizational dynamics. Agile co-creation practices benefit servitizing firms because inter-organizational structures are often complex and unobservable (Henneberg et al., 2006; Vink et al., 2021). These characteristics underscore the importance of an iterative and flexible strategy in understanding and responding to the unpredictable dynamics of inter-organizational constellations (Mouzas et al., 2008). As such, this dissertation illustrates how agile development and lean startup practices facilitate servitization and strategizing in a complex and unpredictable networked context.

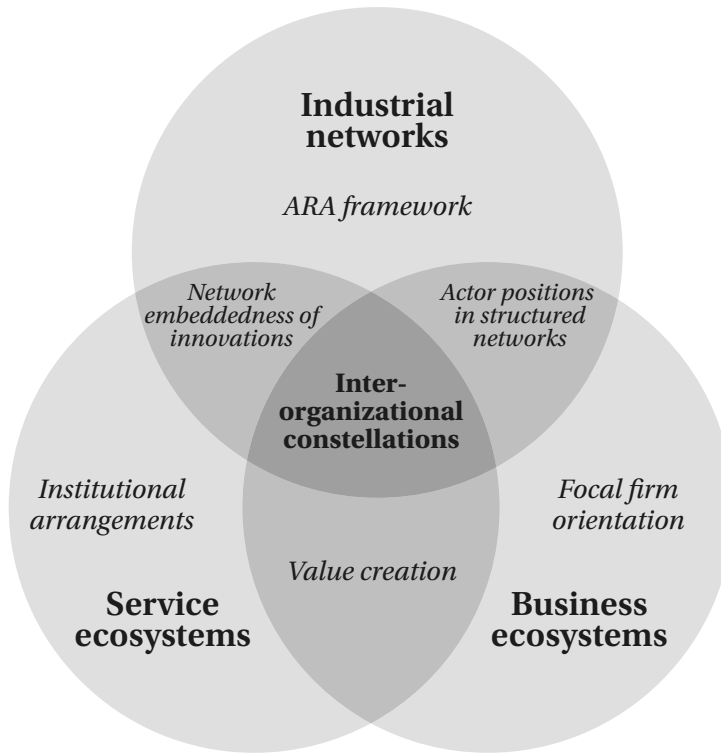
On a more general note, this dissertation highlights that agile co-creation in servitization simultaneously acts as a learning and transformation mechanism. While existing servitization literature predominantly concentrates on the impact of agile co-creation on experiential learning and rapid adaptation to new insights (Linde et al., 2023; Thomson et al., 2023), this dissertation extends this view by demonstrating its transformative effects. Particularly, study four illuminates how stakeholder involvement in the agile co-creation process fosters learning and acts as a catalyst for reflexivity and transformation within the context (Vink & Koskela-Huotari, 2022). In line with Håkansson and Ford (2002), this dynamic interplay illustrates that firms are influenced and influence their external environment. For example, directly co-creating digital services facilitates learning about customer needs but simultaneously generates

new challenges and customer demands (Biggemann et al., 2013). The potential transformative effect of the experimentation process contributes to the literature about the role of agile co-creation in ecosystem alignment (Sjödin et al., 2020), and sheds further light on the decision-making dynamics in servitization (Cui et al., 2019).

This dissertation argues that servitizing firms can benefit from agile co-creation. By providing further empirical evidence, this dissertation deepens our understanding of the processual and structural reconfigurations associated with servitization (Favoretto et al., 2022; Huikkola, Kohtamäki, & Ylimäki, 2022). Together, the four studies indicate that servitization involves organizational changes that are mutually reinforcing. For example, Tronvoll et al. 2020) highlighted that servitization involves shifting from planning to discovery, dematerialization, and hierarchy to partnership. However, studies one and two indicate that as firms move from hierarchy to partnership, they experience more volatility and uncertainty, further increasing the importance of a discovery approach to servitization. Despite the benefits of agile co-creation for servitization, the need for agile innovation approaches may further complicate servitization and increase the risk of experiencing disappointing financial returns. Often, manufacturers are used to adopting plan-based innovation approaches (Huikkola, Kohtamäki, Rabetino, et al., 2022), and adopting agile innovation approaches may be an additional source of resistance to servitization. This indicates that while agile co-creation has substantial benefits, implementing its practices may be associated with organizational challenges. Firms should take this into account when deciding to implement a servitization strategy.

### **6.2.3 A multi-perspective approach for studying servitization in inter-organization constellations**

This dissertation aimed to understand servitization in inter-organizational constellations better. So far, the servitization literature has adopted various network and ecosystem perspectives with overlapping and distinct characteristics to address this issue (Möller et al., 2020; Möller & Halinen, 2017; J. Wang et al., 2023). Because the adoption of a single perspective risks limited framing and sensemaking of a research phenomenon (Ojansivu et al., 2022), this dissertation developed and implemented a multi-perspective approach (Figure 6.1) to study servitization in inter-organizational constellations. Specifically, this dissertation adopted IMP's industrial network approach, SDL's service ecosystem perspective, and the business ecosystem lens from the strategy literature. This section reflects on the implementation of each lens in the various studies and the multi-perspective approach for studying servitization in inter-organizational constellations.



**Figure 6.1** Multi-perspective approach for studying servitization in inter-organizational constellations

The first study illustrated the role of IMP's industrial network approach in understanding the network-embedded nature of strategic decision-making. Utilizing the ARA framework (Håkansson & Snehota, 1995), this approach proved instrumental in analyzing how servitization strategies reshape existing digital and physical resource interfaces and activity structures and impact the status quo (Cenamor et al., 2017; Chakkol et al., 2014). This is exemplified by WinCo's strategy to implement reverse material streams, a move potentially causing significant operational disruptions for one of its suppliers, resulting in tensions. Such instances underscore the necessity for managers, who may not fully grasp their business environment (Abrahamsen et al., 2016; Mouzas et al., 2008), to develop a comprehensive understanding of their firm's activity patterns, resource constellations, and actor networks. The ARA framework can facilitate such network insight and crafting strategies that align with their counterparts' objectives and preemptively address potential conflicts (Kolagar et al., 2022; Parida & Jovanovic, 2021).

Besides its merits, the industrial network approach also has some limitations. First, it focuses on long-term, stable business relationships formed through ongoing interactions. For example, the industrial network approach assumes that resources become by interacting with other resources in interactive business relationships (Bocconcelli et al., 2020). This focus on existing business relationships, however,

renders it somewhat inadequate for exploring the dynamics of novel collaborations, which are increasingly prevalent in servitization contexts involving new partner types like technology firms and software developers (Favoretto et al., 2022; Makkonen et al., 2022). Second, although the IMP framework excels in analyzing resource and activity roles in business relations (Baraldi et al., 2012; Butler & Purchase, 2021), it offers a limited perspective on the broader spectrum of social structures. While acknowledging actor bonds, the ARA framework tends to confine its analysis to specific relationships or networks (Håkansson et al., 2009). As an actor's behavior and perceptions are shaped by various social structures (Koskela-Huotari & Vargo, 2016), the ARA framework risks overseeing the whole of social structures influencing an actor's behavior. Third, the IMP approach attributes relatively limited agency to individual actors, seen primarily as products of their environments (Håkansson & Ford, 2002). Therefore, this perspective provides limited theoretical tools to explore managerial strategies in navigating the complexities of inter-organizational environments.

In exploring the role of institutional complexity in servitization, the second study highlights the added value of the service ecosystem perspective. By emphasizing institutional arrangements as fundamental governance mechanisms in inter-organizational constellations (Vargo & Lusch, 2016), this perspective places servitization in an interconnected web of routinized beliefs and behaviors. These beliefs and behaviors are shaped by written and unwritten norms and rules, the cognitive, normative, and legal dimensions of institutional arrangements (Edvardsson et al., 2014). Focusing on institutional arrangements helps us see how various cultural and regulative prescriptions within existing and novel business relations affect servitization. For instance, in the MetalNet case, the actions and reactions of involved ecosystem actors to decisions related to implementing a servitization strategy were influenced by various cognitive and normative institutional arrangements.

Furthermore, these institutional arrangements operated at both firm and individual levels, illustrating the nested and multi-layered nature of service ecosystems (Koskela-Huotari & Vargo, 2016; Sitaloppi et al., 2016). This multi-level approach aligns well with a similar trend in the servitization literature (Struyf et al., 2021), enabling a more comprehensive understanding of servitization and its associated challenges. Specifically, the ecosystem perspective provides an analytical lens to study the interconnected routines and beliefs influenced by industry best practices, functional backgrounds, and personal beliefs. Such a perspective provides a rich analytical framework to examine how diverse institutional factors collectively shape servitization processes in inter-organizational constellations.

While the service ecosystem perspective offers a valuable theoretical lens for examining servitization in inter-organizational constellations, its comprehensive nature can sometimes lead to losing focus. Due to the wide variety of potential institutions, the researcher may become overwhelmed during the analysis as 'everything becomes important.' This issue is partially rooted in the lack of mid-range theories and analytical tools (Bocconcelli et al., 2020). To mitigate this issue, integrating mid-range theories can help focus attention on the most relevant institutional arrangements for a given context.

In industrial marketing and servitization, IMP's ARA framework may serve as an ideal complement to the service ecosystem perspective. By viewing activity links, resource ties, and actor bonds as manifestations and enactments of underlying social structures, the ARA framework can effectively guide the analysis of these institutional arrangements. For example, the ARA framework could uncover the interdependence between two actors regarding their digital interfaces (i.e., resources) and data-sharing activities. The service ecosystem perspective complements this analysis by revealing the institutional arrangements that underly the (re)enactment of these structures. In the context of the previous example, the interdependencies may be rooted in regulatory and normative frameworks around cyber security and data privacy. These institutional arrangements are essential forces that firms must consider when reconfiguring existing resource and activity structures. The combined application of the ARA framework and the service ecosystem perspectives leverages their complementarity in terms of zooming in on the empirical world and out to the abstract theoretical world (Bocconcelli et al., 2020).

The fourth study of this dissertation demonstrated the added value of a business ecosystem perspective by studying the role of agile innovation approaches in ecosystem strategizing. While the industrial network and service ecosystem perspectives focus on collaboration, a business ecosystem perspective provides more analytical space for a firm's self-interest. This is exemplified in the conceptualization of an ecosystem strategy, defined as "the way in which a focal firm approaches the alignment of partners and secures its role in a competitive ecosystem" (Adner, 2017, p. 47). Whereas the industrial network and service ecosystem perspectives focus on an outside-in perspective of strategy, that is, structure shapes actors (Chandler et al., 2019; Håkansson & Ford, 2002), the business ecosystem perspective enables studying an inside-out perspective. An inside-out perspective refers to the active role of individual firms in adapting to and shaping their external environment, which was embodied by the dynamic capabilities lens in the fourth study. Dynamic capabilities represent an inside-out perspective on inter-organizational constellation by emphasizing the capabilities of individual firms to adapt and transform their external environment (Foss et al., 2023; Teece, 2007). This lens proved helpful in studying managerial practices and approaches to deal with the networked aspects of strategy formulation and implementation (Kamalaldin et al., 2021). In this dissertation, it provided novel insights into how agile innovation approaches enable servitizing firms to deal with inter-organizational complexities and dynamics.

However, despite these insights, the business ecosystem perspective has limitations, particularly in treating structural and temporal dynamics. While it provides a framework for understanding firm-centric strategies within ecosystems, business ecosystems have a relatively static view of activities and resource interfaces (Adner, 2017; Jacobides et al., 2018). Particularly when comparing this perspective to the industrial network and the service ecosystem perspectives, which provide comprehensive analytical tools like the ARA framework, episodic value, and multiple institutional pillars to study structure and time in inter-organizational constellations (Ford, 2011; Håkansson et al., 2009; Vargo & Lusch, 2016). As such, both perspectives complement the business ecosystem lens by providing a more dynamic view and emphasizing the role of resource and social

structures. An integrated view enables a more comprehensive strategy formulation that accounts for past interactions and recognizes the influence of social structures across organizational layers, ensuring ecosystem strategies are adaptive and encompass the broader inter-organizational context.

This dissertation brings forth a multi-perspective for studying servitization in inter-organizational constellations based on the complementarities between the industrial network approach, service ecosystems, and business ecosystems. By approaching inter-organizational constellations from multiple angles, this model mitigates the risk of limited framing and sensemaking of a research phenomenon associated with adopting a single research lens (Ojansivu et al., 2022). In this dissertation, the industrial network approach provided the ARA framework for studying inter-organizational relations and interactions in its multi-dimensionality. In a complementary fashion, service ecosystems provided further depth to this understanding by emphasizing the role of the underlying social structures through institutional arrangements. Finally, business ecosystems added a managerially relevant understanding by turning our attention to the role of the focal firm to navigate in this interconnected context. The multi-perspective approach provided a comprehensive and nuanced understanding of the challenges associated with servitization in inter-organization constellations and valuable insights about mitigation strategies. Hence, it is in line with the pragmatist approach of this dissertation. Overall, the multi-perspective model contributes to the intellectual pluralism of the servitization literature, which is necessary to prevent a limited and one-sided understanding of reality (Cornelissen et al., 2021).

## **6.3 Practical implications**

This dissertation investigated servitization within inter-organizational constellations, particularly examining the role and impact of agile innovation approaches. The findings from the four studies presented in this work collectively reveal that servitization represents a complex and unpredictable journey characterized by considerable interdependencies and tensions at the network level. These insights underscore the critical need for flexible approaches and inter-organizational collaboration when implementing a servitization strategy. The subsequent subchapters will unpack these findings in greater detail, elucidating the practical implications and offering actionable guidance for organizations navigating the challenges and opportunities of servitization in today's rapidly evolving business context.

### **6.3.1 Servitization as a network-embedded transformation**

Managers must recognize that they operate in a networked context and are embedded in multi-dimensional business relationships of interdependent processes, resource interfaces, and social arrangements. Because firms are embedded in these networks, the impact of servitization extends beyond the internal dynamics of their organizations and influences a broader network of external actors, including customers, suppliers, intermediaries, and complementors. For instance, intermediaries might need to enhance their service delivery capabilities to effectively support a regional or global

service network. Conversely, suppliers may be required to elevate their service levels or participate in reverse logistics to complete the service value proposition. In other words, servitization may require reconfiguration of inter-organizational structures and relationships. However, different network actors may have varying goals, perceptions, and expectations. Therefore, managers should be prepared for inter-organizational tensions that might arise. It is essential for firms to deeply understand the underlying motivations and perceptions that could cause resistance to the new service business model. These could range from existing investments that limit adaptability to the new service business model to conflicts in social structures, such as when the new service model challenges the professional identity of an external stakeholder.

As this dissertation reveals, inter-organizational tensions should not only be viewed as challenges to manage but also as catalysts for transformation. Inter-organizational tensions prompt reflexivity among involved actors, encouraging them to critically examine and understand the structures driving their resistance. Consequently, tensions illuminate contradicting network structures and social arrangements, which are usually invisible and enacted unconsciously and facilitate a platform for constructive dialogue and negotiation. Such discussions can lead to reconfiguring existing network structures or adapting the proposed service business model to align it more with existing network structures.

The dissertation underscores the dialectical nature of servitization by highlighting the role of tensions and subsequent transformation through negotiation. Dialectics refers to the driving force behind negotiations and business model adjustments arising from contradictions between the new service model and the existing product-focused context. Inter-organizational tensions, therefore, offer valuable insights into these contradictions and can be the impetus for network reconfiguration. Consequently, managers should acknowledge and embrace these tensions as opportunities for transformation, harnessing their potential to facilitate alignment and adaptation in the servitization journey.

### **6.3.2 An agile innovation approach to servitization**

In addressing the inherent uncertainties and emergent characteristics of servitization, this dissertation recommends that managers adopt an agile innovation approach. This approach involves practices from agile development and lean startup and parallels the principles of abductive research. Abduction is characterized by a dynamic interplay between theory development and theory testing. As such, this process relies on the knowledge manufacturers often possess about their customers, physical products, and supply chains. This knowledge forms a critical foundation for designing effective service business models. However, servitization also frequently propels manufacturers into new, uncharted territories. For example, firms may be inexperienced in digital technology, especially in applications like predictive analytics or developing advanced service contracts tied to product performance. To bridge this gap, firms benefit from integrating their existing knowledge with novel insights. This can be achieved by directly interacting with external stakeholders, including customers, suppliers, and technology

partners. Such interactions can yield essential information about customer needs, technological capabilities, and potential collaborative opportunities, enriching the servitization process by integrating existing knowledge and novel insights.

As firms embark on the servitization journey, they are advised to integrate their knowledge into a preliminary service business model. Tools such as the Business Model Canvas, Value Proposition Canvas, or the Ecosystem Pie Model can be valuable (Osterwalder & Pigneur, 2010; Talmar et al., 2018). These tools provide a systematic framework for ideating crucial elements of a novel service business model, including customer needs, value propositions, technological infrastructure, and network structure. However, managers should recognize that the outcomes of this ideation process are essentially a set of untested assumptions. These plausible assumptions should not be mistaken for validated truths despite being informed by existing knowledge. Firms must critically evaluate the underlying assumptions of their business models to assess the validity of their beliefs. This critical attitude ensures servitizing firms remain agile and open to adapting their service business models in light of new insights and developments. The primary purpose of this initial business model is not to serve as a definitive solution but rather to guide subsequent phases of experimentation and validation in the servitization journey.

Firms are encouraged to align their preliminary service business model with the real-world context through iterative experimentation. This approach, underscored by the findings in this dissertation, highlights the benefits of an iterative and experimental approach in facilitating value co-creation. Iterative experimentation allows managers to align their service business models with the needs of customers and other key stakeholders. By conducting targeted experiments, such as developing wireframes, prototypes, or Minimum Viable Products (MVPs), firms can gather concrete data to validate their assumptions and evaluate the feasibility of their proposed service model. Particularly effective is the integration of these experiments with a sprint-based methodology. This rapid, rhythmic approach to development enables firms to adapt their service models in response to emerging insights quickly. For example, a sprint-based approach allows firms to swiftly adjust their technological solution in response to novel insights regarding customer needs. Consistently testing, iterating, and sharing the results of these experiments not only refines the service business model but also promotes broader acceptance and adoption of the servitization strategy among key internal and external stakeholders.

Besides focusing on customer needs, servitization also requires firms to carefully consider and test the goals and expectations of other external stakeholders. This dissertation stresses the importance of this broader perspective by illustrating that servitization often requires network-level reconfigurations. In this context, network actors like suppliers, intermediaries, or complementors should be approached similarly to customers. For example, firms should develop and validate value propositions specifically targeting these actors to involve and align them with an envisioned servitization strategy. So, servitizing firms should consider the benefits and costs for external actors like suppliers, intermediaries, and complementors involved with adopting and aligning to an envisioned service business model. Like in customer

segmentation, some existing partners might not align with the servitization strategy compared to a traditional product-oriented approach. Therefore, evaluating whether to adjust the value propositions for these current partners or to seek new partnerships better suited to the servitization objectives is essential. This strategic assessment is critical to effectively integrating external actors into servitization.

### **6.3.3 Setting up for agile innovation approach to servitization**

Before firms adopt an agile innovation approach to servitization, they must lay a strategic foundation. Key to this foundation is the development of a long-term vision, or 'north star,' which guides the firm through the inherent uncertainties and complexities of the servitization journey. Such a vision must be specific enough to provide clear direction for servitization-related actions, decisions, and priorities yet broad enough to allow flexibility in adapting to new insights and developments. Take SpaceX as an illustrative example: its overarching goal to make life multi-planetary provides specific guidelines for designing rockets while providing flexibility to adjust its business model, such as launching Starlink to fund Mars mission research. In servitization, a firm's grand vision typically centers on the value-in-use for customers—what the customer aims to achieve by using the focal firm's service offerings. The grand vision should align with the firm's overall objectives and secure executive management support. Without this alignment, servitization initiatives risk being overshadowed and dominated by day-to-day business operations, potentially leading to a misallocation of resources and a lack of internal legitimacy.

In addition to developing a grand vision for servitization, firms are advised to establish an organizational structure in line with agile development principles to facilitate the transformation to service-led business models. The findings of this dissertation advocate for creating a dedicated and diverse team that operates independently from the firm's core business. Depending on the available resources, this could take the form of a distinct business unit or an independent task force. The primary reason for this separation is to foster an environment that facilitates learning and handling the high levels of uncertainty encountered in servitization. Shielding the servitization initiative from traditional business operations is vital, as servitization often involves learning by doing, which includes making and learning from mistakes. Without such autonomy, there's a risk that the traditional business mindset might stifle innovation or prematurely abandon the servitization strategy due to skepticism. Furthermore, this independent team should encompass a comprehensive range of skills and perspectives to function effectively. This includes expertise in engineering, field services, sales, purchasing, and executive management, which are critical for addressing the upcoming challenges. The diversity within this team not only aids in navigating the initial phases of servitization but also lays a solid foundation for the broader adoption of the service business model across the firm in later stages.

Finally, firms are encouraged to cultivate an appropriate cultural and governance context to support successful servitization efforts. This dissertation highlights the importance of an organizational culture prioritizing learning and knowledge sharing. Such a culture

is fundamental for navigating the challenges of servitization and fostering a learning-by-doing mentality. Without this mentality, failures risk being viewed as setbacks that feed skepticism, rather than opportunities for growth and improvement. Additionally, the complexity of servitization, which impacts various organizational aspects such as sales, operations, innovation, and purchasing, necessitates a robust knowledge-sharing environment. Overcoming organizational silos and effective integration of knowledge across these domains is vital to the successful implementation of servitization strategies.

Furthermore, executive management needs to adopt governance principles oriented towards long-term success. These should include metrics that evaluate learning processes and the value-in-use experienced by customers rather than solely focusing on traditional financial metrics. Questions such as “How effectively is the team learning?” and “How well does the service business model align with customer expectations and experiences?” should guide the assessment of servitization initiatives. Metrics centered around these questions enable executive management to monitor and navigate the servitization process effectively, ensuring that a short-term orientation and pressures from the daily business do not hinder the pursuit of innovation and experiential learning.

#### **6.3.4 Inter-organizational collaboration for servitization**

This dissertation underscores the importance of engaging in collaborations beyond a firm’s traditional network when developing and implementing a servitization strategy. For example, as firms develop digitally-enhanced advanced services, they may encounter the need for capabilities not currently present in their supply chain. This necessitates collaborating with new partners specializing in technology or software development, areas often outside the firm’s established domain. Managers, therefore, should not limit their collaborative efforts to existing partners. Instead, they are advised to proactively seek out and establish relationships with novel partners from diverse industries. Expanding the collaboration network is crucial for acquiring the unique capabilities and insights needed to navigate the servitization journey successfully.

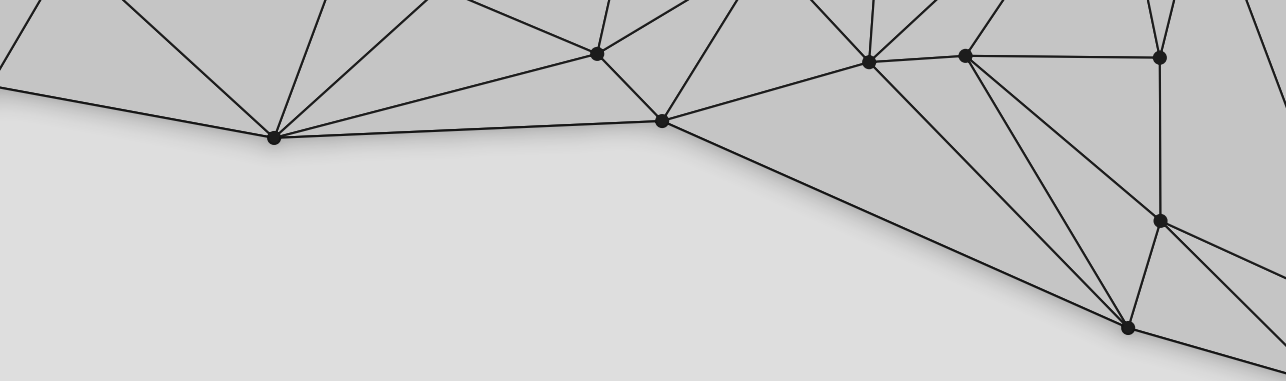
When servitizing firms collaborate with novel partners, they must navigate potential tensions arising from diverging social structures. Even though the substance of existing business relationships may not complicate novel partnerships, they may be complicated by institutional complexity. Partners from varied organization types, such as governmental bodies or industries like software development, often bring distinct expectations and routines to the collaboration. This diversity can lead to misalignments in how these partners perceive collaborative efforts, potentially resulting in tensions. These institutional differences are often hard to recognize as they subconsciously guide behavior and decision-making. Therefore, managers should be aware of the multi-layered reality of organizational behavior and decision-making. Specifically, firms should consider emerging dynamics and tensions regarding cognitive, normative, and legal drivers. In other words, firms should consider the shared beliefs and perceptions, societal norms and values, and formal rules and laws that shape the behavior and thinking of individual actors and organizations.

Besides being aware of these institutional differences, firms can address these challenges through structuring practices that mitigate their impact. One practical approach is to group actors with similar institutional backgrounds, creating committees or teams focused on specific areas like IT or manufacturing. Such clustering fosters a shared language and understanding, streamlining collaboration and easing tensions caused by divergent institutional norms. Additionally, firms can introduce regular progress reporting and accountability mechanisms, such as weekly planning sessions and monthly progress reviews. These practices ensure alignment with shared goals and exert normative pressure on all actors to deliver tangible results despite the diverse backgrounds of the involved actors. Alternatively, directly addressing institutional differences through open discussions about varying perceptions and goals can foster constructive dialogue and reduce disparities. Finally, the COVID-19 pandemic has underscored the value of physical interaction in inter-organizational collaboration. Where possible, facilitating regular face-to-face meetings can significantly enhance reconciling institutional differences and strengthening collaborative relationships among diverse partners.

### **6.3.5 Synthesis of practical implications**

This dissertation underscores the multifaceted and network-embedded nature of servitization and emphasizes the necessity of sprint-based and experimental agile innovation approaches in navigating its complexities. The research reveals that servitization is not just an internal organizational transformation but extends to reshaping network-level interactions demanding flexibility and extensive collaboration with traditional and novel partners. Managers must acknowledge and adequately manage the interdependencies and inter-organizational tensions when implementing a servitization strategy, particularly aligning different organizational structures, expectations, and goals. Establishing a clear long-term vision, coupled with an organizational structure that promotes autonomy and learning and integrates diverse capabilities, is crucial. Also, the dissertation advocates for an innovation culture that fosters learning and knowledge sharing, supported by governance principles that prioritize long-term value and continuous learning over short-term financial metrics. Finally, it highlights the importance of embracing diverse collaborations outside traditional networks, addressing potential institutional complexities through structuring and reconciliation practices, fostering open dialogue, and mitigating tensions. Overall, the dissertation provides a comprehensive guide for organizations to effectively implement and navigate the servitization journey, emphasizing strategic adaptability, collaborative innovation, and institutional alignment.





# 7

## CONCLUSION AND FUTURE RESEARCH

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## 7. Conclusion and future research

This dissertation adopted a multi-perspective approach to explore servitization within inter-organizational constellations. The findings illuminate the network-embedded and complex nature of servitization. As networked contexts are constantly in flux, servitization in inter-organizational constellations is a dynamic interplay of actions and reactions. Especially when firms aim to involve and align external actors with their envisioned servitization strategy, they face a variety of responses and resistance from counterparts, both anticipated and unexpected. This underscores the emergent and unpredictable nature of servitization in such constellations.

Despite the emergent nature of servitization and the inability of firms to control their network relationships, they possess the agency to embark on novel and potentially profitable strategies like servitization. This dissertation highlights the critical role of agile innovation approaches in navigating the unpredictability inherent in inter-organizational settings. Such approaches enable firms to actively engage key stakeholders in the transformation process and offer the flexibility to adapt to unforeseen dynamics in the external environment. While firms may not have control over their external environment, agile innovation approaches enable them to control their response to the uncertainty and dynamism that characterizes these contexts.

Even though the novel insights into servitization within inter-organizational constellations and the role of agile innovation approaches, this dissertation also presents limitations that open avenues for further research. First, an important area not extensively explored in this dissertation is the impact of inter-organizational tensions arising during servitization on the existing product business. Considering that many firms integrate servitization as an extension of their traditional product business (Baines et al., 2020; Vendrell-Herrero et al., 2014), it is imperative to understand how these tensions might adversely affect the product side of the business. Such negative spillover effects warrant investigation, as existing network constellations play a crucial role in the traditional product business of servitizing firms. Future research should delve into how tensions emerging in the servitization process impact the conventional product business and explore ways to mitigate any detrimental effects. This line of inquiry provides a more nuanced understanding of tensions in servitization and their implications for the risks and financial performance at the firm level (Benedettini et al., 2015; E. Fang et al., 2008).

Second, the exploration of agile innovation approaches in this dissertation was conducted through cross-sectional research designs. While this provided valuable insights about the benefits and implications of these approaches in servitization, it offered a limited perspective on its processual aspects. Notably, many manufacturing firms still predominantly employ plan-based approaches (Huikkola, Kohtamäki, & Ylimäki, 2022; Kowalkowski & Ulaga, 2017), and implementing agile innovation approaches requires significant changes. Therefore, future research should delve into how manufacturing firms undertake agile transformations, focusing on implementing agile and lean startup practices before and during the service transformation.

Furthermore, this dissertation identified distinct roles for agile development and lean startup practices in servitization. To deepen our understanding of these roles and their interplay during the servitization journey, future studies should investigate these practices across various stages of servitization. Examining agile innovation approaches in combination with established servitization process models (e.g., Baines et al., 2020; Lütjen et al., 2017) could illuminate their applicability and benefits throughout the servitization journey, offering further insights into the effective integration of agile methodologies in servitization.

Third, this dissertation focused on agile development and lean startup practices in investigating agile innovation approaches. However, the spectrum of innovation approaches is broader and includes approaches that potentially complement these two. For instance, design thinking is frequently considered a valuable addition to agile development and lean startup (Dobrigkeit & de Paula, 2017; Lichtenthaler, 2020). Future research should, therefore, broaden the scope of agile innovation approaches, enriching the conceptual toolkit available to firms when implementing a servitization strategy. Additionally, exploring the integration of seemingly contradicting innovation approaches to harness their relative strengths is essential. An emerging trend in the literature is the integrated agile-stage-gate approach, which combines the benefits of control and certainty with flexibility, customer-centricity, and speed (Bianchi et al., 2020; Cooper & Sommer, 2016a). Such integrated innovation approaches may benefit firms in managing the multifaceted aspects of product, service, and software development in servitization. Investigating such integrative approaches could yield more profound insights into effectively balancing diverse innovation strategies in servitization.

Fourth, while this dissertation has shed light on the role of agile innovation approaches in ecosystem alignment at the firm level, it has not explored these approaches extensively at the network or ecosystem level. In line with Sjödin et al. (2022), our findings suggest that involving external actors in the agile innovation process can benefit the alignment of external actors and servitization outcomes. Hence, future research should focus on the implementation and impact of agile innovation approaches within inter-organizational collaborations. Also, future research should investigate the challenges associated with structuring inter-organizational partnerships in line with agile principles. For example, agile development mainly focuses on the effectiveness and efficiency of autonomous teams, while inter-organization collaborations involve aligning collective and individual goals and inter-team coordination (Kolagar et al., 2022). Therefore, investigating scaled agile development approaches designed for large enterprises to manage complex structures and interdependent teams (Alqudah & Razali, 2016; Paasivaara, 2017), could provide valuable insights. This approach could be particularly beneficial in understanding how to effectively apply agile practices in extensive inter-organizational collaborations to develop novel capabilities for service offerings.





## ABOUT THE AUTHOR

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## About the author

Xander Stegehuis was born on July 10th, 1994, in Enschede, the Netherlands. He embarked on his academic career at the University of Twente, where he earned a bachelor's degree in International Business Administration, followed by a Master's degree in Business Administration. Furthering his education, Stegehuis obtained another Master's degree in Innovation Management, Entrepreneurship, and Sustainability from the Technical University of Berlin.

In September 2018, Xander completed his Master thesis, a project conducted in collaboration with the Fraunhofer Project Center at the UT. The thesis focused on ecosystem transformation in Industry 4.0, marking a key step in his academic path. This work led to his PhD research in the field of digital servitization, funded by the Province of Limburg, the Netherlands. His research concentrated on inter-organizational collaborations, ecosystems, and the role of agile and experiment-driven innovation approaches.

Throughout his PhD studies, Xander was associated with several research groups. He was part of the Business Service Innovation (BSI) group at Fontys University of Applied Science in Venlo, the Industrial Engineering and Business Information Systems (IEBIS) at the University of Twente, and the Entrepreneurship and Technology Management (ETM) research group at the same university. Furthermore, from 2022 to 2023, Xander expanded his research experience as a visiting researcher with the Cambridge Service Alliance at the University of Cambridge. Throughout his research, he put much emphasis on the practical relevance and applications of his research by collaborating closely with companies to investigate servitization-related concepts and ideas.

Xander has presented his work at various industrial marketing and services conferences and his working paper entitled "Strategizing of individual actors in strategic networks" was considered a conference best paper at the Industrial Marketing and Purchasing 2021 conference. Furthermore, his paper entitled "Inter-organizational tensions in servitization: A dialectic process model" has been published in the Industrial Marketing Management journal. His academic journey also included a short episode as a lecturer and coach at the ETM research group, where he shared his knowledge and experience with students.

In 2024, Xander transitioned into the role of a strategy consultant at Berenschot, where he will continue his passion to help companies with complex topics and apply his expertise in strategic decision-making and servitization-related issues.



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## Summary

Manufacturing firms have increasingly recognized the importance of extending and capitalizing on service offerings in their offering portfolio. This awareness has led many firms to embark on a transformative shift to a service-oriented business model, often coined as servitization. Servitization entails the transformation of processes, capabilities, and offerings. It can take many forms including extending the importance of existing service offerings, adding new ones, or offering a physical product as a service (i.e., product-as-a-service). The main strategic imperatives driving manufacturers to engage in servitization are to leverage their product knowledge, installed base, tap in into a new revenue pool with higher margins, and to capitalize on their investments in digital technologies.

Despite its expected benefits, servitization is inherently complex and unpredictable, often leading to disappointing results. These results are partly due to the inter-organizational nature of servitization and the importance of collaboration among a diverse set of actors. Heterogeneity among actors in terms of goals, priorities, and organizational cultures may result in tensions that shape the servitization journey in unpredictable ways. To deal with such complexity and unpredictability, servitizing firms increasingly adopt agile innovation approaches based on incremental service investments, sprint-based working, and learning by doing. However, agile innovation approaches have only recently emerged in the servitization domain and requires further research. To extend our knowledge about how to successfully implement a servitization strategy, this dissertation integrates the narratives on inter-organizational constellations and agile innovation approaches through four studies.

The first study presents a process model to understand servitization in inter-organizational constellations by combining the ARA framework from IMP tradition with dialectic theory. This dialectic process model emphasizes the role of pre-existing network structures and their contradictions with the envisioned service strategy in the emergent nature of servitization. This dialectic process model is illustrated in a case study of WinCo, a construction industry manufacturer, that implemented a new service model called Façade-as-a-Service (FaaS). The case study revealed that introducing FaaS led to tensions due to contradictions between the envisioned servitization strategy and existing network structures. These tensions were not just challenges; they also played a crucial role in the evolution of the servitized offering, demonstrating how inter-organizational tensions can be a driving force in the emergence and development of servitization strategies.

The second study adopts a service ecosystem perspective to investigate institutional complexity in servitization. Institutional complexity refers to conflicting institutional arrangements—cognitive, normative, and regulatory frameworks—among actors in value co-creation. As institutional complexity manifests through different and conflicting perceptions, goals, and prescriptions for behavior, they can lead to tensions in inter-organizational collaborations. A longitudinal case study of an inter-organizational collaboration, MetalNet, which aimed to develop a Spare parts-as-a-Service offering,

exemplifies this. Conducted between September 2019 and July 2023, the study observed tensions stemming from institutional complexity between MetalNet partners. Importantly, the case study also identified strategies used by MetalNet participants to address and alleviate these tensions. The findings offer new insights into managing servitization amidst institutional complexity, especially useful for collaborations across diverse industrial or organizational contexts.

The third study examines the role of agile co-creation in digital servitization, focusing on two distinct methodologies: agile development and lean startup. Agile development, originating from IT, is a project management approach that encourages diverse, autonomous teams and sprint-based working. Additionally, the lean startup method emphasizes experiential learning through iterative experimentation, and validating business models for exploring new opportunities. Employing Partial Least Squares Structural Equation Modeling (PLS-SEM) with data from 114 respondents, this study assesses the impact of these approaches on service business performance. The findings reveal the unique contributions and synergistic effects of agile development and lean startup in the context of servitization.

The fourth study adopts a business ecosystem perspective to examine how agile innovation approaches benefit servitizing firms. Following David Teece's framework of sensing, seizing, and reconfiguring, this study investigates how agile innovation approaches facilitate servitizing firms to respond to and their internal and external environment. Grounded in 31 interviews with 24 respondents, this study clarifies the link between agile co-creation and dynamic capabilities by extending the research scope beyond seizing and illuminates the underlying mechanisms through which agile co-creation fosters dynamic capabilities. The study illuminates that agile practices not only bolster a firm's ability to seize opportunities, as existing literature suggests, but also increase its capacity to sense opportunities and reconfigure resources and existing ecosystem structures.

Altogether, this dissertation extends the servitization and industrial marketing literature by putting forth a multi-perspective approach to study inter-organizational constellations. This approach leverages the unique strengths of IMP's industrial network approach, Service-Dominant Logic's service ecosystem perspective, and the business ecosystem perspective from the strategic management literature. By combining these perspectives, this dissertation sheds further light on the network-embedded nature of servitization and reveals that this transformation unfolds in multiple iterations that are characterized by inter-organizational tensions. These tensions arise from the disruptive nature of servitization, often leading to resistance from both internal and external stakeholders. However, these challenges also present vital opportunities for transformation and innovation. As tensions emerge in an unpredictable fashion, servitizing firms benefit from agile innovation approaches. These approaches provide the tools to work through complexity in inter-organizational constellations and respond to tensions when they emerge.

# Samenvatting

Productiebedrijven hebben steeds meer het belang erkend van diensten in hun aanbodportfolio. Dit bewustzijn heeft veel bedrijven ertoe aangezet om een transformatieve verschuiving naar een op diensten gericht bedrijfsmodel te maken, vaak aangeduid als servitise (servitization). Servitise omvat de transformatie van processen, capaciteiten en aanbiedingen om zo een business model te ontwikkelen dat meer is gefocust op diensten. Bedrijven kunnen op verschillende manieren servitiseren, waaronder het uitbreiden van het bestaande dienstaanbod, het toevoegen van nieuwe diensten, of het aanbieden van een fysiek product als een dienst (i.e., product-as-a-service). De belangrijkste strategische drijvers van servitise bij maakbedrijven zijn het benutten van hun productkennis, installed base, het aanboren van een nieuwe inkomstenbron met hogere marges en het kapitaliseren op hun investeringen in digitale technologieën.

Ondanks de verwachte voordelen is servitise van nature complex en onvoorspelbaar, wat vaak leidt tot teleurstellende resultaten. Deze resultaten zijn deels te wijten aan de interorganisatiele aard van servitise en het belang van samenwerking tussen een diverse set van actoren. Heterogeniteit tussen actoren in termen van doelen, prioriteiten en organisatieculturen kan leiden tot spanningen die het servitiseringsproces op onvoorspelbare manieren vormgeven. Om met dergelijke complexiteit en onvoorspelbaarheid om te gaan, adopteren servitiserende bedrijven in toenemende mate agile innovatiebenaderingen gebaseerd op incrementele investeringen, werken in korte sprints en leren door te doen. Agile innovatiebenaderingen zijn echter pas recentelijk verschenen in het domein van servitise en vereisen verder onderzoek. Om onze kennis over hoe een servitise strategie succesvol te implementeren uit te breiden, integreert deze dissertatie concepten m.b.t. interorganisatiele constellaties en agile innovatiebenaderingen.

De dissertatie bestaat uit vier studies. De eerste studie presenteert een procesmodel om servitise in interorganisatiele constellaties te begrijpen door het ARA-raamwerk uit de IMP-traditie te combineren met dialectische theorie. Dit dialectische procesmodel benadrukt de rol van reeds bestaande netwerkstructuren en hun tegenstellingen met de beoogde dienststrategie in de opkomende aard van servitise. Dit dialectische procesmodel wordt geïllustreerd in een casestudy van WinCo, een fabrikant in de bouwsector, die een nieuw dienstmodel heeft geïmplementeerd genaamd Gevel-as-a-Service. De casestudy onthulde dat het introduceren van FaaS leidde tot spanningen vanwege tegenstellingen tussen de beoogde servitise strategie en bestaande netwerkstructuren. Deze spanningen waren niet alleen uitdagingen; ze speelden ook een cruciale rol in de evolutie van het geservitiseerde aanbod, waarbij werd aangetoond hoe interorganisatiele spanningen een drijvende kracht kunnen zijn in het ontstaan en de ontwikkeling van servitise strategieën.

De tweede studie neemt een service ecosysteem perspectief aan om institutionele complexiteit in servitise te onderzoeken. Institutionele complexiteit verwijst naar conflicterende institutionele regelingen - cognitieve, normatieve en regelgevende

kaders - tussen actoren in waarde cocreatie. Omdat institutionele complexiteit zich manifesteert door verschillende en conflicterende percepties, doelen en voorschriften voor gedrag, kunnen ze leiden tot spanningen in interorganisationale samenwerkingen. Een longitudinale casestudy van een interorganisationale samenwerking, MetalNet, die tot doel had een Reserveonderdelen-als-een-Dienst aanbod te ontwikkelen, illustreert dit. Uitgevoerd tussen september 2019 en juli 2023, observeerde de studie spanningen voortvloeiend uit institutionele complexiteit tussen MetalNet-partners. Belangrijk is dat de casestudy ook strategieën identificeerde die door MetalNet-deelnemers werden gebruikt om deze spanningen aan te pakken en te verlichten. De bevindingen bieden nieuwe inzichten in het beheren van servitatisatie te midden van institutionele complexiteit, vooral nuttig voor samenwerkingen in diverse industriële of organisatorische contexten.

De derde studie onderzoekt de rol van agile cocreatie in digitale servitatisatie, met de focus op twee verschillende methodologieën: agile development en lean startup. Agile development, afkomstig uit de IT wereld, is een projectmanagementbenadering die diverse, autonome teams en sprintgebaseerd werken voorschrijft. Daarnaast benadrukt de lean startup-methode iteratieve experimenten en het valideren van bedrijfsmodellen om nieuwe kansen te verkennen en te leren van eerstehands ervaringen. Met behulp van Partial Least Squares Structural Equation Modeling (PLS-SEM) met gegevens van 114 respondenten, beoordeelt deze studie de impact van deze benaderingen op de prestaties van service bedrijven. De bevindingen onthullen de unieke bijdragen en synergetische effecten van agile development en lean startup in de context van servitatisatie.

De vierde studie neemt een business ecosysteem aan om te onderzoeken hoe agile innovatiebenaderingen servitiserende bedrijven ten goede komen. Volgens het raamwerk van David Teece dat langdurig competitief voordeel verklaart door drie competenties, namelijk, waarnemen (sensing), benutten (seizing), en herconfigureren (reconfiguration), onderzoekt deze studie hoe agile innovatiebenaderingen servitiserende bedrijven helpen te reageren op en hun interne en externe omgeving. Gebaseerd op 31 interviews met 24 respondenten, verheldert deze studie de link tussen agile cocreatie en dynamische capaciteiten (dynamic capabilities) door het onderzoeksbereik uit te breiden voorbij de benutten dimensie en brengt de onderliggende mechanismen in kaart waardoor agile cocreatie dynamische capaciteiten bevordert. De studie brengt aan het licht dat agile praktijken niet alleen het vermogen van een bedrijf versterken om kansen te benutten, zoals bestaande literatuur suggereert, maar ook de capaciteit vergroten om kansen te herkennen en middelen en bestaande ecosysteemstructuren te herconfigureren.

Alles samen genomen breidt deze dissertatie de literatuur over servitatisatie en industriële marketing uit door een multi-perspectiefbenadering voor te stellen om interorganisationale constellaties te bestuderen. Deze benadering maakt gebruik van de unieke sterke punten van IMP's industriële netwerkbenadering, de service ecosysteem perspectie uit de Service-Dominant Logic en het business ecosysteem perspectief uit de strategische managementliteratuur. Door deze perspectieven te combineren, werpt deze dissertatie verder licht op de inter-organisationale aard van servitatisatie en onthult

dat deze transformatie zich ontvouwt in meerdere iteraties die worden gekenmerkt door interorganisationale spanningen. Deze spanningen ontstaan uit de disruptieve aard van servitisatie, wat vaak leidt tot weerstand van zowel interne als externe stakeholders. Echter, deze uitdagingen bieden ook essentiële kansen voor transformatie en innovatie. Omdat spanningen op onvoorspelbare wijze naar voren komen, profiteren servitiserende bedrijven van agile innovatiebenaderingen. Deze benaderingen bieden een conceptuele gereedschapskist om complexiteit in interorganisationale constellaties het hoofd te bieden en te reageren op spanningen wanneer deze zich voordoen.

This dissertation addresses the critical yet underexplored topic of servitization in inter-organizational constellations. Servitization entails the transformation of manufacturing firms from a product orientation to a service orientation and involves radical changes in terms of processes, capabilities, and offerings. Despite the recognized importance of external collaboration in this transformative journey, the dynamics, challenges, and effective management approaches remain underexplored. By investigating servitization in inter-organizational constellations from multiple theoretical angles, this dissertation brings forth a multi-perspective approach. This approach unveils how servitization unfolds across organizational boundaries and how agile innovation approaches can facilitate this complex and unpredictable transformation.

Key findings reveal the importance of an open and flexible managerial mindset, the necessity of embracing inter-organizational tensions stemming from diverging goals, perceptions, and expectations, and the role of agile innovation approaches. Agile methodologies provide a normative innovation framework that promotes the flexibility and open approach required for successful servitization. This work not only advances our theoretical understanding of servitization dynamics but also offers actionable insights for practitioners aiming to navigate the servitization journey effectively.

## Author

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